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Ras inhibitors

Abstract

The disclosure features macrocyclic compounds, and pharmaceutical compositions and protein complexes thereof, capable of inhibiting Ras proteins, and their uses in the treatment of cancers.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of priority to U.S. Application No. 62/930,355, filed on Nov. 4, 2019; U.S. Application No. 62/951,652, filed on Dec. 20, 2019; U.S. Application No. 63/000,357, filed on Mar. 26, 2020; U.S. Application No. 63/011,636, filed on Apr. 17, 2020; and U.S. Application No. 63/043,588, filed on Jun. 24, 2020, all of which are hereby incorporated by reference in their entirety.

BACKGROUND
(1) The vast majority of small molecule drugs act by binding a functionally important pocket on a target protein, thereby modulating the activity of that protein. For example, cholesterol-lowering drugs known as statins bind the enzyme active site of HMG-CoA reductase, thus preventing the enzyme from engaging with its substrates. The fact that many such drug/target interacting pairs are known may have misled some into believing that a small molecule modulator could be discovered for most, if not all, proteins provided a reasonable amount of time, effort, and resources. This is far from the case. Current estimates are that only about 10% of all human proteins are targetable by small molecules. Bojadzic and Buchwald, Curr Top Med Chem 18: 674-699 (2019). The other 90% are currently considered refractory or intractable toward above-mentioned small molecule drug discovery. Such targets are commonly referred to as “undruggable.” These undruggable targets include a vast and largely untapped reservoir of medically important human proteins. Thus, there exists a great deal of interest in discovering new molecular modalities capable of modulating the function of such undruggable targets.

(2) It has been well established in literature that Ras proteins (K-Ras, H-Ras and N-Ras) play an essential role in various human cancers and are therefore appropriate targets for anticancer therapy. Indeed, mutations in Ras proteins account for approximately 30% of all human cancers in the United States, many of which are fatal. Dysregulation of Ras proteins by activating mutations, overexpression or upstream activation is common in human tumors, and activating mutations in Ras are frequently found in human cancer. For example, activating mutations at codon 12 in Ras proteins function by inhibiting both GTPase-activating protein (GAP)—dependent and intrinsic hydrolysis rates of GTP, significantly skewing the population of Ras mutant proteins to the “on” (GTP-bound) state (Ras(ON)), leading to oncogenic MAPK signaling. Notably, Ras exhibits a picomolar affinity for GTP, enabling Ras to be activated even in the presence of low concentrations of this nucleotide. Mutations at codons 13 (e.g., G13D) and 61 (e.g., Q61K) of Ras are also responsible for oncogenic activity in some cancers.
(3) Despite extensive drug discovery efforts against Ras during the last several decades, a drug directly targeting Ras is still not approved. Additional efforts are needed to uncover additional medicines for cancers driven by the various Ras mutations.
SUMMARY
(4) Provided herein are Ras inhibitors. The approach described herein entails formation of a high affinity three-component complex, or conjugate, between a synthetic ligand and two intracellular proteins which do not interact under normal physiological conditions: the target protein of interest (e.g., Ras), and a widely expressed cytosolic chaperone (presenter protein) in the cell (e.g., cyclophilin A). More specifically, in some embodiments, the inhibitors of Ras described herein induce a new binding pocket in Ras by driving formation of a high affinity tri-complex, or conjugate, between the Ras protein and the widely expressed cytosolic chaperone, cyclophilin A (CYPA). Without being bound by theory, the inventors believe that one way the inhibitory effect on Ras is effected by compounds of the invention and the complexes, or conjugates, they form is by steric occlusion of the interaction site between Ras and downstream effector molecules, such as RAF and PI3K, which are required for propagating the oncogenic signal.
(5) As such, in some embodiments, the disclosure features a compound, or pharmaceutically acceptable salt thereof, of structural Formula I:
(6) ##STR00001## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 10-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, a haloacetal, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R.sup.1 is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally

substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl (e.g., methyl).

(7) Also provided are pharmaceutical compositions comprising a compound of Formula I, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable excipient.

(8) Further provided is a conjugate, or salt thereof, comprising the structure of Formula IV:

M-L-P Formula IV wherein L is a linker; P is a monovalent organic moiety; and M has the structure of Formula V:

(9) ##STR00002## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene,

optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-

membered heterocycloalkyl; R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl, or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is H or C.sub.1-C.sub.3 alkyl.
(10) Also provided is a method of treating cancer in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof.
(11) In some embodiments, a method is provided of treating a Ras protein-related disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof.
(12) Further provided is a method of inhibiting a Ras protein in a cell, the method comprising contacting the cell with an effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof.
(13) It is specifically contemplated that any limitation discussed with respect to one embodiment of the invention may apply to any other embodiment of the invention. Furthermore, any compound or composition of the invention may be used in any method of the invention, and any method of the invention may be used to produce or to utilize any compound or composition of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A and FIG. **1**B: These figures illustrate a matched pair analysis of potencies of certain compounds of the present invention (Formula BB) (points on the right) and corresponding compounds of Formula AA (points on the left) wherein a H is replaced with (S)Me in the context of two different cell-based assays. The y axes represent pERK EC50 (FIG. **1**A) or CTG IC50 (FIG. **1**B) as measured in an H358 cell line.
(2) FIG. **2**A and FIG. **2**B: A compound of the present invention, Compound A, drove deep regressions in vivo in a NSCLC (KRAS G12C) xenograft model. Some animals exhibited complete responses (CR)=3 consecutive tumor measurements ≤30 mm3. FIG. **2**A shows Compound A dosed at 100 mg/kg by daily oral gavage led to tumor regression in NCI-H358 KRASG12C xenograft model, which is a sensitive model to KRASG12C inhibition alone. The spaghetti titer plot (FIG. **2**B) displaying individual tumor growth is shown next to the tumor volume plot (FIG. **2**A).
(3) FIG. **3**A and FIG. **3**B: A compound of the present invention, Compound B, drove tumor xenograft regressions in combination with a MEK inhibitor, cobimetinib, in a NSCLC (KRAS G12C) model. FIG. **3**A shows the combination of intermittent intravenous administration of Compound B at 50 mg/kg plus daily oral administration of cobimetinib at 2.5 mg/kg drove tumor regression, whereas each single agent led to tumor growth inhibition. End of study responses were shown as waterfall plots (FIG. **3**B), which indicate 6 out 10 mice had tumor regression in the combination group, whereas no tumor regressions recorded in each single agent group.
(4) FIG. **4**A and FIG. **4**B: A compound of the present invention, Compound C, dosed weekly with daily SHP2 inhibitor, RMC-4550, drove xenograft regressions in a NSCLC (KRAS G12C) model. In FIG. **4**A, the combinatorial activity of once weekly intravenous administration of Compound C at 60 mg/kg plus daily oral administration of SHP2 inhibitor at 30 mg/kg is shown. End of study responses in individual tumors were plotted as a waterfall plot (FIG. **4**B).

(5) FIG. **5**: A compound of the present invention, Compound D, combined with a MEK inhibitor, trametinib, suppressed in vitro growth durably in a long-term cell growth NSCLC (KRAS G12C) model.

DEFINITIONS AND CHEMICAL TERMS

(6) In this application, unless otherwise clear from context, (i) the term "a" means "one or more"; (ii) the term "or" is used to mean "and/or" unless explicitly indicated to refer to alternatives only or the alternative are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and "and/or"; (iii) the terms "comprising" and "including" are understood to encompass itemized components or steps whether presented by themselves or together with one or more additional components or steps; and (iv) where ranges are provided, endpoints are included.

(7) As used herein, the term "about" is used to indicate that a value includes the standard deviation of error for the device or method being employed to determine the value. In certain embodiments, the term "about" refers to a range of values that fall within 25%, 20%, 19%, 18%, 17%, 16%, 15%, 14%, 13%, 12%, 11%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, or less in either direction (greater than or less than) of a stated value, unless otherwise stated or otherwise evident from the context (e.g., where such number would exceed 100% of a possible value).

(8) As used herein, the term "adjacent" in the context of describing adjacent atoms refers to bivalent atoms that are directly connected by a covalent bond.

(9) A "compound of the present invention" and similar terms as used herein, whether explicitly noted or not, refers to Ras inhibitors described herein, including compounds of Formula I and subformula thereof, and compounds of Table 1 and Table 2, as well as salts (e.g., pharmaceutically acceptable salts), solvates, hydrates, stereoisomers (including atropisomers), and tautomers thereof.

(10) The term "wild-type" refers to an entity having a structure or activity as found in nature in a "normal" (as contrasted with mutant, diseased, altered, etc) state or context. Those of ordinary skill in the art will appreciate that wild-type genes and polypeptides often exist in multiple different forms (e.g., alleles).

(11) Those skilled in the art will appreciate that certain compounds described herein can exist in one or more different isomeric (e.g., stereoisomers, geometric isomers, atropisomers, tautomers) or isotopic (e.g., in which one or more atoms has been substituted with a different isotope of the atom, such as hydrogen substituted for deuterium) forms. Unless otherwise indicated or clear from context, a depicted structure can be understood to represent any such isomeric or isotopic form, individually or in combination.

(12) Compounds described herein can be asymmetric (e.g., having one or more stereocenters). All stereoisomers, such as enantiomers and diastereomers, are intended unless otherwise indicated. Compounds of the present disclosure that contain asymmetrically substituted carbon atoms can be isolated in optically active or racemic forms. Methods on how to prepare optically active forms from optically active starting materials are known in the art, such as by resolution of racemic mixtures or by stereoselective synthesis. Many geometric isomers of olefins, C═N double bonds, and the like can also be present in the compounds described herein, and all such stable isomers are contemplated in the present disclosure. Cis and trans geometric isomers of the compounds of the present disclosure are described and may be isolated as a mixture of isomers or as separated isomeric forms.

(13) In some embodiments, one or more compounds depicted herein may exist in different tautomeric forms. As will be clear from context, unless explicitly excluded, references to such compounds encompass all such tautomeric forms. In some embodiments, tautomeric forms result from the swapping of a single bond with an adjacent double bond and the concomitant migration of a proton. In certain embodiments, a tautomeric form may be a prototropic tautomer, which is an isomeric protonation states having the same empirical formula and total charge as a reference form. Examples of moieties with prototropic tautomeric forms are ketone-enol pairs, amide-imidic acid pairs, lactam-lactim pairs, amide-imidic acid pairs, enamine-imine pairs, and annular forms where a proton can occupy two or more positions of a heterocyclic system, such as, 1H- and 3H-imidazole,

1H-, 2H- and 4H-1,2,4-triazole, 1H- and 2H-isoindole, and 1H- and 2H-pyrazole. In some embodiments, tautomeric forms can be in equilibrium or sterically locked into one form by appropriate substitution. In certain embodiments, tautomeric forms result from acetal interconversion.

(14) Unless otherwise stated, structures depicted herein are also meant to include compounds that differ only in the presence of one or more isotopically enriched atoms. Exemplary isotopes that can be incorporated into compounds of the present invention include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorus, sulfur, fluorine, chlorine, and iodine, such as .sup.2H, .sup.3H, .sup.11C, .sup.13C, .sup.14C, .sup.13N, .sup.15N, .sup.15O, .sup.17O, .sup.18O, .sup.32P, .sup.33P, .sup.35S, .sup.18F, .sup.36Cl, .sup.123I and .sup.125I. Isotopically-labeled compounds (e.g., those labeled with .sup.3H and .sup.14C) can be useful in compound or substrate tissue distribution assays. Tritiated (i.e., .sup.3H) and carbon-14 (i.e., .sup.14C) isotopes can be useful for their ease of preparation and detectability. Further, substitution with heavier isotopes such as deuterium (i.e., .sup.2H) may afford certain therapeutic advantages resulting from greater metabolic stability (e.g., increased in vivo half-life or reduced dosage requirements). In some embodiments, one or more hydrogen atoms are replaced by .sup.2H or .sup.3H, or one or more carbon atoms are replaced by .sup.13C- or .sup.14C-enriched carbon. Positron emitting isotopes such as .sup.15O, .sup.13N, .sup.11C, and .sup.18F are useful for positron emission tomography (PET) studies to examine substrate receptor occupancy. Preparations of isotopically labelled compounds are known to those of skill in the art. For example, isotopically labeled compounds can generally be prepared by following procedures analogous to those disclosed for compounds of the present invention described herein, by substituting an isotopically labeled reagent for a non-isotopically labeled reagent.

(15) As is known in the art, many chemical entities can adopt a variety of different solid forms such as, for example, amorphous forms or crystalline forms (e.g., polymorphs, hydrates, solvate). In some embodiments, compounds of the present invention may be utilized in any such form, including in any solid form. In some embodiments, compounds described or depicted herein may be provided or utilized in hydrate or solvate form.

(16) At various places in the present specification, substituents of compounds of the present disclosure are disclosed in groups or in ranges. It is specifically intended that the present disclosure include each and every individual subcombination of the members of such groups and ranges. For example, the term "C.sub.1-C.sub.6 alkyl" is specifically intended to individually disclose methyl, ethyl, C.sub.3 alkyl, C.sub.4 alkyl, C.sub.5 alkyl, and C alkyl. Furthermore, where a compound includes a plurality of positions at which substituents are disclosed in groups or in ranges, unless otherwise indicated, the present disclosure is intended to cover individual compounds and groups of compounds (e.g., genera and subgenera) containing each and every individual subcombination of members at each position.

(17) The term "optionally substituted X" (e.g., "optionally substituted alkyl") is intended to be equivalent to "X, wherein X is optionally substituted" (e.g., "alkyl, wherein said alkyl is optionally substituted"). It is not intended to mean that the feature "X" (e.g., alkyl) per se is optional. As described herein, certain compounds of interest may contain one or more "optionally substituted" moieties. In general, the term "substituted", whether preceded by the term "optionally" or not, means that one or more hydrogens of the designated moiety are replaced with a suitable substituent, e.g., any of the substituents or groups described herein. Unless otherwise indicated, an "optionally substituted" group may have a suitable substituent at each substitutable position of the group, and when more than one position in any given structure may be substituted with more than one substituent selected from a specified group, the substituent may be either the same or different at every position. For example, in the term "optionally substituted C.sub.1-C.sub.6 alkyl-C.sub.2-C.sub.9 heteroaryl," the alkyl portion, the heteroaryl portion, or both, may be optionally substituted. Combinations of substituents envisioned by the present disclosure are preferably those that result in the formation of stable or chemically feasible compounds. The term "stable", as used herein, refers

to compounds that are not substantially altered when subjected to conditions to allow for their production, detection, and, in certain embodiments, their recovery, purification, and use for one or more of the purposes disclosed herein.
(18) Suitable monovalent substituents on a substitutable carbon atom of an “optionally substituted” group may be, independently, deuterium; halogen; —(CH.sub.2).sub.0-4R.sup.∘; —(CH.sub.2).sub.0-4OR.sup.∘; —O(CH.sub.2).sub.0-4R.sup.∘; —O—(CH.sub.2).sub.0-4C(O)OR.sup.∘; —(CH.sub.2).sub.0-4CH(OR.sup.∘).sub.2; —(CH.sub.2).sub.0-4SR.sup.∘; —(CH.sub.2).sub.0-4Ph, which may be substituted with R.sup.∘; —(CH.sub.2).sub.0-4O(CH.sub.2).sub.0-1Ph which may be substituted with R.sup.∘; —CH═CHPh, which may be substituted with R.sup.∘; —(CH.sub.2).sub.0-4O(CH.sub.2).sub.0-1-pyridyl which may be substituted with R.sup.∘; 4-8 membered saturated or unsaturated heterocycloalkyl (e.g., pyridyl); 3-8 membered saturated or unsaturated cycloalkyl (e.g., cyclopropyl, cyclobutyl, or cyclopentyl); —NO.sub.2; —CN; —N.sub.3; —(CH.sub.2).sub.0-4N(R.sup.∘).sub.2; —(CH.sub.2).sub.0-4N(R.sup.∘)C(O)R.sup.∘; —N(R.sup.∘)C(S)R.sup.∘; —(CH.sub.2).sub.0-4N(R.sup.∘)C(O)NR.sup.∘.sub.2; —N(R.sup.∘)C(S)NR.sup.∘.sub.2; —(CH.sub.2).sub.0-4N(R.sup.∘)C(O)OR.sup.∘; —N(R.sup.∘)N(R.sup.∘)C(O)R.sup.∘; —N(R.sup.∘)N(R.sup.∘)C(O)NR.sup.∘.sub.2; —N(R.sup.∘)N(R.sup.∘)C(O)OR.sup.∘; —(CH.sub.2).sub.0-4C(O)R.sup.∘; —C(S)R.sup.∘; —(CH.sub.2).sub.0-4C(O)OR.sup.∘; —(CH.sub.2).sub.0-4—C(O)—N(R.sup.∘).sub.2; —(CH.sub.2).sub.0-4—C(O)—N(R.sup.∘)—S(O).sub.2—R.sup.∘; —C(NCN)NR.sup.∘.sub.2; —(CH.sub.2).sub.0-4C(O)SR.sup.∘; —(CH.sub.2).sub.0-4C(O)OSiR.sup.∘.sub.3; —(CH.sub.2).sub.0-4OC(O)R.sup.∘; —OC(O)(CH.sub.2).sub.0-4SR.sup.∘; —SC(S)SR.sup.∘; —(CH.sub.2).sub.0-4SC(O)R.sup.∘; —(CH.sub.2).sub.0-4C(O)NR.sup.∘.sub.2; —C(S)NR.sup.∘.sub.2; —C(S)SR.sup.∘; —(CH.sub.2).sub.0- 4OC(O) NR.sup.∘.sub.2; —C(O)N(OR.sup.∘)R.sup.∘; —C(O)C(O)R.sup.∘; —C(O)CH.sub.2C(O)R.sup.∘; —C(NOR.sup.∘)R.sup.∘; —(CH.sub.2).sub.0-4SSR.sup.∘; —(CH.sub.2).sub.0- 4S(O).sub.2R.sup.∘; —(CH.sub.2).sub.0-4S(O).sub.2OR.sup.∘; —(CH.sub.2).sub.0-4OS(O).sub.2R.sup.∘; —S(O).sub.2NR.sup.∘.sub.2; —(CH.sub.2).sub.0-4S(O)R.sup.∘; —N(R.sup.∘)S(O).sub.2NR.sup.∘.sub.2; —N(R.sup.∘)S(O).sub.2R.sup.∘; —N(OR.sup.∘)R.sup.∘; —C(NOR.sup.∘)NR.sup.∘.sub.2; —C(NH)NR.sup.∘.sub.2; —P(O).sub.2R.sup.∘; —P(O)R.sup.∘.sub.2; —P(O)(OR.sup.∘).sub.2; —OP(O)R.sup.∘.sub.2; —OP(O)(OR.sup.∘).sub.2; —OP(O)(OR.sup.∘)R.sup.∘, —SiR.sup.∘.sub.3; —(C.sub.1-4 straight or branched alkylene)O—N(R.sup.∘).sub.2; or —(C.sub.1-4 straight or branched alkylene)C(O)O—N(R.sup.∘).sub.2, wherein each R.sup.∘ may be substituted as defined below and is independently hydrogen, —C.sub.1-6 aliphatic, —CH.sub.2Ph, —O(CH.sub.2).sub.0-1Ph, —CH.sub.2-(5-6 membered heteroaryl ring), or a 3-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, notwithstanding the definition above, two independent occurrences of R.sup.∘, taken together with their intervening atom(s), form a 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, which may be substituted as defined below.
(19) Suitable monovalent substituents on R.sup.∘ (or the ring formed by taking two independent occurrences of R.sup.∘ together with their intervening atoms), may be, independently, halogen, —(CH.sub.2).sub.0-2R.sup.•, -(haloR.sup.•), —(CH.sub.2).sub.0-2OH, —(CH.sub.2).sub.0-2OR.sup.•, —(CH.sub.2).sub.0-2CH(OR.sup.•).sub.2; —O(haloR.sup.•), —CN, —N.sub.3, —(CH.sub.2).sub.0-2C(O)R.sup.•, —(CH.sub.2).sub.0-2C(O)OH, —(CH.sub.2).sub.0-2C(O)OR.sup.•, —(CH.sub.2).sub.0-2SR.sup.•, —(CH.sub.2).sub.0-2SH, —(CH.sub.2).sub.0-2NH.sub.2, —(CH.sub.2).sub.0-2NHR.sup.•, —(CH.sub.2).sub.0- 2NR.sup.•.sub.2, —NO.sub.2, —SiR.sup.•.sub.3, —OSiR.sup.•.sub.3, —C(O)SR.sup.•, —(C.sub.1-4 straight or branched alkylene)C(O)OR.sup.•, or —SSR.sup.• wherein each R.sup.• is unsubstituted or where preceded by

“halo” is substituted only with one or more halogens, and is independently selected from C.sub.1-4 aliphatic, —CH.sub.2Ph, —O(CH.sub.2).sub.0-1Ph, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Suitable divalent substituents on a saturated carbon atom of R.sup.∘ include ═O and ═S.
(20) Suitable divalent substituents on a saturated carbon atom of an “optionally substituted” group include the following: ═O, ═S, ═NNR*.sub.2, ═NNHC(O)R*, ═NNHC(O)OR*, ═NNHS(O).sub.2R*, ═NR*, ═NOR*, —O(C(R*.sub.2)).sub.2-3O—, or —S(C(R*.sub.2)).sub.2-3S—, wherein each independent occurrence of R* is selected from hydrogen, C.sub.1-6 aliphatic which may be substituted as defined below, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Suitable divalent substituents that are bound to vicinal substitutable carbons of an “optionally substituted” group include: —O(CR*.sub.2).sub.2-3O—, wherein each independent occurrence of R* is selected from hydrogen, C.sub.1-6 aliphatic which may be substituted as defined below, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.
(21) Suitable substituents on the aliphatic group of R.sup.• include halogen, —R.sup.•, -(haloR.sup.•), —OH, —OR.sup.•, —O(haloR.sup.•), —CN, —C(O)OH, —C(O)OR.sup.•, —NH.sub.2, —NHR.sup.•, —NR.sup.•.sub.2, or —NO.sub.2, wherein each R.sup.• is unsubstituted or where preceded by “halo” is substituted only with one or more halogens, and is independently C.sub.1-4 aliphatic, —CH.sub.2Ph, —O(CH.sub.2).sub.0-1Ph, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.
(22) Suitable substituents on a substitutable nitrogen of an “optionally substituted” group include —R.sup.†, —NR.sup.†.sub.2, —C(O)R.sup.†, —C(O)OR.sup.†, —C(O)C(O)R.sup.†, —C(O)CH.sub.2C(O)R.sup.†, —S(O).sub.2R.sup.†, —S(O).sub.2NR.sup.†.sub.2, —C(S)NR.sup.†.sub.2, —C(NH)NR.sup.†.sub.2, or —N(R.sup.†)S(O).sub.2R.sup.†; wherein each R.sup.† is independently hydrogen, C.sub.1-6 aliphatic which may be substituted as defined below, unsubstituted —OPh, or an unsubstituted 3-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, notwithstanding the definition above, two independent occurrences of R.sup.†, taken together with their intervening atom(s) form an unsubstituted 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.
(23) Suitable substituents on an aliphatic group of R.sup.† are independently halogen, —R.sup.•, -(haloR.sup.•), —OH, —OR.sup.•, —O(haloR.sup.•), —CN, —C(O)OH, —C(O)OR.sup.•, —NH.sub.2, —NHR.sup.•, —NR.sup.•.sub.2, or —NO.sub.2, wherein each R.sup.• is unsubstituted or where preceded by “halo” is substituted only with one or more halogens, and is independently C.sub.1-4 aliphatic, —CH.sub.2Ph, —O(CH.sub.2).sub.0-1Ph, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Suitable divalent substituents on a saturated carbon atom of R.sup.† include ═O and ═S.
(24) The term “acetyl,” as used herein, refers to the group —C(O)CH.sub.3.
(25) The term “alkoxy,” as used herein, refers to a —O—C.sub.1-C.sub.20 alkyl group, wherein the alkoxy group is attached to the remainder of the compound through an oxygen atom.
(26) The term “alkyl,” as used herein, refers to a saturated, straight or branched monovalent hydrocarbon group containing from 1 to 20 (e.g., from 1 to 10 or from 1 to 6) carbons. In some embodiments, an alkyl group is unbranched (i.e., is linear); in some embodiments, an alkyl group is branched. Alkyl groups are exemplified by, but not limited to, methyl, ethyl, n- and iso-propyl, n-, sec-, iso- and tert-butyl, and neopentyl.
(27) The term “alkylene,” as used herein, represents a saturated divalent hydrocarbon group derived from a straight or branched chain saturated hydrocarbon by the removal of two hydrogen atoms, and

is exemplified by methylene, ethylene, isopropylene, and the like. The term “Cr-Cy alkylene” represents alkylene groups having between x and y carbons. Exemplary values for x are 1, 2, 3, 4, 5, and 6, and exemplary values for y are 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, or 20 (e.g., C.sub.1-C.sub.6, C.sub.1-C.sub.10, C.sub.2-C.sub.20, C.sub.2-C.sub.6, C.sub.2-C.sub.10, or C.sub.2-C.sub.20 alkylene). In some embodiments, the alkylene can be further substituted with 1, 2, 3, or 4 substituent groups as defined herein.
(28) The term “alkenyl,” as used herein, represents monovalent straight or branched chain groups of, unless otherwise specified, from 2 to 20 carbons (e.g., from 2 to 6 or from 2 to 10 carbons) containing one or more carbon-carbon double bonds and is exemplified by ethenyl, 1-propenyl, 2-propenyl, 2-methyl-1-propenyl, 1-butenyl, and 2-butenyl. Alkenyls include both cis and trans isomers. The term “alkenylene,” as used herein, represents a divalent straight or branched chain groups of, unless otherwise specified, from 2 to 20 carbons (e.g., from 2 to 6 or from 2 to 10 carbons) containing one or more carbon-carbon double bonds.
(29) The term “alkynyl,” as used herein, represents monovalent straight or branched chain groups from 2 to 20 carbon atoms (e.g., from 2 to 4, from 2 to 6, or from 2 to 10 carbons) containing a carbon-carbon triple bond and is exemplified by ethynyl, and 1-propynyl.
(30) The term “alkynyl sulfone,” as used herein, represents a group comprising the structure
(31) ##STR00003##
wherein R is any chemically feasible substituent described herein.
(32) The term “amino,” as used herein, represents —N(R.sup.•).sub.2, e.g., —NH.sub.2 and —N(CH.sub.3).sub.2.
(33) The term “aminoalkyl,” as used herein, represents an alkyl moiety substituted on one or more carbon atoms with one or more amino moieties.
(34) The term “amino acid,” as described herein, refers to a molecule having a side chain, an amino group, and an acid group (e.g., —CO.sub.2H or —SO.sub.3H), wherein the amino acid is attached to the parent molecular group by the side chain, amino group, or acid group (e.g., the side chain). As used herein, the term “amino acid” in its broadest sense, refers to any compound or substance that can be incorporated into a polypeptide chain, e.g., through formation of one or more peptide bonds. In some embodiments, an amino acid has the general structure H.sub.2N—C(H)(R)—COOH. In some embodiments, an amino acid is a naturally-occurring amino acid. In some embodiments, an amino acid is a synthetic amino acid; in some embodiments, an amino acid is a D-amino acid; in some embodiments, an amino acid is an L-amino acid. “Standard amino acid” refers to any of the twenty standard L-amino acids commonly found in naturally occurring peptides. Exemplary amino acids include alanine, arginine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, histidine, optionally substituted hydroxylnorvaline, isoleucine, leucine, lysine, methionine, norvaline, ornithine, phenylalanine, proline, pyrrolysine, selenocysteine, serine, taurine, threonine, tryptophan, tyrosine, and valine.
(35) The term “aryl,” as used herein, represents a monovalent monocyclic, bicyclic, or multicyclic ring system formed by carbon atoms, wherein the ring attached to the pendant group is aromatic. Examples of aryl groups are phenyl, naphthyl, phenanthrenyl, and anthracenyl. An aryl ring can be attached to its pendant group at any heteroatom or carbon ring atom that results in a stable structure and any of the ring atoms can be optionally substituted unless otherwise specified.
(36) The term “C.sub.0,” as used herein, represents a bond. For example, part of the term —N(C(O)—(C.sub.0-C.sub.5 alkylene-H)— includes —N(C(O)—(C.sub.0 alkylene-H)—, which is also represented by —N(C(O)—H)—.
(37) The terms “carbocyclic” and “carbocyclyl,” as used herein, refer to a monovalent, optionally substituted C.sub.3-C.sub.12 monocyclic, bicyclic, or tricyclic ring structure, which may be bridged, fused or spirocyclic, in which all the rings are formed by carbon atoms and at least one ring is non-aromatic. Carbocyclic structures include cycloalkyl, cycloalkenyl, and cycloalkynyl groups. Examples of carbocyclyl groups are cyclohexyl, cyclohexenyl, cyclooctynyl, 1,2-dihydronaphthyl,

1,2,3,4-tetrahydronaphthyl, fluorenyl, indenyl, indanyl, decalinyl, and the like. A carbocyclic ring can be attached to its pendant group at any ring atom that results in a stable structure and any of the ring atoms can be optionally substituted unless otherwise specified.
(38) The term "carbonyl," as used herein, represents a C(O) group, which can also be represented as C═0.
(39) The term "carboxyl," as used herein, means —CO.sub.2H, (C═O)(OH), COOH, or C(O)OH or the unprotonated counterparts.
(40) The term "cyano," as used herein, represents a —CN group.
(41) The term "cycloalkyl," as used herein, represents a monovalent saturated cyclic hydrocarbon group, which may be bridged, fused or spirocyclic having from three to eight ring carbons, unless otherwise specified, and is exemplified by cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cycloheptyl.
(42) The term "cycloalkenyl," as used herein, represents a monovalent, non-aromatic, saturated cyclic hydrocarbon group, which may be bridged, fused or spirocyclic having from three to eight ring carbons, unless otherwise specified, and containing one or more carbon-carbon double bonds.
(43) The term "diastereomer," as used herein, means stereoisomers that are not mirror images of one another and are non-superimposable on one another.
(44) The term "enantiomer," as used herein, means each individual optically active form of a compound of the invention, having an optical purity or enantiomeric excess (as determined by methods standard in the art) of at least 80% (i.e., at least 90% of one enantiomer and at most 10% of the other enantiomer), preferably at least 90% and more preferably at least 98%.
(45) The term "guanidinyl," refers to a group having the structure:
(46) ##STR00004##
wherein each R is, independently, any chemically feasible substituent described herein.
(47) The term "guanidinoalkyl alkyl," as used herein, represents an alkyl moiety substituted on one or more carbon atoms with one or more guanidinyl moieties.
(48) The term "haloacetyl," as used herein, refers to an acetyl group wherein at least one of the hydrogens has been replaced by a halogen.
(49) The term "haloalkyl," as used herein, represents an alkyl moiety substituted on one or more carbon atoms with one or more of the same of different halogen moieties.
(50) The term "halogen," as used herein, represents a halogen selected from bromine, chlorine, iodine, or fluorine.
(51) The term "heteroalkyl" as used herein, refers to an "alkyl" group, as defined herein, in which at least one carbon atom has been replaced with a heteroatom (e.g., an O, N, or S atom). The heteroatom may appear in the middle or at the end of the radical.
(52) The term "heteroaryl," as used herein, represents a monovalent, monocyclic or polycyclic ring structure that contains at least one fully aromatic ring: i.e., they contain 4n+2 pi electrons within the monocyclic or polycyclic ring system and contains at least one ring heteroatom selected from N, O, or S in that aromatic ring. Exemplary unsubstituted heteroaryl groups are of 1 to 12 (e.g., 1 to 11, 1 to 10, 1 to 9, 2 to 12, 2 to 11, 2 to 10, or 2 to 9) carbons. The term "heteroaryl" includes bicyclic, tricyclic, and tetracyclic groups in which any of the above heteroaromatic rings is fused to one or more, aryl or carbocyclic rings, e.g., a phenyl ring, or a cyclohexane ring. Examples of heteroaryl groups include, but are not limited to, pyridyl, pyrazolyl, benzooxazolyl, benzoimidazolyl, benzothiazolyl, imidazolyl, thiazolyl, quinolinyl, tetrahydroquinolinyl, and 4-azaindolyl. A heteroaryl ring can be attached to its pendant group at any ring atom that results in a stable structure and any of the ring atoms can be optionally substituted unless otherwise specified. In some embodiment, the heteroaryl is substituted with 1, 2, 3, or 4 substituents groups.
(53) The term "heterocycloalkyl," as used herein, represents a monovalent monocyclic, bicyclic or polycyclic ring system, which may be bridged, fused or spirocyclic, wherein at least one ring is non-aromatic and wherein the non-aromatic ring contains one, two, three, or four heteroatoms

independently selected from the group consisting of nitrogen, oxygen, and sulfur. The 5-membered ring has zero to two double bonds, and the 6- and 7-membered rings have zero to three double bonds. Exemplary unsubstituted heterocycloalkyl groups are of 1 to 12 (e.g., 1 to 11, 1 to 10, 1 to 9, 2 to 12, 2 to 11, 2 to 10, or 2 to 9) carbons. The term "heterocycloalkyl" also represents a heterocyclic compound having a bridged multicyclic structure in which one or more carbons or heteroatoms bridges two non-adjacent members of a monocyclic ring, e.g., a quinuclidinyl group. The term "heterocycloalkyl" includes bicyclic, tricyclic, and tetracyclic groups in which any of the above heterocyclic rings is fused to one or more aromatic, carbocyclic, heteroaromatic, or heterocyclic rings, e.g., an aryl ring, a cyclohexane ring, a cyclohexene ring, a cyclopentane ring, a cyclopentene ring, a pyridine ring, or a pyrrolidine ring. Examples of heterocycloalkyl groups are pyrrolidinyl, piperidinyl, 1,2,3,4-tetrahydroquinolinyl, decahydroquinolinyl, dihydropyrrolopyridine, and decahydronapthyridinyl. A heterocycloalkyl ring can be attached to its pendant group at any ring atom that results in a stable structure and any of the ring atoms can be optionally substituted unless otherwise specified.

(54) The term "hydroxy," as used herein, represents a —OH group.

(55) The term "hydroxyalkyl," as used herein, represents an alkyl moiety substituted on one or more carbon atoms with one or more —OH moieties.

(56) The term "isomer," as used herein, means any tautomer, stereoisomer, atropiosmer, enantiomer, or diastereomer of any compound of the invention. It is recognized that the compounds of the invention can have one or more chiral centers or double bonds and, therefore, exist as stereoisomers, such as double-bond isomers (i.e., geometric E/Z isomers) or diastereomers (e.g., enantiomers (i.e., (+) or (−)) or cis/trans isomers). According to the invention, the chemical structures depicted herein, and therefore the compounds of the invention, encompass all the corresponding stereoisomers, that is, both the stereomerically pure form (e.g., geometrically pure, enantiomerically pure, or diastereomerically pure) and enantiomeric and stereoisomeric mixtures, e.g., racemates. Enantiomeric and stereoisomeric mixtures of compounds of the invention can typically be resolved into their component enantiomers or stereoisomers by well-known methods, such as chiral-phase gas chromatography, chiral-phase high performance liquid chromatography, crystallizing the compound as a chiral salt complex, or crystallizing the compound in a chiral solvent. Enantiomers and stereoisomers can also be obtained from stereomerically or enantiomerically pure intermediates, reagents, and catalysts by well-known asymmetric synthetic methods.

(57) As used herein, the term "linker" refers to a divalent organic moiety connecting moiety B to moiety W in a compound of Formula I, such that the resulting compound is capable of achieving an IC50 of 2 uM or less in the Ras-RAF disruption assay protocol provided in the Examples below, and provided here:

(58) The purpose of this biochemical assay is to measure the ability of test compounds to facilitate ternary complex formation between a nucleotide-loaded Ras isoform and cyclophilin A; the resulting ternary complex disrupts binding to a BRAF.sup.RBD construct, inhibiting Ras signaling through a RAF effector.

(59) In assay buffer containing 25 mM HEPES pH 7.3, 0.002% Tween20, 0.1% BSA, 100 mM NaCl and 5 mM MgCl.sub.2, tagless Cyclophilin A, His6-K-Ras-GMPPNP (or other Ras variant), and GST-BRAF.sup.RBD are combined in a 384-well assay plate at final concentrations of 25 μM, 12.5 nM and 50 nM, respectively. Compound is present in plate wells as a 10-point 3-fold dilution series starting at a final concentration of 30 μM. After incubation at 25° C. for 3 hours, a mixture of Anti-His Eu—W1024 and anti-GST allophycocyanin is then added to assay sample wells at final concentrations of 10 nM and 50 nM, respectively, and the reaction incubated for an additional 1.5 hours. TR-FRET signal is read on a microplate reader (Ex 320 nm, Em 665/615 nm). Compounds that facilitate disruption of a Ras:RAF complex are identified as those eliciting a decrease in the TR-FRET ratio relative to DMSO control wells.

(60) In some embodiments, the linker comprises 20 or fewer linear atoms. In some embodiments,

the linker comprises 15 or fewer linear atoms. In some embodiments, the linker comprises 10 or fewer linear atoms. In some embodiments, the linker has a molecular weight of under 500 g/mol. In some embodiments, the linker has a molecular weight of under 400 g/mol. In some embodiments, the linker has a molecular weight of under 300 g/mol. In some embodiments, the linker has a molecular weight of under 200 g/mol. In some embodiments, the linker has a molecular weight of under 100 g/mol. In some embodiments, the linker has a molecular weight of under 50 g/mol.

(61) As used herein, a “monovalent organic moiety” is less than 500 kDa. In some embodiments, a “monovalent organic moiety” is less than 400 kDa. In some embodiments, a “monovalent organic moiety” is less than 300 kDa. In some embodiments, a “monovalent organic moiety” is less than 200 kDa. In some embodiments, a “monovalent organic moiety” is less than 100 kDa. In some embodiments, a “monovalent organic moiety” is less than 50 kDa. In some embodiments, a “monovalent organic moiety” is less than 25 kDa. In some embodiments, a “monovalent organic moiety” is less than 20 kDa. In some embodiments, a “monovalent organic moiety” is less than 15 kDa. In some embodiments, a “monovalent organic moiety” is less than 10 kDa. In some embodiments, a “monovalent organic moiety” is less than 1 kDa. In some embodiments, a “monovalent organic moiety” is less than 500 g/mol. In some embodiments, a “monovalent organic moiety” ranges between 500 g/mol and 500 kDa.

(62) The term “stereoisomer,” as used herein, refers to all possible different isomeric as well as conformational forms which a compound may possess (e.g., a compound of any formula described herein), in particular all possible stereochemically and conformationally isomeric forms, all diastereomers, enantiomers or conformers of the basic molecular structure, including atropisomers. Some compounds of the present invention may exist in different tautomeric forms, all of the latter being included within the scope of the present invention.

(63) The term “sulfonyl,” as used herein, represents an —S(O).sub.2— group.

(64) The term “thiocarbonyl,” as used herein, refers to a —C(S)— group.

(65) The term “vinyl ketone,” as used herein, refers to a group comprising a carbonyl group directly connected to a carbon-carbon double bond.

(66) The term “vinyl sulfone,” as used herein, refers to a group comprising a sulfonyl group directed connected to a carbon-carbon double bond.

(67) The term “ynone,” as used herein, refers to a group comprising the structure,

(68) ##STR00005##

wherein R is any chemically feasible substituent described herein.

(69) Those of ordinary skill in the art, reading the present disclosure, will appreciate that certain compounds described herein may be provided or utilized in any of a variety of forms such as, for example, salt forms, protected forms, pro-drug forms, ester forms, isomeric forms (e.g., optical or structural isomers), isotopic forms, etc. In some embodiments, reference to a particular compound may relate to a specific form of that compound. In some embodiments, reference to a particular compound may relate to that compound in any form. In some embodiments, for example, a preparation of a single stereoisomer of a compound may be considered to be a different form of the compound than a racemic mixture of the compound; a particular salt of a compound may be considered to be a different form from another salt form of the compound; a preparation containing one conformational isomer ((Z) or (E)) of a double bond may be considered to be a different form from one containing the other conformational isomer ((E) or (Z)) of the double bond; a preparation in which one or more atoms is a different isotope than is present in a reference preparation may be considered to be a different form.

DETAILED DESCRIPTION

(70) Compounds

(71) Provided herein are Ras inhibitors. The approach described herein entails formation of a high affinity three-component complex, or conjugate, between a synthetic ligand and two intracellular proteins which do not interact under normal physiological conditions: the target protein of interest

(e.g., Ras), and a widely expressed cytosolic chaperone (presenter protein) in the cell (e.g., cyclophilin A). More specifically, in some embodiments, the inhibitors of Ras described herein induce a new binding pocket in Ras by driving formation of a high affinity tri-complex, or conjugate, between the Ras protein and the widely expressed cytosolic chaperone, cyclophilin A (CYPA). Without being bound by theory, the inventors believe that one way the inhibitory effect on Ras is effected by compounds of the invention and the complexes, or conjugates, they form is by steric occlusion of the interaction site between Ras and downstream effector molecules, such as RAF, which are required for propagating the oncogenic signal.
(72) Without being bound by theory, the inventors postulate that both covalent and non-covalent interactions of a compound of the present invention with Ras and the chaperone protein (e.g., cyclophilin A) may contribute to the inhibition of Ras activity. In some embodiments, a compound of the present invention forms a covalent adduct with a side chain of a Ras protein (e.g., a sulfhydryl side chain of the cysteine at position 12 or 13 of a mutant Ras protein). Covalent adducts may also be formed with other side chains of Ras. In addition, or alternatively, non-covalent interactions may be at play: for example, van der Waals, hydrophobic, hydrophilic and hydrogen bond interactions, and combinations thereof, may contribute to the ability of the compounds of the present invention to form complexes and act as Ras inhibitors. Accordingly, a variety of Ras proteins may be inhibited by compounds of the present invention (e.g., K-Ras, N-Ras, H-Ras, and mutants thereof at positions 12, 13 and 61, such as G12C, G12D, G12V, G12S, G13C, G13D, and Q61L, and others described herein).
(73) Methods of determining covalent adduct formation are known in the art. One method of determining covalent adduct formation is to perform a “cross-linking” assay, such as under these conditions (Note—the following protocol describes a procedure for monitoring cross-linking of K-Ras G12C (GMP-PNP) to a compound of the invention. This protocol may also be executed substituting other Ras proteins or nucleotides).
(74) The purpose of this biochemical assay is to measure the ability of test compounds to covalently label nucleotide-loaded K-Ras isoforms. In assay buffer containing 12.5 mM HEPES pH 7.4, 75 mM NaCl, 1 mM MgCl.sub.2, 1 mM BME, 5 μM Cyclophilin A and 2 μM test compound, a 5 μM stock of GMP-PNP-loaded K-Ras (1-169) G12C is diluted 10-fold to yield a final concentration of 0.5 μM; with final sample volume being 100 μL.
(75) The sample is incubated at 25° C. for a time period of up to 24 hours prior to quenching by the addition of 10 μL of 5% Formic Acid. Quenched samples are centrifuged at 15000 rpm for 15 minutes in a benchtop centrifuge before injecting a 10 μL aliquot onto a reverse phase C.sub.4 column and eluting into the mass spectrometer with an increasing acetonitrile gradient in the mobile phase. Analysis of raw data may be carried out using Waters MassLynx MS software, with % bound calculated from the deconvoluted protein peaks for labeled and unlabeled K-Ras.
(76) Accordingly, provided herein is a compound, or pharmaceutically acceptable salt thereof, having the structure of Formula I:
(77) ##STR00006## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 10-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene,

or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, a haloacetal, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl, or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or

C.sub.1-C.sub.3 alkyl; and R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl (e.g., methyl).
(78) In some embodiments, R.sup.9 is H, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl.
(79) In some embodiments, R.sup.21 is hydrogen.
(80) In some embodiments, provided herein is a compound, or pharmaceutically acceptable salt thereof, having the structure of Formula Ia:
(81) ##STR00007## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 10-membered heteroarylene; B is —CH(R.sup.9)— or >C═CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, a haloacetal, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form

C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl, or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; and R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl.

(82) In some embodiments, the disclosure features a compound, or pharmaceutically acceptable salt thereof, of structural Formula Ib:

(83) ##STR00008## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, a haloacetal, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 and Y.sup.6 are, independently, CH or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with

1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; and R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl.
(84) In some embodiments of compounds of the present invention, G is optionally substituted C.sub.1-C.sub.4 heteroalkylene.
(85) In some embodiments, a compound having the structure of Formula Ic is provided, or a pharmaceutically acceptable salt thereof:
(86) ##STR00009## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 and Y.sup.6 are, independently, CH or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3

combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; and R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl.

(87) In some embodiments of compounds of the present invention, X.sup.2 is NH. In some embodiments, X.sup.3 is CH. In some embodiments, R.sup.11 is hydrogen. In some embodiments, R.sup.11 is C.sub.1-C.sub.3 alkyl. In some embodiments, R.sup.11 is methyl.

(88) In some embodiments, a compound of the present invention has the structure of Formula Id, or a pharmaceutically acceptable salt thereof:

(89) ##STR00010## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 and Y.sup.6 are, independently, CH or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally

substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; and R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl.

(90) In some embodiments of a compound of the present invention, X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene. In some embodiments, X.sup.1 is methylene. In some embodiments, X.sup.1 is methylene substituted with a C.sub.1-C.sub.6 alkyl group or a halogen. In some embodiments, X.sup.1 is —CH(Br)—. In some embodiments, X.sup.1 is —CH(CH.sub.3)—. In some embodiments, R.sup.5 is hydrogen. In some embodiments, R.sup.5 is C.sub.1-C.sub.4 alkyl optionally substituted with halogen. In some embodiments, R.sup.5 is methyl. In some embodiments, Y.sup.4 is C. In some embodiments, R.sup.4 is hydrogen. In some embodiments, Y.sup.5 is CH.

(91) In some embodiments, Y.sup.6 is CH. In some embodiments, Y.sup.1 is C. In some embodiments, Y.sup.2 is C. In some embodiments, Y.sup.3 is N. In some embodiments, R.sup.3 is absent. In some embodiments, Y.sup.7 is C.

(92) In some embodiments, a compound of the present invention has the structure of Formula Ie, or a pharmaceutically acceptable salt thereof:

(93) ##STR00011## wherein A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a

cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; and R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl.

(94) In some embodiments of a compound of the present invention, R.sup.6 is hydrogen. In some embodiments, R.sup.2 is hydrogen, cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 6-membered heterocycloalkyl. In some embodiments, R.sup.2 is optionally substituted C.sub.1-C.sub.6 alkyl. In some embodiments, R.sub.2 is fluoroalkyl. In some embodiments, R.sup.2 is ethyl. In some embodiments, R.sub.2 is —CH.sub.2CF.sub.3. In some embodiments, R.sub.2 is C.sub.2-C.sub.6 alkynyl. In some embodiments, R.sub.2 is —CHC≡CH. In some embodiments, R.sub.2 is —CH.sub.2C≡CCH.sub.3. In some embodiments, R.sup.7 is optionally substituted C.sub.1-C.sub.3 alkyl. In some embodiments, R.sup.7 is C.sub.1-C.sub.3 alkyl. In some embodiments, R.sup.8 is optionally substituted C.sub.1-C.sub.3 alkyl. In some embodiments, R.sup.8 is C.sub.1-C.sub.3 alkyl.

(95) In some embodiments, a compound of the present invention has the structure of Formula If, or a pharmaceutically acceptable salt thereof:

(96) ##STR00012## wherein A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)—

where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is C.sub.1-C.sub.6 alkyl or 3 to 6-membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl.
(97) In some embodiments of a compound of the present invention, R.sup.1 is optionally substituted 6 to 10-membered aryl, optionally substituted 3 to 6-membered cycloalkenyl, or optionally substituted 5 to 10-membered heteroaryl. In some embodiments, R.sup.1 is optionally substituted 6-membered aryl, optionally substituted 6-membered cycloalkenyl, or optionally substituted 6-membered heteroaryl.
(98) In some embodiments of a compound of the present invention, R.sub.1 is
(99) ##STR00013##
or a stereoisomer (e.g., atropisomer) thereof.
(100) In some embodiments of a compound of the present invention, R.sub.1 is
(101) ##STR00014##
or a stereoisomer (e.g., atropisomer) thereof. In some embodiments of a compound of the present invention, R.sub.1 is
(102) ##STR00015##
(103) In some embodiments, a compound of the present invention has the structure of Formula Ig, or a pharmaceutically acceptable salt thereof:
(104) ##STR00016## wherein A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; R.sup.2 is C.sub.1-C.sub.6 alkyl, C.sub.1-C.sub.6 fluoroalkyl, or 3 to 6-membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl X.sup.e and X.sup.f are, independently, N or CH; and R.sup.12 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, or optionally substituted 3 to 6-membered heterocycloalkylene.
(105) In some embodiments of a compound of the present invention, X.sup.e is N and X.sup.f is CH. In some embodiments, X.sup.e is CH and X.sup.f is N.
(106) In some embodiments of a compound of the present invention, R.sup.12 is optionally substituted C.sub.1-C.sub.6 heteroalkyl. In some embodiments, R.sup.12 is
(107) ##STR00017##
In some embodiments, R.sup.12
(108) ##STR00018##
(109) In some embodiments, a compound of the present invention has the structure of Formula VI,

or a pharmaceutically acceptable salt thereof:

(110) ##STR00019## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene (e.g., phenyl or phenol), or optionally substituted 5 to 10-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, a haloacetal, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl;

R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl (e.g., methyl); and X.sup.e and X.sup.f are, independently, N or CH.
(111) In some embodiments, a compound of the present invention has the structure of Formula VIa, or a pharmaceutically acceptable salt thereof:
(112) ##STR00020## wherein A optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene (e.g., phenyl or phenol), or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; R.sup.2 is C.sub.1-C.sub.6 alkyl, C.sub.1-C.sub.6 fluoroalkyl, or 3 to 6-membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; X.sup.e and X.sup.f are, independently, N or CH; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl.
(113) In some embodiments of a compound of the present invention, X.sup.e is N and X.sup.f is CH. In some embodiments, X.sup.e is CH and X.sup.f is N.
(114) In some embodiments, a compound of the present invention has the structure of Formula VIb, or a pharmaceutically acceptable salt thereof:
(115) ##STR00021## wherein A optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene (e.g., phenyl or phenol), or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; L is absent or a linker; and W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone. In some embodiments of a compound of the present invention, A is optionally substituted 6-membered arylene.
(116) In some embodiments, a compound of the present invention has the structure of Formula VIc (corresponding for Formula BB of FIG. **1**A and FIG. **1**B), or a pharmaceutically acceptable salt thereof:
(117) ##STR00022## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene,

optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene (e.g., phenyl or phenol), or optionally substituted 5 to 10-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, a haloacetal, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or

optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl (e.g., methyl).
(118) In some embodiments, A has the structure:
(119) ##STR00023##
wherein R.sup.13 is hydrogen, halo, hydroxy, amino, optionally substituted C.sub.1-C.sub.6 alkyl, or optionally substituted C.sub.1-C.sub.6 heteroalkyl; and R.sup.13a is hydrogen or halo. In some embodiments, R.sup.13 is hydrogen. In some embodiments, R.sup.13 and R.sup.13a are each hydrogen. In some embodiments, R.sup.13 is hydroxy, methyl, fluoro, or difluoromethyl.
(120) In some embodiments, A is optionally substituted 5 to 6-membered heteroarylene. In some embodiments, A is:
(121) ##STR00024## ##STR00025##
(122) In some embodiments, A is optionally substituted C.sub.1-C.sub.4 heteroalkylene. In some embodiments, A is:
(123) ##STR00026##
In some embodiments, A is optionally substituted 3 to 6-membered heterocycloalkylene. In some embodiments, A is:
(124) ##STR00027##
In some embodiments, A is
(125) ##STR00028##
(126) In some embodiments of a compound of the present invention, B is —CHR.sup.9—. In some embodiments, R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl. In some embodiments, R.sup.9 is:
(127) ##STR00029##
In some embodiments, R.sup.9 is:
(128) ##STR00030##
(129) In some embodiments, R.sup.9 is H, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl.
(130) In some embodiments of a compound of the present invention, B is optionally substituted 6-membered arylene. In some embodiments, B is 6-membered arylene. In some embodiments, B is:
(131) ##STR00031##
(132) In some embodiments of a compound of the present invention, R.sup.7 is methyl.
(133) In some embodiments of a compound of the present invention, R.sup.8 is methyl.
(134) In some embodiments, R.sup.21 is hydrogen.
(135) In some embodiments of a compound of the present invention, the linker is the structure of Formula II:
A.sup.1-(B.sup.1).sub.f—(C.sup.1).sub.g—(B.sup.2).sub.h-(D.sup.1)-(B.sup.3).sub.i—(C.sup.2).sub.j—(B.sup.4).sub.k-A.sup.2 Formula II
where A.sup.1 is a bond between the linker and B; A.sup.2 is a bond between W and the linker; B.sup.1, B.sup.2, B.sup.3, and B.sup.4 each, independently, is selected from optionally substituted C.sub.1-C.sub.2 alkylene, optionally substituted C.sub.1-C.sub.3 heteroalkylene, O, S, and NR.sup.N; R.sup.N is hydrogen, optionally substituted C.sub.1-4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted C.sub.1-C.sub.7 heteroalkyl; C.sup.1 and C.sup.2 are each, independently, selected from carbonyl, thiocarbonyl, sulphonyl, or phosphoryl; f, g, h, i, j, and k are each, independently, 0 or 1;

and D.sup.1 is optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.2-C.sub.10 alkenylene, optionally substituted C.sub.2-C.sub.10 alkynylene, optionally substituted 3 to 14-membered heterocycloalkylene, optionally substituted 5 to 10-membered heteroarylene, optionally substituted 3 to 8-membered cycloalkylene, optionally substituted 6 to 10-membered arylene, optionally substituted C.sub.2-C.sub.10 polyethylene glycolene, or optionally substituted C.sub.1-C.sub.10 heteroalkylene, or a chemical bond linking A.sup.1-(B.sup.1).sub.f—(C.sup.1).sub.g—(B.sup.2).sub.h— to —(B.sup.3).sub.i—(C.sup.2).sub.j—(B.sup.4).sub.k-A.sup.2. In some embodiments, the linker is acyclic. In some embodiments, linker has the structure of Formula IIa:
(136) ##STR00032## wherein X.sup.a is absent or N; R.sup.14 is absent, hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; and L.sup.2 is absent, —SO.sub.2—, optionally substituted C.sub.1-C.sub.4 alkylene or optionally substituted C.sub.1-C.sub.4 heteroalkylene, wherein at least one of X.sup.a, R.sup.14, or L.sup.2 is present. In some embodiments, the linker has the structure:
(137) ##STR00033##
(138) In some embodiments, the linker is or comprises a cyclic moiety. In some embodiments, the linker has the structure of Formula IIb:
(139) ##STR00034## wherein o is 0 or 1; R.sup.15 is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted 3 to 8-membered cycloalkylene, or optionally substituted 3 to 8-membered heterocycloalkylene; X.sup.4 is absent, optionally substituted C.sub.1-C.sub.4 alkylene, O, NCH.sub.3, or optionally substituted C.sub.1-C.sub.4 heteroalkylene; Cy is optionally substituted 3 to 8-membered cycloalkylene, optionally substituted 3 to 8-membered heterocycloalkylene, optionally substituted 6-10 membered arylene, or optionally substituted 5 to 10-membered heteroarylene; and L.sup.3 is absent, —SO.sub.2—, optionally substituted C.sub.1-C.sub.4 alkylene or optionally substituted C.sub.1-C.sub.4 heteroalkylene.
(140) In some embodiments, the linker has the structure of Formula IIb-1:
(141) ##STR00035## wherein o is 0 or 1; R.sup.15 is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted 3 to 8-membered cycloalkylene, or optionally substituted 3 to 8-membered heterocycloalkylene; Cy is optionally substituted 3 to 8-membered cycloalkylene, optionally substituted 3 to 8-membered heterocycloalkylene, optionally substituted 6-10 membered arylene, or optionally substituted 5 to 10-membered heteroarylene; and L.sup.3 is absent, —SO.sub.2—, optionally substituted C.sub.1-C.sub.4 alkylene or optionally substituted C.sub.1-C.sub.4 heteroalkylene.
(142) In some embodiments, the linker has the structure of Formula IIc:
(143) ##STR00036## wherein R.sup.15 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted 3 to 8-membered cycloalkylene, or optionally substituted 3 to 8-membered heterocycloalkylene; and R.sup.15a, R.sup.15b, R.sup.15c, R.sup.15d, R.sup.15e, R.sup.15f, and R.sup.15g are, independently, hydrogen, halo, hydroxy, cyano, amino, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, or, or R.sup.15b and R.sup.15d combine with the carbons to which they are attached to form an optionally substituted 3 to 8-membered cycloalkylene, or optionally substituted 3 to 8-membered heterocycloalkylene.
(144) In some embodiments, the linker has the structure:
(145) ##STR00037## ##STR00038## ##STR00039## ##STR00040##
(146) In some embodiments, the linker has the structure:
(147) ##STR00041## ##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047## ##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054## ##STR00055## ##STR00056##
(148) In some embodiments, the linker has the structure
(149) ##STR00057##
(150) In some embodiments, the linker has the structure
(151) ##STR00058##

In some embodiments of a compound of the present invention, W is a cross-linking group comprising a vinyl ketone. In some embodiments, W has the structure of Formula IIIa:
(152) ##STR00059## wherein R.sup.16a, R.sup.16b, and R.sup.16c are, independently, hydrogen, —CN, halogen, or —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl. In some embodiments, W is:
(153) ##STR00060## ##STR00061##
In some embodiments, W is a cross-linking group comprising an ynone. In some embodiments, W has the structure of Formula IIIb:
(154) ##STR00062## wherein R.sup.17 is hydrogen, —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl, or a 4 to 7-membered saturated heterocycloalkyl. In some embodiments, W is:
(155) ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067## ##STR00068##
In some embodiments, W is
(156) ##STR00069##
(157) In some embodiments, W is a cross-linking group comprising a vinyl sulfone. In some embodiments, W has the structure of Formula IIIc:
(158) ##STR00070## wherein R.sup.18a, R.sup.18b, and R.sup.18c are, independently, hydrogen, —CN, or —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl. In some embodiments, W is:
(159) ##STR00071## In some embodiments, W is a cross-linking group comprising an alkynyl sulfone. In some embodiments, W has the structure of Formula IIId:
(160) ##STR00072## wherein R.sup.19 is hydrogen, —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl, or a 4 to 7-membered saturated heterocycloalkyl. In some embodiments, W is:
(161) ##STR00073## In some embodiments, W has the structure of Formula IIIe: wherein X.sup.e is a halogen; and
(162) ##STR00074## R.sup.20 is hydrogen, —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl. In some embodiments, W is haloacetal. In some embodiments, W is not haloacetal.
(163) In some embodiments, a compound of the present invention is selected from Table 1, or a pharmaceutically acceptable salt or stereoisomer thereof. In some embodiments, a compound of the present invention is selected from Table 1, or a pharmaceutically acceptable salt or atropisomer thereof.
(164) TABLE-US-00001 TABLE 1 Certain Compounds of the Present Invention Ex# Structure A1 embedded image A2 embedded image A3 embedded image A4 embedded image A5 embedded image A6 0embedded image A7 embedded image A8 embedded image A9 embedded image A10 embedded image A11 embedded image A12 embedded image A13 embedded image A14 embedded image A15 embedded image A16 0embedded image A17 embedded image A18 embedded image A19 embedded image A20 embedded image A21 embedded image A22 embedded image A23 embedded image A24 embedded image A25

embedded image A26 00 embedded image A27 01 embedded image A28 02 embedded image A29 03 embedded image A30 04 embedded image A31 05 embedded image A32 06 embedded image A33 07 embedded image A34 08 embedded image A35 09 embedded image A36 0 embedded image A37 embedded image A38 embedded image A39 embedded image A40 embedded image A41 embedded image A42 embedded image A43 embedded image A44 embedded image A45 embedded image A46 0 embedded image A47 embedded image A48 embedded image A49 embedded image A50 embedded image A51 embedded image A52 embedded image A53 embedded image A54 embedded image A55 embedded image A56 0 embedded image A57 embedded image A58 embedded image A59 embedded image A60 embedded image A61 embedded image A62 embedded image A63 embedded image A64 embedded image A65 embedded image A66 0 embedded image A67 embedded image A68 embedded image A69 embedded image A70 embedded image A71 embedded image A72 embedded image A73 embedded image A74 embedded image A75 embedded 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A725 embedded image A726 embedded image A727 00embedded image A728 01 embedded image A729 02embedded image A730 03embedded image A731 04 embedded image A732 05embedded image A733 06embedded image A734 07 embedded image A735 08embedded image A736 09embedded image A737 0 embedded image A738 embedded image A739 embedded image A740 embedded image A741 embedded image Note that some compounds are shown with bonds as flat or wedged. In some instances, the relative stereochemistry of stereoisomers has been determined; in some instances, the absolute stereochemistry has been determined. In some instances, a single Example number corresponds to a mixture of stereoisomers. All stereoisomers of the compounds of the foregoing table are contemplated by the present invention. In particular embodiments, an atropisomer of a compound of the foregoing table is contemplated. Brackets are to be ignored. *The activity of this stereoisomer may, in fact, be attributable to the presence of a small amount of the stereoisomer with the (S) configuration at the —NC(O)—**C**H(CH.sub.3).sub.2—N(CH.sub.3)— position.

(165) In some embodiments, a compound of Table 2 is provided, or a pharmaceutically acceptable salt thereof. In some embodiments, a compound of the present invention is selected from Table 2, or a pharmaceutically acceptable salt or atropisomer thereof.

(166) TABLE-US-00002 TABLE 2 Certain Compounds of the Present Invention Ex # Structure B1 embedded image B2 embedded image B3 embedded image B4 embedded image B5 embedded image B6 0embedded image B7 embedded image B11 embedded image B12 embedded image B13 embedded image B18 embedded image B21 embedded image B22 embedded image B25 embedded image B27 embedded image B28 0embedded image B29 embedded image B30 embedded image B32 embedded image B34 embedded image B38 embedded image B47 embedded image B64 embedded image B65 embedded image B66 embedded image B70 0embedded image B73 embedded image B74 embedded image B75 embedded image B76 embedded image B77 embedded image B81 embedded image B83 embedded image B85 embedded image B86 embedded image B87 0embedded image B88 embedded image B89 embedded image B90 embedded image B91 embedded image B96 embedded image B97 embedded image B102 embedded image B103 embedded image B104 embedded image B106 0 embedded image B107 embedded image B109 embedded image B111 embedded image B112 embedded image B113 embedded image B115 embedded image B116 embedded image B117 embedded image B118 embedded image B119 0embedded image B120 embedded image B121 embedded image B122 embedded image B123 embedded image B124 embedded image B126 embedded image B127 embedded image B128 embedded image B129 embedded image B130 0embedded image B131 embedded image B132 embedded image B139 embedded image B140 embedded image B141 embedded image B142 embedded image B143 embedded image B144 embedded image B145 embedded image B146 0embedded image B147 embedded image B148 embedded image B149 embedded image B150 embedded image B161 embedded image B162 embedded image B163 embedded image B164 embedded image B165 embedded image B167 00embedded image B168 01embedded image B169 02 embedded image B170 03embedded image B171 04embedded image B172 05 embedded image B173 06embedded image B174 07embedded image B175 08 embedded image B176 09embedded image B177 0embedded image B178 embedded image B179 embedded image B180 embedded image B181 embedded image B182 embedded image B183 embedded image B184 embedded image B185 embedded image B186 embedded image B187 0embedded image B188 embedded image B189 embedded image B190 embedded image B191 embedded image B192 embedded image

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(167) In some embodiments, a compound of the present invention is or acts as a prodrug, such as with respect to administration to a cell or to a subject in need thereof.
(168) Also provided are pharmaceutical compositions comprising a compound of the present invention, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable excipient.
(169) Further provided is a conjugate, or salt thereof, comprising the structure of Formula IV:
M-L-P Formula IV wherein L is a linker; P is a monovalent organic moiety; and M has the structure of Formula Va:
(170) ##STR01157## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′,

S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is H, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl, or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; and R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl.

(171) In some embodiments the conjugate, or salt thereof, comprises the structure of Formula IV: M-L-P Formula IV wherein L is a linker; P is a monovalent organic moiety; and M has the structure of Formula Vb:

(172) ##STR01158## wherein the dotted lines represent zero, one, two, three, or four non-adjacent

double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— or >C═CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or

R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl, or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a a is hydrogen or halo; and R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl.

(173) In some embodiments, the conjugate has the structure of Formula IV:

M-L-P   Formula IV wherein L is a linker; P is a monovalent organic moiety; and M has the structure of Formula Vc:

(174) ##STR01159## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 and Y.sup.6 are, independently, CH or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8

combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; and R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl.
(175) In some embodiments, a compound of the present invention has the structure of Formula IV: M-L-P Formula IV wherein L is a linker; P is a monovalent organic moiety; and M has the structure of Formula Vd:
(176) ##STR01160## wherein A optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene (e.g., phenyl or phenol), or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; R.sup.2 is C.sub.1-C.sub.6 alkyl, C.sub.1-C.sub.6 fluoroalkyl, or 3 to 6-membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; X.sup.e and X.sup.f are, independently, N or CH; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl.
(177) In some embodiments of a compound of the present invention, X.sup.e is N and X.sup.f is CH. In some embodiments, X.sup.e is CH and X.sup.f is N.
(178) In some embodiments, a compound of the present invention has the structure of Formula IV: M-L-P Formula IV wherein L is a linker; P is a monovalent organic moiety; and M has the structure of Formula Ve:
(179) ##STR01161## wherein A is optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene (e.g., phenyl or phenol), or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl.
(180) In some embodiments of a conjugate of the present invention, the linker has the structure of Formula II:
A.sup.1-(B.sup.1).sub.f—(C.sup.1).sub.g—(B.sup.2).sub.h-(D.sup.1)-(B.sup.3).sub.i—(C.sup.2).sub.j—(B.sup.4).sub.k-A.sup.2 Formula II where A.sup.1 is a bond between the linker

and B; A.sup.2 is a bond between P and the linker; B.sup.1, B.sup.2, B.sup.3, and B.sup.4 each, independently, is selected from optionally substituted C.sub.1-C.sub.2 alkylene, optionally substituted C.sub.1-C.sub.3 heteroalkylene, O, S, and NR.sup.N; R.sup.N is hydrogen, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted C.sub.1-C.sub.7 heteroalkyl; C.sub.1 and C.sub.2 are each, independently, selected from carbonyl, thiocarbonyl, sulphonyl, or phosphoryl; f, g, h, i, j, and k are each, independently, 0 or 1; and D.sup.1 is optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.2-C.sub.10 alkenylene, optionally substituted C.sub.2-C.sub.10 alkynylene, optionally substituted 3 to 14-membered heterocycloalkylene, optionally substituted 5 to 10-membered heteroarylene, optionally substituted 3 to 8-membered cycloalkylene, optionally substituted 6 to 10-membered arylene, optionally substituted C.sub.2-C.sub.10 polyethylene glycolene, or optionally substituted C.sub.1-C.sub.10 heteroalkylene, or a chemical bond linking A.sup.1-(B.sup.1).sub.f—(C.sup.1).sub.g—(B.sup.2).sub.h— to —(B.sup.3).sub.i—(C.sup.2).sub.j—(B.sup.4).sub.k-A.sup.2.
(181) In some embodiments of a conjugate of the present invention, the monovalent organic moiety is a protein, such as a Ras protein. In some embodiments, the Ras protein is K-Ras G12C, K-Ras G13C, H-Ras G12C, H-Ras G13C, N-Ras G12C, or N-Ras G13C. Other Ras proteins are described herein. In some embodiments, the linker is bound to the monovalent organic moiety through a bond to a sulfhydryl group of an amino acid residue of the monovalent organic moiety. In some embodiments, the linker is bound to the monovalent organic moiety through a bond to a carboxyl group of an amino acid residue of the monovalent organic moiety.
(182) Further provided is a method of treating cancer in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof. The cancer may, for example, be pancreatic cancer, colorectal cancer, non-small cell lung cancer, acute myeloid leukemia, multiple myeloma, thyroid gland adenocarcinoma, a myelodysplastic syndrome, or squamous cell lung carcinoma. In some embodiments, the cancer comprises a Ras mutation, such as K-Ras G12C, K-Ras G13C, H-Ras G12C, H-Ras G13C, N-Ras G12C, or N-Ras G13C. Other Ras mutations are described herein.
(183) Further provided is a method of treating a Ras protein-related disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof.
(184) Further provided is a method of inhibiting a Ras protein in a cell, the method comprising contacting the cell with an effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof. For example, the Ras protein is K-Ras G12C, K-Ras G13C, H-Ras G12C, H-Ras G13C, N-Ras G12C, or N-Ras G13C. Other Ras proteins are described herein. The cell may be a cancer cell, such as a pancreatic cancer cell, a colorectal cancer cell, a non-small cell lung cancer cell, an acute myeloid leukemia cell, a multiple myeloma cell, a thyroid gland adenocarcinoma cell, a myelodysplastic syndrome cell, or a squamous cell lung carcinoma cell. Other cancer types are described herein. The cell may be in vivo or in vitro.
(185) With respect to compounds of the present invention, one stereoisomer may exhibit better inhibition than another stereoisomer. For example, one atropisomer may exhibit inhibition, whereas the other atropisomer may exhibit little or no inhibition.
(186) In some embodiments, a method or use described herein further comprises administering an additional anti-cancer therapy. In some embodiments, the additional anti-cancer therapy is a HER2 inhibitor, an EGFR inhibitor, a second Ras inhibitor, a SHP2 inhibitor, an SOS1 inhibitor, a Raf inhibitor, a MEK inhibitor, an ERK inhibitor, a PI3K inhibitor, a PTEN inhibitor, an AKT inhibitor, an mTORC1 inhibitor, a BRAF inhibitor, a PD-L1 inhibitor, a PD-1 inhibitor, a CDK4/6 inhibitor, or a combination thereof. In some embodiments, the additional anticancer therapy is a SHP2 inhibitor.

Other additional anti-cancer therapies are described herein.

Methods of Synthesis

(187) The compounds described herein may be made from commercially available starting materials or synthesized using known organic, inorganic, or enzymatic processes.

(188) The compounds of the present invention can be prepared in a number of ways well known to those skilled in the art of organic synthesis. By way of example, compounds of the present invention can be synthesized using the methods described in the Schemes below, together with synthetic methods known in the art of synthetic organic chemistry, or variations thereon as appreciated by those skilled in the art. These methods include but are not limited to those methods described in the Schemes below.

(189) ##STR01162## ##STR01163## ##STR01164##

(190) A general synthesis of macrocyclic esters is outlined in Scheme 1. An appropriately substituted aryl-3-(5-bromo-1-ethyl-1H-indol-3-yl)-2,2-dimethylpropan-1-ol (1) can be prepared in three steps starting from protected 3-(5-bromo-2-iodo-1H-indol-3-yl)-2,2-dimethylpropan-1-ol and appropriately substituted boronic acid, including palladium mediated coupling, alkylation, and de-protection reactions. Methyl-amino-hexahydropyridazine-3-carboxylate-boronic ester (2) can be prepared in three steps, including protection, iridium catalyst mediated borylation, and coupling with methyl methyl (S)-hexahydropyridazine-3-carboxylate.

(191) An appropriately substituted acetylpyrrolidine-3-carbonyl-N-methyl-L-valine (or an alternative aminoacid derivative (4) can be made by coupling of methyl-L-valinate and protected (S)-pyrrolidine-3-carboxylic acid, followed by deprotection, coupling with a carboxylic acid containing an appropriately substituted Michael acceptor, and a hydrolysis step.

(192) The final macrocyclic esters can be made by coupling of methyl-amino-hexahydropyridazine-3-carboxylate-boronic ester (2) and aryl-3-(5-bromo-1-ethyl-1H-indol-3-yl)-2,2-dimethylpropan-1-ol (1) in the presence of a Pd catalyst followed by hydrolysis and macrolactonization steps to result in an appropriately protected macrocyclic intermediate (5). Deprotection and coupling with an appropriately substituted intermediate 4 results in a macrocyclic product. Additional deprotection and/or functionalization steps can be required to produce the final compound.

(193) ##STR01165##

Alternatively, macrocyclic ester can be prepared as described in Scheme 2. An appropriately protected bromo-indolyl (6) coupled in the presence of a Pd catalyst with boronic ester (3), followed by iodination, deprotection, and ester hydrolysis. Subsequent coupling with methyl (S)-hexahydropyridazine-3-carboxylate, followed by hydrolysis and macrolactonization can result in iodo intermediate (7). Coupling in the presence of a Pd catalyst with an appropriately substituted boronic ester and alkyllation can yield fully protected macrocycle (5). Additional deprotection or functionalization steps are required to produce the final compound.

(194) In addition, compounds of the disclosure can be synthesized using the methods described in the Examples below, together with synthetic methods known in the art of synthetic organic chemistry, or variations thereon as appreciated by those skilled in the art. These methods include but are not limited to those methods described in the Examples below. For example, a person of skill in the art would be able to install into a macrocyclic ester a desired —B-L-W group of a compound of Formula (I), where B, L and W are defined herein, including by using methods exemplified in the Example section herein.

(195) Compounds of Table 1 herein were prepared using methods disclosed herein or were prepared using methods disclosed herein combined with the knowledge of one of skill in the art. Compounds of Table 2 may be prepared using methods disclosed herein or may be prepared using methods disclosed herein combined with the knowledge of one of skill in the art.

(196) ##STR01166## ##STR01167##

(197) An alternative general synthesis of macrocyclic esters is outlined in Scheme 3. An appropriately substituted indolyl boronic ester (8) can be prepared in four steps starting from

protected 3-(5-bromo-2-iodo-1H-indol-3-yl)-2,2-dimethylpropan-1-ol and appropriately substituted boronic acid, including Palladium mediated coupling, alkylation, de-protection, and Palladium mediated borylation reactions.
(198) Methyl-amino-3-(4-bromothiazol-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (10) can be prepared via coupling of (S)-2-amino-3-(4-bromothiazol-2-yl)propanoic acid (9) with methyl (S)-hexahydropyridazine-3-carboxylate.
(199) The final macrocyclic esters can be made by coupling of Methyl-amino-3-(4-bromothiazol-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (10) and an appropriately substituted indolyl boronic ester (8) in the presence of Pd catalyst followed by hydrolysis and macrolactonization steps to result in an appropriately protected macrocyclic intermediate (11). Deprotection and coupling with an appropriately substituted intermediate 4 can result in a macrocyclic product. Additional deprotection or functionalization steps could be required to produce a final compound 13 or 14.
(200) ##STR01168## ##STR01169##
(201) An alternative general synthesis of macrocyclic esters is outlined in Scheme 4. An appropriately substituted morpholine or an alternative herecyclic intermediate (15) can be coupled with appropriately protected Intermediate 1 via Palladium mediated coupling. Subsequent ester hydrolysis, and coupling with piperazoic ester results in intermediate 16.
(202) The macrocyclic esters can be made by hydrolysis, deprotection and macrocyclization sequence. Subsequent deprotection and coupling with Intermediate 4 (or analogs) result in an appropriately substituted final macrocyclic products. Additional deprotection or functionalization steps could be required to produce a final compound 17.
(203) ##STR01170## ##STR01171##
(204) An alternative general synthesis of macrocyclic esters is outlined in Scheme 5. An appropriately substituted macrocycle (20) can be prepared starting from an appropriately protected boronic ester 18 and bromo indolyl intermediate (19), including Palladium mediated coupling, hydrolysis, coupling with piperazoic ester, hydrolysis, de-protection, and macrocyclizarion steps. Subsequent coupling with an appropriately substituted protected aminoacid followed by palladium mediated coupling yiels intermediate 21. Additional deprotection and derivatization steps, including alkyllation may be required at this point.
(205) The final macrocyclic esters can be made by coupling of intermediate (22) and an appropriately substituted carboxylic acid intermediate (23). Additional deprotection or functionalization steps could be required to produce a final compound (24).
(206) In addition, compounds of the disclosure can be synthesized using the methods described in the Examples below, together with synthetic methods known in the art of synthetic organic chemistry, or variations thereon as appreciated by those skilled in the art. These methods include but are not limited to those methods described in the Examples below. For example, a person of skill in the art would be able to install into a macrocyclic ester a desired —B-L-W group of a compound of Formula (I), where B, L and W are defined herein, including by using methods exemplified in the Example section herein.
(207) Pharmaceutical Compositions and Methods of Use
(208) Pharmaceutical Compositions and Methods of Administration
(209) The compounds with which the invention is concerned are Ras inhibitors, and are useful in the treatment of cancer. Accordingly, one embodiment of the present invention provides pharmaceutical compositions containing a compound of the invention or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable excipient, as well as methods of using the compounds of the invention to prepare such compositions.
(210) As used herein, the term "pharmaceutical composition" refers to a compound, such as a compound of the present invention, or a pharmaceutically acceptable salt thereof, formulated together with a pharmaceutically acceptable excipient.
(211) In some embodiments, a compound is present in a pharmaceutical composition in unit dose

amount appropriate for administration in a therapeutic regimen that shows a statistically significant probability of achieving a predetermined therapeutic effect when administered to a relevant population. In some embodiments, pharmaceutical compositions may be specially formulated for administration in solid or liquid form, including those adapted for the following: oral administration, for example, drenches (aqueous or non-aqueous solutions or suspensions), tablets, e.g., those targeted for buccal, sublingual, and systemic absorption, boluses, powders, granules, pastes for application to the tongue; parenteral administration, for example, by subcutaneous, intramuscular, intravenous or epidural injection as, for example, a sterile solution or suspension, or sustained-release formulation; topical application, for example, as a cream, ointment, or a controlled-release patch or spray applied to the skin, lungs, or oral cavity; intravaginally or intrarectally, for example, as a pessary, cream, or foam; sublingually; ocularly; transdermally; or nasally, pulmonary, and to other mucosal surfaces.

(212) A "pharmaceutically acceptable excipient," as used herein, refers any inactive ingredient (for example, a vehicle capable of suspending or dissolving the active compound) having the properties of being nontoxic and non-inflammatory in a subject. Typical excipients include, for example: antiadherents, antioxidants, binders, coatings, compression aids, disintegrants, dyes (colors), emollients, emulsifiers, fillers (diluents), film formers or coatings, flavors, fragrances, glidants (flow enhancers), lubricants, preservatives, printing inks, sorbents, suspensing or dispersing agents, sweeteners, or waters of hydration. Excipients include, but are not limited to: butylated optionally substituted hydroxyltoluene (BHT), calcium carbonate, calcium phosphate (dibasic), calcium stearate, croscarmellose, crosslinked polyvinyl pyrrolidone, citric acid, crospovidone, cysteine, ethylcellulose, gelatin, optionally substituted hydroxylpropyl cellulose, optionally substituted hydroxylpropyl methylcellulose, lactose, magnesium stearate, maltitol, mannitol, methionine, methylcellulose, methyl paraben, microcrystalline cellulose, polyethylene glycol, polyvinyl pyrrolidone, povidone, pregelatinized starch, propyl paraben, retinyl palmitate, shellac, silicon dioxide, sodium carboxymethyl cellulose, sodium citrate, sodium starch glycolate, sorbitol, starch (corn), stearic acid, stearic acid, sucrose, talc, titanium dioxide, vitamin A, vitamin E, vitamin C, and xylitol. Those of ordinary skill in the art are familiar with a variety of agents and materials useful as excipients. See, e.g., e.g., Ansel, et al., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems. Philadelphia: Lippincott, Williams & Wilkins, 2004; Gennaro, et al., Remington: The Science and Practice of Pharmacy. Philadelphia: Lippincott, Williams & Wilkins, 2000; and Rowe, Handbook of Pharmaceutical Excipients. Chicago, Pharmaceutical Press, 2005. In some embodiments, a composition includes at least two different pharmaceutically acceptable excipients.

(213) Compounds described herein, whether expressly stated or not, may be provided or utilized in salt form, e.g., a pharmaceutically acceptable salt form, unless expressly stated to the contrary. The term "pharmaceutically acceptable salt," as use herein, refers to those salts of the compounds described herein that are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and other animals without undue toxicity, irritation, allergic response and the like, and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well known in the art. For example, pharmaceutically acceptable salts are described in: Berge et al., *J. Pharmaceutical Sciences* 66:1-19, 1977 and in *Pharmaceutical Salts: Properties, Selection, and Use*, (Eds. P. H. Stahl and C. G. Wermuth), Wiley-VCH, 2008. The salts can be prepared in situ during the final isolation and purification of the compounds described herein or separately by reacting the free base group with a suitable organic acid.

(214) The compounds of the invention may have ionizable groups so as to be capable of preparation as pharmaceutically acceptable salts. These salts may be acid addition salts involving inorganic or organic acids or the salts may, in the case of acidic forms of the compounds of the invention, be prepared from inorganic or organic bases. In some embodiments, the compounds are prepared or used as pharmaceutically acceptable salts prepared as addition products of pharmaceutically acceptable acids or bases. Suitable pharmaceutically acceptable acids and bases are well-known in

the art, such as hydrochloric, sulfuric, hydrobromic, acetic, lactic, citric, or tartaric acids for forming acid addition salts, and potassium hydroxide, sodium hydroxide, ammonium hydroxide, caffeine, various amines, and the like for forming basic salts. Methods for preparation of the appropriate salts are well-established in the art.

(215) Representative acid addition salts include acetate, adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, fumarate, glucoheptonate, glycerophosphate, hemisulfate, heptonate, hexanoate, hydrobromide, hydrochloride, hydroiodide, 2-optionally substituted hydroxyl-ethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, picrate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, toluenesulfonate, undecanoate, valerate salts and the like. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, magnesium and the like, as well as nontoxic ammonium, quaternary ammonium, and amine cations, including, but not limited to ammonium, tetramethylammonium, tetraethylammonium, methylamine, dimethylamine, trimethylamine, triethylamine, ethylamine, and the like.

(216) As used herein, the term “subject” refers to any member of the animal kingdom. In some embodiments, “subject” refers to humans, at any stage of development. In some embodiments, “subject” refers to a human patient. In some embodiments, “subject” refers to non-human animals. In some embodiments, the non-human animal is a mammal (e.g., a rodent, a mouse, a rat, a rabbit, a monkey, a dog, a cat, a sheep, cattle, a primate, or a pig). In some embodiments, subjects include, but are not limited to, mammals, birds, reptiles, amphibians, fish, or worms. In some embodiments, a subject may be a transgenic animal, genetically-engineered animal, or a clone.

(217) As used herein, the term “dosage form” refers to a physically discrete unit of a compound (e.g., a compound of the present invention) for administration to a subject. Each unit contains a predetermined quantity of compound. In some embodiments, such quantity is a unit dosage amount (or a whole fraction thereof) appropriate for administration in accordance with a dosing regimen that has been determined to correlate with a desired or beneficial outcome when administered to a relevant population (i.e., with a therapeutic dosing regimen). Those of ordinary skill in the art appreciate that the total amount of a therapeutic composition or compound administered to a particular subject is determined by one or more attending physicians and may involve administration of multiple dosage forms.

(218) As used herein, the term “dosing regimen” refers to a set of unit doses (typically more than one) that are administered individually to a subject, typically separated by periods of time. In some embodiments, a given therapeutic compound (e.g., a compound of the present invention) has a recommended dosing regimen, which may involve one or more doses. In some embodiments, a dosing regimen comprises a plurality of doses each of which are separated from one another by a time period of the same length; in some embodiments, a dosing regimen comprises a plurality of doses and at least two different time periods separating individual doses. In some embodiments, all doses within a dosing regimen are of the same unit dose amount. In some embodiments, different doses within a dosing regimen are of different amounts. In some embodiments, a dosing regimen comprises a first dose in a first dose amount, followed by one or more additional doses in a second dose amount different from the first dose amount. In some embodiments, a dosing regimen comprises a first dose in a first dose amount, followed by one or more additional doses in a second dose amount same as the first dose amount. In some embodiments, a dosing regimen is correlated with a desired or beneficial outcome when administered across a relevant population (i.e., is a therapeutic dosing regimen).

(219) A “therapeutic regimen” refers to a dosing regimen whose administration across a relevant population is correlated with a desired or beneficial therapeutic outcome.

(220) The term “treatment” (also “treat” or “treating”), in its broadest sense, refers to any administration of a substance (e.g., a compound of the present invention) that partially or completely alleviates, ameliorates, relieves, inhibits, delays onset of, reduces severity of, or reduces incidence of one or more symptoms, features, or causes of a particular disease, disorder, or condition. In some embodiments, such treatment may be administered to a subject who does not exhibit signs of the relevant disease, disorder or condition or of a subject who exhibits only early signs of the disease, disorder, or condition. Alternatively, or additionally, in some embodiments, treatment may be administered to a subject who exhibits one or more established signs of the relevant disease, disorder, or condition. In some embodiments, treatment may be of a subject who has been diagnosed as suffering from the relevant disease, disorder, or condition. In some embodiments, treatment may be of a subject known to have one or more susceptibility factors that are statistically correlated with increased risk of development of the relevant disease, disorder, or condition.
(221) The term “therapeutically effective amount” means an amount that is sufficient, when administered to a population suffering from or susceptible to a disease, disorder, or condition in accordance with a therapeutic dosing regimen, to treat the disease, disorder, or condition. In some embodiments, a therapeutically effective amount is one that reduces the incidence or severity of, or delays onset of, one or more symptoms of the disease, disorder, or condition. Those of ordinary skill in the art will appreciate that the term “therapeutically effective amount” does not in fact require successful treatment be achieved in a particular individual. Rather, a therapeutically effective amount may be that amount that provides a particular desired pharmacological response in a significant number of subjects when administered to patients in need of such treatment. It is specifically understood that particular subjects may, in fact, be “refractory” to a “therapeutically effective amount.” In some embodiments, reference to a therapeutically effective amount may be a reference to an amount as measured in one or more specific tissues (e.g., a tissue affected by the disease, disorder or condition) or fluids (e.g., blood, saliva, serum, sweat, tears, urine). Those of ordinary skill in the art will appreciate that, in some embodiments, a therapeutically effective amount may be formulated or administered in a single dose. In some embodiments, a therapeutically effective amount may be formulated or administered in a plurality of doses, for example, as part of a dosing regimen.
(222) For use as treatment of subjects, the compounds of the invention, or a pharmaceutically acceptable salt thereof, can be formulated as pharmaceutical or veterinary compositions. Depending on the subject to be treated, the mode of administration, and the type of treatment desired, e.g., prevention, prophylaxis, or therapy, the compounds, or a pharmaceutically acceptable salt thereof, are formulated in ways consonant with these parameters. A summary of such techniques may be found in *Remington: The Science and Practice of Pharmacy*, 21.sup.st Edition, Lippincott Williams & Wilkins, (2005); and *Encyclopedia of Pharmaceutical Technology*, eds. J. Swarbrick and J. C. Boylan, 1988-1999, Marcel Dekker, New York, each of which is incorporated herein by reference.
(223) Compositions can be prepared according to conventional mixing, granulating or coating methods, respectively, and the present pharmaceutical compositions can contain from about 0.1% to about 99%, from about 5% to about 90%, or from about 1% to about 20% of a compound of the present invention, or pharmaceutically acceptable salt thereof, by weight or volume. In some embodiments, compounds, or a pharmaceutically acceptable salt thereof, described herein may be present in amounts totaling 1-95% by weight of the total weight of a composition, such as a pharmaceutical composition.
(224) The composition may be provided in a dosage form that is suitable for intraarticular, oral, parenteral (e.g., intravenous, intramuscular), rectal, cutaneous, subcutaneous, topical, transdermal, sublingual, nasal, vaginal, intravesicular, intraurethral, intrathecal, epidural, aural, or ocular administration, or by injection, inhalation, or direct contact with the nasal, genitourinary, reproductive or oral mucosa. Thus, the pharmaceutical composition may be in the form of, e.g., tablets, capsules, pills, powders, granulates, suspensions, emulsions, solutions, gels including

hydrogels, pastes, ointments, creams, plasters, drenches, osmotic delivery devices, suppositories, enemas, injectables, implants, sprays, preparations suitable for iontophoretic delivery, or aerosols. The compositions may be formulated according to conventional pharmaceutical practice.

(225) As used herein, the term "administration" refers to the administration of a composition (e.g., a compound, or a preparation that includes a compound as described herein) to a subject or system. Administration to an animal subject (e.g., to a human) may be by any appropriate route. For example, in some embodiments, administration may be bronchial (including by bronchial instillation), buccal, enteral, interdermal, intra-arterial, intradermal, intragastric, intramedullary, intramuscular, intranasal, intraperitoneal, intrathecal, intravenous, intraventricular, mucosal, nasal, oral, rectal, subcutaneous, sublingual, topical, tracheal (including by intratracheal instillation), transdermal, vaginal, or vitreal.

(226) Formulations may be prepared in a manner suitable for systemic administration or topical or local administration. Systemic formulations include those designed for injection (e.g., intramuscular, intravenous or subcutaneous injection) or may be prepared for transdermal, transmucosal, or oral administration. A formulation will generally include a diluent as well as, in some cases, adjuvants, buffers, preservatives and the like. Compounds, or a pharmaceutically acceptable salt thereof, can be administered also in liposomal compositions or as microemulsions.

(227) For injection, formulations can be prepared in conventional forms as liquid solutions or suspensions or as solid forms suitable for solution or suspension in liquid prior to injection or as emulsions. Suitable excipients include, for example, water, saline, dextrose, glycerol and the like. Such compositions may also contain amounts of nontoxic auxiliary substances such as wetting or emulsifying agents, pH buffering agents and the like, such as, for example, sodium acetate, sorbitan monolaurate, and so forth.

(228) Various sustained release systems for drugs have also been devised. See, for example, U.S. Pat. No. 5,624,677.

(229) Systemic administration may also include relatively noninvasive methods such as the use of suppositories, transdermal patches, transmucosal delivery and intranasal administration. Oral administration is also suitable for compounds of the invention, or a pharmaceutically acceptable salt thereof. Suitable forms include syrups, capsules, and tablets, as is understood in the art.

(230) Each compound, or a pharmaceutically acceptable salt thereof, as described herein, may be formulated in a variety of ways that are known in the art. For example, the first and second agents of the combination therapy may be formulated together or separately. Other modalities of combination therapy are described herein.

(231) The individually or separately formulated agents can be packaged together as a kit. Non-limiting examples include, but are not limited to, kits that contain, e.g., two pills, a pill and a powder, a suppository and a liquid in a vial, two topical creams, etc. The kit can include optional components that aid in the administration of the unit dose to subjects, such as vials for reconstituting powder forms, syringes for injection, customized IV delivery systems, inhalers, etc. Additionally, the unit dose kit can contain instructions for preparation and administration of the compositions. The kit may be manufactured as a single use unit dose for one subject, multiple uses for a particular subject (at a constant dose or in which the individual compounds, or a pharmaceutically acceptable salt thereof, may vary in potency as therapy progresses); or the kit may contain multiple doses suitable for administration to multiple subjects ("bulk packaging"). The kit components may be assembled in cartons, blister packs, bottles, tubes, and the like.

(232) Formulations for oral use include tablets containing the active ingredient(s) in a mixture with non-toxic pharmaceutically acceptable excipients. These excipients may be, for example, inert diluents or fillers (e.g., sucrose, sorbitol, sugar, mannitol, microcrystalline cellulose, starches including potato starch, calcium carbonate, sodium chloride, lactose, calcium phosphate, calcium sulfate, or sodium phosphate); granulating and disintegrating agents (e.g., cellulose derivatives including microcrystalline cellulose, starches including potato starch, croscarmellose sodium,

alginates, or alginic acid); binding agents (e.g., sucrose, glucose, sorbitol, acacia, alginic acid, sodium alginate, gelatin, starch, pregelatinized starch, microcrystalline cellulose, magnesium aluminum silicate, carboxymethylcellulose sodium, methylcellulose, optionally substituted hydroxylpropyl methylcellulose, ethylcellulose, polyvinylpyrrolidone, or polyethylene glycol); and lubricating agents, glidants, and antiadhesives (e.g., magnesium stearate, zinc stearate, stearic acid, silicas, hydrogenated vegetable oils, or talc). Other pharmaceutically acceptable excipients can be colorants, flavoring agents, plasticizers, humectants, buffering agents, and the like.

(233) Two or more compounds may be mixed together in a tablet, capsule, or other vehicle, or may be partitioned. In one example, the first compound is contained on the inside of the tablet, and the second compound is on the outside, such that a substantial portion of the second compound is released prior to the release of the first compound.

(234) Formulations for oral use may also be provided as chewable tablets, or as hard gelatin capsules wherein the active ingredient is mixed with an inert solid diluent (e.g., potato starch, lactose, microcrystalline cellulose, calcium carbonate, calcium phosphate or kaolin), or as soft gelatin capsules wherein the active ingredient is mixed with water or an oil medium, for example, peanut oil, liquid paraffin, or olive oil. Powders, granulates, and pellets may be prepared using the ingredients mentioned above under tablets and capsules in a conventional manner using, e.g., a mixer, a fluid bed apparatus or a spray drying equipment.

(235) Dissolution or diffusion-controlled release can be achieved by appropriate coating of a tablet, capsule, pellet, or granulate formulation of compounds, or by incorporating the compound, or a pharmaceutically acceptable salt thereof, into an appropriate matrix. A controlled release coating may include one or more of the coating substances mentioned above or, e.g., shellac, beeswax, glycowax, castor wax, carnauba wax, stearyl alcohol, glyceryl monostearate, glyceryl distearate, glycerol palmitostearate, ethylcellulose, acrylic resins, dl-polylactic acid, cellulose acetate butyrate, polyvinyl chloride, polyvinyl acetate, vinyl pyrrolidone, polyethylene, polymethacrylate, methylmethacrylate, 2-optionally substituted hydroxylmethacrylate, methacrylate hydrogels, 1,3 butylene glycol, ethylene glycol methacrylate, or polyethylene glycols. In a controlled release matrix formulation, the matrix material may also include, e.g., hydrated methylcellulose, carnauba wax and stearyl alcohol, carbopol 934, silicone, glyceryl tristearate, methyl acrylate-methyl methacrylate, polyvinyl chloride, polyethylene, or halogenated fluorocarbon.

(236) The liquid forms in which the compounds, or a pharmaceutically acceptable salt thereof, and compositions of the present invention can be incorporated for administration orally include aqueous solutions, suitably flavored syrups, aqueous or oil suspensions, and flavored emulsions with edible oils such as cottonseed oil, sesame oil, coconut oil, or peanut oil, as well as elixirs and similar pharmaceutical vehicles.

(237) Generally, when administered to a human, the oral dosage of any of the compounds of the invention, or a pharmaceutically acceptable salt thereof, will depend on the nature of the compound, and can readily be determined by one skilled in the art. A dosage may be, for example, about 0.001 mg to about 2000 mg per day, about 1 mg to about 1000 mg per day, about 5 mg to about 500 mg per day, about 100 mg to about 1500 mg per day, about 500 mg to about 1500 mg per day, about 500 mg to about 2000 mg per day, or any range derivable therein.

(238) In some embodiments, the pharmaceutical composition may further comprise an additional compound having antiproliferative activity. Depending on the mode of administration, compounds, or a pharmaceutically acceptable salt thereof, will be formulated into suitable compositions to permit facile delivery. Each compound, or a pharmaceutically acceptable salt thereof, of a combination therapy may be formulated in a variety of ways that are known in the art. For example, the first and second agents of the combination therapy may be formulated together or separately. Desirably, the first and second agents are formulated together for the simultaneous or near simultaneous administration of the agents.

(239) It will be appreciated that the compounds and pharmaceutical compositions of the present

invention can be formulated and employed in combination therapies, that is, the compounds and pharmaceutical compositions can be formulated with or administered concurrently with, prior to, or subsequent to, one or more other desired therapeutics or medical procedures. The particular combination of therapies (therapeutics or procedures) to employ in a combination regimen will take into account compatibility of the desired therapeutics or procedures and the desired therapeutic effect to be achieved. It will also be appreciated that the therapies employed may achieve a desired effect for the same disorder, or they may achieve different effects (e.g., control of any adverse effects).

(240) Administration of each drug in a combination therapy, as described herein, can, independently, be one to four times daily for one day to one year, and may even be for the life of the subject. Chronic, long-term administration may be indicated.

(241) Methods of Use

(242) In some embodiments, the invention discloses a method of treating a disease or disorder that is characterized by aberrant Ras activity due to a Ras mutant. In some embodiments, the disease or disorder is a cancer.

(243) Accordingly, also provided is a method of treating cancer in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition comprising such a compound or salt. In some embodiments, the cancer is colorectal cancer, non-small cell lung cancer, small-cell lung cancer, pancreatic cancer, appendiceal cancer, melanoma, acute myeloid leukemia, small bowel cancer, ampullary cancer, germ cell cancer, cervical cancer, cancer of unknown primary origin, endometrial cancer, esophagogastric cancer, GI neuroendocrine cancer, ovarian cancer, sex cord stromal tumor cancer, hepatobiliary cancer, or bladder cancer. In some embodiments, the cancer is appendiceal, endometrial or melanoma. Also provided is a method of treating a Ras protein-related disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof, or a pharmaceutical composition comprising such a compound or salt.

(244) In some embodiments, the compounds of the present invention or pharmaceutically acceptable salts thereof, pharmaceutical compositions comprising such compounds or salts, and methods provided herein may be used for the treatment of a wide variety of cancers including tumors such as lung, prostate, breast, brain, skin, cervical carcinomas, testicular carcinomas, etc. More particularly, cancers that may be treated by the compounds or salts thereof, pharmaceutical compositions comprising such compounds or salts, and methods of the invention include, but are not limited to tumor types such as astrocytic, breast, cervical, colorectal, endometrial, esophageal, gastric, head and neck, hepatocellular, laryngeal, lung, oral, ovarian, prostate and thyroid carcinomas and sarcomas. Other cancers include, for example: Cardiac, for example: sarcoma (angiosarcoma, fibrosarcoma, rhabdomyosarcoma, liposarcoma), myxoma, rhabdomyoma, fibroma, lipoma, and teratoma; Lung, for example: bronchogenic carcinoma (squamous cell, undifferentiated small cell, undifferentiated large cell, adenocarcinoma), alveolar (bronchiolar) carcinoma, bronchial adenoma, sarcoma, lymphoma, chondromatous hamartoma, mesothelioma; Gastrointestinal, for example: esophagus (squamous cell carcinoma, adenocarcinoma, leiomyosarcoma, lymphoma), stomach (carcinoma, lymphoma, leiomyosarcoma), pancreas (ductal adenocarcinoma, insulinoma, glucagonoma, gastrinoma, carcinoid tumors, vipoma), small bowel (adenocarcinoma, lymphoma, carcinoid tumors, Kaposi's sarcoma, leiomyoma, hemangioma, lipoma, neurofibroma, fibroma), large bowel (adenocarcinoma, tubular adenoma, villous adenoma, hamartoma, leiomyoma); Genitourinary tract, for example: kidney (adenocarcinoma, Wilm's tumor (nephroblastoma), lymphoma, leukemia), bladder and urethra (squamous cell carcinoma, transitional cell carcinoma, adenocarcinoma), prostate (adenocarcinoma, sarcoma), testis (seminoma, teratoma, embryonal carcinoma, teratocarcinoma, choriocarcinoma, sarcoma, interstitial cell carcinoma, fibroma,

fibroadenoma, adenomatoid tumors, lipoma); Liver, for example: hepatoma (hepatocellular carcinoma), cholangiocarcinoma, hepatoblastoma, angiosarcoma, hepatocellular adenoma, hemangioma; Biliary tract, for example: gall bladder carcinoma, ampullary carcinoma, cholangiocarcinoma; Bone, for example: osteogenic sarcoma (osteosarcoma), fibrosarcoma, malignant fibrous histiocytoma, chondrosarcoma, Ewing's sarcoma, malignant lymphoma (reticulum cell sarcoma), multiple myeloma, malignant giant cell tumor chordoma, osteochronfroma (osteocartilaginous exostoses), benign chondroma, chondroblastoma, chondromyxofibroma, osteoid osteoma, and giant cell tumors; Nervous system, for example: skull (osteoma, hemangioma, granuloma, xanthoma, osteitis deformans), meninges (meningioma, meningiosarcoma, gliomatosis), brain (astrocytoma, medulloblastoma, glioma, ependymoma, germinoma (pinealoma), glioblastoma multiform, oligodendrogliorna, schwannoma, retinoblastoma, congenital tumors), spinal cord neurofibroma, neurofibromatosis type 1, meningioma, glioma, sarcoma); Gynecological, for example: uterus (endometrial carcinoma, uterine carcinoma, uterine corpus endometrial carcinoma), cervix (cervical carcinoma, pre-tumor cervical dysplasia), ovaries (ovarian carcinoma (serous cystadenocarcinoma, mucinous cystadenocarcinoma, unclassified carcinoma), granulosa-thecal cell tumors, Sertoli-Leydig cell tumors, dysgerminoma, malignant teratoma), vulva (squamous cell carcinoma, intraepithelial carcinoma, adenocarcinoma, fibrosarcoma, melanoma), vagina (clear cell carcinoma, squamous cell carcinoma, botryoid sarcoma (embryonal rhabdomyosarcoma), fallopian tubes (carcinoma); Hematologic, for example: blood (myeloid leukemia (acute and chronic), acute lymphoblastic leukemia, chronic lymphocytic leukemia, myeloproliferative diseases, multiple myeloma, myelodysplastic syndrome), Hodgkin's disease, non-Hodgkin's lymphoma (malignant lymphoma); Skin, for example: malignant melanoma, basal cell carcinoma, squamous cell carcinoma, Kaposi's sarcoma, moles dysplastic nevi, lipoma, angioma, dermatofibroma, keloids, psoriasis; and Adrenal glands, for example: neuroblastoma.

(245) In some embodiments, the Ras protein is wild-type (Ras.sup.WT). Accordingly, in some embodiments, a compound of the present invention is employed in a method of treating a patient having a cancer comprising a Ras.sup.WT (e.g., K-Ras.sup.WT, H-Ras.sup.WT or N-Ras.sup.WT). In some embodiments, the Ras protein is Ras amplification (e.g., K-Ras.sup.amp). Accordingly, in some embodiments, a compound of the present invention is employed in a method of treating a patient having a cancer comprising a Ras.sup.amp (K-Ras.sup.amp, H-Ras.sup.amp or N-Ras.sup.amp). In some embodiments, the cancer comprises a Ras mutation, such as a Ras mutation described herein. In some embodiments, a mutation is selected from: (a) the following K-Ras mutants: G12D, G12V, G12C, G13D, G12R, G12A, Q61H, G12S, A146T, G13C, Q61L, Q61R, K117N, A146V, G12F, Q61K, L19F, Q22K, V14I, A59T, A146P, G13R, G12L, or G13V, and combinations thereof; (b) the following H-Ras mutants: Q61R, G13R, Q61K, G12S, Q61L, G12D, G13V, G13D, G12C, K117N, A59T, G12V, G13C, Q61H, G13S, A18V, D119N, G13N, A146T, A66T, G12A, A146V, G12N, or G12R, and combinations thereof; and (c) the following N-Ras mutants: Q61R, Q61K, G12D, Q61L, Q61H, G13R, G13D, G12S, G12C, G12V, G12A, G13V, G12R, P185S, G13C, A146T, G60E, Q61P, A59D, E132K, E49K, T50I, A146V, or A59T, and combinations thereof;

or a combination of any of the foregoing. In some embodiments, the cancer comprises a K-Ras mutation selected from the group consisting of G12C, G12D, G13C, G12V, G13D, G12R, G12S, Q61H, Q61K and Q61 L. In some embodiments, the cancer comprises an N-Ras mutation selected from the group consisting of G12C, Q61H, Q61K, Q61L, Q61P and Q61R. In some embodiments, the cancer comprises an H-Ras mutation selected from the group consisting of Q61H and Q61 L. In some embodiments, the cancer comprises a Ras mutation selected from the group consisting of G12C, G13C, G12A, G12D, G13D, G12S, G13S, G12V and G13V. In some embodiments, the cancer comprises at least two Ras mutations selected from the group consisting of G12C, G13C, G12A, G12D, G13D, G12S, G13S, G12V and G13V. In some embodiments, a compound of the present invention inhibits more than one Ras mutant. For example, a compound may inhibit both K-

Ras G12C and K-Ras G13C. A compound may inhibit both N-Ras G12C and K-Ras G12C. In some embodiments, a compound may inhibit both K-Ras G12C and K-Ras G12D. In some embodiments, a compound may inhibit both K-Ras G12V and K-Ras G12C. In some embodiments, a compound may inhibit both K-Ras G12V and K-Ras G12S. In some embodiments, a compound of the present invention inhibits Ras.sup.WT in addition to one or more additional Ras mutations (e.g., K-, H- or N-Ras.sup.WT and K-Ras G12D, G12V, G12C, G13D, G12R, G12A, Q61H, G12S, A146T, G13C, Q61L, Q61R, K117N, A146V, G12F, Q61K, L19F, Q22K, V14I, A59T, A146P, G13R, G12L, or G13V; K, H or N-Ras.sup.WT and H-Ras Q61R, G13R, Q61K, G12S, Q61L, G12D, G13V, G13D, G12C, K117N, A59T, G12V, G13C, Q61H, G13S, A18V, D119N, G13N, A146T, A66T, G12A, A146V, G12N, or G12R; or K, H or N-Ras.sup.WT and N-Ras Q61R, Q61K, G12D, Q61L, Q61H, G13R, G13D, G12S, G12C, G12V, G12A, G13V, G12R, P185S, G13C, A146T, G60E, Q61P, A59D, E132K, E49K, T50I, A146V, or A59T). In some embodiments, a compound of the present invention inhibits Ras.sup.amp in addition to one or more additional Ras mutations (e.g., K-, H- or N-Ras.sup.amp and K-Ras G12D, G12V, G12C, G13D, G12R, G12A, Q61H, G12S, A146T, G13C, Q61L, Q61R, K117N, A146V, G12F, Q61K, L19F, Q22K, V14I, A59T, A146P, G13R, G12L, or G13V; K, H or N-Ras.sup.amp and H-Ras Q61R, G13R, Q61K, G12S, Q61L, G12D, G13V, G13D, G12C, K117N, A59T, G12V, G13C, Q61H, G13S, A18V, D119N, G13N, A146T, A66T, G12A, A146V, G12N, or G12R; or K, H or N-Ras.sup.amp and N-Ras Q61R, Q61K, G12D, Q61L, Q61H, G13R, G13D, G12S, G12C, G12V, G12A, G13V, G12R, P185S, G13C, A146T, G60E, Q61P, A59D, E132K, E49K, T50I, A146V, or A59T).

(246) Methods of detecting Ras mutations are known in the art. Such means include, but are not limited to direct sequencing, and utilization of a high-sensitivity diagnostic assay (with CE-IVD mark), e.g., as described in Domagala, et al., Pol J Pathol 3: 145-164 (2012), incorporated herein by reference in its entirety, including TheraScreen PCR; AmoyDx; PNACIamp; RealQuality; EntroGen; LightMix; StripAssay; Hybcell plexA; Devyser; Surveyor; Cobas; and TheraScreen Pyro. See, also, e.g., WO 2020/106640.

(247) In some embodiments, the cancer is non-small cell lung cancer and the Ras mutation comprises a K-Ras mutation, such as K-Ras G12C, K-Ras G12V or K-Ras G12D. In some embodiments, the cancer is colorectal cancer and the Ras mutation comprises a K-Ras mutation, such as K-Ras G12C, K-Ras G12V or K-Ras G12D. In some embodiments, the cancer is pancreatic cancer and the Ras mutation comprises an K-Ras mutation, such as K-Ras G12D or K-Ras G12V. In some embodiments, the cancer is pancreatic cancer and the Ras mutation comprises an N-Ras mutation, such as N-Ras G12D. In some embodiments, the cancer is melanoma and the Ras mutation comprises an N-Ras mutation, such as N-Ras Q61R or N-Ras Q61K. In some embodiments, the cancer is non-small cell lung cancer and the Ras protein is K-Ras.sup.amP. In any of the foregoing if not already specified, a compound may inhibit Ras.sup.WT (e.g., K-, H- or N-Ras.sup.WT) or Ras.sup.amp (e.g., K-, H- or N-Ras.sup.amp) as well.

(248) In some embodiments, a cancer comprises a Ras mutation and an STK11.sup.LOF, a KEAP1, an EPHA5 or an NF1 mutation. In some embodiments, the cancer is non-small cell lung cancer and comprises a K-Ras G12C mutation. In some embodiments, the cancer is non-small cell lung cancer and comprises a K-Ras G12C mutation and an STK11.sup.LOF mutation. In some embodiments, the cancer is non-small cell lung cancer and comprises a K-Ras G12C mutation and an STK11.sup.LOF mutation. In some embodiments, a cancer comprises a K-Ras G13C Ras mutation and an STK11.sup.LOF, a KEAP1, an EPHA5 or an NF1 mutation. In some embodiments, the cancer is non-small cell lung cancer and comprises a K-Ras G12D mutation. In some embodiments, the cancer is non-small cell lung cancer and comprises a K-Ras G12V mutation. In some embodiments, the cancer is colorectal cancer and comprises a K-Ras G12C mutation. In some embodiments, the cancer is pancreatic cancer and comprises a K-Ras G12C or K-Ras G12D mutation. In some embodiments, the cancer is pancreatic cancer and comprises a K-Ras G12V mutation. In some embodiments, the cancer is endometrial cancer, ovarian cancer, cholangiocarcinoma, or mucinous

appendiceal cancer and comprises a K-Ras G12C mutation. In some embodiments, the cancer is gastric cancer and comprises a K-Ras G12C mutation. In any of the foregoing, a compound may inhibit Ras.sup.WT (e.g., K-, H- or N-Ras.sup.WT) or Ras.sup.amp (e.g., K-, H- or N-Ras.sup.amp) as well.

(249) Also provided is a method of inhibiting a Ras protein in a cell, the method comprising contacting the cell with an effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof. A method of inhibiting RAF-Ras binding, the method comprising contacting the cell with an effective amount of a compound of the present invention, or a pharmaceutically acceptable salt thereof, is also provided. The cell may be a cancer cell. The cancer cell may be of any type of cancer described herein. The cell may be in vivo or in vitro.

(250) Combination Therapy

(251) The methods of the invention may include a compound of the invention used alone or in combination with one or more additional therapies (e.g., non-drug treatments or therapeutic agents). The dosages of one or more of the additional therapies (e.g., non-drug treatments or therapeutic agents) may be reduced from standard dosages when administered alone. For example, doses may be determined empirically from drug combinations and permutations or may be deduced by isobolographic analysis (e.g., Black et al., *Neurology* 65:S3-S6 (2005)).

(252) A compound of the present invention may be administered before, after, or concurrently with one or more of such additional therapies. When combined, dosages of a compound of the invention and dosages of the one or more additional therapies (e.g., non-drug treatment or therapeutic agent) provide a therapeutic effect (e.g., synergistic or additive therapeutic effect). A compound of the present invention and an additional therapy, such as an anti-cancer agent, may be administered together, such as in a unitary pharmaceutical composition, or separately and, when administered separately, this may occur simultaneously or sequentially. Such sequential administration may be close or remote in time.

(253) In some embodiments, the additional therapy is the administration of side-effect limiting agents (e.g., agents intended to lessen the occurrence or severity of side effects of treatment. For example, in some embodiments, the compounds of the present invention can also be used in combination with a therapeutic agent that treats nausea. Examples of agents that can be used to treat nausea include: dronabinol, granisetron, metoclopramide, ondansetron, and prochlorperazine, or pharmaceutically acceptable salts thereof.

(254) In some embodiments, the one or more additional therapies includes a non-drug treatment (e.g., surgery or radiation therapy). In some embodiments, the one or more additional therapies includes a therapeutic agent (e.g., a compound or biologic that is an anti-angiogenic agent, signal transduction inhibitor, antiproliferative agent, glycolysis inhibitor, or autophagy inhibitor). In some embodiments, the one or more additional therapies includes a non-drug treatment (e.g., surgery or radiation therapy) and a therapeutic agent (e.g., a compound or biologic that is an anti-angiogenic agent, signal transduction inhibitor, antiproliferative agent, glycolysis inhibitor, or autophagy inhibitor). In other embodiments, the one or more additional therapies includes two therapeutic agents. In still other embodiments, the one or more additional therapies includes three therapeutic agents. In some embodiments, the one or more additional therapies includes four or more therapeutic agents.

(255) In this Combination Therapy section, all references are incorporated by reference for the agents described, whether explicitly stated as such or not.

(256) Non-Drug Therapies

(257) Examples of non-drug treatments include, but are not limited to, radiation therapy, cryotherapy, hyperthermia, surgery (e.g., surgical excision of tumor tissue), and T cell adoptive transfer (ACT) therapy.

(258) In some embodiments, the compounds of the invention may be used as an adjuvant therapy after surgery. In some embodiments, the compounds of the invention may be used as a neo-adjuvant

therapy prior to surgery.
(259) Radiation therapy may be used for inhibiting abnormal cell growth or treating a hyperproliferative disorder, such as cancer, in a subject (e.g., mammal (e.g., human)). Techniques for administering radiation therapy are known in the art Radiation therapy can be administered through one of several methods, or a combination of methods, including, without limitation, external-beam therapy, internal radiation therapy, implant radiation, stereotactic radiosurgery, systemic radiation therapy, radiotherapy, and permanent or temporary interstitial brachy therapy. The term “brachy therapy,” as used herein, refers to radiation therapy delivered by a spatially confined radioactive material inserted into the body at or near a tumor or other proliferative tissue disease site. The term is intended, without limitation, to include exposure to radioactive isotopes (e.g., At-211, I-131, I-125, Y-90, Re-186, Re-188, Sm-153, Bi-212, P-32, and radioactive isotopes of Lu). Suitable radiation sources for use as a cell conditioner of the present invention include both solids and liquids. By way of non-limiting example, the radiation source can be a radionuclide, such as I-125, I-131, Yb-169, Ir-192 as a solid source, 1-125 as a solid source, or other radionuclides that emit photons, beta particles, gamma radiation, or other therapeutic rays. The radioactive material can also be a fluid made from any solution of radionuclide(s), e.g., a solution of 1-125 or 1-131, or a radioactive fluid can be produced using a slurry of a suitable fluid containing small particles of solid radionuclides, such as Au-198, or Y-90. Moreover, the radionuclide(s) can be embodied in a gel or radioactive micro spheres.
(260) In some embodiments, the compounds of the present invention can render abnormal cells more sensitive to treatment with radiation for purposes of killing or inhibiting the growth of such cells. Accordingly, this invention further relates to a method for sensitizing abnormal cells in a mammal to treatment with radiation which comprises administering to the mammal an amount of a compound of the present invention, which amount is effective to sensitize abnormal cells to treatment with radiation. The amount of the compound in this method can be determined according to the means for ascertaining effective amounts of such compounds described herein. In some embodiments, the compounds of the present invention may be used as an adjuvant therapy after radiation therapy or as a neo-adjuvant therapy prior to radiation therapy.
(261) In some embodiments, the non-drug treatment is a T cell adoptive transfer (ACT) therapy. In some embodiments, the T cell is an activated T cell. The T cell may be modified to express a chimeric antigen receptor (CAR). CAR modified T (CAR-T) cells can be generated by any method known in the art. For example, the CAR-T cells can be generated by introducing a suitable expression vector encoding the CAR to a T cell. Prior to expansion and genetic modification of the T cells, a source of T cells is obtained from a subject. T cells can be obtained from a number of sources, including peripheral blood mononuclear cells, bone marrow, lymph node tissue, cord blood, thymus tissue, tissue from a site of infection, ascites, pleural effusion, spleen tissue, and tumors. In certain embodiments of the present invention, any number of T cell lines available in the art may be used. In some embodiments, the T cell is an autologous T cell. Whether prior to or after genetic modification of the T cells to express a desirable protein (e.g., a CAR), the T cells can be activated and expanded generally using methods as described, for example, in U.S. Pat. Nos. 6,352,694; 6,534,055; 6,905,680; 6,692,964; 5,858,358; 6,887,466; 6,905,681; 7,144,575; 7,067,318; 7,172,869; 7,232,566; 7,175,843; 7,572,631; 5,883,223; 6,905,874; 6,797,514; and 6,867,041.
(262) Therapeutic Agents
(263) A therapeutic agent may be a compound used in the treatment of cancer or symptoms associated therewith.
(264) For example, a therapeutic agent may be a steroid. Accordingly, in some embodiments, the one or more additional therapies includes a steroid. Suitable steroids may include, but are not limited to, 21-acetoxypregnenolone, alclometasone, algestone, amcinonide, beclomethasone, betamethasone, budesonide, chloroprednisone, clobetasol, clocortolone, cloprednol, corticosterone, cortisone, cortivazol, deflazacort, desonide, desoximetasone, dexamethasone, diflorasone, diflucortolone,

difuprednate, enoxolone, fluazacort, fiucloronide, flumethasone, flunisolide, fluocinolone acetonide, fluocinonide, fluocortin butyl, fluocortolone, fluorometholone, fluperolone acetate, fluprednidene acetate, fluprednisolone, flurandrenolide, fluticasone propionate, formocortal, halcinonide, halobetasol propionate, halometasone, hydrocortisone, loteprednol etabonate, mazipredone, medrysone, meprednisone, methylprednisolone, mometasone furoate, paramethasone, prednicarbate, prednisolone, prednisolone 25-diethylaminoacetate, prednisolone sodium phosphate, prednisone, prednival, prednylidene, rimexolone, tixocortol, triamcinolone, triamcinolone acetonide, triamcinolone benetonide, triamcinolone hexacetonide, and salts or derivatives thereof.

(265) Further examples of therapeutic agents that may be used in combination therapy with a compound of the present invention include compounds described in the following patents: U.S. Pat. Nos. 6,258,812, 6,630,500, 6,515,004, 6,713,485, 5,521,184, 5,770,599, 5,747,498, 5,990,141, 6,235,764, and 8,623,885, and International Patent Applications WO01/37820, WO01/32651, WO02/68406, WO02/66470, WO02/55501, WO04/05279, WO04/07481, WO04/07458, WO04/09784, WO02/59110, WO99/45009, WO00/59509, WO99/61422, WO00/12089, and WO00/02871.

(266) A therapeutic agent may be a biologic (e.g., cytokine (e.g., interferon or an interleukin such as IL-2)) used in treatment of cancer or symptoms associated therewith. In some embodiments, the biologic is an immunoglobulin-based biologic, e.g., a monoclonal antibody (e.g., a humanized antibody, a fully human antibody, an Fc fusion protein, or a functional fragment thereof) that agonizes a target to stimulate an anti-cancer response or antagonizes an antigen important for cancer. Also included are antibody-drug conjugates.

(267) A therapeutic agent may be a T-cell checkpoint inhibitor. In one embodiment, the checkpoint inhibitor is an inhibitory antibody (e.g., a monospecific antibody such as a monoclonal antibody). The antibody may be, e.g., humanized or fully human. In some embodiments, the checkpoint inhibitor is a fusion protein, e.g., an Fc-receptor fusion protein. In some embodiments, the checkpoint inhibitor is an agent, such as an antibody, that interacts with a checkpoint protein. In some embodiments, the checkpoint inhibitor is an agent, such as an antibody, that interacts with the ligand of a checkpoint protein. In some embodiments, the checkpoint inhibitor is an inhibitor (e.g., an inhibitory antibody or small molecule inhibitor) of CTLA-4 (e.g., an anti-CTLA-4 antibody or fusion a protein). In some embodiments, the checkpoint inhibitor is an inhibitor or antagonist (e.g., an inhibitory antibody or small molecule inhibitor) of PD-1. In some embodiments, the checkpoint inhibitor is an inhibitor or antagonist (e.g., an inhibitory antibody or small molecule inhibitor) of PDL-1. In some embodiments, the checkpoint inhibitor is an inhibitor or antagonist (e.g., an inhibitory antibody or Fc fusion or small molecule inhibitor) of PDL-2 (e.g., a PDL-2/lg fusion protein). In some embodiments, the checkpoint inhibitor is an inhibitor or antagonist (e.g., an inhibitory antibody or small molecule inhibitor) of B7-H3, B7-H4, BTLA, HVEM, TIM3, GAL9, LAG3, VISTA, KIR, 2B4, CD160, CGEN-15049, CHK 1, CHK2, A2aR, B-7 family ligands, or a combination thereof. In some embodiments, the checkpoint inhibitor is pembrolizumab, nivolumab, PDR001 (NVS), REGN.sub.2810 (Sanofi/Regeneron), a PD-L1 antibody such as, e.g., avelumab, durvalumab, atezolizumab, pidilizumab, JNJ-63723283 (JNJ), BGB-A317 (BeiGene & Celgene) or a checkpoint inhibitor disclosed in Preusser, M. et al. (2015) Nat. Rev. Neurol., including, without limitation, ipilimumab, tremelimumab, nivolumab, pembrolizumab, AMP224, AMP514/MEDI0680, BMS936559, MEDI4736, MPDL3280A, MSB0010718C, BMS986016, IMP321, lirilumab, IPH2101, 1-7F9, and KW-6002.

(268) A therapeutic agent may be an anti-TIGIT antibody, such as MBSA43, BMS-986207, MK-7684, COM902, AB154, MTIG7192A or OMP-313M32 (etigilimab).

(269) A therapeutic agent may be an agent that treats cancer or symptoms associated therewith (e.g., a cytotoxic agent, non-peptide small molecules, or other compound useful in the treatment of cancer or symptoms associated therewith, collectively, an “anti-cancer agent”). Anti-cancer agents can be, e.g., chemotherapeutics or targeted therapy agents.

(270) Anti-cancer agents include mitotic inhibitors, intercalating antibiotics, growth factor inhibitors, cell cycle inhibitors, enzymes, topoisomerase inhibitors, biological response modifiers, alkylating agents, antimetabolites, folic acid analogs, pyrimidine analogs, purine analogs and related inhibitors, *vinca* alkaloids, epipodopyyllotoxins, antibiotics, L-Asparaginase, topoisomerase inhibitors, interferons, platinum coordination complexes, anthracenedione substituted urea, methyl hydrazine derivatives, adrenocortical suppressant, adrenocorticosteroides, progestins, estrogens, antiestrogen, androgens, antiandrogen, and gonadotropin-releasing hormone analog. Further anti-cancer agents include leucovorin (LV), irenotecan, oxaliplatin, capecitabine, paclitaxel, and doxetaxel. In some embodiments, the one or more additional therapies includes two or more anti-cancer agents. The two or more anti-cancer agents can be used in a cocktail to be administered in combination or administered separately. Suitable dosing regimens of combination anti-cancer agents are known in the art and described in, for example, Saltz et al., *Proc. Am. Soc. Clin. Oncol.* 18:233a (1999), and Douillard et al., Lancet 355(9209):1041-1047 (2000).
(271) Other non-limiting examples of anti-cancer agents include Gleevec® (Imatinib Mesylate); Kyprolis® (carfilzomib); Velcade® (bortezomib); Casodex (bicalutamide); Iressa® (gefitinib); alkylating agents such as thiotepa and cyclosphosphamide; alkyl sulfonates such as busulfan, improsulfan and piposulfan; aziridines such as benzodopa, carboquone, meturedopa, and uredopa; ethylenimines and methylamelamines including altretamine, triethylenemelamine, triethylenephosphoramide, triethiylenethiophosphoramide and trimethylolomelamine; acetogenins (especially bullatacin and bullatacinone); a camptothecin (including the synthetic analogue topotecan); bryostatin; callystatin; CC-1065 (including its adozelesin, carzelesin and bizelesin synthetic analogues); cryptophycins (particularly cryptophycin 1 and cryptophycin 8); dolastatin; duocarmycin (including the synthetic analogues, KW-2189 and CB1-TM1); eleutherobin; pancratistatin; sarcodictyin A; spongistatin; nitrogen mustards such as chlorambucil, chlornaphazine, cholophosphamide, estramustine, ifosfamide, mechlorethamine, mechlorethamine oxide hydrochloride, melphalan, novembichin, phenesterine, prednimustine, trofosfamide, uracil mustard; nitrosureas such as carmustine, chlorozotocin, fotemustine, lomustine, nimustine, and ranimustine; antibiotics such as the enediyne antibiotics (e.g., calicheamicin, such as calicheamicin gammall and calicheamicin omegall (see, e.g., Agnew, *Chem. Intl. Ed Engl.* 33:183-186 (1994)); dynemicin such as dynemicin A; bisphosphonates such as clodronate; an esperamicin; neocarzinostatin chromophore and related chromoprotein enediyne antiobiotic chromophores, aclacinomysins, actinomycin, authramycin, azaserine, bleomycins, cactinomycin, calicheamicin, carabicin, caminomycin, carminomycin, carzinophilin, chromomycins, dactinomycin, daunorubicin, detorubicin, 6-diazo-5-oxo-L-norleucine, adriamycin (doxorubicin), morpholino-doxorubicin, cyanomorpholino-doxorubicin, 2-pyrrolino-doxorubicin, deoxydoxorubicin, epirubicin, esorubicin, idarubicin, marcellomycin, mitomycins such as mitomycin C, mycophenolic acid, nogalamycin, olivomycins, peplomycin, potfiromycin, puromycin, quelamycin, rodorubicin, streptonigrin, streptozocin, tubercidin, ubenimex, zinostatin, zorubicin; anti-metabolites such as methotrexate and 5-fluorouracil (5-FU); folic acid analogues such as denopterin, pteropterin, trimetrexate; purine analogs such as fludarabine, 6-mercaptopurine, thiamiprine, thioguanine; pyrimidine analogs such as ancitabine, azacitidine, 6-azauridine, carmofur, cytarabine, dideoxyuridine, doxifluridine, enocitabine, floxuridine; androgens such as calusterone, dromostanolone propionate, epitiostanol, mepitiostane, testolactone; anti-adrenals such as aminoglutethimide, mitotane, trilostane; folic acid replenishers such as frolinic acid; aceglatone; aldophosphamide glycoside; aminolevulinic acid; eniluracil; amsacrine; bestrabucil; bisantrene; edatraxate; defofamine; demecolcine; diaziquone; elfomithine; elliptinium acetate; an epothilone such as epothilone B; etoglucid; gallium nitrate; hydroxyurea; lentinan; lonidamine; maytansinoids such as maytansine and ansamitocins; mitoguazone; mitoxantrone; mopidamol; nitracrine; pentostatin; phenamet; pirarubicin; losoxantrone; podophyllinic acid; 2-ethylhydrazide; procarbazine; PSK® polysaccharide complex (JHS Natural Products, Eugene, OR); razoxane; rhizoxin; sizofiran; spirogermanium; tenuazonic acid; triaziquone;

2,2′,2″-trichlorotriethylamine; trichothecenes such as T-2 toxin, verracurin A, roridin A and anguidine; urethane; vindesine; dacarbazine; mannomustine; mitobronitol; mitolactol; pipobroman; gacytosine; arabinoside ("Ara-C"); cyclophosphamide; thiotepa; taxoids, e.g., Taxol® (paclitaxel), Abraxane® (cremophor-free, albumin-engineered nanoparticle formulation of paclitaxel), and Taxotere® (doxetaxel); chloranbucil; tamoxifen (Nolvadex™); raloxifene; aromatase inhibiting 4(5)-imidazoles; 4-hydroxytamoxifen; trioxifene; keoxifene; LY 117018; onapristone; toremifene (Fareston®); flutamide, nilutamide, bicalutamide, leuprolide, goserelin; chlorambucil; Gemzar® gemcitabine; 6-thioguanine; mercaptopurine; platinum coordination complexes such as cisplatin, oxaliplatin and carboplatin; vinblastine; platinum; etoposide (VP-16); ifosfamide; mitoxantrone; vincristine; Navelbine® (vinorelbine); novantrone; teniposide; edatrexate; daunomycin; aminopterin; ibandronate; irinotecan (e.g., CPT-11); topoisomerase inhibitor RFS 2000; difluoromethylornithine (DMFO); retinoids such as retinoic acid; esperamicins; capecitabine (e.g., Xeloda®); and pharmaceutically acceptable salts of any of the above.

(272) Additional non-limiting examples of anti-cancer agents include trastuzumab (Herceptin®), bevacizumab (Avastin®), cetuximab (Erbitux®), rituximab (Rituxan®), Taxol®, Arimidex®, ABVD, avicine, abagovomab, acridine carboxamide, adecatumumab, 17-N-allylamino-17-demethoxygeldanamycin, alpharadin, alvocidib, 3-aminopyridine-2-carboxaldehyde thiosemicarbazone, amonafide, anthracenedione, anti-CD22 immunotoxins, antineoplastics (e.g., cell-cycle nonspecific antineoplastic agents, and other antineoplastics described herein), antitumorigenic herbs, apaziquone, atiprimod, azathioprine, belotecan, bendamustine, BIBW 2992, biricodar, brostallicin, bryostatin, buthionine sulfoximine, CBV (chemotherapy), calyculin, dichloroacetic acid, discodermolide, elsamitrucin, enocitabine, eribulin, exatecan, exisulind, ferruginol, forodesine, fosfestrol, ICE chemotherapy regimen, IT-101, imexon, imiquimod, indolocarbazole, irofulven, laniquidar, larotaxel, lenalidomide, lucanthone, lurtotecan, mafosfamide, mitozolomide, nafoxidine, nedaplatin, olaparib, ortataxel, PAC-1, pawpaw, pixantrone, proteasome inhibitors, rebeccamycin, resiquimod, rubitecan, SN-38, salinosporamide A, sapacitabine, Stanford V, swainsonine, talaporfin, tariquidar, tegafur-uracil, temodar, tesetaxel, triplatin tetranitrate, tris(2-chloroethyl)amine, troxacitabine, uramustine, vadimezan, vinflunine, ZD6126, and zosuquidar.

(273) Further non-limiting examples of anti-cancer agents include natural products such as vinca alkaloids (e.g., vinblastine, vincristine, and vinorelbine), epidipodophyllotoxins (e.g., etoposide and teniposide), antibiotics (e.g., dactinomycin (actinomycin D), daunorubicin, and idarubicin), anthracyclines, mitoxantrone, bleomycins, plicamycin (mithramycin), mitomycin, enzymes (e.g., L-asparaginase which systemically metabolizes L-asparagine and deprives cells which do not have the capacity to synthesize their own asparagine), antiplatelet agents, antiproliferative/antimitotic alkylating agents such as nitrogen mustards (e.g., mechlorethamine, cyclophosphamide and analogs, melphalan, and chlorambucil), ethylenimines and methylmelamines (e.g., hexaamethylmelaamine and thiotepa), CDK inhibitors (e.g., a CDK4/6 inhibitor such as abemaciclib, ribociclib, palbociclib; seliciclib, UCN-01, P1446A-05, PD-0332991, dinaciclib, P27-00, AT-7519, RGB286638, and SCH727965), alkyl sulfonates (e.g., busulfan), nitrosoureas (e.g., carmustine (BCNU) and analogs, and streptozocin), trazenes-dacarbazinine (DTIC), antiproliferative/antimitotic antimetabolites such as folic acid analogs, pyrimidine analogs (e.g., fluorouracil, floxuridine, and cytarabine), purine analogs and related inhibitors (e.g., mercaptopurine, thioguanine, pentostatin, and 2-chlorodeoxyadenosine), aromatase inhibitors (e.g., anastrozole, exemestane, and letrozole), and platinum coordination complexes (e.g., cisplatin and carboplatin), procarbazine, hydroxyurea, mitotane, aminoglutethimide, histone deacetylase (HDAC) inhibitors (e.g., trichostatin, sodium butyrate, apicidan, suberoyl anilide hydroamic acid, vorinostat, LBH 589, romidepsin, ACY-1215, and panobinostat), mTOR inhibitors (e.g., vistusertib, temsirolimus, everolimus, ridaforolimus, and sirolimus), KSP(Eg5) inhibitors (e.g., Array 520), DNA binding agents (e.g., Zalypsis®), PI3K inhibitors such as PI3K delta inhibitor (e.g., GS-1101 and TGR-1202), PI3K delta and gamma inhibitor (e.g., CAL-130), copanlisib, alpelisib and idelalisib; multi-kinase inhibitor (e.g., TG02 and

sorafenib), hormones (e.g., estrogen) and hormone agonists such as leutinizing hormone releasing hormone (LHRH) agonists (e.g., goserelin, leuprolide and triptorelin), BAFF-neutralizing antibody (e.g., LY2127399), IKK inhibitors, p38MAPK inhibitors, anti-IL-6 (e.g., CNT0328), telomerase inhibitors (e.g., GRN 163L), aurora kinase inhibitors (e.g., MLN8237), cell surface monoclonal antibodies (e.g., anti-CD38 (HUMAX-CD38), anti-CSI (e.g., elotuzumab), HSP90 inhibitors (e.g., 17 AAG and KOS 953), PI3K/Akt inhibitors (e.g., perifosine), Akt inhibitors (e.g., GSK-2141795), PKC inhibitors (e.g., enzastaurin), FTIs (e.g., Zarnestra™), anti-CD138 (e.g., BT062), Torcl/2 specific kinase inhibitors (e.g., INK128), ER/UPR targeting agents (e.g., MKC-3946), cFMS inhibitors (e.g., ARRY-382), JAK1/2 inhibitors (e.g., CYT387), PARP inhibitors (e.g., olaparib and veliparib (ABT-888)), and BCL-2 antagonists.

(274) In some embodiments, an anti-cancer agent is selected from mechlorethamine, camptothecin, ifosfamide, tamoxifen, raloxifene, gemcitabine, Navelbine®, sorafenib, or any analog or derivative variant of the foregoing.

(275) In some embodiments, the anti-cancer agent is a HER2 inhibitor. Non-limiting examples of HER2 inhibitors include monoclonal antibodies such as trastuzumab (Herceptin®) and pertuzumab (Perjeta®); small molecule tyrosine kinase inhibitors such as gefitinib (Iressa®), erlotinib (Tarceva®), pilitinib, CP-654577, CP-724714, canertinib (CI 1033), HKI-272, lapatinib (GW-572016; Tykerb®), PKI-166, AEE788, BMS-599626, HKI-357, BIBW 2992, ARRY-334543, and JNJ-26483327.

(276) In some embodiments, an anti-cancer agent is an ALK inhibitor. Non-limiting examples of ALK inhibitors include ceritinib, TAE-684 (NVP-TAE694), PF02341066 (crizotinib or 1066), alectinib; brigatinib; entrectinib; ensartinib (X-396); lorlatinib; ASP3026; CEP-37440; 4SC-203; TL-398; PLB1003; TSR-011; CT-707; TPX-0005, and AP26113. Additional examples of ALK kinase inhibitors are described in examples 3-39 of WO05016894.

(277) In some embodiments, an anti-cancer agent is an inhibitor of a member downstream of a Receptor Tyrosine Kinase (RTK)/Growth Factor Receptor (e.g., a SHP2 inhibitor (e.g., SHP099, TNO155, RMC-4550, RMC-4630, JAB-3068, RLY-1971), a SOS1 inhibitor (e.g., BI-1701963, BI-3406), a Raf inhibitor, a MEK inhibitor, an ERK inhibitor, a PI3K inhibitor, a PTEN inhibitor, an AKT inhibitor, or an mTOR inhibitor (e.g., mTORC1 inhibitor or mTORC2 inhibitor). In some embodiments, the anti-cancer agent is JAB-3312. In some embodiments, an anti-cancer agent is an additional Ras inhibitor (e.g., AMG 510, MRTX1257, MRTX849, JNJ-74699157 (ARS-3248), LY3499446, ARS-853, or ARS-1620), or a Ras vaccine, or another therapeutic modality designed to directly or indirectly decrease the oncogenic activity of Ras. Other examples of Ras inhibitors that may be combined with a Ras inhibitor of the present invention are provided in the following, incorporated herein by reference in their entireties: WO 2020050890, WO 2020047192, WO 2020035031, WO 2020028706, WO 2019241157, WO 2019232419, WO 2019217691, WO 2019217307, WO 2019215203, WO 2019213526, WO 2019213516, WO 2019155399, WO 2019150305, WO 2019110751, WO 2019099524, WO 2019051291, WO 2018218070, WO 2018217651, WO 2018218071, WO 2018218069, WO 2018206539, WO 2018143315, WO 2018140600, WO 2018140599, WO 2018140598, WO 2018140514, WO 2018140513, WO 2018140512, WO 2018119183, WO 2018112420, WO 2018068017, WO 2018064510, WO 2017201161, WO 2017172979, WO 2017100546, WO 2017087528, WO 2017058807, WO 2017058805, WO 2017058728, WO 2017058902, WO 2017058792, WO 2017058768, WO 2017058915, WO 2017015562, WO 2016168540, WO 2016164675, WO 2016049568, WO 2016049524, WO 2015054572, WO 2014152588, WO 2014143659 and WO 2013155223.

(278) In some embodiments, a therapeutic agent that may be combined with a compound of the present invention is an inhibitor of the MAP kinase (MAPK) pathway (or “MAPK inhibitor”). MAPK inhibitors include, but are not limited to, one or more MAPK inhibitor described in Cancers (Basel) 2015 September; 7(3): 1758-1784. For example, the MAPK inhibitor may be selected from one or more of trametinib, binimetinib, selumetinib, cobimetinib, LErafAON (NeoPharm), ISIS

5132; vemurafenib, pimasertib, TAK733, RO4987655 (CH4987655); CI-1040; PD-0325901; CH5126766; MAP855; AZD6244; refametinib (RDEA 119/BAY 86-9766); GDC-0973/XL581; AZD8330 (ARRY-424704/ARRY-704); R05126766 (Roche, described in PLoS One. 2014 Nov. 25; 9(11)); and GSK1120212 (or JTP-74057, described in Clin Cancer Res. 2011 Mar. 1; 17(5):989-1000). The MAPK inhibitor may be PLX8394, LXH254, GDC-5573, or LY3009120.
(279) In some embodiments, an anti-cancer agent is a disrupter or inhibitor of the RAS-RAF-ERK or PI3K-AKT-TOR or PI3K-AKT signaling pathways. The PI3K/AKT inhibitor may include, but is not limited to, one or more PI3K/AKT inhibitor described in Cancers (Basel) 2015 September; 7(3): 1758-1784. For example, the PI3K/AKT inhibitor may be selected from one or more of NVP-BEZ235; BGT226; XL765/SAR245409; SF1126; GDC-0980; PI-103; PF-04691502; PKI-587; GSK2126458.
(280) In some embodiments, an anti-cancer agent is a PD-1 or PD-L1 antagonist.
(281) In some embodiments, additional therapeutic agents include ALK inhibitors, HER2 inhibitors, EGFR inhibitors, IGF-1R inhibitors, MEK inhibitors, PI3K inhibitors, AKT inhibitors, TOR inhibitors, MCL-1 inhibitors, BCL-2 inhibitors, SHP2 inhibitors, proteasome inhibitors, and immune therapies. In some embodiments, a therapeutic agent may be a pan-RTK inhibitor, such as afatinib.
(282) IGF-1R inhibitors include linsitinib, or a pharmaceutically acceptable salt thereof.
(283) EGFR inhibitors include, but are not limited to, small molecule antagonists, antibody inhibitors, or specific antisense nucleotide or siRNA. Useful antibody inhibitors of EGFR include cetuximab (Erbitux®), panitumumab (Vectibix®), zalutumumab, nimotuzumab, and matuzumab. Further antibody-based EGFR inhibitors include any anti-EGFR antibody or antibody fragment that can partially or completely block EGFR activation by its natural ligand. Non-limiting examples of antibody-based EGFR inhibitors include those described in Modjtahedi et al., Br. J. Cancer 1993, 67:247-253; Teramoto et al., Cancer 1996, 77:639-645; Goldstein et al., Clin. Cancer Res. 1995, 1:1311-1318; Huang et al., 1999, Cancer Res. 15:59(8):1935-40; and Yang et al., Cancer Res. 1999, 59:1236-1243. The EGFR inhibitor can be monoclonal antibody Mab E7.6.3 (Yang, 1999 supra), or Mab C225 (ATCC Accession No. HB-8508), or an antibody or antibody fragment having the binding specificity thereof.
(284) Small molecule antagonists of EGFR include gefitinib (Iressa®), erlotinib (Tarceva®), and lapatinib (TykerB®). See, e.g., Yan et al., Pharmacogenetics and Pharmacogenomics In Oncology Therapeutic Antibody Development, BioTechniques 2005, 39(4):565-8; and Paez et al., EGFR Mutations In Lung Cancer Correlation With Clinical Response To Gefitinib Therapy, Science 2004, 304(5676):1497-500. In some embodiments, the EGFR inhibitor is osimertinib (Tagrisso®). Further non-limiting examples of small molecule EGFR inhibitors include any of the EGFR inhibitors described in the following patent publications, and all pharmaceutically acceptable salts of such EGFR inhibitors: EP 0520722; EP 0566226; WO96/33980; U.S. Pat. No. 5,747,498; WO96/30347; EP 0787772; WO97/30034; WO97/30044; WO97/38994; WO97/49688; EP 837063; WO98/02434; WO97/38983; WO95/19774; WO95/19970; WO97/13771; WO98/02437; WO98/02438; WO97/32881; DE 19629652; WO98/33798; WO97/32880; WO97/32880; EP 682027; WO97/02266; WO97/27199; WO98/07726; WO97/34895; WO96/31510; WO98/14449; WO98/14450; WO98/14451; WO95/09847; WO97/19065; WO98/17662; U.S. Pat. Nos. 5,789,427; 5,650,415; 5,656,643; WO99/35146; WO99/35132; WO99/07701; and WO92/20642. Additional non-limiting examples of small molecule EGFR inhibitors include any of the EGFR inhibitors described in Traxler et al., Exp. Opin. Ther. Patents 1998, 8(12):1599-1625. In some embodiments, an EGFR inhibitor is an ERBB inhibitor. In humans, the ERBB family contains HER1 (EGFR, ERBB1), HER2 (NEU, ERBB2), HER3 (ERBB3), and HER (ERBB4).
(285) MEK inhibitors include, but are not limited to, pimasertib, selumetinib, cobimetinib (Cotellic®), trametinib (Mekinist®), and binimetinib (Mektovi®). In some embodiments, a MEK inhibitor targets a MEK mutation that is a Class I MEK1 mutation selected from D67N; P124L; P124S; and L177V. In some embodiments, the MEK mutation is a Class II MEK1 mutation selected

from ΔE51-Q58; ΔF53-Q58; E203K; L177M; C121S; F53L; K57E; Q56P; and K57N.
(286) PI3K inhibitors include, but are not limited to, wortmannin; 17-hydroxywortmannin analogs described in WO06/044453; 4-[2-(1H-Indazol-4-yl)-6-[[4-(methylsulfonyl)piperazin-1-yl]methyl]thieno[3,2-d]pyrimidin-4-yl]morpholine (also known as pictilisib or GDC-0941 and described in WO09/036082 and WO09/055730); 2-methyl-2-[4-[3-methyl-2-oxo-8-(quinolin-3-yl)-2,3-dihydroimidazo[4,5-c]quinolin-1-yl]phenyl]propionitrile (also known as BEZ 235 or NVP-BEZ 235, and described in WO06/122806); (S)—I-(4-((2-(2-aminopyrimidin-5-yl)-7-methyl-4-morpholinothieno[3,2-d]pyrimidin-6-yl)methyl)piperazin-1-yl)-2-hydroxypropan-1-one (described in WO08/070740); LY294002 (2-(4-morpholinyl)-8-phenyl-4H-1-benzopyran-4-one (available from Axon Medchem); PI 103 hydrochloride (3-[4-(4-morpholinylpyrido-[3′,2′:4,5]furo[3,2-d]pyrimidin-2-yl]phenol hydrochloride (available from Axon Medchem); PIK 75 (2-methyl-5-nitro-2-[(6-bromoimidazo[1,2-a]pyridin-3-yl)methylene]-1-methylhydrazide-benzenesulfonic acid, monohydrochloride) (available from Axon Medchem); PIK 90 (N-(7,8-dimethoxy-2,3-dihydro-imidazo[1,2-c]quinazolin-5-yl)-nicotinamide (available from Axon Medchem); AS-252424 (5-[1-[5-(4-fluoro-2-hydroxy-phenyl)-furan-2-yl]-meth-(Z)-ylidene]-thiazolidine-2,4-dione (available from Axon Medchem); TGX-221 (7-methyl-2-(4-morpholinyl)-9-[1-(phenylamino)ethyl]-4H-pyrido-[1,2-a]pyrirnidin-4-one (available from Axon Medchem); XL-765; and XL-147. Other PI3K inhibitors include demethoxyviridin, perifosine, CAL101, PX-866, BEZ235, SF1126, INK1117, IPI-145, BKM120, XL147, XL765, Palomid 529, GSK1059615, ZSTK474, PWT33597, IC87114, TGI 00-115, CAL263, PI-103, GNE-477, CUDC-907, and AEZS-136.
(287) AKT inhibitors include, but are not limited to, Akt-1-1 (inhibits Aktl) (Barnett et al., Biochem. J. 2005, 385(Pt. 2): 399-408); Akt-1-1,2 (inhibits Akl and 2) (Barnett et al., Biochem. J. 2005, 385(Pt. 2): 399-408); API-59CJ-Ome (e.g., Jin et al., Br. J. Cancer 2004, 91:1808-12); 1-H-imidazo[4,5-c]pyridinyl compounds (e.g., WO 05/011700); indole-3-carbinol and derivatives thereof (e.g., U.S. Pat. No. 6,656,963; Sarkar and Li J Nutr. 2004, 134(12 Suppl):3493S-3498S); perifosine (e.g., interferes with Akt membrane localization; Dasmahapatra et al. Clin. Cancer Res. 2004, 10(15):5242-52); phosphatidylinositol ether lipid analogues (e.g., Gills and Dennis Expert. Opin. Investig. Drugs 2004, 13:787-97); and triciribine (TCN or API-2 or NCI identifier: NSC 154020; Yang et al., Cancer Res. 2004, 64:4394-9).
(288) mTOR inhibitors include, but are not limited to, ATP-competitive mTORC1/mTORC2 inhibitors, e.g., PI-103, PP242, PP30; Torin 1; FKBP12 enhancers; 4H-1-benzopyran-4-one derivatives; and rapamycin (also known as sirolimus) and derivatives thereof, including: temsirolimus (Torisel®); everolimus (Afinitor®; WO94/09010); ridaforolimus (also known as deforolimus or AP23573); rapalogs, e.g., as disclosed in WO98/02441 and WO01/14387, e.g. AP23464 and AP23841; 40-(2-hydroxyethyl)rapamycin; 40-[3-hydroxy(hydroxymethyl)methylpropanoate]-rapamycin (also known as CC1779); 40-epi-(tetrazolyt)-rapamycin (also called ABT578); 32-deoxorapamycin; 16-pentynyloxy-32(S)-dihydrorapanycin; derivatives disclosed in WO05/005434; derivatives disclosed in U.S. Pat. Nos. 5,258,389, 5,118,677, 5,118,678, 5,100,883, 5,151,413, 5,120,842, and 5,256,790, and in WO94/090101, WO92/05179, WO93/111130, WO94/02136, WO94/02485, WO95/14023, WO94/02136, WO95/16691, WO96/41807, WO96/41807, and WO2018204416; and phosphorus-containing rapamycin derivatives (e.g., WO05/016252). In some embodiments, the mTOR inhibitor is a bisteric inhibitor (see, e.g., WO2018204416, WO2019212990 and WO2019212991), such as RMC-5552.
(289) BRAF inhibitors that may be used in combination with compounds of the invention include, for example, vemurafenib, dabrafenib, and encorafenib. A BRAF may comprise a Class 3 BRAF mutation. In some embodiments, the Class 3 BRAF mutation is selected from one or more of the following amino acid substitutions in human BRAF: D287H; P367R; V459L; G466V; G466E; G466A; S467L; G469E; N581 S; N581 I; D594N; D594G; D594A; D594H; F595L; G596D; G596R and A762E.

(290) MCL-1 inhibitors include, but are not limited to, AMG-176, MIK665, and S63845. The myeloid cell leukemia-1 (MCL-1) protein is one of the key anti-apoptotic members of the B-cell lymphoma-2 (BCL-2) protein family. Over-expression of MCL-1 has been closely related to tumor progression as well as to resistance, not only to traditional chemotherapies but also to targeted therapeutics including BCL-2 inhibitors such as ABT-263.
(291) In some embodiments, the additional therapeutic agent is a SHP2 inhibitor. SHP2 is a non-receptor protein tyrosine phosphatase encoded by the PTPN11 gene that contributes to multiple cellular functions including proliferation, differentiation, cell cycle maintenance and migration. SHP2 has two N-terminal Src homology 2 domains (N—SH2 and C—SH2), a catalytic domain (PTP), and a C-terminal tail. The two SH2 domains control the subcellular localization and functional regulation of SHP2. The molecule exists in an inactive, self-inhibited conformation stabilized by a binding network involving residues from both the N—SH2 and PTP domains. Stimulation by, for example, cytokines or growth factors acting through receptor tyrosine kinases (RTKs) leads to exposure of the catalytic site resulting in enzymatic activation of SHP2.
(292) SHP2 is involved in signaling through the RAS-mitogen-activated protein kinase (MAPK), the JAK-STAT or the phosphoinositol 3-kinase-AKT pathways. Mutations in the PTPN11 gene and subsequently in SHP2 have been identified in several human developmental diseases, such as Noonan Syndrome and Leopard Syndrome, as well as human cancers, such as juvenile myelomonocytic leukemia, neuroblastoma, melanoma, acute myeloid leukemia and cancers of the breast, lung and colon. Some of these mutations destabilize the auto-inhibited conformation of SHP2 and promote autoactivation or enhanced growth factor driven activation of SHP2. SHP2, therefore, represents a highly attractive target for the development of novel therapies for the treatment of various diseases including cancer. A SHP2 inhibitor (e.g., RMC-4550 or SHP099) in combination with a RAS pathway inhibitor (e.g., a MEK inhibitor) have been shown to inhibit the proliferation of multiple cancer cell lines in vitro (e.g., pancreas, lung, ovarian and breast cancer). Thus, combination therapy involving a SHP2 inhibitor with a RAS pathway inhibitor could be a general strategy for preventing tumor resistance in a wide range of malignancies.
(293) Non-limiting examples of such SHP2 inhibitors that are known in the art, include: Chen et al. *Mol Pharmacol.* 2006, 70, 562; Sarver et al., *J. Med. Chem.* 2017, 62, 1793; Xie et al., *J. Med. Chem.* 2017, 60, 113734; and Igbe et al., *Oncotarget*, 2017, 8, 113734; and PCT applications: WO2015107493; WO2015107494; WO201507495; WO2016203404; WO2016203405; WO2016203406; WO2011022440; WO2017156397; WO2017079723; WO2017211303; WO2012041524; WO2017211303; WO2019051084; WO2017211303; US20160030594; US20110281942; WO2010011666; WO2014113584; WO2014176488; WO2017100279; WO2019051469; U.S. Pat. No. 8,637,684; WO2007117699; WO2015003094; WO2005094314; WO2008124815; WO2009049098; WO2009135000; WO2016191328; WO2016196591; WO2017078499; WO2017210134; WO2018013597; WO2018129402; WO2018130928; WO20181309928; WO2018136264; WO2018136265; WO2018160731; WO2018172984; and WO2010121212, each of which is incorporated herein by reference.
(294) In some embodiments, a SHP2 inhibitor binds in the active site. In some embodiments, a SHP2 inhibitor is a mixed-type irreversible inhibitor. In some embodiments, a SHP2 inhibitor binds an allosteric site e.g., a non-covalent allosteric inhibitor. In some embodiments, a SHP2 inhibitor is a covalent SHP2 inhibitor, such as an inhibitor that targets the cysteine residue (C333) that lies outside the phosphatase's active site. In some embodiments a SHP2 inhibitor is a reversible inhibitor. In some embodiments, a SHP2 inhibitor is an irreversible inhibitor. In some embodiments, the SHP2 inhibitor is SHP099. In some embodiments, the SHP2 inhibitor is TNO155. In some embodiments, the SHP2 inhibitor is RMC-4550. In some embodiments, the SHP2 inhibitor is RMC-4630. In some embodiments, the SHP2 inhibitor is JAB-3068. In some embodiments, the SHP2 inhibitor is RLY-1971.
(295) In some embodiments, the additional therapeutic agent is selected from the group consisting of

a MEK inhibitor, a HER2 inhibitor, a SHP2 inhibitor, a CDK4/6 inhibitor, an mTOR inhibitor, a SOS1 inhibitor, and a PD-L1 inhibitor. In some embodiments, the additional therapeutic agent is selected from the group consisting of a MEK inhibitor, a SHP2 inhibitor, and a PD-L1 inhibitor. See, e.g., Hallin et al., Cancer Discovery, DOI: 10.1158/2159-8290 (Oct. 28, 2019) and Canon et al., Nature, 575:217 (2019). In some embodiments, a Ras inhibitor of the present invention is used in combination with a MEK inhibitor and a SOS1 inhibitor. In some embodiments, a Ras inhibitor of the present invention is used in combination with a PDL-1 inhibitor and a SOS1 inhibitor. In some embodiments, a Ras inhibitor of the present invention is used in combination with a PDL-1 inhibitor and a SHP2 inhibitor. In some embodiments, a Ras inhibitor of the present invention is used in combination with a MEK inhibitor and a SHP2 inhibitor. In some embodiments, the cancer is colorectal cancer and the treatment comprises administration of a Ras inhibitor of the present invention in combination with a second or third therapeutic agent.

(296) Proteasome inhibitors include, but are not limited to, carfilzomib (Kyprolis®), bortezomib (Velcade®), and oprozomib.

(297) Immune therapies include, but are not limited to, monoclonal antibodies, immunomodulatory imides (IMiDs), GITR agonists, genetically engineered T-cells (e.g., CAR-T cells), bispecific antibodies (e.g., BiTEs), and anti-PD-1, anti-PDL-1, anti-CTLA4, anti-LAGI, and anti-OX40 agents).

(298) Immunomodulatory agents (IMiDs) are a class of immunomodulatory drugs (drugs that adjust immune responses) containing an imide group. The IMiD class includes thalidomide and its analogues (lenalidomide, pomalidomide, and apremilast).

(299) Exemplary anti-PD-1 antibodies and methods for their use are described by Goldberg et al., Blood 2007, 110(1):186-192; Thompson et al., Clin. Cancer Res. 2007, 13(6):1757-1761; and WO06/121168 A1), as well as described elsewhere herein.

(300) GITR agonists include, but are not limited to, GITR fusion proteins and anti-GITR antibodies (e.g., bivalent anti-GITR antibodies), such as, a GITR fusion protein described in U.S. Pat. Nos. 6,111,090, 8,586,023, WO2010/003118 and WO2011/090754; or an anti-GITR antibody described, e.g., in U.S. Pat. No. 7,025,962, EP 1947183, U.S. Pat. Nos. 7,812,135, 8,388,967, 8,591,886, 7,618,632, EP 1866339, and WO2011/028683, WO2013/039954, WO05/007190, WO07/133822, WO05/055808, WO99/40196, WO01/03720, WO99/20758, WO06/083289, WO05/115451, and WO2011/051726.

(301) Another example of a therapeutic agent that may be used in combination with the compounds of the invention is an anti-angiogenic agent. Anti-angiogenic agents are inclusive of, but not limited to, in vitro synthetically prepared chemical compositions, antibodies, antigen binding regions, radionuclides, and combinations and conjugates thereof. An anti-angiogenic agent can be an agonist, antagonist, allosteric modulator, toxin or, more generally, may act to inhibit or stimulate its target (e.g., receptor or enzyme activation or inhibition), and thereby promote cell death or arrest cell growth. In some embodiments, the one or more additional therapies include an anti-angiogenic agent.

(302) Anti-angiogenic agents can be MMP-2 (matrix-metalloproteinase 2) inhibitors, MMP-9 (matrix-metalloprotienase 9) inhibitors, and COX-II (cyclooxygenase 11) inhibitors. Non-limiting examples of anti-angiogenic agents include rapamycin, temsirolimus (CCI-779), everolimus (RAD001), sorafenib, sunitinib, and bevacizumab. Examples of useful COX-II inhibitors include alecoxib, valdecoxib, and rofecoxib. Examples of useful matrix metalloproteinase inhibitors are described in WO96/33172, WO96/27583, WO98/07697, WO98/03516, WO98/34918, WO98/34915, WO98/33768, WO98/30566, WO90/05719, WO99/52910, WO99/52889, WO99/29667, WO99007675, EP0606046, EP0780386, EP1786785, EP1181017, EP0818442, EP1004578, and US20090012085, and U.S. Pat. Nos. 5,863,949 and 5,861,510. Preferred MMP-2 and MMP-9 inhibitors are those that have little or no activity inhibiting MMP-1. More preferred, are those that selectively inhibit MMP-2 or AMP-9 relative to the other matrix-metalloproteinases (i.e., MAP-1,

MMP-3, MMP-4, MMP-5, MMP-6, MMP-7, MMP-8, MMP-10, MMP-11, MMP-12, and MMP-13). Some specific examples of MMP inhibitors are AG-3340, RO 32-3555, and RS 13-0830.

(303) Further exemplary anti-angiogenic agents include KDR (kinase domain receptor) inhibitory agents (e.g., antibodies and antigen binding regions that specifically bind to the kinase domain receptor), anti-VEGF agents (e.g., antibodies or antigen binding regions that specifically bind VEGF (e.g., bevacizumab), or soluble VEGF receptors or a ligand binding region thereof) such as VEGF-TRAP™, and anti-VEGF receptor agents (e.g., antibodies or antigen binding regions that specifically bind thereto), EGFR inhibitory agents (e.g., antibodies or antigen binding regions that specifically bind thereto) such as Vectibix® (panitumumab), erlotinib (Tarceva®), anti-Angl and anti-Ang2 agents (e.g., antibodies or antigen binding regions specifically binding thereto or to their receptors, e.g., Tie2/Tek), and anti-Tie2 kinase inhibitory agents (e.g., antibodies or antigen binding regions that specifically bind thereto). Other anti-angiogenic agents include Campath, IL-8, B-FGF, Tek antagonists (US2003/0162712; U.S. Pat. No. 6,413,932), anti-TWEAK agents (e.g., specifically binding antibodies or antigen binding regions, or soluble TWEAK receptor antagonists; see U.S. Pat. No. 6,727,225), ADAM distintegrin domain to antagonize the binding of integrin to its ligands (US 2002/0042368), specifically binding anti-eph receptor or anti-ephrin antibodies or antigen binding regions (U.S. Pat. Nos. 5,981,245; 5,728,813; 5,969,110; 6,596,852; 6,232,447; 6,057,124 and patent family members thereof), and anti-PDGF-BB antagonists (e.g., specifically binding antibodies or antigen binding regions) as well as antibodies or antigen binding regions specifically binding to PDGF-BB ligands, and PDGFR kinase inhibitory agents (e.g., antibodies or antigen binding regions that specifically bind thereto). Additional anti-angiogenic agents include: SD-7784 (Pfizer, USA); cilengitide (Merck KGaA, Germany, EPO 0770622); pegaptanib octasodium, (Gilead Sciences, USA); Alphastatin, (BioActa, UK); M-PGA, (Celgene, USA, U.S. Pat. No. 5,712,291); ilomastat, (Arriva, USA, U.S. Pat. No. 5,892,112); emaxanib, (Pfizer, USA, U.S. Pat. No. 5,792,783); vatalanib, (Novartis, Switzerland); 2-methoxyestradiol (EntreMed, USA); TLC ELL-12 (Elan, Ireland); anecortave acetate (Alcon, USA); alpha-D148 Mab (Amgen, USA); CEP-7055 (Cephalon, USA); anti-Vn Mab (Crucell, Netherlands), DACantiangiogenic (ConjuChem, Canada); Angiocidin (InKine Pharmaceutical, USA); KM-2550 (Kyowa Hakko, Japan); SU-0879 (Pfizer, USA); CGP-79787 (Novartis, Switzerland, EP 0970070); ARGENT technology (Ariad, USA); YIGSR-Stealth (Johnson & Johnson, USA); fibrinogen-E fragment (BioActa, UK); angiogenic inhibitor (Trigen, UK); TBC-1635 (Encysive Pharmaceuticals, USA); SC-236 (Pfizer, USA); ABT-567 (Abbott, USA); Metastatin (EntreMed, USA); maspin (Sosei, Japan); 2-methoxyestradiol (Oncology Sciences Corporation, USA); ER-68203-00 (IV AX, USA); BeneFin (Lane Labs, USA); Tz-93 (Tsumura, Japan); TAN-1120 (Takeda, Japan); FR-111142 (Fujisawa, Japan, JP 02233610); platelet factor 4 (RepliGen, USA, EP 407122); vascular endothelial growth factor antagonist (Borean, Denmark); bevacizumab (pINN) (Genentech, USA); angiogenic inhibitors (SUGEN, USA); XL 784 (Exelixis, USA); XL 647 (Exelixis, USA); MAb, alpha5beta3 integrin, second generation (Applied Molecular Evolution, USA and MedImmune, USA); enzastaurin hydrochloride (Lilly, USA); CEP 7055 (Cephalon, USA and Sanofi-Synthelabo, France); BC 1 (Genoa Institute of Cancer Research, Italy); rBPI 21 and BPI-derived antiangiogenic (XOMA, USA); PI 88 (Progen, Australia); cilengitide (Merck KGaA, German; Munich Technical University, Germany, Scripps Clinic and Research Foundation, USA); AVE 8062 (Ajinomoto, Japan); AS 1404 (Cancer Research Laboratory, New Zealand); SG 292, (Telios, USA); Endostatin (Boston Childrens Hospital, USA); ATN 161 (Attenuon, USA); 2-methoxyestradiol (Boston Childrens Hospital, USA); ZD 6474, (AstraZeneca, UK); ZD 6126, (Angiogene Pharmaceuticals, UK); PPI 2458, (Praecis, USA); AZD 9935, (AstraZeneca, UK); AZD 2171, (AstraZeneca, UK); vatalanib (pINN), (Novartis, Switzerland and Schering AG, Germany); tissue factor pathway inhibitors, (EntreMed, USA); pegaptanib (Pinn), (Gilead Sciences, USA); xanthorrhizol, (Yonsei University, South Korea); vaccine, gene-based, VEGF-2, (Scripps Clinic and Research Foundation, USA); SPV5.2, (Supratek, Canada); SDX 103, (University of California at San Diego, USA); PX 478, (ProIX, USA); METASTATIN, (EntreMed,

USA); troponin I, (Harvard University, USA); SU 6668, (SUGEN, USA); OXI 4503, (OXiGENE, USA); o-guanidines, (Dimensional Pharmaceuticals, USA); motuporamine C, (British Columbia University, Canada); CDP 791, (Celltech Group, UK); atiprimod (pINN), (GlaxoSmithKline, UK); E 7820, (Eisai, Japan); CYC 381, (Harvard University, USA); AE 941, (Aeterna, Canada); vaccine, angiogenic, (EntreMed, USA); urokinase plasminogen activator inhibitor, (Dendreon, USA); oglufanide (pINN), (Melmotte, USA); HIF-lalfa inhibitors, (Xenova, UK); CEP 5214, (Cephalon, USA); BAY RES 2622, (Bayer, Germany); Angiocidin, (InKine, USA); A6, (Angstrom, USA); KR 31372, (Korea Research Institute of Chemical Technology, South Korea); GW 2286, (GlaxoSmithKline, UK); EHT 0101, (ExonHit, France); CP 868596, (Pfizer, USA); CP 564959, (OSI, USA); CP 547632, (Pfizer, USA); 786034, (GlaxoSmithKline, UK); KRN 633, (Kirin Brewery, Japan); drug delivery system, intraocular, 2-methoxyestradiol; anginex (Maastricht University, Netherlands, and Minnesota University, USA); ABT 510 (Abbott, USA); AAL 993 (Novartis, Switzerland); VEGI (ProteomTech, USA); tumor necrosis factor-alpha inhibitors; SU 11248 (Pfizer, USA and SUGEN USA); ABT 518, (Abbott, USA); YH16 (Yantai Rongchang, China); S-3APG (Boston Childrens Hospital, USA and EntreMed, USA); MAb, KDR (ImClone Systems, USA); MAb, alpha5 beta (Protein Design, USA); KDR kinase inhibitor (Celltech Group, UK, and Johnson & Johnson, USA); GFB 116 (South Florida University, USA and Yale University, USA); CS 706 (Sankyo, Japan); combretastatin A4 prodrug (Arizona State University, USA); chondroitinase AC (IBEX, Canada); BAY RES 2690 (Bayer, Germany); AGM 1470 (Harvard University, USA, Takeda, Japan, and TAP, USA); AG 13925 (Agouron, USA); Tetrathiomolybdate (University of Michigan, USA); GCS 100 (Wayne State University, USA) CV 247 (Ivy Medical, UK); CKD 732 (Chong Kun Dang, South Korea); irsogladine, (Nippon Shinyaku, Japan); RG 13577 (Aventis, France); WX 360 (Wilex, Germany); squalamine, (Genaera, USA); RPI 4610 (Sirna, USA); heparanase inhibitors (InSight, Israel); KL 3106 (Kolon, South Korea); Honokiol (Emory University, USA); ZK CDK (Schering AG, Germany); ZK Angio (Schering AG, Germany); ZK 229561 (Novartis, Switzerland, and Schering AG, Germany); XMP 300 (XOMA, USA); VGA 1102 (Taisho, Japan); VE-cadherin-2 antagonists (ImClone Systems, USA); Vasostatin (National Institutes of Health, USA); Flk-1 (ImClone Systems, USA); TZ 93 (Tsumura, Japan); TumStatin (Beth Israel Hospital, USA); truncated soluble FLT 1 (vascular endothelial growth factor receptor 1) (Merck & C.sub.0, USA); Tie-2 ligands (Regeneron, USA); and thrombospondin 1 inhibitor (Allegheny Health, Education and Research Foundation, USA).

(304) Further examples of therapeutic agents that may be used in combination with compounds of the invention include agents (e.g., antibodies, antigen binding regions, or soluble receptors) that specifically bind and inhibit the activity of growth factors, such as antagonists of hepatocyte growth factor (HGF, also known as Scatter Factor), and antibodies or antigen binding regions that specifically bind its receptor, c-Met.

(305) Another example of a therapeutic agent that may be used in combination with compounds of the invention is an autophagy inhibitor. Autophagy inhibitors include, but are not limited to chloroquine, 3-methyladenine, hydroxychloroquine (Plaquenil™), bafilomycin A1, 5-amino-4-imidazole carboxamide riboside (AICAR), okadaic acid, autophagy-suppressive algal toxins which inhibit protein phosphatases of type 2A or type 1, analogues of cAMP, and drugs which elevate cAMP levels such as adenosine, LY204002, N6-mercaptopurine riboside, and vinblastine. In addition, antisense or siRNA that inhibits expression of proteins including but not limited to ATG5 (which are implicated in autophagy), may also be used. In some embodiments, the one or more additional therapies include an autophagy inhibitor.

(306) Another example of a therapeutic agent that may be used in combination with compounds of the invention is an anti-neoplastic agent. In some embodiments, the one or more additional therapies include an anti-neoplastic agent. Non-limiting examples of anti-neoplastic agents include acemannan, aclarubicin, aldesleukin, alemtuzumab, alitretinoin, altretamine, amifostine, aminolevulinic acid, amrubicin, amsacrine, anagrelide, anastrozole, ancer, ancestim, arglabin,

arsenic trioxide, BAM-002 (Novelos), bexarotene, bicalutamide, broxuridine, capecitabine, celmoleukin, cetrorelix, cladribine, clotrimazole, cytarabine ocfosfate, DA 3030 (Dong-A), daclizumab, denileukin diftitox, deslorelin, dexrazoxane, dilazep, docetaxel, docosanol, doxercalciferol, doxifluridine, doxorubicin, bromocriptine, carmustine, cytarabine, fluorouracil, HIT diclofenac, interferon alfa, daunorubicin, doxorubicin, tretinoin, edelfosine, edrecolomab, eflornithine, emitefur, epirubicin, epoetin beta, etoposide phosphate, exemestane, exisulind, fadrozole, filgrastim, finasteride, fludarabine phosphate, formestane, fotemustine, gallium nitrate, gemcitabine, gemtuzumab zogamicin, gimeracil/oteracil/tegafur combination, glycopine, goserelin, heptaplatin, human chorionic gonadotropin, human fetal alpha fetoprotein, ibandronic acid, idarubicin, (imiquimod, interferon alfa, interferon alfa, natural, interferon alfa-2, interferon alfa-2a, interferon alfa-2b, interferon alfa-NI, interferon alfa-n3, interferon alfacon-1, interferon alpha, natural, interferon beta, interferon beta-la, interferon beta-Ib, interferon gamma, natural interferon gamma-Ia, interferon gamma-Ib, interleukin-1 beta, iobenguane, irinotecan, irsogladine, Ianreotide, LC 9018 (Yakult), leflunomide, lenograstim, lentinan sulfate, letrozole, leukocyte alpha interferon, leuprorelin, levamisole+fluorouracil, liarozole, lobaplatin, Ionidamine, lovastatin, masoprocol, melarsoprol, metoclopramide, mifepristone, miltefosine, mirimostim, mismatched double stranded RNA, mitoguazone, mitolactol, mitoxantrone, molgramostim, nafarelin, naloxone+pentazocine, nartograstim, nedaplatin, nilutamide, noscapine, novel erythropoiesis stimulating protein, NSC 631570 octreotide, oprelvekin, osaterone, oxaliplatin, paclitaxel, pamidronic acid, pegaspargase, peginterferon alfa-2b, pentosan polysulfate sodium, pentostatin, picibanil, pirarubicin, rabbit antithymocyte polyclonal antibody, polyethylene glycol interferon alfa-2a, porfimer sodium, raloxifene, raltitrexed, rasburiembodiment, rhenium Re 186 etidronate, RII retinamide, rituximab, romurtide, samarium (153 Sm) lexidronam, sargramostim, sizofiran, sobuzoxane, sonermin, strontium-89 chloride, suramin, tasonermin, tazarotene, tegafur, temoporfin, temozolomide, teniposide, tetrachlorodecaoxide, thalidomide, thymalfasin, thyrotropin alfa, topotecan, toremifene, tositumomab-iodine 131, trastuzumab, treosulfan, tretinoin, trilostane, trimetrexate, triptorelin, tumor necrosis factor alpha, natural, ubenimex, bladder cancer vaccine, Maruyama vaccine, melanoma lysate vaccine, valrubicin, verteporfin, vinorelbine, virulizin, zinostatin stimalamer, or zoledronic acid; abarelix; AE 941 (Aeterna), ambamustine, antisense oligonucleotide, bcl-2 (Genta), APC 8015 (Dendreon), decitabine, dexaminoglutethimide, diaziquone, EL 532 (Elan), EM 800 (Endorecherche), eniluracil, etanidazole, fenretinide, filgrastim SDO1 (Amgen), fulvestrant, galocitabine, gastrin 17 immunogen, HLA-B7 gene therapy (Vical), granulocyte macrophage colony stimulating factor, histamine dihydrochloride, ibritumomab tiuxetan, ilomastat, IM 862 (Cytran), interleukin-2, iproxifene, LDI 200 (Milkhaus), leridistim, lintuzumab, CA 125 MAb (Biomira), cancer MAb (Japan Pharmaceutical Development), HER-2 and Fc MAb (Medarex), idiotypic 105AD7 MAb (CRC Technology), idiotypic CEA MAb (Trilex), LYM-1-iodine 131 MAb (Techni clone), polymorphic epithelial mucin-yttrium 90 MAb (Antisoma), marimastat, menogaril, mitumomab, motexafin gadolinium, MX 6 (Galderma), nelarabine, nolatrexed, P 30 protein, pegvisomant, pemetrexed, porfiromycin, prinomastat, RL 0903 (Shire), rubitecan, satraplatin, sodium phenylacetate, sparfosic acid, SRL 172 (SR Pharma), SU 5416 (SUGEN), TA 077 (Tanabe), tetrathiomolybdate, thaliblastine, thrombopoietin, tin ethyl etiopurpurin, tirapazamine, cancer vaccine (Biomira), melanoma vaccine (New York University), melanoma vaccine (Sloan Kettering Institute), melanoma oncolysate vaccine (New York Medical College), viral melanoma cell lysates vaccine (Royal Newcastle Hospital), or valspodar.
(307) Additional examples of therapeutic agents that may be used in combination with compounds of the invention include ipilimumab (Yervoy®); tremelimumab; galiximab; nivolumab, also known as BMS-936558 (Opdivo®); pembrolizumab (Keytruda®); avelumab (Bavencio®); AMP224; BMS-936559; MPDL3280A, also known as RG7446; MEDI-570; AMG557; MGA271; IMP321; BMS-663513; PF-05082566; CDX-1127; anti-OX40 (Providence Health Services); huMAbOX40L; atacicept; CP-870893; lucatumumab; dacetuzumab; muromonab-CD3; ipilumumab; MEDI4736

(Imfinzi®); MSB0010718C; AMP 224; adalimumab (Humira®); ado-trastuzumab emtansine (Kadcyla®); aflibercept (Eylea®); alemtuzumab (Campath®); basiliximab (Simulect®); belimumab (Benlysta®); basiliximab (Simulect®); belimumab (Benlysta®); brentuximab vedotin (Adcetris®); canakinumab (Ilaris®); certolizumab pegol (Cimzia®); daclizumab (Zenapax®); daratumumab (Darzalex®); denosumab (Prolia®); eculizumab (Soliris®); efalizumab (Raptiva®); gemtuzumab ozogamicin (Mylotarg®); golimumab (Simponi®); ibritumomab tiuxetan (Zevalin®); infliximab (Remicade®); motavizumab (Numax®); natalizumab (Tysabri®); obinutuzumab (Gazyva®); ofatumumab (Arzerra®); omalizumab (Xolair®); palivizumab (Synagis®); pertuzumab (Perjeta®); pertuzumab (Perjeta®); ranibizumab (Lucentis®); raxibacumab (Abthrax®); tocilizumab (Actemra®); tositumomab; tositumomab-i-131; tositumomab and tositumomab-i-131 (Bexxar®); ustekinumab (Stelara®); AMG 102; AMG 386; AMG 479; AMG 655; AMG 706; AMG 745; and AMG 951.

(308) The compounds described herein can be used in combination with the agents disclosed herein or other suitable agents, depending on the condition being treated. Hence, in some embodiments the one or more compounds of the disclosure will be co-administered with other therapies as described herein. When used in combination therapy, the compounds described herein may be administered with the second agent simultaneously or separately. This administration in combination can include simultaneous administration of the two agents in the same dosage form, simultaneous administration in separate dosage forms, and separate administration. That is, a compound described herein and any of the agents described herein can be formulated together in the same dosage form and administered simultaneously. Alternatively, a compound of the invention and any of the therapies described herein can be simultaneously administered, wherein both the agents are present in separate formulations. In another alternative, a compound of the present disclosure can be administered and followed by any of the therapies described herein, or vice versa. In some embodiments of the separate administration protocol, a compound of the invention and any of the therapies described herein are administered a few minutes apart, or a few hours apart, or a few days apart.

(309) In some embodiments of any of the methods described herein, the first therapy (e.g., a compound of the invention) and one or more additional therapies are administered simultaneously or sequentially, in either order. The first therapeutic agent may be administered immediately, up to 1 hour, up to 2 hours, up to 3 hours, up to 4 hours, up to 5 hours, up to 6 hours, up to 7 hours, up to, 8 hours, up to 9 hours, up to 10 hours, up to 11 hours, up to 12 hours, up to 13 hours, 14 hours, up to hours 16, up to 17 hours, up 18 hours, up to 19 hours up to 20 hours, up to 21 hours, up to 22 hours, up to 23 hours, up to 24 hours, or up to 1-7, 1-14, 1-21 or 1-30 days before or after the one or more additional therapies.

(310) The invention also features kits including (a) a pharmaceutical composition including an agent (e.g., a compound of the invention) described herein, and (b) a package insert with instructions to perform any of the methods described herein. In some embodiments, the kit includes (a) a pharmaceutical composition including an agent (e.g., a compound of the invention) described herein, (b) one or more additional therapies (e.g., non-drug treatment or therapeutic agent), and (c) a package insert with instructions to perform any of the methods described herein.

(311) As one aspect of the present invention contemplates the treatment of the disease or symptoms associated therewith with a combination of pharmaceutically active compounds that may be administered separately, the invention further relates to combining separate pharmaceutical compositions in kit form.

(312) The kit may comprise two separate pharmaceutical compositions: a compound of the present invention, and one or more additional therapies. The kit may comprise a container for containing the separate compositions such as a divided bottle or a divided foil packet. Additional examples of containers include syringes, boxes, and bags. In some embodiments, the kit may comprise directions for the use of the separate components. The kit form is particularly advantageous when the separate components are preferably administered in different dosage forms (e.g., oral and parenteral), are

administered at different dosage intervals, or when titration of the individual components of the combination is desired by the prescribing health care professional.

NUMBERED EMBODIMENTS

(313) [1] A compound, or pharmaceutically acceptable salt thereof, having the structure of Formula I:

(314) ##STR01172## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 10-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl;

R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl, or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is H or C.sub.1-C.sub.3 alkyl. [2] The compound, or pharmaceutically acceptable salt thereof, of paragraph [1], wherein G is optionally substituted C.sub.1-C.sub.4 heteroalkylene. [3] The compound, or pharmaceutically acceptable salt thereof, of paragraph [1] or [2], wherein the compound has the structure of Formula Ic:

(315) ##STR01173## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 and Y.sup.6 are, independently, CH or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered

cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; and R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl. [4] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [3], wherein X.sup.2 is NH. [5] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [4], wherein X.sup.3 is CH. [6] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [5], wherein R.sup.11 is hydrogen. [7] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [5], wherein R.sup.11 is C.sub.1-C.sub.3 alkyl. [8] The compound, or pharmaceutically acceptable salt thereof, of paragraph [7], wherein R.sup.11 is methyl. [9] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [6], wherein the compound has the structure of Formula Id:

(316) ##STR01174## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 and Y.sup.6 are, independently, CH or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an

optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; and R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl. [10] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [9] wherein X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene. [11] The compound, or pharmaceutically acceptable salt thereof, of paragraph [10], wherein X.sup.1 is methylene. [12] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [11], wherein R.sup.5 is hydrogen. [13] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [11], wherein R.sup.5 is C.sub.1-C.sub.4 alkyl optionally substituted with halogen. [14] The compound, or pharmaceutically acceptable salt thereof, of paragraph [13], wherein R.sup.5 is methyl. [15] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [14], wherein Y.sup.4 is C. [16] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [15], wherein R.sup.4 is hydrogen. [17] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [16], wherein Y.sup.5 is CH. [18] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [17], wherein Y.sup.6 is CH. [19] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [18], wherein Y.sup.1 is C. [20] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [19] wherein Y.sup.2 is C. [21] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [20] wherein Y.sup.3 is N. [22] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [21], wherein R.sup.3 is absent. [23] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [22], wherein Y.sup.7 is C. [24] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [6] or [9] to [23], wherein the compound has the structure of Formula Ie:

(317) ##STR01175## wherein A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-

membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; and R.sup.10 is hydrogen, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl. [25] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [3] to [24], wherein R.sup.6 is hydrogen. [26] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [25], wherein R.sup.2 is hydrogen, cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 6-membered heterocycloalkyl. [27] The compound, or pharmaceutically acceptable salt thereof, of paragraph [26], wherein R.sup.2 is optionally substituted C.sub.1-C.sub.6 alkyl. [28] The compound, or pharmaceutically acceptable salt thereof, of paragraph [27], wherein R.sup.2 is ethyl. [29] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [28], wherein R.sup.7 is optionally substituted C.sub.1-C.sub.3 alkyl. [30] The compound, or pharmaceutically acceptable salt thereof, of paragraph 29, wherein R.sup.7 is C.sub.1-C.sub.3 alkyl. [31] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to 30, wherein R.sup.8 is optionally substituted C.sub.1-C.sub.3 alkyl. [32] The compound, or pharmaceutically acceptable salt thereof, of paragraph

[31], wherein R.sup.8 is C.sub.1-C.sub.3 alkyl. [33] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [32], wherein the compound has the structure of Formula If:
(318) ##STR01176## wherein A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl; R.sup.2 is C.sub.1-C.sub.6 alkyl or 3 to 6-membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl. [34] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [33], wherein R.sup.1 is optionally substituted 6 to 10-membered aryl, optionally substituted 3 to 6-membered cycloalkenyl, or optionally substituted 5 to 10-membered heteroaryl. [35] The compound, or pharmaceutically acceptable salt thereof, of paragraph [34], wherein R.sup.1 is optionally substituted 6-membered aryl, optionally substituted 6-membered cycloalkenyl, or optionally substituted 6-membered heteroaryl. [36] The compound, or pharmaceutically acceptable salt thereof, of paragraph [35], wherein R.sup.1 is
(319) ##STR01177## [37] The compound, or pharmaceutically acceptable salt thereof, of paragraph [36], wherein R.sup.1 is
(320) ##STR01178## [38] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [37], wherein the compound has the structure of Formula Ig:
(321) ##STR01179## wherein A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; R.sup.2 is C.sub.1-C.sub.6 alkyl, C.sub.1-C.sub.6 fluoroalkyl, or 3 to 6-membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl X.sup.e and X.sup.f are, independently, N or CH; and R.sup.12 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl. [39] The compound, or pharmaceutically acceptable salt thereof, of paragraph [38], wherein X.sup.e is N and X.sup.f is CH. [40] The compound, or pharmaceutically acceptable salt thereof, of paragraph [38], wherein X.sup.e is CH and X.sup.f is N. [41] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [38] to [40], wherein R.sup.12 is optionally substituted C.sub.1-C.sub.6 heteroalkyl. [42] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [38] to [41], wherein R.sup.12 is

(322) ##STR01180## [43] The compound, or pharmaceutically acceptable salt thereof, of paragraph [1] or [2], wherein the compound has the structure of Formula VI:

(323) ##STR01181## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 10-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, a haloacetal, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally

substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl (e.g., methyl); and X.sup.e and X.sup.f are, independently, N or CH. [44] The compound, or pharmaceutically acceptable salt thereof, of paragraph [43], wherein the compound has the structure of Formula VIa:

(324) ##STR01182## wherein A is optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; L is absent or a linker; W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; R.sup.2 is C.sub.1-C.sub.6 alkyl, C.sub.1-C.sub.6 fluoroalkyl, or 3 to 6-membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; X.sup.e and X.sup.1 are, independently, N or CH; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl. [45] The compound, or pharmaceutically acceptable salt thereof, of paragraph [43] or [44], wherein the compound has the structure of Formula VIb:

(325) ##STR01183## wherein A is optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; L is absent or a linker; and W is a cross-linking group comprising a vinyl ketone, a vinyl sulfone, an ynone, or an alkynyl sulfone. [46] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [45], wherein A is optionally substituted 6-membered arylene. [47] The compound, or pharmaceutically acceptable salt thereof, of paragraph [46], wherein A has the structure:

(326) ##STR01184## wherein R.sup.13 is hydrogen, halo, hydroxy, amino, optionally substituted C.sub.1-C.sub.6 alkyl, or optionally substituted C.sub.1-C.sub.6 heteroalkyl; and R.sup.13a is hydrogen or halo. [48] The compound, or pharmaceutically acceptable salt thereof, of paragraph [47], wherein R.sup.13 and R.sup.13a are each hydrogen. [49] The compound, or pharmaceutically acceptable salt thereof, of paragraph [47], wherein R.sup.13 is hydroxy, methyl, fluoro, or difluoromethyl. [50] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [45], wherein A is optionally substituted 5 to 6-membered heteroarylene. [51] The

compound, or pharmaceutically acceptable salt thereof, of paragraph [50], wherein A is:
(327) ##STR01185## ##STR01186## [52] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [45], wherein A is optionally substituted C_1-C_4 heteroalkylene. [53] The compound, or pharmaceutically acceptable salt thereof, of paragraph [52], wherein A is:
(328) ##STR01187## [54] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [45], wherein A is optionally substituted 3 to 6-membered heterocycloalkylene. [55] The compound, or pharmaceutically acceptable salt thereof, of paragraph [54], wherein A is:
(329) ##STR01188## [56] The compound, or pharmaceutically acceptable salt thereof, of paragraph [55], wherein A is
(330) ##STR01189## [57] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [56], wherein B is —CHR^9—. [58] The compound, or pharmaceutically acceptable salt thereof, of paragraph [57], wherein R^9 is F, optionally substituted C_1-C_6 alkyl, optionally substituted C_1-C_6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl. [59] The compound, or pharmaceutically acceptable salt thereof, of paragraph [58], wherein R^9 is:
(331) ##STR01190## [60] The compound, or pharmaceutically acceptable salt thereof, of paragraph [59], wherein R^9 is:
(332) ##STR01191## [61] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [56], wherein B is optionally substituted 6-membered arylene. [62] The compound, or pharmaceutically acceptable salt thereof, of paragraph [61], wherein B is 6-membered arylene. [63] The compound, or pharmaceutically acceptable salt thereof, of paragraph [61], wherein B is:
(333) ##STR01192## [64] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [63], wherein R^7 is methyl. [65] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [64], wherein R^8 is methyl. [66] The compound, or pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [65], wherein the linker is the structure of Formula II:
A^1-$(B^1)_f$—$(C^1)_g$—$(B^2)_h$-(D^1)-$(B^3)_i$—$(C^2)_j$—$(B^4)_k$-A^2 Formula II where A^1 is a bond between the linker and B; A^2 is a bond between W and the linker; B^1, B^2, B^3, and B^4 each, independently, is selected from optionally substituted C_1-C_2 alkylene, optionally substituted C_1-C_3 heteroalkylene, O, S, and NR^N; R^N is hydrogen, optionally substituted C_1-C_4 alkyl, optionally substituted C_2-C_4 alkenyl, optionally substituted C_2-C_4 alkynyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted C_1-C_7 heteroalkyl; C_1 and C^2 are each, independently, selected from carbonyl, thiocarbonyl, sulphonyl, or phosphoryl; f, g, h, i, j, and k are each, independently, 0 or 1; and D^1 is optionally substituted C_1-C_{10} alkylene, optionally substituted C_2-C_{10} alkenylene, optionally substituted C_2-C_{10} alkynylene, optionally substituted 3 to 14-membered heterocycloalkylene, optionally substituted 5 to 10-membered heteroarylene, optionally substituted 3 to 8-membered cycloalkylene, optionally substituted 6 to 10-membered arylene, optionally substituted C_2-C_{10} polyethylene glycolene, or optionally substituted C_1-C_{10} heteroalkylene, or a chemical bond linking A^1-$(B^1)_f$—$(C^1)_g$—$(B^2)_h$— to —$(B^3)_i$—$(C^2)_j$—$(B^4)_k$-A^2. [67] The compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [66], wherein the linker is acyclic. [68] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [67], wherein the linker has the structure of Formula IIa:
(334) ##STR01193## wherein X^a is absent or N; R^{14} is absent, hydrogen or optionally substituted C_1-C_6 alkyl; and L^2 is absent, —SO_2—, optionally substituted C_1-C_4 alkylene or optionally substituted C_1-C_4 heteroalkylene, wherein at least

one of X.sup.a, R.sup.14, or L.sup.2 is present. [69] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [68], wherein the linker has the structure:

(335) ##STR01194## [70] The compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [66], wherein the linker is or comprises a cyclic moiety. [71] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [70], wherein the linker has the structure of Formula IIb:

(336) ##STR01195## wherein o is 0 or 1; R.sup.15 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted 3 to 8-membered cycloalkylene, or optionally substituted 3 to 8-membered heterocycloalkylene; X.sup.4 is absent, optionally substituted C.sub.1-C.sub.4 alkylene, 0, NCH.sub.3, or optionally substituted C.sub.1-C.sub.4 heteroalkylene; Cy is optionally substituted 3 to 8-membered cycloalkylene, optionally substituted 3 to 8-membered heterocycloalkylene, optionally substituted 6-10 membered arylene, or optionally substituted 5 to 10-membered heteroarylene; and L.sup.3 is absent, —SO.sub.2—, optionally substituted C.sub.1-C.sub.4 alkylene or optionally substituted C.sub.1-C.sub.4 heteroalkylene. [72] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [71], wherein the linker has the structure:

(337) ##STR01196## wherein R.sup.15 is hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted 3 to 8-membered cycloalkylene, or optionally substituted 3 to 8-membered heterocycloalkylene; and R.sup.15a, R.sup.15b, R.sup.15c, R.sup.15d, R.sup.15e, R.sup.15f, and R.sup.15g are, independently, hydrogen, halo, hydroxy, cyano, amino, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, or, or R.sup.15b and R.sup.15d combine with the carbons to which they are attached to form an optionally substituted 3 to 8-membered cycloalkylene, or optionally substituted 3 to 8-membered heterocycloalkylene. [73] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [72], wherein the linker has the structure:

(338) ##STR01197## ##STR01198## ##STR01199## ##STR01200## [74] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [71], wherein the linker has the structure:

(339) ##STR01201## ##STR01202## ##STR01203## ##STR01204## ##STR01205## ##STR01206## ##STR01207## ##STR01208## ##STR01209## ##STR01210## ##STR01211## ##STR01212## [75] The compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [74], wherein W is a cross-linking group comprising a vinyl ketone. [76]. The compound, or a pharmaceutically acceptable salt thereof, of paragraph [75], wherein W has the structure of Formula IIIa:

(340) ##STR01213## wherein R.sup.16a, R.sup.16b, and R.sup.16c are, independently, hydrogen, —CN, halogen, or —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl. [77] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [76], wherein W is:

(341) ##STR01214## ##STR01215## [78] The compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [74], wherein W is a cross-linking group comprising an ynone. [79] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [78], wherein W has the structure of Formula IIIb:

(342) ##STR01216## wherein R.sup.17 is hydrogen, —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated cycloalkyl, or a 4 to 7-membered saturated heterocycloalkyl. [80] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [79], wherein W is:

(343) ##STR01217## ##STR01218## ##STR01219## ##STR01220## [81] The compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [74], wherein W is a cross-linking group comprising a vinyl sulfone. [82] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [81], wherein W has the structure of Formula IIIc:

(344) ##STR01221## wherein R.sup.18a, R.sup.18b, and R.sup.18c are, independently, hydrogen, —CN, or —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl. [83] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [82], wherein W is:
(345) ##STR01222## [84] The compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [74], wherein W is a cross-linking group comprising an alkynyl sulfone. [85] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [84], wherein W has the structure of Formula IIId:
(346) ##STR01223## wherein R.sup.19 is hydrogen, —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl, or a 4 to 7-membered saturated heterocycloalkyl. [86] The compound, or a pharmaceutically acceptable salt thereof, of paragraph [85], wherein W is:
(347) ##STR01224## [87] The compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [74], wherein W has the structure of Formula IIIe:
(348) ##STR01225## wherein X.sup.e is a halogen; and R.sup.20 is hydrogen, —C.sub.1-C.sub.3 alkyl optionally substituted with one or more substituents independently selected from —OH, —O—C.sub.1-C.sub.3 alkyl, —NH.sub.2, —NH(C.sub.1-C.sub.3 alkyl), —N(C.sub.1-C.sub.3 alkyl).sub.2, or a 4 to 7-membered saturated heterocycloalkyl. [88] A compound, or a pharmaceutically acceptable salt thereof, selected from Table 1 or Table 2. [89] A pharmaceutical composition comprising a compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [88], and a pharmaceutically acceptable excipient. [90] A conjugate, or salt thereof, comprising the structure of Formula IV:
M-L-P Formula IV wherein L is a linker; P is a monovalent organic moiety; and M has the structure of Formula V:
(349) ##STR01226## wherein the dotted lines represent zero, one, two, three, or four non-adjacent double bonds; A is —N(H or CH.sub.3)C(O)—(CH.sub.2)— where the amino nitrogen is bound to the carbon atom of —CH(R.sup.10)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is absent, —CH(R.sup.9)—, >C═CR.sup.9R.sup.9′, or >CR.sup.9R.sup.9′ where the carbon is bound to the carbonyl carbon of —N(R.sup.11)C(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; G is optionally substituted C.sub.1-C.sub.4 alkylene, optionally substituted C.sub.1-C.sub.4 alkenylene, optionally substituted C.sub.1-C.sub.4 heteroalkylene, —C(O)O—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, —C(O)NH—CH(R.sup.6)— where C is bound to —C(R.sup.7R.sup.8)—, optionally substituted C.sub.1-C.sub.4 heteroalkylene, or 3 to 8-membered heteroarylene; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; X.sup.3 is N or CH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; Y.sup.1 is C, CH, or N; Y.sup.2, Y.sup.3, Y.sup.4, and Y.sup.7 are, independently, C or N; Y.sup.5 is CH, CH.sub.2, or N; Y.sup.6 is C(O), CH, CH.sub.2, or N; R.sup.1 is cyano, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 6-membered cycloalkenyl, optionally substituted 3 to 6-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted 5 to 10-membered heteroaryl, or R.sup.1 and R.sup.2 combine with the atoms to which they are attached to form an optionally

substituted 3 to 14-membered heterocycloalkyl; R.sup.2 is absent, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 6-membered cycloalkyl, optionally substituted 3 to 7-membered heterocycloalkyl, optionally substituted 6-membered aryl, optionally substituted 5 or 6-membered heteroaryl; R.sup.3 is absent, or R.sup.2 and R.sup.3 combine with the atom to which they are attached to form an optionally substituted 3 to 8-membered cycloalkyl or optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.4 is absent, hydrogen, halogen, cyano, or methyl optionally substituted with 1 to 3 halogens; R.sup.5 is hydrogen, C.sub.1-C.sub.4 alkyl optionally substituted with halogen, cyano, hydroxy, or C.sub.1-C.sub.4 alkoxy, cyclopropyl, or cyclobutyl; R.sup.6 is hydrogen or methyl; R.sup.7 is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl, or R.sup.6 and R.sup.7 combine with the carbon atoms to which they are attached to form an optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.8 is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7 and R.sup.8 combine with the carbon atom to which they are attached to form C═CR.sup.7′R.sup.8′; C═N(OH), C═N(O—C.sub.1-C.sub.3 alkyl), C═O, C═S, C═NH, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.7a and R.sup.8a are, independently, hydrogen, halo, optionally substituted C.sub.1-C.sub.3 alkyl, or combine with the carbon to which they are attached to form a carbonyl; R.sup.7′ is hydrogen, halogen, or optionally substituted C.sub.1-C.sub.3 alkyl; R.sup.8′ is hydrogen, halogen, hydroxy, cyano, optionally substituted C.sub.1-C.sub.3 alkoxy, optionally substituted C.sub.1-C.sub.3 alkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted 3 to 8-membered cycloalkyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 5 to 10-membered heteroaryl, or optionally substituted 6 to 10-membered aryl, or R.sup.7′ and R.sup.8′ combine with the carbon atom to which they are attached to form optionally substituted 3 to 6-membered cycloalkyl or optionally substituted 3 to 7-membered heterocycloalkyl; R.sup.9 is H, F, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl, or R.sup.9 and L combine with the atoms to which they are attached to form an optionally substituted 3 to 14-membered heterocycloalkyl; R.sup.9′ is hydrogen or optionally substituted C.sub.1-C.sub.6 alkyl; or R.sup.9 and R.sup.9′, combined with the atoms to which they are attached, form a 3 to 6-membered cycloalkyl or a 3 to 6-membered heterocycloalkyl; R.sup.10 is hydrogen, halo, hydroxy, C.sub.1-C.sub.3 alkoxy, or C.sub.1-C.sub.3 alkyl; R.sup.10a is hydrogen or halo; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is H or C.sub.1-C.sub.3 alkyl. [91] The conjugate of paragraph [90], or salt thereof, wherein M has the structure of Formula Vd:

(350) ##STR01227## wherein A is optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; X.sup.1 is optionally substituted C.sub.1-C.sub.2 alkylene, NR, O, or S(O).sub.n; X.sup.2 is O or NH; n is 0, 1, or 2; R is hydrogen, cyano, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, C(O)R′, C(O)OR′, C(O)N(R′).sub.2, S(O)R′, S(O).sub.2R′, or S(O).sub.2N(R′).sub.2; each R′ is, independently, H or optionally substituted C.sub.1-C.sub.4 alkyl; R.sup.2 is C.sub.1-C.sub.6 alkyl, C.sub.1-C.sub.6 fluoroalkyl, or 3 to 6-

membered cycloalkyl; R.sup.7 is C.sub.1-C.sub.3 alkyl; R.sup.8 is C.sub.1-C.sub.3 alkyl; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl; X.sup.e and X.sup.1 are, independently, N or CH; R.sup.11 is hydrogen or C.sub.1-C.sub.3 alkyl; and R.sup.21 is hydrogen or C.sub.1-C.sub.3 alkyl. [92] The conjugate of paragraph [91], or salt thereof, wherein M has the structure of Formula Ve:

(351) ##STR01228## wherein A is optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or optionally substituted 5 to 6-membered heteroarylene; B is —CH(R.sup.9)— where the carbon is bound to the carbonyl carbon of —NHC(O)—, optionally substituted 3 to 6-membered cycloalkylene, optionally substituted 3 to 6-membered heterocycloalkylene, optionally substituted 6-membered arylene, or 5 to 6-membered heteroarylene; and R.sup.9 is optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6-membered cycloalkyl, or optionally substituted 3 to 7-membered heterocycloalkyl. [93] The conjugate, or salt thereof, of any one of paragraphs [90] to [92], wherein the linker has the structure of Formula II:

A.sup.1-(B.sup.1).sub.f—(C.sup.1).sub.g—(B.sup.2).sub.h-(D.sup.1)-(B.sup.3).sub.i—(C.sup.2).sub.j—(B.sup.4).sub.k-A.sup.2 Formula II where A.sup.1 is a bond between the linker and B; A.sup.2 is a bond between W and the linker; B.sup.1, B.sup.2, B.sup.3, and B.sup.4 each, independently, is selected from optionally substituted C.sub.1-C.sub.2 alkylene, optionally substituted C.sub.1-C.sub.3 heteroalkylene, O, S, and NR.sup.N; R.sup.N is hydrogen, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.2-C.sub.4 alkenyl, optionally substituted C.sub.2-C.sub.4 alkynyl, optionally substituted 3 to 14-membered heterocycloalkyl, optionally substituted 6 to 10-membered aryl, or optionally substituted C.sub.1-C.sub.7 heteroalkyl; C.sup.1 and C.sup.2 are each, independently, selected from carbonyl, thiocarbonyl, sulphonyl, or phosphoryl; f, g, h, i, j, and k are each, independently, 0 or 1; and D.sup.1 is optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.2-C.sub.10 alkenylene, optionally substituted C.sub.2-C.sub.10 alkynylene, optionally substituted 3 to 14-membered heterocycloalkylene, optionally substituted 5 to 10-membered heteroarylene, optionally substituted 3 to 8-membered cycloalkylene, optionally substituted 6 to 10-membered arylene, optionally substituted C.sub.2-C.sub.10 polyethylene glycolene, or optionally substituted C.sub.1-C.sub.10 heteroalkylene, or a chemical bond linking A.sup.1-(B.sup.1).sub.f—(C.sup.1).sub.g—(B.sup.2).sub.h— to —(B.sup.3).sub.i—(C.sup.2).sub.j—(B.sup.4).sub.k-A.sup.2. [94] The conjugate, or salt thereof, of any one of paragraphs [90] to [93], wherein the monovalent organic moiety is a protein. [95] The conjugate, or salt thereof, of paragraph [94], wherein the protein is a Ras protein. [96] The conjugate, or salt thereof, of paragraph [95], wherein the Ras protein is K-Ras G12C, K-Ras G13C, H-Ras G12C, H-Ras G13C, N-Ras G12C, or N-Ras G13C. [97] The conjugate, or salt thereof, of any one of paragraphs [93] to [96], wherein the linker is bound to the monovalent organic moiety through a bond to a sulfhydryl group of an amino acid residue of the monovalent organic moiety. [98] A method of treating cancer in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [88] or a pharmaceutical composition of paragraph [89]. [99] The method of paragraph [98], wherein the cancer is pancreatic cancer, colorectal cancer, non-small cell lung cancer, or endometrial cancer. [100] The method of paragraph [98] or [99], wherein the cancer comprises a Ras mutation. [101] The method of paragraph [100], wherein the Ras mutation is K-Ras G12C, K-Ras G13C, H-Ras G12C, H-Ras G13C, N-Ras G12C, or N-Ras G13C. [102] A method of treating a Ras protein-related disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [88] or a pharmaceutical composition of paragraph [89]. [103] A method of inhibiting a Ras protein in a cell, the method comprising contacting the cell with an effective amount of a compound, or a pharmaceutically acceptable salt thereof, of any one of paragraphs [1] to [88] or a pharmaceutical composition of paragraph [89]. [104] The method of paragraph [102] or [103], wherein the Ras protein is K-Ras G12C, K-Ras G13C, H-Ras G12C, H-Ras G13C, N-Ras G12C, or N-Ras G13C. [105] The method of paragraph [103] or [104], wherein the cell is a cancer cell. [106] The method of paragraph [105], wherein the cancer cell is a pancreatic cancer cell, a colorectal cancer cell, a non-small cell lung cancer cell, or an endometrial cancer cell. [107] The method or use of any one of paragraphs [98] to [106], wherein the method or use further comprises administering an additional anti-cancer therapy. [108] The method of paragraph [107], wherein the additional anti-cancer therapy is an EGFR inhibitor, a second Ras inhibitor, a SHP2 inhibitor, a SOS1 inhibitor, a Raf inhibitor, a MEK inhibitor, an ERK inhibitor, a PI3K inhibitor, a PTEN inhibitor, an AKT inhibitor, an mTORC1 inhibitor, a BRAF inhibitor, a PD-L1 inhibitor, a PD-1 inhibitor, a CDK4/6 inhibitor, a HER2 inhibitor, or a combination thereof. [109] The method of paragraph [107] or [108], wherein the additional anti-cancer therapy is a SHP2 inhibitor.

EXAMPLES

(352) The disclosure is further illustrated by the following examples and synthesis examples, which are not to be construed as limiting this disclosure in scope or spirit to the specific procedures herein described. It is to be understood that the examples are provided to illustrate certain embodiments and that no limitation to the scope of the disclosure is intended thereby. It is to be further understood that resort may be had to various other embodiments, modifications, and equivalents thereof which may suggest themselves to those skilled in the art without departing from the spirit of the present disclosure or scope of the appended claims.

(353) Chemical Syntheses

(354) Definitions used in the following examples and elsewhere herein are:

(355) TABLE-US-00003 CH.sub.2Cl.sub.2, Methylene chloride, Dichloromethane DCM CH.sub.3CN, Acetonitrile MeCN CuI Copper (I) iodide DIPEA Diisopropylethyl amine DMF N,N-Dimethylformamide EtOAc Ethyl acetate h hour H.sub.2O Water HCl Hydrochloric acid K.sub.3PO.sub.4 Potassium phosphate (tribasic) MeOH Methanol Na.sub.2SO.sub.4 Sodium sulfate NMP N-methyl pyrrolidone Pd(dppf)Cl.sub.2 [1,1′-Bis(diphenylphosphino)ferrocene]dichloropalladium(II)

Instrumentation

(356) Mass spectrometry data collection took place with a Shimadzu LCMS-2020, an Agilent 1260LC-6120/6125MSD, a Shimadzu LCMS-2010EV, or a Waters Acquity UPLC, with either a QDa detector or SQ Detector 2. Samples were injected in their liquid phase onto a C-18 reverse phase. The compounds were eluted from the column using an acetonitrile gradient and fed into the mass analyzer. Initial data analysis took place with either Agilent ChemStation, Shimadzu LabSolutions, or Waters MassLynx. NMR data was collected with either a Bruker AVANCE III HD 400 MHz, a Bruker Ascend 500 MHz instrument, or a Varian 400 MHz, and the raw data was analyzed with either TopSpin or Mestrelab Mnova.

(357) Synthesis of Intermediates

Intermediate 1. Synthesis of 3-(5-bromo-1-ethyl-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]indol-3-yl)-2,2-dimethylpropan-1-ol

(358) ##STR01229## ##STR01230##

Step 1.

(359) To a mixture of 3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropanoyl chloride (65 g, 137 mmol, crude) in DCM (120 mL) at 0° C. under an atmosphere of N.sub.2 was added 1M SnCl.sub.4 in DCM (137 mL, 137 mmol) slowly. The mixture was stirred at 0° C. for 30 min, then a solution of 5-bromo-1H-indole (26.8 g, 137 mmol) in DCM (40 mL) was added dropwise. The mixture was stirred at 0° C. for 45 min, then diluted with EtOAc (300 mL), washed with brine (100 mL×4), dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the

residue was purified by silica gel column chromatography to give 1-(5-bromo-1H-indol-3-yl)-3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropan-1-one (55 g, 75% yield). LCMS (ESI): m/z [M+Na] calc'd for $C_{29}H_{32}BrNO_2SiNa$ 556.1; found 556.3.

(360) Step 2.

(361) To a mixture of 1-(5-bromo-1H-indol-3-yl)-3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropan-1-one (50 g, 93.6 mmol) in THF (100 mL) at 0° C. under an atmosphere of N_2 was added $LiBH_4$ (6.1 g, 281 mmol). The mixture was heated to 60° C. and stirred for 20 h, then MeOH (10 mL) and EtOAc (100 mL) were added and the mixture washed with brine (50 mL), dried over Na_2SO_4, filtered, and the filtrate concentrated under reduced pressure. The residue was diluted with DCM (50 mL), cooled to 10° C. and diludine (9.5 g, 37.4 mmol) and TsOH.Math.H_2O (890 mg, 4.7 mmol) added. The mixture was stirred at 10° C. for 2 h, filtered, the filtrate concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 1-(5-bromo-1H-indol-3-yl)-3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropan-1-one (41 g, 84% yield). LCMS (ESI): m/z [M+H] calc'd for $C_{29}H_{34}BrNOSi$ 519.2; found 520.1; ^1H NMR (400 MHz, $CDCl_3$) δ 7.96 (s, 1H), 7.75-7.68 (m, 5H), 7.46-7.35 (m, 6H), 7.23-7.19 (m, 2H), 6.87 (d, J=2.1 Hz, 1H), 3.40 (s, 2H), 2.72 (s, 2H), 1.14 (s, 9H), 0.89 (s, 6H).

(362) Step 3.

(363) To a mixture of 1-(5-bromo-1H-indol-3-yl)-3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropan-1-one (1.5 g, 2.9 mmol) and 12 (731 mg, 2.9 mmol) in THF (15 mL) at rt was added AgOTf (888 mg, 3.5 mmol). The mixture was stirred at rt for 2 h, then diluted with EtOAc (200 mL) and washed with saturated $Na_2S_2O_3$ (100 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-iodo-1H-indole (900 mg, 72% yield) as a solid. ^1H NMR (400 MHz, DMSO-d_6) δ 11.70 (s, 1H), 7.68 (d, J=1.3 Hz, 1H), 7.64-7.62 (m, 4H), 7.46-7.43 (m, 6H), 7.24-7.22 (d, 1H), 7.14-7.12 (dd, J=8.6, 1.6 Hz, 1H), 3.48 (s, 2H), 2.63 (s, 2H), 1.08 (s, 9H), 0.88 (s, 6H).

(364) Step 4.

(365) To a stirred mixture of HCOOH (66.3 g, 1.44 mol) in TEA (728 g, 7.2 mol) at 0° C. under an atmosphere of Ar was added (4S,5S)-2-chloro-2-methyl-1-(4-methylbenzenesulfonyl)-4,5-diphenyl-1,3-diaza-2-ruthenacyclopentane cymene (3.9 g, 6.0 mmol) portion-wise. The mixture was heated to 40° C. and stirred for 15 min, then cooled to rt and 1-(3-bromopyridin-2-yl)ethanone (120 g, 600 mmol) added in portions. The mixture was heated to 40° C. and stirred for an additional 2 h, then the solvent was concentrated under reduced pressure. Brine (2 L) was added to the residue, the mixture was extracted with EtOAc (4×700 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (1 S)-1-(3-bromopyridin-2-yl)ethanol (100 g, 74% yield) a an oil. LCMS (ESI): m/z [M+H] calc'd for C_7H_8BrNO 201.1; found 201.9.

(366) Step 5.

(367) To a stirred mixture of (1 S)-1-(3-bromopyridin-2-yl)ethanol (100 g, 495 mmol) in DMF (1 L) at 0° C. was added NaH, 60% dispersion in oil (14.25 g, 594 mmol) in portions. The mixture was stirred at 0° C. for 1 h. Mel (140.5 g, 990 mmol) was added dropwise at 0° C. and the mixture was allowed to warm to rt and stirred for 2 h. The mixture was cooled to 0° C. and saturated NH_4Cl (5 L) was added. The mixture was extracted with EtOAc (3×1.5 L), dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 3-bromo-2-[(1S)-1-methoxyethyl]pyridine (90 g, 75% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_8H_{10}BrNO$ 215.0; found 215.9.

(368) Step 6.

(369) To a stirred mixture of 3-bromo-2-[(1S)-1-methoxyethyl]pyridine (90 g, 417 mmol) and Pd(dppf)Cl.sub.2 (30.5 g, 41.7 mmol) in toluene (900 mL) at rt under an atmosphere of Ar was added bis(pinacolato)diboron (127 g, 500 mmol) and KOAc (81.8 g, 833 mmol) in portions. The mixture was heated to 100° C. and stirred for 3 h. The filtrate was concentrated under reduced pressure and the residue was purified by Al.sub.2O.sub.3 column chromatography to give 2-[(1 S)-1-methoxyethyl]-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine (100 g, 63% yield) as a semi-solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.14H.sub.22BNO.sub.3 263.2; found 264.1.
(370) Step 7.
(371) To a stirred mixture of 5-bromo-3-[3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl]-2-iodo-1H-indole (140 g, 217 mmol) and 2-[(1 S)-1-methoxyethyl]-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine (100 g, 380 mmol) in 1,4-dioxane (1.4 L) at rt under an atmosphere of Ar was added K.sub.2CO.sub.3 (74.8 g, 541 mmol), Pd(dppf)Cl.sub.2 (15.9 g, 21.7 mmol), and H.sub.2O (280 mL) in portions. The mixture was heated to 85° C. and stirred for 4 h, then cooled, H.sub.2O (5 L) added, and the mixture extracted with EtOAc (3×2 L). The combined organic layers were washed with brine (2×1 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 5-bromo-3-[3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl]-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]-1H-indole (71 g, 45% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.37H.sub.43BrN.sub.2O.sub.2Si 654.2; found 655.1.
(372) Step 8.
(373) To a stirred mixture of 5-bromo-3-[3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl]-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]-1H-indole (71 g, 108 mmol) in DMF (0.8 L) at 0° C. under an atmosphere of N.sub.2 was added Cs.sub.2CO.sub.3 (70.6 g, 217 mmol) and Etl (33.8 g, 217 mmol) in portions. The mixture was warmed to rt and stirred for 16 h then H.sub.2O (4 L) added and the mixture extracted with EtOAc (3×1.5 L). The combined organic layers were washed with brine (2×1 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 5-bromo-3-[3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl]-1-ethyl-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]indole (66 g, 80% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.47BrN.sub.2O.sub.2Si 682.3; found 683.3.
(374) Step 9.
(375) To a stirred mixture of TBAF (172.6 g, 660 mmol) in THF (660 mL) at rt under an atmosphere of N.sub.2 was added 5-bromo-3-[3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl]-1-ethyl-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indole (66 g, 97 mmol) in portions. The mixture was heated to 50° C. and stirred for 16 h, cooled, diluted with H.sub.2O (5 L), and extracted with EtOAc (3×1.5 L). The combined organic layers were washed with brine (2×1 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. After filtration, the filtrate was concentrated under reduced pressure. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 3-(5-bromo-1-ethyl-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-3-yl)-2,2-dimethylpropan-1-ol (30 g, 62% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.29BrN.sub.2O.sub.2 444.1; found 445.1.
Intermediate 1. Alternative Synthesis Through Fisher Indole Route
(376) ##STR01231## ##STR01232##
Step 1.
(377) To a mixture of i-PrMgCl (2M in in THF, 0.5 L) at −10° C. under an atmosphere of N.sub.2 was added n-BuLi, 2.5 M in hexane (333 mL, 833 mmol) dropwise over 15 min. The mixture was stirred for 30 min at −10° C. then 3-bromo-2-[(1 S)-1-methoxyethyl]pyridine (180 g, 833 mmol) in THF (0.5 L) added dropwise over 30 min at −10° C. The resulting mixture was warmed to −5° C. and stirred for 1 h, then 3,3-dimethyloxane-2,6-dione (118 g, 833 mmol) in THF (1.2 L) was added dropwise over 30 min at −5° C. The mixture was warmed to 0° C. and stirred for 1.5 h, then

quenched with the addition of pre-cooled 4M HCl in 1,4-dioxane (0.6 L) at 0° C. to adjust pH˜5. The mixture was diluted with ice-water (3 L) and extracted with EtOAc (3×2.5 L). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered, the filtrate was concentrated under reduced pressure, and the residue was purified by silica gel column chromatography to give 5-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]-2,2-dimethyl-5-oxopentanoic acid (87 g, 34% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.15H.sub.21NO.sub.4 279.2; found 280.1.
(378) Step 2.
(379) To a mixture of 5-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]-2,2-dimethyl-5-oxopentanoic acid (78 g, 279 mmol) in EtOH (0.78 L) at rt under an atmosphere of N.sub.2 was added (4-bromophenyl)hydrazine HCl salt (68.7 g, 307 mmol) in portions. The mixture was heated to 85° C. and stirred for 2 h, cooled to rt, then 4M HCl in 1,4-dioxane (69.8 mL, 279 mmol) added dropwise. The mixture was heated to 85° C. and stirred for an additional 3 h, then concentrated under reduced pressure, and the residue was dissolved in TFA (0.78 L). The mixture was heated to 60° C. and stirred for 1.5 h, concentrated under reduced pressure, and the residue adjusted to pH˜5 with saturated NaHCO.sub.3, then extracted with EtOAc (3×1.5 L). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered, the filtrate concentrated under reduced pressure, and the residue was purified by silica gel column chromatography to give 3-(5-bromo-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]-1H-indol-3-yl)-2,2-dimethylpropanoic acid and ethyl (S)-3-(5-bromo-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-indol-3-yl)-2,2-dimethylpropanoate (78 g, crude). LCMS (ESI): m/z [M+H] calc'd for C.sub.21H.sub.23BrN.sub.2O.sub.3 430.1 and C.sub.23H.sub.27BrN.sub.2O.sub.3 458.1; found 431.1 and 459.1.
(380) Step 3.
(381) To a mixture of 3-(5-bromo-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]-1H-indol-3-yl)-2,2-dimethylpropanoic acid and ethyl (S)-3-(5-bromo-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-indol-3-yl)-2,2-dimethylpropanoate (198 g, 459 mmol) in DMF (1.8 L) at 0° C. under an atmosphere of N.sub.2 was added Cs.sub.2CO.sub.3 (449 g, 1.38 mol) in portions. Etl (215 g, 1.38 mmol) in DMF (200 mL) was then added dropwise at 0° C. The mixture was warmed to rt and stirred for 4 h then diluted with brine (5 L) and extracted with EtOAc (3×2.5 L). The combined organic layers were washed with brine (2×1.5 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give ethyl 3-(5-bromo-1-ethyl-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-3-yl)-2,2-dimethylpropanoate (160 g, 57% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.25H.sub.31BrN.sub.2O.sub.3 486.2; found 487.2.
(382) Step 4.
(383) To a mixture of ethyl 3-(5-bromo-1-ethyl-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-3-yl)-2,2-dimethylpropanoate (160 g, 328 mmol) in THF (1.6 L) at 0° C. under an atmosphere of N.sub.2 was added LiBH.sub.4 (28.6 g, 1.3 mol). The mixture was heated to 60° C. for 16 h, cooled, and quenched with pre-cooled (0° C.) aqueous NH.sub.4Cl (5 L). The mixture was extracted with EtOAc (3×2 L) and the combined organic layers were washed with brine (2×1 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give to two atropisomers (as single atropisomers) of 3-(5-bromo-1-ethyl-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-3-yl)-2,2-dimethylpropan-1-ol (60 g, 38% yield) and (40 g, 26% yield) both as solids. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.29BrN.sub.2O.sub.2 444.1; found 445.2.
Intermediate 2 and Intermediate 4. Synthesis of (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)oxy)phenyl)propanoyl)hexahydropyridazine-3-carboxylate
(384) ##STR01233##
Step 1.
(385) To a mixture of (S)-methyl 2-(tert-butoxycarbonylamino)-3-(3-hydroxyphenyl)propanoate

(10.0 g, 33.9 mmol) in DCM (100 mL) was added imidazole (4.6 g, 67.8 mmol) and TIPSCI (7.8 g, 40.7 mmol). The mixture was stirred at rt overnight then diluted with DCM (200 mL) and washed with H.sub.2O (150 mL×3). The organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered, concentrated under reduced pressure, and the residue was purified by silica gel column chromatography to give (S)-methyl 2-(tert-butoxycarbonylamino)-3-(3-(triisopropylsilyloxy)phenyl)-propanoate (15 g, 98% yield) as an oil. LCMS (ESI): m/z [M+Na] calc'd for C.sub.24H.sub.41NO.sub.5SiNa 474.3; found 474.2.
(386) Step 2.
(387) A mixture of (S)-methyl 2-(tert-butoxycarbonylamino)-3-(3-(triisopropylsilyloxy)phenyl)-propanoate (7.5 g, 16.6 mmol), PinB.sub.2 (6.3 g, 24.9 mmol), [Ir(OMe)(COD)].sub.2 (1.1 g, 1.7 mmol), and 4-tert-butyl-2-(4-tert-butyl-2-pyridyl)pyridine (1.3 g, 5.0 mmol) was purged with Ar (×3), then THF (75 mL) was added and the mixture placed under an atmosphere of Ar and sealed. The mixture was heated to 80° C. and stirred for 16 h, concentrated under reduced pressure, and the residue was purified by silica gel column chromatography to give (S)-methyl 2-(tert-butoxycarbonylamino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-(triisopropylsilyloxy)phenyl)-propanoate (7.5 g, 78% yield) as a solid. LCMS (ESI): m/z [M+Na] calc'd for C.sub.30H.sub.52BNO.sub.7SiNa 600.4; found 600.4; .sup.1H NMR (300 MHz, CD.sub.3OD) δ 7.18 (s, 1H), 7.11 (s, 1H), 6.85 (s, 1H), 4.34 (m, 1H), 3.68 (s, 3H), 3.08 (m, 1H), 2.86 (m, 1H), 1.41-1.20 (m, 26H), 1.20-1.01 (m, 22H), 0.98-0.79 (m, 4H).
(388) Step 3.
(389) To a mixture of triisopropylsilyl (S)-2-((tert-butoxycarbonyl)amino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)oxy)phenyl)propanoate (4.95 g, 6.9 mmol) in MeOH (53 mL) at 0° C. was added LiOH (840 mg, 34.4 mmol) in H.sub.2O (35 mL). The mixture was stirred at 0° C. for 2 h, then acidified to pH˜5 with 1M HCl and extracted with EtOAc (250 mL×2). The combined organic layers were washed with brine (100 mL×3), dried over anhydrous Na.sub.2SO.sub.4, filtered, and the filtrate concentrated under reduced pressure to give (S)-2-((tert-butoxycarbonyl)amino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)oxy)phenyl)propanoic acid (3.7 g, 95% yield), which was used directly in the next step without further purification. LCMS (ESI): m/z [M+NH.sub.4] calc'd for C.sub.29H.sub.50BNO.sub.7SiNH.sub.4 581.4; found 581.4.
(390) Step 4.
(391) To a mixture of methyl (S)-hexahydropyridazine-3-carboxylate (6.48 g, 45.0 mmol) in DCM (200 mL) at 0° C. was added NMM (41.0 g, 405 mmol), (S)-2-((tert-butoxycarbonyl)amino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)oxy)phenyl)propanoic acid (24 g, 42.6 mmol) in DCM (50 mL) then HOBt (1.21 g, 9.0 mmol) and EDCI HCl salt (12.9 g, 67.6 mmol). The mixture was warmed to rt and stirred for 16 h, then diluted with DCM (200 mL) and washed with H.sub.2O (3×150 mL). The organic layer was dried over anhydrous Na.sub.2SO, filtered, the filtrate concentrated under reduced pressure, and the residue was purified by silica gel column chromatography to give methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-((triisopropylsilyl)oxy)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (22 g, 71% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.35H.sub.60BN.sub.3O.sub.8Si 689.4; found 690.5.
Intermediate 3. Synthesis of N—((S)-1-acryloylpyrrolidine-3-carbonyl)-N-methyl-L-valine
(392) ##STR01234##
Step 1.
(393) To a mixture of (S)-1-(tert-butoxycarbonyl)pyrrolidine-3-carboxylic acid (2.2 g, 10.2 mmol) in DMF (10 mL) at rt was added HATU (7.8 g, 20.4 mmol) and DIPEA (5 mL). After stirring at rt for 10 min, tert-butyl methyl-L-valinate (3.8 g, 20.4 mmol) in DMF (10 mL) was added. The mixture was stirred at rt for 3 h, then diluted with DCM (40 mL) and H.sub.2O (30 mL). The aqueous and organic layers were separated and the organic layer was washed with H.sub.2O (3×30 mL), brine (30

mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (S)-tert-butyl 3-(((S)-1-(tert-butoxy)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)pyrrolidine-1-carboxylate (3.2 g, 82% yield) as an oil. LCMS (ESI): m/z [M+Na] calc'd for C.sub.20H.sub.36N.sub.2O.sub.5Na 407.3; found 407.2.
(394) Step 2.
(395) A mixture of (S)-tert-butyl 3-(((S)-1-(tert-butoxy)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)pyrrolidine-1-carboxylate (3.2 g, 8.4 mmol) in DCM (13 mL) and TFA (1.05 g, 9.2 mmol) was stirred at rt for 5 h. The mixture was concentrated under reduced pressure to give (S)-tert-butyl 3-methyl-2-((S)—N-methylpyrrolidine-3-carboxamido)butanoate (2.0 g, 84% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.15H.sub.28N.sub.2O.sub.3 284.2; found 285.2.
(396) Step 3.
(397) To a mixture of (S)-tert-butyl 3-methyl-2-((S)—N-methylpyrrolidine-3-carboxamido)butanoate (600 mg, 2.1 mmol) in DCM (6 mL) at 0° C. was added TEA (342 mg, 3.36 mmol). After stirring at 0° C. for 10 mins, acryloyl chloride (284 mg, 3.2 mmol) in DCM (10 mL) was added. The mixture was warmed to rt and stirred for 24 h, then diluted with DCM (30 mL) and H.sub.2O (30 mL). The aqueous and organic layers were separated and the organic layer was washed with H.sub.2O (3×30 mL), brine (30 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N—((S)-1-acryloylpyrrolidine-3-carbonyl)-N-methyl-L-valinate (500 mg, 70% yield) as an oil.
(398) Step 4.
(399) To a mixture of tert-butyl N—((S)-1-acryloylpyrrolidine-3-carbonyl)-N-methyl-L-valinate (100 mg, 0.29 mmol) in DCM (3.0 mL) at 15° C. was added TFA (0.3 mL). The mixture was warmed to rt and stirred for 5 h, then the mixture was concentrated under reduced pressure to give N—((S)-1-acryloylpyrrolidine-3-carbonyl)-N-methyl-L-valine (150 mg) as a solid. The crude product was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.14H.sub.22N.sub.2O.sub.4 282.2; found 283.2.
Intermediate 5. Synthesis of tert-butyl ((63S,4S)-11-ethyl-12-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-25-((triisopropylsilyl)oxy)-61,62,63,64,65,66-hexahydro-11H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate
(400) ##STR01235##
Step 1.
(401) To a stirred mixture of 3-(5-bromo-1-ethyl-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-3-yl)-2,2-dimethylpropan-1-ol (30 g, 67 mmol) and methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylate (55.8 g, 80.8 mmol) in 1,4-dioxane (750 mL) at rt under an atmosphere of Ar was added Na.sub.2CO.sub.3 (17.9 g, 168.4 mmol), Pd(DtBPF)Cl.sub.2 (4.39 g, 6.7 mmol), and H.sub.2O (150.00 mL) in portions. The mixture was heated to 85° C. and stirred for 3 h, cooled, diluted with H.sub.2O (2 L), and extracted with EtOAc (3×1 L). The combined organic layers were washed with brine (2×500 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylate (50 g, 72% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.52H.sub.77N.sub.5O.sub.8Si 927.6; found 928.8.
(402) Step 2.
(403) To a stirred mixture of methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-5-

[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylate (50 g, 54 mmol) in DCE (500 mL) at rt was added trimethyltin hydroxide (48.7 g, 269 mmol) in portion. The mixture was heated to 65° C. and stirred for 16 h, then filtered and the filter cake washed with DCM (3×150 mL). The filtrate was concentrated under reduced pressure to give (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid (70 g, crude), which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.75N.sub.5O.sub.8Si 913.5; found 914.6.

(404) Step 3.

(405) To a stirred mixture of (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid (70 g) in DCM (5 L) at 0° C. under an atmosphere of N.sub.2 was added DIPEA (297 g, 2.3 mol), HOBT (51.7 g, 383 mmol) and EDCI (411 g, 2.1 mol) in portions. The mixture was warmed to rt and stirred for 16 h, then diluted with DCM (1 L), washed with brine (3×1 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (36 g, 42% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.73N.sub.5O.sub.7Si 895.5; found 896.5.

Intermediate 6. Synthesis of tert-butyl N-[(8S,14S)-21-iodo-18,18-dimethyl-9,15-dioxo-4-[(triisopropylsilyl)oxy]-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate

(406) ##STR01236## ##STR01237##

Step 1.

(407) This reaction was undertaken on 5-batches in parallel on the scale illustrated below.

(408) Into a 2 L round-bottom flasks each were added 5-bromo-3-[3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl]-1H-indole (100 g, 192 mmol) and TBAF (301.4 g, 1.15 mol) in THF (1.15 L) at rt. The resulting mixture was heated to 50° C. and stirred for 16 h, then the mixture was concentrated under reduced pressure. The combined residues were diluted with H.sub.2O (5 L) and extracted with EtOAc (3×2 L). The combined organic layers were washed with brine (2×1.5 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 3-(5-bromo-1H-indol-3-yl)-2,2-dimethylpropan-1-ol (310 g, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.13H.sub.16BrNO 281.0 and 283.0; found 282.1 and 284.1.

(409) Step 2.

(410) This reaction was undertaken on two batches in parallel on the scale illustrated below.

(411) To a stirred mixture of 3-(5-bromo-1H-indol-3-yl)-2,2-dimethylpropan-1-ol (135 g, 478 mmol) and TEA (145.2 g, 1.44 mol) in DCM (1.3 L) at 0° C. under an atmosphere of N.sub.2 was added Ac.sub.2O (73.3 g, 718 mmol) and DMAP (4.68 g, 38.3 mmol) in portions. The resulting mixture was stirred for 10 min at 0° C., then washed with H.sub.2O (3×2 L). The organic layers from each experiment were combined and washed with brine (2×1 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by column chromatography to give 3-(5-bromo-1H-indol-3-yl)-2,2-dimethylpropyl acetate (304 g, 88% yield) as a solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.16-11.11 (m, 1H), 7.69 (d, J=2.0 Hz, 1H), 7.32 (d, J=8.6 Hz, 1H), 7.19-7.12 (m, 2H), 3.69 (s, 2H), 2.64 (s, 2H), 2.09 (s, 3H), 0.90 (s, 6H).

(412) Step 3.
(413) This reaction was undertaken on four batches in parallel on the scale illustrated below.
(414) Into a 2 L round-bottom flasks were added methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-5-[(triisopropylsilyl)oxy]phenyl]propanoate (125 g, 216 mmol), 1,4-dioxane (1 L), H.sub.2O (200 mL), 3-(5-bromo-1H-indol-3-yl)-2,2-dimethylpropyl acetate (73.7 g, 227 mmol), K2CO.sub.3 (59.8 g, 433 mmol), and Pd(DtBPF)Cl.sub.2 (7.05 g, 10.8 mmol) at rt under an atmosphere of Ar. The resulting mixture was heated to 65° C. and stirred for 2 h, then diluted with H.sub.2O (10 L) and extracted with EtOAc (3×3 L). The combined organic layers were washed with brine (2×2 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by column chromatography to give methyl (2S)-3-(3-[3-[3-(acetyloxy)-2,2-dimethylpropyl]-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (500 g, 74% yield) as an oil. LCMS (ESI): m/z [M+Na] calc'd for C.sub.39H.sub.58N.sub.2O.sub.7SiNa 717.4; found 717.3.
(415) Step 4.
(416) This reaction was undertaken on three batches in parallel on the scale illustrated below.
(417) To a stirred mixture of methyl (2S)-3-(3-[3-[3-(acetyloxy)-2,2-dimethylpropyl]-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (150 g, 216 mmol) and NaHCO.sub.3 (21.76 g, 259 mmol) in THF (1.5 L) was added AgOTf (66.5 g, 259 mmol) in THF dropwise at 0° C. under an atmosphere of nitrogen. 12 (49.3 g, 194 mmol) in THF was added dropwise over 1 h at 0° C. and the resulting mixture was stirred for an additional 10 min at 0° C. The combined experiments were diluted with aqueous Na.sub.2S.sub.2O.sub.3 (5 L) and extracted with EtOAc (3×3 L). The combined organic layers were washed with brine (2×1.5 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by column chromatography to give methyl (2S)-3-(3-[3-[3-(acetyloxy)-2,2-dimethylpropyl]-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (420 g, 71% yield) as an oil. LCMS (ESI): m/z [M+Na] calc'd for C.sub.39H.sub.57IN.sub.2O.sub.7SiNa, 843.3; found 842.9.
(418) Step 5.
(419) This reaction was undertaken on three batches in parallel on the scale illustrated below.
(420) To a 2 L round-bottom flask were added methyl (2S)-3-(3-[3-[3-(acetyloxy)-2,2-dimethylpropyl]-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (140 g, 171 mmol), MeOH (1.4 L) and K.sub.3PO.sub.4 (108.6 g, 512 mmol) at 0° C. The mixture was warmed to rt and stirred for 1 h, then the combined experiments were diluted with H.sub.2O (9 L) and extracted with EtOAc (3×3 L). The combined organic layers were washed with brine (2×2 L), dried over anhydrous Na.sub.2SO.sub.4, filtered, and the filtrate was concentrated under reduced pressure to give methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoate (438 g, crude) as a solid. LCMS (ESI): m/z [M+Na] calc'd for C.sub.37H.sub.55IN.sub.2O.sub.6SiNa 801.3; found 801.6.
(421) Step 6.
(422) This reaction was undertaken on three batches in parallel on the scale illustrated below.
(423) To a stirred mixture of methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoate (146 g, 188 mmol) in THF (1.46 L) was added LiOH (22.45 g, 937 mmol) in H.sub.2O (937 mL) dropwise at 0° C. The resulting mixture was warmed to rt and stirred for 1.5 h [note: LCMS showed 15% de-TIPS product]. The mixture was acidified to pH 5 with 1M HCl (1M) and the combined experiments were extracted with EtOAc (3×3 L). The combined organic layers were washed with brine (2×2 L), dried over anhydrous Na.sub.2SO.sub.4, filtered, and the filtrate was concentrated under reduced pressure to give (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoic acid (402 g, crude) as a solid. LCMS (ESI):

m/z [M+Na] calc'd for C.sub.36H.sub.53IN.sub.2O.sub.6SiNa 787.3; found 787.6.
(424) Step 7.
(425) To a stirred mixture of (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoic acid (340 g, 445 mmol) and methyl (3S)-1,2-diazinane-3-carboxylate (96.1 g, 667 mmol) in DCM (3.5 L) was added NMM (225 g, 2.2 mol), EDCI (170 g, 889 mmol), and HOBT (12.0 g, 88.9 mmol) portionwise at 0° C. The mixture was warmed to rt and stirred for 16 h, then washed with H.sub.2O (3×2.5 L), brine (2×1 L), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by column chromatography to give methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylate (310 g, 62% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.63IN.sub.4O.sub.7Si 890.4; found 890.8.
(426) Step 8.
(427) This reaction was undertaken on three batches in parallel on the scale illustrated below.
(428) To a stirred mixture of methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylate (85.0 g, 95.4 mmol) in THF (850 mL) each added LiOH (6.85 g, 286 mmol) in H.sub.2O (410 mL) dropwise at 0° C. under an atmosphere of N.sub.2. The mixture was stirred at 0° C. for 1.5 h [note: LCMS showed 15% de-TIPS product], then acidified to pH 5 with 1M HCl and the combined experiments extracted with EtOAc (3×2 L). The combined organic layers were washed with brine (2×1.5 L), dried over anhydrous Na.sub.2SO.sub.4, filtered, and the filtrate was concentrated under reduced pressure to give (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid (240 g, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.41H.sub.61IN.sub.4O.sub.7Si 876.3; found 877.6.
(429) Step 9.
(430) This reaction was undertaken on two batches in parallel on the scale illustrated below.
(431) To a stirred mixture of (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid (120 g, 137 mmol) in DCM (6 L) was added DIPEA (265 g, 2.05 mol), EDCI (394 g, 2.05 mol), and HOBT (37 g, 274 mmol) in portions at 0° C. under an atmosphere of N.sub.2. The mixture was warmed to rt and stirred overnight, then the combined experiments were washed with H.sub.2O (3×6 L), brine (2×6 L), dried over anhydrous Na.sub.2SO.sub.4, and filtered. After filtration, the filtrate was concentrated under reduced pressure. The filtrate was concentrated under reduced pressure and the residue was purified by column chromatography to give tert-butyl N-[(8S,14S)-21-iodo-18,18-dimethyl-9,15-dioxo-4-[(triisopropylsilyl)oxy]-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (140 g, 50% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.41H.sub.59IN.sub.4O.sub.6Si 858.9; found 858.3.
Intermediate 7. Synthesis of (S)-methyl 2-((tert-butoxycarbonyl)amino)-3-(4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-2-yl)propanoate
(432) ##STR01238## Step 1. Zn dust (28 g, 428 mmol) was added to a 1 L, three necked, round bottomed flask, purged with N.sub.2, and heated with a heat gun for 10 min under vacuum. The mixture was cooled to rt, and a solution of 1,2-dibromoethane (1.85 mL, 21.5 mmol) in DMF (90 mL) was added dropwise over 10 min. The mixture was heated at 90° C. for 30 min and re-cooled to rt. TMSCI (0.55 mL, 4.3 mmol) was added, and the mixture was stirred for 30 min at rt, then a mixture of (R)-methyl 2-((tert-butoxycarbonyl)amino)-3-iodopropanoate (22.5 g, 71.4 mmol) in DMF (200 mL) was added dropwise over a period of 10 min. The mixture was heated at 35° C. and

stirred for 2 h, then cooled to rt, and 2,4-dichloropyridine (16 g, 109 mmol) and $Pd(PPh_3)_2Cl_2$ (4 g, 5.7 mmol) added. The mixture was heated at 45° C. and stirred for 2 h, cooled, and filtered, then H_2O (1 L) and EtOAc (0.5 L) were added to the filtrate. The organic and aqueous layers were separated, and the aqueous layer was extracted with EtOAc (2×500 mL). The organic layers were dried over anhydrous Na_2SO_4, filtered, and concentrated under reduced pressure. The crude residue was purified by silica gel column chromatography to give (S)-methyl 2-((tert-butoxycarbonyl)amino)-3-(4-chloropyridin-2-yl)propanoate (6.5 g, 29% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{14}H_{19}ClN_2O_4$ 314.1; found 315.1.

Step 2.

(433) To a mixture of (S)-methyl 2-((tert-butoxycarbonyl)amino)-3-(4-chloropyridin-2-yl)propanoate (6.5 g, 20.6 mmol) in 1,4-dioxane (80 mL) at rt under an atmosphere of N_2 was added bis(pinacolato)diboron (6.3 g, 24.7 mmol), KOAc (8.1 g, 82.4 mmol), and $Pd(PCy_3)_2Cl_2$ (1.9 g, 2.5 mmol). The mixture was heated to 100° C. and stirred for 3 h, then H_2O (100 mL) added and the mixture extracted with EtOAc (3×200 mL). The organic layers were combined, washed with brine (2×100 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (S)-methyl 2-((tert-butoxycarbonyl)amino)-3-(4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-2-yl)propanoate (6 g, 72% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{20}H_{31}BN_2O_6$ 406.2; found 407.3.

(434) Synthesis of Intermediate 8.

(435) ##STR01239##

Step 1.

(436) To a mixture of 4-(dimethylamino)but-2-ynoic acid (900 mg, 7.0 mmol) in DMF (20 mL) at −5° C. was added tert-butyl N-methyl-N—((S)-pyrrolidine-3-carbonyl)-L-valinate (1.0 g, 3.5 mmol), DIPEA (2.2 g, 17.6 mmol) and HATU (2.7 g, 7.0 mmol) in portions. The mixture was stirred between −5 to 5° C. for 1 h, then diluted with EtOAc (100 mL) and ice-H_2O (100 mL). The aqueous and organic layers were separated and the organic layer was washed with H_2O (3×100 mL), brine (100 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N—((S)-1-(4-(dimethylamino)but-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate (900 mg, 55% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{21}H_{35}N_3O_4$ 393.5; found 394.3.

(437) Step 2.

(438) To a mixture of tert-butyl N—((S)-1-(4-(dimethylamino)but-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate (260 mg, 0.66 mmol) in DCM (6 mL) was added TFA (3 mL) at rt. The mixture was stirred at rt for 2 h, then the solvent was concentrated under reduced pressure to give (2S)-2-{1-[(3S)-1-[4-(dimethylamino)but-2-ynoyl]pyrrolidin-3-yl]-N-methylformamido}-3-methylbutanoic acid (280 mg) as an impure oil. The crude product was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{17}H_{27}N_3O_4$ 337.2; found 338.3.

(439) Synthesis of Intermediate 9.

(440) ##STR01240##

Step 1.

(441) To a mixture of tert-butyl N-methyl-N—((S)-pyrrolidine-3-carbonyl)-L-valinate (500 mg, 1.8 mmol) in DCM (8 mL) at 5° C. was added TEA (533 mg, 5.3 mmol) followed by dropwise addition of 2-chloroethane-1-sulfonyl chloride (574 mg, 3.5 mmol) in DCM (2 mL) The mixture was stirred at 5° C. for 1 h, then diluted with H_2O (20 mL) and extracted with EtOAC (3×10 mL). The combined organic layers were washed with brine (10 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N-methyl-N—((S)-1-(vinylsulfonyl)pyrrolidine-

3-carbonyl)-L-valinate (300 mg, 45% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.17H.sub.30N.sub.2O.sub.5S 374.2; found 375.2.

(442) Step 2.

(443) To a mixture of tert-butyl N-methyl-N—((S)-1-(vinylsulfonyl)pyrrolidine-3-carbonyl)-L-valinate (123 mg, 0.33 mmol) in DCM (3 mL) at rt was added TFA (1 mL). The mixture was stirred at rt for 1 h, then concentrated under reduced pressure to give N-methyl-N—((S)-1-(vinylsulfonyl)pyrrolidine-3-carbonyl)-L-valine (130 mg, crude) as a solid, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.13H.sub.22N.sub.2O.sub.5S 318.1; found 319.1.

(444) Synthesis of Intermediate 10.

(445) ##STR01241## ##STR01242##

Step 1.

(446) A mixture of 5-chloro-1H-pyrrolo[3,2-b]pyridine-3-carbaldehyde (8.5 g, 47.1 mmol) and ethyl 2-(triphenylphosphoranylidene)propionate (2.56 g, 70.7 mmol) in 1,4-dioxane (120 mL) was stirred at reflux for 4 h, then concentrated under reduced pressure. EtOAc (200 mL) was added and the mixture was washed with brine, dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give ethyl (E)-3-(5-chloro-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylacrylate (7.5 g, 60% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.13H.sub.13ClN.sub.2O.sub.2 264.1; found 265.1.

(447) Step 2.

(448) To a mixture of ethyl (E)-3-(5-chloro-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylacrylate (7.5 g, 28.3 mmol) and NiCl.sub.2 (4.8 g, 28.3 mmol) in 1:1 THF/MeOH (300 mL) was added NaBH.sub.4 (21.5 g, 566 mmol) in 20 portions every 25 minutes. After complete addition, the mixture was stirred at rt for 30 min, then diluted with EtOAc (500 mL) and washed with brine, dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give ethyl 3-(5-chloro-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (3.4 g, 45% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.13H.sub.15ClN.sub.2O.sub.2 266.1; found 267.1.

(449) Step 3.

(450) To a mixture of ethyl 3-(5-chloro-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (7.0 g, 26.2 mmol) and AgOTf (6.7 g, 26.2 mmol) in THF (50 mL) at 0° C. was added 12 (6.65 g, 26.2 mol). The mixture was stirred at 0° C. for 30 min then diluted with EtOAc (100 mL), washed with Na.sub.2SO.sub.3 (50 mL), brine (50 mL), dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give ethyl 3-(5-chloro-2-iodo-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (6 g, 58% yield) as white solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.13H.sub.14ClIN.sub.2O.sub.2 392.0; found 393.0.

(451) Step 4.

(452) To a mixture of ethyl 3-(5-chloro-2-iodo-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (6.0 g, 15.3 mmol) and 2-(2-(2-methoxyethyl)phenyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (5.6 g, 21.4 mmol) and K2CO.sub.3 (6.3 g, 45.9 mmol) in 1,4-dioxane (150 mL) and H.sub.2O (30 mL) under an atmosphere of N.sub.2 was added Pd(dppf)Cl.sub.2.DCM (1.3 g, 3.1 mmol). The mixture was heated to 80° C. and stirred for 4 h, then diluted with EtOAc (500 mL), washed with brine, dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 3-(5-chloro-2-(2-(2-methoxyethyl)phenyl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (5.5 g, 50% yield) as a viscous oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.22H.sub.25ClN.sub.2O.sub.3 400.2; found 401.2.

(453) Step 5.

(454) A mixture of ethyl 3-(5-chloro-2-(2-(2-methoxyethyl)phenyl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (5.5 g, 13.8 mmol), Cs.sub.2CO.sub.3 (8.9 g, 27.5 mmol), and Etl (3.5 g, 27.5 mmol) in DMF (30 mL) at rt was stirred for 10 h. The mixture was diluted with EtOAc (100 mL), washed with brine (20 mL×4), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give ethyl 3-(5-chloro-1-ethyl-2-(2-(2-methoxyethyl)phenyl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (5.6 g, 95% yield) as a viscous oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.25H.sub.31ClN.sub.2O.sub.3 428.2; found 429.2.
(455) Step 6.
(456) To a mixture of ethyl 3-(5-chloro-1-ethyl-2-(2-(2-methoxyethyl)phenyl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (5.4 g, 12.6 mmol) in THF (50 mL) at −65° C. was added 2M LDA (25 mL, 50 mmol) and stirred at −65° C. for 1 h. Mel (3.6 g, 25 mmol) was added and the mixture was stirred at −65° C. for 2.5 h, then aqueous NH.sub.4Cl and EtOAc (50 mL) were added. The aqueous and organic layers were separated and the organic layer was washed with brine (30 mL), dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give ethyl 3-(5-chloro-1-ethyl-2-(2-(2-methoxyethyl)phenyl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (3.2 g, 57% yield) as a viscous oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.25H.sub.31ClN.sub.2O.sub.3 442.2; found 443.2.
(457) Step 7.
(458) To a mixture of ethyl 3-(5-chloro-1-ethyl-2-(2-(2-methoxyethyl)phenyl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (1.0 g, 2.3 mmol) in THF (10 mL) at 5° C. was added LiBH.sub.4 (196 mg, 9.0 mmol). The mixture was heated to 65° C. and stirred for 2 h then aqueous NH.sub.4Cl and EtOAc (50 mL) added. The aqueous and organic layers were separated and the organic layer was washed with brine (30 mL), dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 3-(5-chloro-1-ethyl-2-(2-(2-methoxyethyl)phenyl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropan-1-ol (0.75 g, 82% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.29ClN.sub.2O.sub.2 400.2; found 401.2.
Intermediate 11: Methyl (3S)-1-{(2S)-2-(tert-butoxycarbonyl)amino-3-[3-fluoro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]propanoyl}-1,2-diazinane-3-carboxylate
(459) ##STR01243##
Step 1.
(460) To a stirred solution of methyl (2R)-2-{[(tert-butoxy)carbonyl]amino}-3-(iodozincio)propanoate (12 g, 30 mmol, 1.2 eq) in DMF (100 mL) was added 1-bromo-3-fluoro-5-iodobenzene (7.5 g, 25 mmol, 1 eq) and Pd(PPh.sub.3).sub.2Cl.sub.2 (1.7 g, 2.5 mmol, 0.1 equiv) at 20° C. under N.sub.2 atmosphere. The resulting mixture was stirred for 2 hrs at 65° C. under N.sub.2 atmosphere. The reaction mixture was quenched with water and extracted with EA (200 mL×2). The organic phase was washed with water (200 mL×1) and brine (100 mL×1) and concentrated to dryness to give a residue. The residue was purified by prep-TLC (PE/EA=10/1) to afford methyl 3-(3-bromo-5-fluorophenyl)-2-{[(tert-butoxy)carbonyl]amino}propanoate (6 g, 58% yield) as a colorless oil. LCMS (ESI) m/z=398.1 [M+Na].sup.+, calculated for C.sub.15H.sub.19BrFNO.sub.4: 375.0
(461) Step 2.
(462) To a solution of methyl 3-(3-bromo-5-fluorophenyl)-2-{[(tert-butoxy)carbonyl]amino}propanoate (3.2 g, 8.5 mmol, 1 eq) in THF (50 mL) was added Lithium hydroxide (610.7 mg, 25.5 mmol, 3 eq) in H.sub.2O (10 mL). Then the reaction mixture was stirred at 20° C. for 1 h. The mixture was adjusted to pH=5.0 with 1 M HCl aqueous solution. The mixture was quenched with H.sub.2O (150 mL) and extracted with EA (200 mL×3). The combined organic layers was washed bine (50 mL), dried over Na.sub.2SO.sub.4 and concentrated to afford 3-(3-

bromo-5-fluorophenyl)-2-{[(tert-butoxy)carbonyl]amino}propanoic acid (2.65 g, 68% yield) as a white solid. LCMS (ESI) m/z=384.1 [M+Na].sup.+, calculated for C.sub.14H.sub.15BrFNO4 MW: 361.0
(463) Step 3.
(464) To a mixture of 3-(3-bromo-5-fluorophenyl)-2-{[(tert-butoxy)carbonyl]amino}propanoic acid (2.3 g, 6.4 mmol, 1 eq) and methyl (3S)-1,2-diazinane-3-carboxylate (1.66 g, 11.5 mmol, 1.8 eq) in DMF (150 mL) was added HATU (4.9 g, 12.8 mmol, 2 eq) and DIEA (16.5 g, 128 mmol, 20 eq) in DMF (50 mL) at 0° C. Then the reaction mixture was stirred at 0° C. for 1 h. The mixture was quenched with H.sub.2O (100 mL) and extracted with EA (300 mL×3). The combined organic layers was washed bine (50 mL), dried over Na.sub.2SO.sub.4 and concentrated to give the residue, which was purified by Pre-HPLC eluting with acetonitrile in water (0.1% FA) from 60% to 70% in 10 minutes to give methyl (3S)-1-[(2S)-3-(3-bromo-5-fluorophenyl)-2-{[(tert-butoxy)carbonyl]amino}propanoyl]-1,2-diazinane-3-carboxylate (2.7 g, 78% yield) as a pale yellow solid. LCMS (ESI) m/z=510.1 [M+Na].sup.+,calculated for C.sub.20H.sub.27BrFNO.sub.5: 487.1.
(465) Step 4.
(466) A mixture of methyl (S)-1-((S)-3-(3-bromo-5-fluorophenyl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (3 g, 6.16 mmol, 1 eq), 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (1.9 g, 7.4 mmol, 1.2 eq), KOAc (900 mg, 9.24 mmol, 1.5 eq) and Pd(dppf)Cl.sub.2DCM (0.3 g, 0.37 mmol, 0.05 eq) in dioxane (50 mL) was heated at 100° C. for 17 h under N.sub.2 atmosphere. The mixture was concentrated and purified by column chromatography (DCM/MeOH=100/1 to 40/1) to give methyl (3S)-1-(2S)-2-{(tert-butoxycarbonyl)amino-3-[3-fluoro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]propanoyl}-1,2-diazinane-3-carboxylate (2.6 g, 79% yield) as a yellow oil. LCMS (ESI) m/z=536.2 [M+H]+, calculated for C.sub.26H.sub.39BFNO.sub.7: 535.3.
(467) Compounds A341 and A342 may be prepared using methods disclosed herein via Intermediate 11.
Example A75. Synthesis of two atropisomers of (2S)—N-[(8S,14S,20M)-22-ethyl-4-hydroxy-21-{2-[(1S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide
(468) ##STR01244##
Step 1.
(469) To a stirred mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (18.0 g, 20.1 mmol) in THF (180 mL) at 0° C. was added a 1M solution of TBAF in THF (24.1 mL, 24.1 mmol). The mixture was stirred at 0° C. for 1 h, then diluted with brine (1.5 L) and extracted with EtOAc (3×1 L). The combined organic layers were washed with brine (2×500 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (11.5 g, 69% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.53N.sub.5O.sub.7 739.4; found 740.4.
(470) Step 2.
(471) To a stirred mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-

pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (11.5 g, 15.5 mmol) in DCM (120 mL) at 0° C. was added TFA (60 mL, 808 mmol). The mixture was stirred at 0° C. for 1 h, then concentrated under reduced pressure and the residue again concentrated under reduced pressure with toluene (20 mL; repeated ×3) to give (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (12 g, crude), which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.37H.sub.45N.sub.5O.sub.5 639.3; found 640.6.

(472) Step 3.

(473) To a stirred mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (11.9 g, 18.6 mmol) in DMF (240 mL) at 0° C. under an atmosphere of N.sub.2 was added DIPEA (48.1 g, 372 mmol), (2S)-3-methyl-2-[N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido]butanoic acid (9.45 g, 33.5 mmol) and COMU (11.95 g, 27.9 mmol) in portions. The mixture was stirred ay 0° C. for 90 min, then diluted with brine (1.5 L) and extracted with EtOAc (3×1 L). The combined organic layers were washed with brine (2×500 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. After filtration, the filtrate was concentrated under reduced pressure. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography (×2) to give two atropisomers of (2S)—N-[(8S,14S,20M)-22-ethyl-4-hydroxy-21-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide (2.7 g, 15.5%, yield) and (4.2 g, 24.7% yield) both as solids. LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.65N.sub.7O.sub.8 903.5; found 904.7; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.35-9.27 (m, 1H), 8.77 (dd, J=4.7, 1.7 Hz, 1H), 7.95 (dq, J=6.2, 2.0 Hz, 2H), 7.55 (ddd, J=28.0, 8.2, 4.3 Hz, 3H), 7.08 (dd, J=37.9, 6.2 Hz, 2H), 6.69-6.48 (m, 2H), 6.17 (ddt, J=16.7, 7.2, 2.3 Hz, 1H), 5.74-5.62 (m, 1H), 5.43-5.34 (m, 1H), 5.12-5.00 (m, 1H), 4.25 (d, J=12.3 Hz, 1H), 4.17-3.99 (m, 3H), 3.89-3.65 (m, 4H), 3.66-3.45 (m, 3H), 3.12 (s, 4H), 2.95-2.70 (m, 6H), 2.41-2.06 (m, 5H), 1.99-1.88 (m, 1H), 1.82 (d, J=12.1 Hz, 2H), 1.54 (t, J=12.0 Hz, 1H), 1.21 (dd, J=6.3, 2.5 Hz, 3H), 1.11 (t, J=7.1 Hz, 3H), 0.99-0.88 (m, 6H), 0.79 (ddd, J=27.8, 6.7, 2.1 Hz, 3H), 0.48 (d, J=3.7 Hz, 3H) and LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.65N.sub.7O.sub.8 903.5; found 904.7; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.34-9.27 (m, 1H), 8.77 (dd, J=4.7, 1.7 Hz, 1H), 8.17-7.77 (m, 3H), 7.64-7.43 (m, 3H), 7.33 (d, J=13.7 Hz, 1H), 7.05-6.94 (m, 1H), 6.69-6.41 (m, 2H), 6.26-5.94 (m, 1H), 5.73-5.63 (m, 1H), 5.50-5.20 (m, 2H), 4.40-4.15 (m, 3H), 4.00-3.40 (m, 9H), 3.11 (d, J=4.4 Hz, 3H), 2.93-2.60 (m, 8H), 2.29-2.01 (m, 3H), 1.99 (s, 1H), 1.87-1.75 (m, 2H), 1.73-1.47 (m, 2H), 1.40 (d, J=6.0 Hz, 3H), 1.01-0.88 (m, 6H), 0.85-0.65 (m, 7H), 0.56 (s, 3H).

Example A89. Synthesis of (2S)—N-[(8S,14S)-22-ethyl-4-hydroxy-18,18-dimethyl-21-[6-(4-methylpiperazin-1-yl)pyridin-3-yl]-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide

(474) ##STR01245## ##STR01246##

Step 1.

(475) To a mixture of tert-butyl ((6.sup.3S,4S)-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (240.00 mg, 0.279 mmol, 1.00 equiv) and Cs.sub.2CO.sub.3 (182 mg, 0.558 mmol, 2 equiv) in DMF (5.00 mL) was

added ethyl iodide (113.45 mg, 0.727 mmol, 2.60 equiv) dropwise at 0° C. The reaction was stirred for 16 h at 25° C. The resulting mixture was diluted with water (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (3×10 mL), and dried over anhydrous Na.sub.2SO.sub.4. The filtrate was concentrated under reduced pressure and the remaining residue was purified by silica gel column chromatography to afford tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (190 mg, 77% yield) as a yellow solid.

(476) Step 2.

(477) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (500 mg, 0.54 mmol), 1-methyl-4-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-2-yl]piperazine (257 mg, 0.8 mmol), Pd(dppf)Cl.sub.2 (83 mg, 0.11 mmol) and K.sub.2CO.sub.3 (156 mg, 1.1 mmol) in 1,4-dioxane (25 mL) and H.sub.2O (5 mL) under an atmosphere of Ar was stirred at 80° C. for 2 h. The mixture was concentrated under reduced pressure and the residue was purified by prep-TLC to afford tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-10,10-dimethyl-1.sup.2-(6-(4-methylpiperazin-1-yl)pyridin-3-yl)-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (400 mg, 76% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.53H.sub.77N.sub.7O.sub.6Si 935.6; found 936.6.

(478) Step 3.

(479) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-10,10-dimethyl-1.sup.2-(6-(4-methylpiperazin-1-yl)pyridin-3-yl)-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (350 mg, 0.36 mmol) and 1M TBAF in THF (0.4 mL, 0.4 mmol) in THF (5 mL) was stirred at 0° C. for 1 h. The mixture was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-2.sup.5-hydroxy-10,10-dimethyl-1.sup.2-(6-(4-methylpiperazin-1-yl)pyridin-3-yl)-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (290 mg, 100% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.44H.sub.57N.sub.7O.sub.6 779.4; found 780.4.

(480) Step 4.

(481) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-2.sup.5-hydroxy-10,10-dimethyl-1.sup.2-(6-(4-methylpiperazin-1-yl)pyridin-3-yl)-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (300 mg, 0.37 mmol) in TFA (5 mL) and DCM (5 mL) was stirred at rt for 1 h. The mixture was concentrated under reduced pressure to give (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-10,10-dimethyl-1.sup.2-(6-(4-methylpiperazin-1-yl)pyridin-3-yl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (300 mg, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.49N.sub.7O.sub.4 679.4; found 680.3.

(482) Step 5.

(483) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-10,10-dimethyl-1.sup.2-(6-(4-methylpiperazin-1-yl)pyridin-3-yl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (300 mg, 0.36 mmol) in DMF (3 mL) at 0° C. under an atmosphere of N.sub.2 was added DIPEA (0.96

mL, 5.4 mmol) and (2S)-3-methyl-2-[N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido]butanoic acid (213 mg, 0.72 mmol), followed by dropwise addition of COMU (243 mg, 0.56 mmol). H_2O was added at 0° C. and the mixture was extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (3×10 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the crude residue was purified by Prep-HPLC to give (2S)—N-[(8S,14S)-22-ethyl-4-hydroxy-18,18-dimethyl-21-[6-(4-methylpiperazin-1-yl)pyridin-3-yl]-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide (45 mg, 13.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{53}H_{69}N_9O_7$ 943.5; found 944.8; ^1H NMR (400 MHz, DMSO-d_6) δ 9.39-9.23 (m, 1H), 8.64-8.60 (m, 1H), 8.19-8.16 (m, 1H), 8.15 (d, J=6.2 Hz, 1H), 7.86 (s, 1H), 7.66-7.62 (m, 1H), 7.56-7.54 (m, 1H), 7.50-7.43 (m, 1H), 7.13-7.11 (m, 1H), 7.03-6.95 (m, 1H), 6.70-6.47 (m, 2H), 6.17 (ddt, J=16.8, 6.4, 2.8 Hz, 1H), 5.76-5.63 (m, 1H), 5.45-5.33 (m, 1H), 5.11 (m, 1H), 4.75-4.72 (m, 1H), 4.28-4.24 (m, 1H), 4.11-3.98 (m, 4H), 3.91-3.76 (m, 1H), 3.73-3.71 (m, 1H), 3.59-3.56 (m, 7H), 3.51-3.40 (m, 2H), 3.08-2.94 (m, 1H), 2.94-2.92 (m, 2H), 2.92-2.87 (m, 2H), 2.86-2.83 (m, 2H), 2.80-2.65 (m, 2H), 2.83-2.82 (m, 3H), 2.28-2.25 (m, 3H), 2.08-2.05 (m, 2H), 2.02-1.96 (m, 1H), 1.87-1.78 (m, 1H), 1.74-1.66 (m, 1H), 1.56-1.48 (m, 1H), 1.11-1.08 (m, 4H), 0.99-0.92 (m, 2H), 0.89-0.87 (m, 5H), 0.82-0.73 (m, 2H).

Example A115. Synthesis of two atropisomers of (2S)—N-[(8S,14S,20P)-22-ethyl-21-{4-[(1S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide

(484) ##STR01247## ##STR01248##

Step 1.

(485) A 1 L round-bottom flask was charged with tert-butyl ((6.sup.3S,4S)-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (22.00 g, 32.042 mmol, 1.00 equiv), toluene (300.00 mL), $Pd_2(dba)_3$ (3.52 g, 3.845 mmol, 0.12 equiv), S-Phos (3.95 g, 9.613 mmol, 0.30 equiv), and KOAc (9.43 g, 96.127 mmol, 3.00 equiv) at room temperature. To the mixture was added 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (26.66 g, 208.275 mmol, 6.50 equiv) dropwise with stirring at room temperature. The resulting solution was stirred for 3 h at 60° C. The resulting mixture was filtered, and the filter cake was washed with EtOAc. The filtrate was concentrated under reduced pressure and the remaining residue was purified by silica gel column chromatography to afford tert-butyl ((6.sup.3S,4S)-10,10-dimethyl-5,7-dioxo-12-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (22 g, 90%) as a light yellow solid. ESI-MS m/z=687.3 $[M+H]^+$; Calculated MW: 686.4

(486) Step 2.

(487) A mixture of tert-butyl ((6.sup.3S,4S)-10,10-dimethyl-5,7-dioxo-1.sup.2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (2.0 g, 2.8 mmol), 3-bromo-2-[(1S)-1-methoxyethyl]pyridine (0.60 g, 2.8 mmol), Pd(dppf)Cl_2 (0.39 g, 0.5 mmol), and K_3PO_4 (1.2 g, 6.0 mmol) in 1,4-dioxane (50 mL) and H_2O (10 mL) under an atmosphere of N_2 was heated to 70° C. and stirred for 2 h. The mixture was diluted with H_2O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (3×50 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-

yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (1.5 g, 74% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.40H.sub.49N.sub.5O.sub.6695.4; found 696.5.

(488) Step 3.

(489) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (1.5 g, 2.1 mmol), Cs.sub.2CO.sub.3 (2.1 g, 6.3 mmol), and ethyl iodide (0.43 mL, 5.1 mmol) in DMF (50 mL) was stirred at 0° C. for 16 h. The mixture was quenched at 0° C. with H.sub.2O and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (3×50 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (1.5 g, 99% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.53N.sub.5O.sub.6 723.4; found 724.6.

(490) Step 4.

(491) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (1.3 g, 1.7 mmol) in TFA (10 mL) and DCM (20 mL) was stirred at 0° C. for 2 h. The mixture was concentrated under reduced pressure to afford (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (1.30 g, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.37H.sub.45N.sub.5O.sub.4 623.3; found 624.4.

(492) Step 5.

(493) Into a 40-mL vial purged and maintained with an inert atmosphere of Ar, was placed (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (250 mg, 0.4 mmol), (2S)-3-methyl-2-[N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido]butanoic acid (226 mg, 0.8 mmol), DIPEA (774 mg, 6.0 mmol), and DMF (3 mL). A solution of COMU (257 mg, 0.6 mmol) in DMF (2 mL) was added at 0° C. and the resulting mixture was stirred at 0° C. for 1 h. The mixture was filtered, the filtrate was concentrated under reduced pressure, and the residue was purified by prep-HPLC to give two atropisomers of (2S)—N-[(8S,14S,20P)-22-ethyl-21-{4-[(1 S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide (56 mg, 15% yield) and (46 mg, 13% yield) both as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.65N.sub.7O.sub.7 887.5; found 888.4; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.81 (s, 1H), 8.07 (s, 1H), 8.05-7.96 (m, 1H), 7.78-7.45 (m, 5H), 7.41-7.08 (m, 2H), 6.66-6.58 (m, 1H), 6.18 (d, J=17.0 Hz, 1H), 5.75-5.67 (m, 1H), 5.46-5.31 (m, 1H), 5.16-5.04 (m, 1H), 4.75 (dd, J=10.9, 4.5 Hz, 1H), 4.31-4.21 (m, 2H), 4.11-3.95 (m, 3H), 3.87-3.71 (m, 5H), 3.74-3.54 (m, 3H), 3.11 (s, 4H), 2.95 (d, J=9.7 Hz, 2H), 2.85-2.72 (m, 3H), 2.31-2.04 (m, 3H), 1.88-1.47 (m, 2H), 1.24-1.21 (m, 3H), 1.16-1.08 (m, 3H), 1.03-0.91 (m, 6H), 0.85-0.74 (m, 3H), 0.51-0.46 (m, 3H) and LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.65N.sub.7O.sub.7 887.5; found 888.4; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.77 (s, 1H), 8.71-8.63 (m, 0.5H), 8.23-8.17 (m, 0.5H), 8.00 (s, 1H), 7.85 (t, J=9.9 Hz, 2H), 7.77-

7.62 (m, 3H), 7.57-7.50 (m, 1H), 7.33-7.22 (m, 1H), 7.15-7.06 (m, 1H), 6.73-6.56 (m, 1H), 6.17 (ddd, J=16.7, 6.1, 2.7 Hz, 1H), 5.76-5.64 (m, 1H), 5.49-5.29 (m, 2H), 4.70 (dd, J=10.8, 3.5 Hz, 1H), 4.33-4.22 (m, 3H), 4.14-3.95 (m, 2H), 3.86-3.77 (m, 1H), 3.72-3.65 (m, 2H), 3.61 (t, J=10.6 Hz, 3H), 3.46-3.42 (m, 1H), 3.13 (d, J=4.8 Hz, 3H), 2.99 (d, J=14.4 Hz, 1H), 2.95-2.70 (m, 6H), 2.24-1.99 (m, 4H), 1.95-1.44 (m, 4H), 1.40 (d, J=6.1 Hz, 3H), 0.98-0.87 (m, 6H), 0.86-0.64 (m, 6H), 0.64-0.54 (m, 3H).
Example A2. Synthesis of (2S)—N-[(8S,14S)-4-amino-22-ethyl-21-[2-(2-methoxyethyl)phenyl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide
(494) ##STR01249## ##STR01250##
Step 1.
(495) Into a 25 mL sealed tube were added 3-[1-ethyl-2-[2-(methoxymethyl)phenyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]-2,2-dimethylpropan-1-ol (590 mg, 1.2 mmol), methyl (2S)-3-(3-bromo-5-nitrophenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (747 mg, 1.9 mmol), XPhos Pd G3 (105 mg, 0.12 mmol), XPhos (71 mg, 0.15 mmol), K.sub.2CO.sub.3 (427 mg, 3.1 mmol), and 1,4-dioxane (2 mL) under an atmosphere of N.sub.2 at rt. The mixture was heated to 60° C. and stirred overnight, then cooled and H.sub.2O added. The mixture was extracted with EtOAc (3×20 mL) and the combined organic layers were washed with brine (1×20 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)phenyl]indol-5-yl]-5-nitrophenyl]propanoic acid (500 mg, 61% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.37H.sub.45N.sub.3O.sub.8 659.3; found 660.4.
(496) Step 2.
(497) A mixture of (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)phenyl]indol-5-yl]-5-nitrophenyl]propanoic acid (500 mg, 0.79 mmol), methyl (3S)-1,2-diazinane-3-carboxylate (164 mg, 1.1 mmol), DCM (6 mL), DIPEA (294 mg, 2.3 mmol) and HATU (432 mg, 1.1 mmol) was stirred at 0° C. for 1 h under an atmosphere of air. H.sub.2O was added and the mixture was extracted with DCM (3×20 mL), then the combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)phenyl]indol-5-yl]-5-nitrophenyl]propanoyl]-1,2-diazinane-3-carboxylate (520 mg, 87% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.43H.sub.55N.sub.5O.sub.9 785.4; found 786.8.
(498) Step 3.
(499) Into a 40 mL sealed tube were added methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)phenyl]indol-5-yl]-5-nitrophenyl]propanoyl]-1,2-diazinane-3-carboxylate (510 mg, 0.65 mmol), DCE (5 mL) and trimethyltin hydroxide (587 mg, 3.3 mmol) at rt under an atmosphere of air. The mixture was heated to 60° C. and stirred overnight, cooled, and diluted with DCM (20 mL). The mixture was washed with 0.1 N KHSO.sub.4 (3×20 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and the filtrate was concentrated under reduced pressure to give (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)phenyl]indol-5-yl]-5-nitrophenyl]propanoyl]-1,2-diazinane-3-carboxylic acid (500 mg, 100%) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.53N.sub.5O.sub.9 771.4; found 772.7.
(500) Step 4.
(501) A mixture of (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-

dimethylpropyl)-2-[2-(methoxymethyl)phenyl]indol-5-yl]-5-nitrophenyl]propanoyl]-1,2-diazinane-3-carboxylic acid (490 mg, 0.64 mmol), DCM (100 mL), DIPEA (2.5 g, 19.0 mmol), HOBT (429 mg, 3.2 mmol), and EDCI (3.65 g, 19.0 mmol) at room temperature was stirred at rt overnight under an atmosphere of air. H.sub.2O was added and the mixture was extracted with DCM (3×60 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)phenyl)-10,10-dimethyl-2.sup.5-nitro-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (350 mg, 73% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.51N.sub.5O.sub.8 753.4; found 754.2.
(502) Step 5.
(503) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)phenyl)-10,10-dimethyl-2.sup.5-nitro-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)- benzenacycloundecaphane-4-yl)carbamate (200 mg, 0.27 mmol), MeOH (4 mL), and Pd on carbon (20 mg) was stirred at rt for 2 h under an atmosphere of H.sub.2. The mixture was filtered, the filter cake was washed with MeOH (3×5 mL), and the filtrate was concentrated under reduced pressure to give tert-butyl ((6.sup.3S,4S)-2.sup.5-amino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)phenyl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (60 mg, 31% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.53N.sub.5O.sub.6 723.4; found 724.4.
(504) Step 6.
(505) Into an 8 mL vial were added tert-butyl ((6.sup.3S,4S)-2.sup.5-amino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)phenyl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (50 mg, 0.07 mmol), DCM (1 mL), and TFA (158 mg, 1.4 mmol) at 0° C. under an atmosphere of air. The mixture was stirred for at 0° C. for 2 h then concentrated under reduced pressure to give (6.sup.3S,4S)-2.sup.5,4-diamino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)phenyl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (45 mg) as a solid, which was used directly in the next step directly without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.37H.sub.45N.sub.5O.sub.4 623.3; found 624.4.
(506) Step 7.
(507) Into an 8 mL vial were added (6.sup.3S,4S)-2.sup.5,4-diamino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)phenyl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (40 mg, 0.06 mmol), DMF (1 mL), DIPEA (75 mg, 0.58 mmol), and COMU (41 mg, 0.1 mmol) at 0° C. under an atmosphere of air. The mixture was stirred at 0° C. for 1 h, then H.sub.2O added. The mixture was extracted with EtOAc (3×30 mL), the combined organic layers were concentrated under reduced pressure, and purified by prep-HPLC to give (2S)—N-[(8S,14S)-4-amino-22-ethyl-21-[2-(2-methoxyethyl)phenyl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide (2.5 mg, 4.4% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.65N.sub.7O.sub.7 887.5; found 888.6; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.74-8.55 (m, 1H), 7.89 (d, J=9.6 Hz, 1H), 7.66-7.53 (m, 1H), 7.57-7.47 (m, 6H), 7.32 (t, J=6.4 Hz, 1H), 6.85 (d, J=8.4 Hz, 2H), 6.70-6.55 (m, 1H), 6.24-6.12 (m, 1H), 5.69 (ddd, J=14.8, 8.0, 3.9 Hz, 1H), 5.41 (s, 1H), 5.09-4.80 (m, 2H), 4.26 (d, J=10.1 Hz, 2H), 4.19 (s, 2H),

4.17-4.06 (m, 1H), 4.02 (dd, J=12.0, 3.9 Hz, 1H), 3.92 (d, J=8.0 Hz, 3H), 3.78 (d, J=8.7 Hz, 5H), 3.29 (s, 2H), 3.14 (d, J=1.9 Hz, 1H), 2.98-2.92 (m, 1H), 2.87-2.68 (m, 3H), 2.62 (d, J=12.5 Hz, 3H), 2.15-1.99 (m, 4H), 1.80 (s, 1H), 1.68-1.53 (m, 2H), 1.08 (t, J=7.1 Hz, 1H), 0.98-0.88 (m, 6H), 0.82 (dd, J=23.3, 16.4 Hz, 3H), 0.74 (t, J=7.2 Hz, 3H), 0.44 (s, 2H), 0.43 (s, 3H).

Example A118. Synthesis of (2S)—N-[(7S,13S)-21-ethyl-20-[2-(methoxymethyl)pyridin-3-yl]-17,17-dimethyl-8,14-dioxo-15-oxa-3-thia-9,21,27,28-tetraazapentacyclo[17.5.2.1.SUP.2.,.SUP.5..1.SUP.9.,.SUP.13..0.SUP.22.,.SUP.26.]octacosa-1(25),2(28),4,19,22(26),23-hexaen-7-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide

(508) ##STR01251## ##STR01252##

Step 1.

(509) A mixture of methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-(1,3-thiazol-4-yl)propanoate (2.08 g, 7.26 mmol) and mCPBA (1.88 g, 10.9 mmol) in DCE (15 mL) at 0° C. under an atmosphere of N.sub.2 was diluted with DCM (100 mL). The mixture was allowed to warm to rt and stirred for 16 h, then diluted with DCM, washed with H.sub.2O (1×30 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 4-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-methoxy-3-oxopropyl]-1,3-thiazol-3-ium-3-olate (1.15 g, 47% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.12H.sub.18N.sub.2O.sub.5S 302.1; found 303.2.

(510) Step 2.

(511) To a mixture of 4-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-methoxy-3-oxopropyl]-1,3-thiazol-3-ium-3-olate (1.15 g, 3.8 mmol) in THF at 0° C. under an atmosphere of N.sub.2 was added NBS (0.74 g, 4.2 mmol) dropwise. The mixture was allowed to warm to rt and stirred for 2 h, then diluted with H.sub.2O (500 mL) and extracted with EtOAc (3×500 mL). The combined organic layers were washed with water (2×30 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 2-bromo-4-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-methoxy-3-oxopropyl]-1,3-thiazol-3-ium-3-olate (1.2 g, 74% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.12H.sub.17BrN.sub.2O.sub.5S 380.0; found 381.0.

(512) Step 3.

(513) To a stirred mixture of 2-bromo-4-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-methoxy-3-oxopropyl]-1,3-thiazol-3-ium-3-olate (1.2 g, 3.2 mmol) and 4,4,5,5-tetramethyl-2-(tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (1.04 g, 4.1 mmol) in MeCN at 70° C. under an atmosphere of N.sub.2 was added ethane-1,2-diamine (1.89 g, 31.5 mmol) in portions. The mixture was cooled to 60° C. and the mixture was stirred overnight, then diluted with water (500 mL) and extracted with EtOAc (3×400 mL). The combined organic layers were washed with brine (1×50 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (2S)-3-(2-bromo-1,3-thiazol-4-yl)-2-[(tert-butoxycarbonyl)amino]propanoate (653 mg, 54% yield) as a solid.

(514) Step 4.

(515) A 50 mL sealed tube was charged with 3-[1-ethyl-2-[2-(methoxymethyl)pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]-2,2-dimethylpropan-1-ol (1.00 g, 2.1 mmol), K.sub.2CO.sub.3 (727 mg, 5.2 mmol), Pd(dppf)Cl.sub.2 (153 mg, 0.21 mmol), and 2,4-dibromo-1,3-thiazole (1.0 g, 4.2 mmol) at rt under an atmosphere of N.sub.2, then 1,4-dioxane (1.0 mL) and H.sub.2O (0.20 mL) were added. The mixture was heated to 70° C. and stirred for 4 h, then cooled, diluted with H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (3×100 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 3-[5-(4-bromo-1,3-thiazol-2-yl)-1-ethyl-2-[2-

(methoxymethyl)pyridin-3-yl]indol-3-yl]-2,2-dimethylpropan-1-ol (727 mg, 67% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.34H.sub.44N.sub.4O.sub.6S 636.3; found 637.3.
(516) Step 5.
(517) To a stirred mixture of methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoate (636 mg, 1.0 mmol) and LiOH.Math.H.sub.2O (126 mg, 3.0 mmol) in THF at 0° C. under an atmosphere of N.sub.2 was added H.sub.2O (1.24 mL) portionwise. The mixture was allowed to warm to rt and stirred for 1 h, then diluted with water (300 mL) and extracted with EtOAc (3×300 mL). The combined organic layers were washed with brine (1×100 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and the filtrate concentrated under reduced pressure to give (2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoic acid (622 mg, crude), which was used in the next step directly without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.33H.sub.42N.sub.4O.sub.6S 622.3; found 623.2.
(518) Step 6.
(519) To a stirred mixture of (2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoic acid (622 mg, 1.0 mmol) and methyl (3S)-1,2-diazinane-3-carboxylate (288 mg, 2.0 mmol) in DMF at 0° C. under an atmosphere of N.sub.2 was added HATU (570 mg, 1.5 mmol). The mixture was stirred at 0° C. for 1 h, then diluted with EtOAc and washed with H.sub.2O (1×10 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and the filtrate concentrated under reduced pressure to give methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoyl]-1,2-diazinane-3-carboxylate (550 mg, 62% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.52N.sub.6O.sub.7S 748.4; found 749.6.
(520) Step 7.
(521) (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoyl]-1,2-diazinane-3-carboxylic acid was synthesized in a manner similar to (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid except methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylate was substituted with methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoyl]-1,2-diazinane-3-carboxylate. LCMS (ESI): m/z [M+H] calc'd for C.sub.38H.sub.50N.sub.6O.sub.7S 734.3; found 735.3.
(522) Step 8.
(523) tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate was synthesized in a manner similar to tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate except (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid was substituted with (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoyl]-1,2-diazinane-3-carboxylic acid. LCMS (ESI): m/z [M+H] calc'd for

C.sub.38H.sub.48N.sub.6O.sub.6S 716.3; found 717.4.
(524) Step 9.
(525) To a stirred mixture of tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-thiazola-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)carbamate (253 mg) in DCM at 0° C. under an atmosphere of N.sub.2 was added TFA (1.0 mL) dropwise. The mixture was stirred at 0° C. for 1 h, then concentrated under reduced pressure and then repeated using toluene (20 mL×3) to give (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (253 mg, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.33H.sub.40N.sub.6O.sub.4S 616.3; found 617.3.
(526) Step 10.
(527) (2S)—N-[(7S,13S)-21-ethyl-20-[2-(methoxymethyl)pyridin-3-yl]-17,17-dimethyl-8,14-dioxo-15-oxa-3-thia-9,21,27,28-tetraazapentacyclo[17.5.2.1.sup.2,.sup.5.1.sup.9,.sup.13.0.sup.22,.sup.26]octacosa-1(25),2(28),4,19,22(26),23-hexaen-7-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide was synthesized in a manner similar to (2S)—N-[(8S,14S)-4-amino-22-ethyl-21-[2-(2-methoxyethyl)phenyl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide except (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione was substituted with (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione. LCMS (ESI): m/z [M+H] calc'd for C.sub.47H.sub.60N.sub.8O.sub.7S 880.4; found 881.6; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.75 (m, 1H), 8.55 (d, J=6.7 Hz, 1H), 8.32 (d, J=8.3 Hz, 1H), 7.99 (d, J=7.7 Hz, 1H), 7.65-7.51 (m, 3H), 7.11-6.92 (m, 1H), 6.72-6.56 (m, 1H), 6.18 (dd, J=16.8, 2.9 Hz, 1H), 5.82-5.65 (m, 1H), 5.61-5.46 (m, 1H), 5.02 (dd, J=24.2, 12.2 Hz, 1H), 4.69 (d, J=10.9 Hz, 1H), 4.37-4.11 (m, 5H), 4.05-3.79 (m, 4H), 3.76-3.50 (m, 6H), 3.47 (s, 2H), 3.08 (s, 3H), 3.04 (s, 1H), 2.98 (d, J=1.9 Hz, 1H), 2.95 (d, J=3.6 Hz, 2H), 2.83 (d, J=2.0 Hz, 2H), 2.24-2.03 (m, 4H), 1.81 (s, 2H), 1.56 (s, 1H), 1.11 (t, J=7.0 Hz, 2H), 1.02-0.87 (m, 8H), 0.80 (dd, J=24.6, 6.6 Hz, 3H), 0.41 (s, 2H), 0.31 (s, 1H).
Example A194. Synthesis of (2S)—N-[(7S,13S)-21-ethyl-20-[2-(methoxymethyl)pyridin-3-yl]-17,17-dimethyl-8,14-dioxo-15-oxa-4-thia-9,21,27,28-tetraazapentacyclo[17.5.2.1.SUP.2.,.SUP.5..1.SUP.9.,.SUP.13..0.SUP.22.,.SUP.26.]octacosa-1(25),2,5(28),19,22(26),23-hexaen-7-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide
(528) ##STR01253## ##STR01254##
Step 1.
(529) A mixture of Zn (1.2 g, 182 mmol) and 1,2-dibromoethane (1.71 g, 9.1 mmol) and DMF (50 mL) was stirred for 30 min at 90° C. under an atmosphere of Ar. The mixture was allowed to rt, then TMSCI (198 mg, 1.8 mmol) was added dropwise over 30 min at rt. Methyl (2R)-2-[(tert-butoxycarbonyl) amino]-3-iodopropanoate (10.0 g, 30.4 mmol) in DMF (100 mL) was added dropwise over 10 min at rt. The mixture was heated to 35° C. and stirred for 2 h, then a mixture of 2,5-dibromo-1,3-thiazole (1.48 g, 60.8 mmol) and Pd(PPh.sub.3).sub.2Cl.sub.2 (2.1 g, 3.0 mmol) in DMF (100 mL) was added dropwise. The mixture was heated to 70° C. and stirred for 2 h, then filtered and the filtrate diluted with EtOAc (1 L) and washed with H.sub.2O (3×1 L), dried with

anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (2S)-3-(5-bromo-1,3-thiazol-2-yl)-2-[(tert-butoxycarbonyl)amino]propanoate (3 g, 27% yield) as a semi-solid. LCMS (ESI): m/z [M+H] calc'd for $C_{12}H_{17}BrN_2O_4S$ 364.0; found 365.1.
(530) Step 2.
(531) Into a 20 mL sealed tube were added 3-[1-ethyl-2-[2-(methoxymethyl)pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]-2,2-dimethylpropan-1-ol (100 mg, 0.21 mmol), K_3PO_4 (111 mg, 0.52 mmol), Pd(dppf)Cl_2 (15 mg, 0.02 mmol), methyl (2S)-3-(4-bromo-1,3-thiazol-2-yl)-2-[(tert-butoxycarbonyl)amino]propanoate (153 mg, 0.42 mmol), toluene (1 mL), and H_2O (0.2 mL) at rt under an atmosphere of N_2. The mixture was heated to 60° C. and stirred for 3 h, cooled, diluted with H_2O (10 mL) and extracted with EtOAc (10 mL×3). The combined organic layers were washed with brine (3×10 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-[4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-2-yl]propanoate (72 mg, 54% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{34}H_{44}N_4O_6S$ 636.3; found 637.2.
(532) Step 3.
(533) A mixture of methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-[4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-2-yl]propanoate (40 mg, 0.06 mmol) and LiOH.Math.H_2O (unspecified) in THF (1 mL) and H_2O (0.2 mL) was stirred at rt under an atmosphere of N_2 for 2 h. The mixture was acidified to pH 5 with aqueous $NaHSO_4$ and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (3×10 mL), dried over anhydrous Na_2SO_4, filtered, and concentrated under reduced pressure to give (2S)-2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-(methoxymethyl)pyridin-3-yl)-1H-indol-5-yl)thiazol-2-yl)propanoic acid. The crude product was used in the next step directly without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{33}H_{42}N_4O_6S$ 622.3; found 623.3.
(534) Step 4.
(535) Methyl (3S)-1-((2S)-2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-(methoxymethyl)pyridin-3-yl)-1H-indol-5-yl)thiazol-2-yl)propanoyl)hexahydropyridazine-3-carboxylate was synthesized in a manner similar to methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoyl]-1,2-diazinane-3-carboxylate except (2S)-2-[(tert-butoxycarbonyl)amino]-3-[2-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-4-yl]propanoic acid was substituted with (2S)-2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-(methoxymethyl)pyridin-3-yl)-1H-indol-5-yl)thiazol-2-yl)propanoic acid. LCMS (ESI): m/z [M+H] calc'd for $C_{39}H_{52}N_6O_7S$ 748.4; found 749.4.
(536) Step 5.
(537) (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-2-yl]propanoyl]-1,2-diazinane-3-carboxylic acid was synthesized in a manner similar to (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid except methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylate was substituted with methyl (3S)-1-((2S)-2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-(methoxymethyl)pyridin-3-yl)-1H-indol-5-yl)thiazol-2-yl)propanoyl)hexahydropyridazine-3-carboxylate. LCMS (ESI): m/z [M+H] calc'd for

C.sub.38H.sub.50N.sub.6O.sub.7S 734.3; found 735.4.
(538) Step 6.
(539) Tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-12-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate was synthesized in a manner similar to tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.5-((triisopropylsilyl)oxy)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate except (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-5-[(triisopropylsilyl)oxy]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid was substituted with (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-(methoxymethyl)pyridin-3-yl]indol-5-yl]-1,3-thiazol-2-yl]propanoyl]-1,2-diazinane-3-carboxylic acid. LCMS (ESI): m/z [M+H] calc'd for C.sub.38H.sub.48N.sub.6O.sub.6S 716.3; found 717.3.
(540) Step 7.
(541) (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione was synthesized in a manner similar to (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione except tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate was substituted with tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-12-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate. LCMS (ESI): m/z [M+Na] calc'd for C.sub.33H.sub.40N.sub.6O.sub.4SNa 639.3; found 640.3.
(542) Step 8.
(543) (2S)—N-[(7S,13S)-21-ethyl-20-[2-(methoxymethyl)pyridin-3-yl]-17,17-dimethyl-8,14-dioxo-15-oxa-4-thia-9,21,27,28-tetraazapentacyclo[17.5.2.1.sup.2,.sup.5.1.sup.9,.sup.13.0.sup.22,.sup.26]octacosa-1(25),2,5(28),19,22(26),23-hexaen-7-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide was synthesized in a manner similar to (2S)—N-[(7S,13S)-21-ethyl-20-[2-(methoxymethyl)pyridin-3-yl]-17,17-dimethyl-8,14-dioxo-15-oxa-3-thia-9,21,27,28-tetraazapentacyclo[17.5.2.1.sup.2,.sup.5.1.sup.9,.sup.13.0.sup.22,.sup.26]octacosa-1(25),2(28),4,19,22(26),23-hexaen-7-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide except (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane- 5,7-dione was substituted with (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione. LCMS (ESI): m/z [M+H] calc'd for C.sub.47H.sub.60N.sub.8O.sub.7S 880.4; found 881.5; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.70 (dt, J=16.2, 8.1 Hz, 1H), 8.54 (ddd, J=6.6, 4.7, 1.7 Hz, 1H), 8.50 (m, 1H), 7.96 (d, J=7.8 Hz, 1H), 7.88 (t, J=2.1 Hz, 2H), 6.70-6.57 (m, 2H), 6.24-6.13 (m, 2H), 5.75 (m, 1H), 5.55 (t, J=7.3 Hz, 1H), 5.46 (d, J=8.5 Hz, 1H), 5.14 (d, J=13.0 Hz, 1H), 4.84-4.75 (m, 1H), 4.35 (d, J=10.7 Hz, 1H), 4.28-4.19 (m, 4H), 3.91 (s, 3H), 3.87 (dd, J=10.4, 8.1 Hz, 1H), 3.78-3.70 (m, 2H), 3.63 (t, J=8.8 Hz,

2H), 3.61-3.49 (m, 2H), 2.87 (d, J=1.1 Hz, 2H), 2.79 (s, 1H), 2.38 (s, 1H), 2.18 (s, 1H), 2.13 (d, J=10.7 Hz, 4H), 1.96 (s, 2H), 1.81 (s, 1H), 1.53 (s, 2H), 1.11 (t, J=7.1 Hz, 2H), 0.99-0.89 (m, 7H), 0.93-0.81 (m, 2H), 0.78 (d, J=6.6 Hz, 2H), 0.28 (s, 3H).
Example A71. Synthesis of (2S)-2-(1-{1-[(2E)-4-(dimethylamino)but-2-enoyl]azetidin-3-yl}-N-methylformamido)-N-[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methylbutanamide
(544) ##STR01255##
Step 1.
(545) To a mixture of tert-butyl N-(azetidine-3-carbonyl)-N-methyl-L-valinate (350 mg, 1.3 mmol) and (2E)-4-(dimethylamino)but-2-enoic acid (201 mg, 1.56 mmol) in DCM (8 mL) at 5° C. was added a solution of T3P, 50% in EtOAc (827 mg, 2.6 mmol) and DIPEA (1.7 g, 13 mmol) in DCM (2 mL). The mixture was stirred for 1 h, then diluted with EtOAc (20 mL) and H.sub.2O (20 mL). The aqueous and organic layers were separated and the organic layer was washed with H.sub.2O (3×10 mL), brine (10 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by prep-HPLC to give tert-butyl (E)-N-(1-(4-(dimethylamino)but-2-enoyl)azetidine-3-carbonyl)-N-methyl-L-valinate (200 mg, 39% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.20H.sub.35N.sub.3O.sub.4 381.3; found 382.3.
(546) Step 2.
(547) To a mixture of tert-butyl (E)-N-(1-(4-(dimethylamino)but-2-enoyl)azetidine-3-carbonyl)-N-methyl-L-valinate (190 mg, 0.32 mmol) in DCM (3 mL) at rt was added TFA (1 mL). The mixture was stirred at rt for 1 h, then concentrated under reduced pressure to give (E)-N-(1-(4-(dimethylamino)but-2-enoyl)azetidine-3-carbonyl)-N-methyl-L-valine (190 mg, 90%) as a solid, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.16H.sub.27N.sub.3O.sub.4 325.2; found 326.2.
(548) Step 3.
(549) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (172 mg, 0.27 mmol) and (E)-N-(1-(4-(dimethylamino)but-2-enoyl)azetidine-3-carbonyl)-N-methyl-L-valine (105 mg, 0.32 mmol) in DMF (2 mL) at 5° C. was added a mixture of HATU (133 mg, 0.297 mmol) and DIPEA (348 mg, 2.7 mmol) in DMF (1 mL). The mixture was stirred for 1 h, then diluted with EtOAc (20 mL) and H.sub.2O (20 mL). The aqueous and organic layers were separated, and the organic layer was washed with H.sub.2O (3×10 mL), brine (10 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by prep-TLC to give (2S)-2-(1-{1-[(2E)-4-(dimethylamino)but-2-enoyl]azetidin-3-yl}-N-methylformamido)-N-[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methylbutanamide (4.8 mg, 2% yield over 2 steps) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.52H.sub.68N.sub.8O.sub.8 932.5; found 933.5; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.71 (d, J=3.2 Hz, 1H), 8.50 (s, 1.5H), 8.08-7.85 (m, 2H), 7.65-7.44 (m, 3H), 7.32-7.14 (m, 1H), 7.07-6.95 (m, 1H), 6.80 (dt, J=22.1, 6.8 Hz, 1H), 6.55 (d, J=35.8 Hz, 1H), 6.30 (d, J=15.4 Hz, 1H), 5.56 (dd, J=13.8, 6.7 Hz, 1H), 4.76 (dd, J=19.8, 10.5 Hz, 1H), 4.54 (dd, J=15.9, 7.5 Hz, 2H), 4.48-4.38 (m, 2H), 4.36-4.23 (m, 3H), 4.22-4.14 (m, 1H), 3.96 (qd, J=15.6, 7.9 Hz, 3H), 3.77 (ddd, J=25.8, 23.4, 11.9 Hz, 2H), 3.58 (dd, J=17.2, 8.3 Hz, 2H), 3.38 (s, 2H), 3.25-3.11 (m, 3H), 3.05-2.94 (m, 1H), 2.94-2.81 (m, 4H), 2.73 (dd, J=20.9, 11.0 Hz, 1H), 2.45 (d, J=6.9 Hz, 5H), 2.32-2.07 (m, 3H), 1.92 (d, J=13.2 Hz, 1H), 1.72 (s, 1H), 1.64-1.51 (m, 1H),

1.18 (t, J=7.0 Hz, 2H), 1.00 (ddd, J=14.6, 11.8, 8.5 Hz, 6H), 0.92-0.81 (m, 4H), 0.55-0.41 (m, 3H).
Example A67. Synthesis of (2E)-4-(dimethylamino)-N-(6-{[(1S)-1-{[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamoyl}-2-methylpropyl](methyl)carbamoyl}pyridin-3-yl)but-2-enamide
(550) ##STR01256##
Step 1.
(551) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione TFA salt (225 mg, 0.28 mmol) and (E)-N-(5-(4-(dimethylamino)but-2-enamido)picolinoyl)-N-methyl-L-valine TFA salt (260 mg crude, 0.56 mmol) in DMF (5 mL) at 0° C. were added DIPEA (0.46 mL, 2.8 mmol) followed by HATU (140 mg, 0.36 mmol). The mixture was stirred at 0-10° C. for 1 h, then concentrated under reduced pressure and the residue was purified by prep-HPLC to give (2E)-4-(dimethylamino)-N-(6-{[(1S)-1-{[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamoyl}-2-methylpropyl](methyl)carbamoyl}pyridin-3-yl)but-2-enamide TFA salt (23.3 mg, 8% yield over 2 steps) as a solid. LCMS (ESI): m/z [M+Na] calc'd for C.sub.54H.sub.67N.sub.9O.sub.8Na 992.5; found 992.4; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 9.05 (d, J=2.5 Hz, 1H), 8.85-8.71 (m, 1H), 8.43 (ddd, J=33.3, 18.0, 2.6 Hz, 2H), 8.01-7.87 (m, 2H), 7.83-7.70 (m, 1H), 7.60-7.47 (m, 2H), 7.31-7.19 (m, 1H), 7.07-6.90 (m, 2H), 6.70-6.36 (m, 3H), 5.81-5.61 (m, 1H), 4.50-4.20 (m, 4H), 4.01-3.68 (m, 3H), 3.64-3.35 (m, 5H), 3.27-3.08 (m, 3H), 3.04-2.44 (m, 11H), 2.36-2.10 (m, 3H), 1.93 (d, J=13.0 Hz, 1H), 1.61 (dd, J=34.3, 21.6 Hz, 3H), 1.39-1.16 (m, 3H), 1.12-0.81 (m, 6H), 0.78-0.45 (m, 6H).
Example A54. Synthesis of (2S)-2-{1-[(3S)-1-[(2E)-4-(dimethylamino)but-2-enoyl]pyrrolidin-3-yl]-N-methylformamido}-N-[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methylbutanamide
(552) ##STR01257##
Step 1.
(553) To a mixture of tert-butyl N-methyl-N—((S)-pyrrolidine-3-carbonyl)-L-valinate (210 mg, 0.73 mmol) in DMF (4 mL) at rt were added 4-(dimethylamino)-4-methylpent-2-ynoic acid (450 mg, 2.9 mmol), DIPEA (1.2 mL, 7.3 mmol), and HATU (332 mg, 0.88 mmol). The mixture was stirred at rt for 1 h then diluted with EtOAc, and the mixture washed with H.sub.2O, brine, dried over Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N—((S)-1-(4-(dimethylamino)-4-methylpent-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate (140 mg, 45% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.39N.sub.3O.sub.4 421.3; found 422.3.
(554) Step 2.
(555) A mixture of tert-butyl N—((S)-1-(4-(dimethylamino)-4-methylpent-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate (130 mg, 0.31 mmol) in DCM (2 mL) and TFA (1 mL) was stirred at rt for 90 min. The mixture was concentrated under reduced pressure to give N—((S)-1-(4-(dimethylamino)-4-methylpent-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valine TFA salt (150 mg) as an oil, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.19H.sub.31N.sub.3O.sub.4 365.2; found 366.2.
(556) Step 3.

(557) (3S)-1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-N-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide TFA salt was synthesized in a manner similar to 1-acryloyl-N-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylazetidine-3-carboxamide except (2S)-2-{1-[(3S)-1-[(2E)-4-(dimethylamino)but-2-enoyl]pyrrolidin-3-yl]-N-methylformamido}-N-[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methylbutanamide TFA salt. (120 mg, 54% yield over 2 steps) as a solid. .sup.1H-NMR (400 MHz, CD.sub.3OD) δ 8.76-8.68 (m, 1H), 8.44 (s, 1H), 8.02-7.94 (m, 1H), 7.94-7.84 (m, 1H), 7.65-7.43 (m, 3H), 7.27-7.14 (m, 1H), 7.06-6.96 (m, 1H), 6.65-6.48 (m, 1H), 5.62-5.46 (m, 1H), 4.81-4.57 (m, 1H), 4.46-4.22 (m, 3H), 4.10-3.35 (m, 11H), 3.26-2.93 (m, 6H), 2.91-2.51 (m, 4H), 2.42-2.09 (m, 9H), 1.95-1.87 (m, 1H), 1.85-1.40 (m, 6H), 1.38-1.10 (m, 6H), 1.07-0.81 (m, 9H), 0.56-0.38 (m, 3H). LCMS (ESI): m/z [M+H] C.sub.52H.sub.68N.sub.8O.sub.8 found 947.7.

Example A95. Synthesis of (2S)—N-[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-[4-(morpholin-4-yl)but-2-ynoyl]pyrrolidin-3-yl]formamido}butanamide

(558) ##STR01258##

Step 1.

(559) A mixture of tert-butyl (2S)-3-methyl-2-[N-methyl-1-(3S)-pyrrolidin-3-ylformamido]butanoate (500 mg, 1.8 mmol), 4-(morpholin-4-yl)but-2-ynoic acid (1.49 g, 8.8 mmol), DIPEA (682 mg, 5.3 mmol) and CIP (635 mg, 2.3 mmol) in DMF (5 mL) was stirred at 0° C. for 2 h. The mixture was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N-methyl-N—((S)-1-(4-morpholinobut-2-ynoyl)pyrrolidine-3-carbonyl)-L-valinate (150 mg, 19% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.37N.sub.3O.sub.5 435.3; found 436.5.

(560) Step 2.

(561) A mixture of tert-butyl N-methyl-N—((S)-1-(4-morpholinobut-2-ynoyl)pyrrolidine-3-carbonyl)-L-valinate (250 mg, 0.57 mmol) in DCM (5 mL) and TFA (2.5 mL) was stirred at rt for 2 h. The mixture was concentrated under reduced pressure to give (2S)-3-methyl-2-[N-methyl-1-[(3S)-1-[4-(morpholin-4-yl)but-2-ynoyl]pyrrolidin-3-yl]formamido]butanoic acid (310 mg, crude) as an oil, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.19H.sub.29N.sub.3O.sub.5 379.2; found 380.2.

(562) Step 3.

(563) A mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-2.sup.5-hydroxy-1.sup.2-(2-(methoxymethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (250 mg, 0.4 mmol), DIPEA (516 mg, 4.0 mmol), (2S)-3-methyl-2-[N-methyl-1-[(3S)-1-[4-(morpholin-4-yl)but-2-ynoyl]pyrrolidin-3-yl]formamido]butanoic acid (182 mg, 0.48 mmol), and COMU (205 mg, 0.48 mmol) in DMF (3 mL) was stirred at −20° C. for 2 h. The mixture was diluted with H.sub.2O (10 mL), then extracted with EtOAc (3×10 mL) and the combined organic layers were washed with brine (3×10 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The mixture was concentrated under reduced pressure and the residue was purified by reverse-phase

silica gel column chromatography to give (2S)—N-[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-[4-(morpholin-4-yl)but-2-ynoyl]pyrrolidin-3-yl]formamido}butanamide (207 mg, 53% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{55}H_{70}N_8O_9$ 986.5; found 987.8; ^{1}H NMR (400 MHz, DMSO-d_6) δ 9.39-9.28 (m, 1H), 8.74 (t, J=4.8, 1H), 8.70-8.04 (m, 1H), 7.98-7.90 (m, 1.5H), 7.82 (d, J=7.7 Hz, 0.5H), 7.63-7.46 (m, 3H), 7.26-7.10 (m, 1H), 7.03 (s, 1H), 6.58-6.43 (m, 1H), 5.44-5.30 (m, 1H), 5.06 (q, 0.5H), 4.72 (t, J=11.0, 0.5H), 4.39-4.20 (m, 3H), 4.15 (d, J=11.1 Hz, 1H), 4.09-3.85 (m, 4H), 3.66 (s, 2H), 3.65-3.58 (m, 4H), 3.58-3.55 (m, 2H), 3.55-3.48 (m, 3H), 3.47-3.41 (m, 3H), 3.31 (s, 2H), 3.10 (s, 2H), 2.92 (s, 1H), 2.89-2.65 (m, 5H), 2.68 (s, 1H), 2.45-2.38 (m, 1H), 2.29-2.24 (m, 1H), 2.23-1.99 (m, 3H), 1.82 (d, J=12.1 Hz, 1H), 1.76-1.62 (m, 1H), 1.61-1.45 (m, 1H), 1.14-1.04 (m, 2H), 1.02-0.92 (m, 3H), 0.91-0.86 (m, 3H), 0.83-0.77 (m, 3H), 0.77-0.70 (m, 2H), 0.50-0.35 (m, 3H).

Example A145. Synthesis of two atropisomers of (2S)-2-{1-[(3S)-1-(but-2-ynoyl)pyrrolidin-3-yl]-N-methylformamido}-N-[(8S,14S,20M)-22-ethyl-4-hydroxy-21-{2-[(1S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.SUP.2.,.SUP.6..1.SUP.10.,.SUP.14..0.SUP.23.,.SUP.27.]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methylbutanamide

(564) ##STR01259##

Step 1.

(565) To a mixture of but-2-ynoic acid (222 mg) and CIP (588 mg) in ACN (8 mL) at 0° C. under PGR-1230,C.sub.3 an atmosphere of Ar was added DIPEA (681 mg). The mixture was stirred at 0° C. then tert-butyl N-methyl-N—((S)-pyrrolidine-3-carbonyl)-L-valinate (500 mg) in ACN (3 mL) was added dropwise and the mixture stirred at 0° C. for 2 h. EtOAc was added and the mixture was washed with brine (3×20 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N—((S)-1-(but-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{19}H_{30}N_2O_4$ 350.2; found 352.1.

(566) Step 2.

(567) A mixture of tert-butyl N—((S)-1-(but-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate (200 mg) in DCM (4 mL) and TFA (2 mL) was stirred at 0° C. for 2 h. The mixture was concentrated under reduced pressure with azeotropic removal of H_2O using toluene (4 mL×2) to give N—((S)-1-(but-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{15}H_{22}N_2O_4$ 294.2; found 295.2.

(568) Step 3.

(569) Two atropisomers of (2S)-2-{1-[(3S)-1-(but-2-ynoyl)pyrrolidin-3-yl]-N-methylformamido}-N-[(8S,14S,20M)-22-ethyl-4-hydroxy-21-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methylbutanamide was synthesized in a manner similar to (2S)—N-[(8S,14S)-22-ethyl-4-hydroxy-21-[2-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1.sup.2,.sup.6.1.sup.10,.sup.14.0.sup.23,.sup.27]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-[4-(morpholin-4-yl)but-2-ynoyl]pyrrolidin-3-yl]formamido}butanamide except (2S)-3-methyl-2-[N-methyl-1-[(3S)-1-[4-(morpholin-4-yl)but-2-ynoyl]pyrrolidin-3-yl]formamido]butanoic acid was substituted with N—((S)-1-(but-2-ynoyl)pyrrolidine-3-carbonyl)-N-methyl-L-valinate. (43.3 mg, 12% yield) and (33 mg, 9% yield) both as solids. LCMS (ESI): m/z [M+H] calc'd for

C.sub.52H.sub.65N.sub.7O.sub.8 915.5; found 916.7; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.34-9.27 (m, 1H), 8.78 (t, J=2.5 Hz, 1H), 8.68 (t, J=8.5 Hz, 0.5H), 8.20-8.11 (m, 0.6H), 7.95 (ddt, J=5.4, 3.5, 1.7 Hz, 2H), 7.63-7.60 (m, 1H), 7.61-7.49 (m, 2H), 7.13 (s, 1H), 7.03 (d, J=6.2 Hz, 1H), 6.60-6.49 (d, J=35.5 Hz, 1H), 5.43-5.39 (m, 1H), 5.12-5.00 (m, 0.7H), 4.74 (d, J=10.6 Hz, 0.4H), 4.32-4.25 (m, 1H), 4.18-3.85 (m, 5H), 3.81-3.45 (m, 8H), 3.18-3.02 (m, 5H), 2.93-2.80 (m, 4H), 2.80-2.70 (m, 2H), 2.42-2.36 (m, 1H), 2.31-2.20 (m, 1H), 2.18-1.96 (m, 6H), 1.85-1.74 (m, 1H), 1.74-1.63 (m, 1H), 1.62-1.42 (m, 1H), 1.32-1.16 (m, 4H), 1.15-1.05 (t, J=6.3 Hz, 4H), 1.04-0.95 (m, 2H), 0.95-0.85 (m, 5H), 0.68-0.52 (m, 4H), 0.52-0.37 (m, 4H). and LCMS (ESI): m/z [M+H] calc'd for C.sub.52H.sub.65N.sub.7O.sub.8 915.5; found 916.7; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 9.36-9.28 (m, 1H), 8.77 (dd, J=4.7, 1.8 Hz, 1H), 8.62-8.57 (m, 0.5H), 8.15-8.07 (m, 0.5H), 7.95 (s, 1H), 7.87-7.81 (m, 1H), 7.65-7.51 (m, 3H), 7.37-7.25 (m, 1H), 7.10-7.03 (m, 1H), 6.54 (d, J=35.5 Hz, 1H), 5.52-5.21 (m, 2H), 4.78-4.66 (m, 0.5H), 4.34-4.20 (m, 3H), 4.15-3.85 (m, 4H), 3.85-3.42 (m, 7H), 3.22-3.11 (m, 3H), 2.97-2.72 (m, 7H), 2.62-2.54 (m, 1H), 2.28-1.96 (m, 7H), 1.95-1.74 (m, 2H), 1.73-1.44 (m, 2H), 1.42-1.37 (m, 3H), 1.28-1.14 (m, 1H), 1.03-0.85 (m, 6H), 0.83-0.72 (m, 7H), 0.71-0.55 (m, 3H).

Example A28. Synthesis of (2S)—N-[(8S,14S)-22-ethyl-4-hydroxy-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-[N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido]butanamide

(570) ##STR01260##

Step 1.

(571) To a mixture of 3-bromo-4-(methoxymethyl)pyridine (1.00 g, 5.0 mmol), 4,4,5,5-tetramethyl-2-(tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (1.51 g, 5.9 mmol) and KOAc (1.21 g, 12.3 mmol) in toluene (10 mL) at rt under an atmosphere of Ar was added Pd(dppf)Cl.sub.2 (362 mg, 0.5 mmol). The mixture was heated to 110° C. and stirred overnight, then concentrated under reduced pressure to give 4-(methoxymethyl)-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine, which was used directly in the next step directly without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.13H.sub.20BNO.sub.3 249.2; found 250.3.

(572) Step 2.

(573) To a mixture of 4-(methoxymethyl)-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine (290 mg, 1.16 mmol), K.sub.3PO.sub.4 (371 mg, 1.75 mmol) and tert-butyl N-[(8S,14S)-21-iodo-18,18-dimethyl-9,15-dioxo-4-[(triisopropylsilyl)oxy]-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (500 mg, 0.58 mmol) in 1,4-dioxane (5 mL) and H.sub.2O (1 mL) at rt under an atmosphere of Ar was added Pd(dppf)Cl.sub.2 (43 mg, 0.06 mmol). The mixture was heated to 70° C. and stirred for 2 h, then H.sub.2O added and the mixture extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N-[(8S,14S)-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-4-[(triisopropylsilyl)oxy]-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (370 mg, 74% yield) as a foam. LCMS (ESI): m/z [M+H] calc'd for C.sub.48H.sub.67N.sub.5O.sub.7Si 853.6; found 854.6.

(574) Step 3.

(575) A mixture of tert-butyl N-[(8S,14S)-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-4-[(triisopropylsilyl)oxy]-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-

1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (350 mg, 0.41 mmol), Cs_2CO_3 (267 mg, 0.82 mmol) and Etl (128 mg, 0.82 mmol) in DMF (4 mL) was stirred at 35° C. overnight. H_2O was added and the mixture was extracted with EtOAc (2×15 mL). The combined organic layers were washed with brine (15 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N-[(8S,14S)-22-ethyl-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-4-[(triisopropylsilyl)oxy]-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (350 mg, 97% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{50}H_{71}N_5O_7Si$ 881.5; found 882.6.

(576) Step 4.

(577) A mixture of tert-butyl N-[(8S,14S)-22-ethyl-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-4-[(triisopropylsilyl)oxy]-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (350 mg, 0.4 mmol) and 1M TBAF in THF (0.48 mL, 0.480 mmol) in THF (3 mL) at 0° C. under an atmosphere of Ar was stirred for 1 h. The mixture was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl N-[(8S,14S)-22-ethyl-4-hydroxy-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (230 mg, 80% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{41}H_{51}N_5O_7$ 725.4; found 726.6.

(578) Step 5.

(579) To a mixture of tert-butyl N-[(8S,14S)-22-ethyl-4-hydroxy-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]carbamate (200 mg, 0.28 mmol) in 1,4-dioxane (2 mL) at 0° C. under an atmosphere of Ar was added 4M HCl in 1,4-dioxane (2 mL, 8 mmol). The mixture was allowed to warm to rt and was stirred overnight, then concentrated under reduced pressure to give (8S,14S)-8-amino-22-ethyl-4-hydroxy-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaene-9,15-dione (200 mg). LCMS (ESI): m/z [M+H] calc'd for $C_{36}H_{43}N_5O_5$ 625.3; found 626.5.

(580) Step 6.

(581) To a mixture of (2S)-3-methyl-2-[N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido]butanoic acid (108 mg, 0.38 mmol) and (8S,14S)-8-amino-22-ethyl-4-hydroxy-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaene-9,15-dione (200 mg, 0.32 mmol) in DCM (3 mL) at 0° C. was added DIPEA (165 mg, 1.3 mmol) and COMU (274 mg, 0.64 mmol) in portions. The mixture was stirred at 0° C. for 2 h, H_2O added and extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over anhydrous Na_2SO_4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography then prep-HPLC to give (2S)—N-[(8S,14S)-22-ethyl-4-hydroxy-21-[4-(methoxymethyl)pyridin-3-yl]-18,18-dimethyl-9,15-dioxo-16-oxa-10,22,28-triazapentacyclo[18.5.2.1{circumflex over ( )}[2,6].1{circumflex over ( )}[10,14].0{circumflex over ( )}[23,27]]nonacosa-1(26),2,4,6(29),20,23(27),24-heptaen-8-yl]-3-methyl-2-[N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido]butanamide (16 mg, 5.6% yield) as a solid. LCMS

(ESI): m/z [M+H] calc'd for $C_{50}H_{63}N_7O_8$ 889.5; found 890.6; ^1H NMR (400 MHz, DMSO-d_6) δ 9.33 (dd, J=9.1, 6.9 Hz, 1H), 8.79-8.46 (m, 2H), 7.93 (s, 1H), 7.68-7.58 (m, 2H), 7.53 (t, J=8.5 Hz, 1H), 7.26-6.98 (m, 2H), 6.71-6.47 (m, 2H), 6.24-6.07 (m, 1H), 5.80-5.60 (m, 1H), 5.49-5.18 (m, 1H), 4.45-4.07 (m, 4H), 4.08-3.87 (m, 3H), 3.87-3.64 (m, 4H), 3.64-3.40 (m, 5H), 3.34 (s, 2H), 3.30 (s, 2H), 3.23 (d, J=1.8 Hz, 1H), 2.94-2.74 (m, 6H), 2.16-2.01 (m, 3H), 1.82-1.47 (m, 3H), 1.08 (q, J=8.9, 8.0 Hz, 1H), 1.00-0.88 (m, 6H), 0.82 (d, J=10.8 Hz, 4H), 0.76-0.66 (m, 2H), 0.44 (d, J=14.2 Hz, 3H).

Example A316. Synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6^3S,4S)-1^1-ethyl-1^2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6^1, 6^2,6^3,6^4,6^5,6^6-hexahydro-1^1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methylbutanamide

(582) ##STR01261##

Step 1.

(583) To a mixture of 2-(((1-((benzyloxy)carbonyl)azetidin-3-yl)oxy)methyl)-3-methylbutanoic acid (650 mg, 2 mmol) and di-tert-butyl dicarbonate (883 mg, 4 mmol) in ^{t}BuOH (10 mL) was added 4-dimethylaminopyridine (124 mg, 1 mmol). The mixture was heated to 30° C. and stirred for 1 h, then diluted with H_2O (50 mL) and extracted with EtOAc (50 mL×3). The combined organic layers were concentrated under reduced pressure and the residue was purified by silica gel column chromatography to afford benzyl 3-(2-(tert-butoxycarbonyl)-3-methylbutoxy)azetidine-1-carboxylate (450 mg, 56% yield) as an oil. LCMS (ESI): m/z [M+Na] calc'd for $C_{21}H_{31}NO_5Na$ 400.2; found 400.2.

(584) Step 2.

(585) A mixture of benzyl 3-(2-(tert-butoxycarbonyl)-3-methylbutoxy)azetidine-1-carboxylate (450 mg, 1.19 mmol) and Pd/C (50 mg) in THF (30 mL) was stirred for 2 h under an atmosphere of H_2 (15 psi). The mixture was filtered and the filtrate was concentrated under reduced pressure to give tert-butyl 2-((azetidin-3-yloxy)methyl)-3-methylbutanoate (300 mg, 100% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{13}H_{26}NO_3$243.2; found 244.2; ^1H NMR (400 MHz, $CDCl_3$) δ 4.35-4.25 (m, 1H), 3.71-3.63 (m, 2H), 3.63-3.56 (m, 2H), 3.50 (t, J=8.0 Hz, 1H), 3.43 (dd, J=9.0, 4.0 Hz, 1H), 2.37-2.26 (m, 1H), 2.21 (br. s, 1H), 1.92-1.81 (m, 1H), 1.47 (s, 9H), 0.93 (d, J=6.8 Hz, 6H).

(586) Step 3.

(587) To a mixture of tert-butyl 2-((azetidin-3-yloxy)methyl)-3-methylbutanoate (270 mg, 1.11 mmol), 4-(dimethylamino)-4-methylpent-2-ynoic acid (860 mg, 5.55 mmol) and DIPEA (1.56 g, 11.1 mmol) in DMF (20 mL) at 0° C. was added T3P (2.12 g, 6.7 mmol). The mixture was stirred at 0° C. for 1 h, diluted with EtOAc (200 mL), then washed with H_2O (30 mL×5), brine (30 mL), dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl 2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-3-methylbutanoate (200 mg, 47% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{21}H_{36}N_2O_4$ 380.3; found 381.3.

(588) Step 4.

(589) To a mixture of tert-butyl 2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-3-methylbutanoate (190 mg, 0.5 mmol) in DCM (4 mL) was added TFA (2 mL). The mixture was stirred for 1 h, then concentrated under reduced pressure to give 2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-3-methylbutanoic acid (162 mg, 100% yield) as an oil, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{17}H_{28}N_2O_4$ 324.2; found 325.3.

(590) Step 5.

(591) To a solution of (2S)-3-(3-bromophenyl)-2-[(tert-butoxycarbonyl)amino]propanoic acid (100

g, 290 mmol) in DMF (1 L) at room temperature was added $NaHCO_3$ (48.8 g, 581.1 mmol) and MeI (61.9 g, 435.8 mmol). The reaction mixture was stirred for 16 h and was then quenched with H_2O (1 L) and extracted with EtOAc (3×1 L). The combined organic layers were washed with brine (3×500 mL), dried over Na_2SO_4, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography (13% EtOAc/pet. ether) to give methyl (S)-3-(3-bromophenyl)-2-((tert-butoxycarbonyl)amino)propanoate (109 g, crude). LCMS (ESI): m/z [M+Na] calc'd for $C_{15}H_{20}BrNO_4$ 380.05; found 380.0.

(592) Step 6.

(593) To a stirred solution of methyl (2S)-3-(3-bromophenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (108 g, 301.5 mmol) and bis(pinacolato)diboron (99.53 g, 391.93 mmol) in 1,4-dioxane (3.2 L) was added KOAc (73.97 g, 753.70 mmol) and Pd(dppf)Cl_2 (22.06 g, 30.15 mmol). The reaction mixture was heated to 90° C. for 3 h and was then cooled to room temperature and extracted with EtOAc (2×3 L). The combined organic layers were washed with brine (3×800 mL), dried over Na_2SO_4, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography (5% EtOAc/pet. ether) to give methyl (S)-2-((tert-butoxycarbonyl)amino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)propanoate (96 g, 78.6% yield). LCMS (ESI): m/z [M+Na] calc'd for $C_{21}H_{32}BNO_6$ 428.22; found 428.1.

(594) Step 7.

(595) To a mixture of methyl (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]propanoate (94 g, 231.9 mmol) and 3-(5-bromo-1H-indol-3-yl)-2,2-dimethylpropyl acetate (75.19 g, 231.93 mmol) in 1,4-dioxane (1.5 L) and H_2O (300 mL) was added K_2CO_3 (64.11 g, 463.85 mmol) and Pd(DtBPF)Cl_2 (15.12 g, 23.19 mmol). The reaction mixture was heated to 70° C. and stirred for 4 h. The reaction mixture was extracted with EtOAc (2×2 L) and the combined organic layers were washed with brine (3×600 mL), dried over Na_2SO_4, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography (20% EtOAc/pet. ether) to give methyl (S)-3-(3-(3-(3-acetoxy-2,2-dimethylpropyl)-1H-indol-5-yl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoate (130 g, crude). LCMS (ESI): m/z [M+H] calc'd for $C_{30}H_{38}N_2O_6$ 523.28; found 523.1.

(596) Step 8.

(597) To a solution of methyl (2S)-3-(3-[3-[3-(acetyloxy)-2,2-dimethylpropyl]-1H-indol-5-yl]phenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (95.0 g, 181.8 mmol) and iodine (36.91 g, 145.41 mmol) in THF (1 L) at −10° C. was added AgOTf (70.0 g, 272.7 mmol) and $NaHCO_3$ (22.9 g, 272.65 mmol). The reaction mixture was stirred for 30 min and was then quenched by the addition of sat. $Na_2S_2O_3$ (100 mL) at 0° C. The resulting mixture was extracted with EtOAc (3×1 L) and the combined organic layers were washed with brine (3×500 mL), dried over Na_2SO_4, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography (50% EtOAc/pet. ether) to give methyl (S)-3-(3-(3-(3-acetoxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoate (49.3 g, 41.8% yield). LCMS (ESI) m/z: [M+H] calcd for $C_{30}H_{37}IN_2O_6$: 649.18; found 649.1.

(598) Step 9.

(599) To a solution of methyl (2S)-3-(3-[3-[3-(acetyloxy)-2,2-dimethylpropyl]-2-iodo-1H-indol-5-yl]phenyl)-2-[(tert-butoxycarbonyl)amino]propanoate (60 g, 92.5 mmol) in THF (600 mL) was added a solution of LiOH.Math.H_2O (19.41 g, 462.5 mmol) in H_2O (460 mL). The resulting solution was stirred overnight and then the pH was adjusted to 6 with HCl (1 M). The resulting solution was extracted with EtOAc (2×500 mL) and the combined organic layers was washed with sat. brine (2×500 mL), dried over Na_2SO_4, filtered, and concentrated under reduced pressure to give (S)-2-((tert-butoxycarbonyl)amino)-3-(3-(3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl)phenyl)propanoic acid (45 g, 82.1% yield). LCMS (ESI): m/z

[M+Na] calc'd for C.sub.27H.sub.33IN.sub.2O.sub.6 615.13; found 615.1.
(600) Step 10.
(601) To a solution of (2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]phenyl]propanoic acid (30 g, 50.6 mmol) and methyl (3S)-1,2-diazinane-3-carboxylate (10.9 g, 75.9 mmol) in DCM (400 mL) was added NMM (40.97 g, 405.08 mmol), HOBT (2.05 g, 15.19 mmol), and EDCI (19.41 g, 101.27 mmol). The reaction mixture was stirred overnight and then the mixture was washed with sat. NH.sub.4Cl (2×200 mL) and sat. brine (2×200 mL), and the mixture was dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure to give methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (14 g, 38.5% yield). LCMS (ESI): m/z [M+H] calc'd for C.sub.33H.sub.43IN.sub.4O.sub.6 718.23; found 719.4.
(602) Step 11.
(603) To a solution of methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (92 g, 128.0 mmol) in THF (920 mL) at 0° C. was added a solution of LiOH.Math.H.sub.2O (26.86 g, 640.10 mmol) in H.sub.2O (640 mL). The reaction mixture was stirred for 2 h and was then concentrated under reduced pressure to give (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylic acid (90 g, crude). LCMS (ESI): m/z [M+H] calc'd for C.sub.32H.sub.41IN.sub.4O.sub.6 705.22; found 705.1.
(604) Step 12.
(605) To a solution of (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[3-[3-(3-hydroxy-2,2-dimethylpropyl)-2-iodo-1H-indol-5-yl]phenyl]propanoyl]-1,2-diazinane-3-carboxylic acid (90 g, 127.73 mmol) in DCM (10 L) at 0° C. was added HOBt (34.52 g, 255.46 mmol), DIPEA (330.17 g, 2554.62 mmol) and EDCI (367.29 g, 1915.96 mmol). The reaction mixture was stirred for 16 h and was then concentrated under reduced pressure. The mixture was extracted with DCM (2×2 L) and the combined organic layers were washed with brine (3×1 L), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography (50% EtOAc/pet. ether) to give tert-butyl ((6.sup.3S,4S)-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (70 g, 79.8% yield). LCMS (ESI): m/z [M+H] calc'd for C.sub.32H.sub.39IN.sub.4O.sub.5 687.21; found 687.1.
(606) Step 13.
(607) A1 L round-bottom flask was charged with tert-butyl ((6.sup.3S,4S)-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (22.0 g, 32.042 mmol), toluene (300.0 mL), Pd.sub.2(dba).sub.3 (3.52 g, 3.845 mmol), S-Phos (3.95 g, 9.613 mmol), and KOAc (9.43 g, 96.127 mmol) at room temperature. To the mixture was added 4,4,5,5-tetramethyl-1,3,2-dioxaborolane (26.66 g, 208.275 mmol) dropwise with stirring at room temperature. The resulting solution was stirred for 3 h at 60° C. The resulting mixture was filtered, and the filter cake was washed with EtOAc. The filtrate was concentrated under reduced pressure and the remaining residue was purified by silica gel column chromatography to afford tert-butyl ((6.sup.3S,4S)-10,10-dimethyl-5,7-dioxo-12-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (22 g, 90% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.38H.sub.51BN.sub.4O.sub.7 687.3; found 687.4.
(608) Step 14.
(609) A mixture of tert-butyl ((6.sup.3S,4S)-10,10-dimethyl-5,7-dioxo-1.sup.2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-

1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (2.0 g, 2.8 mmol), 3-bromo-2-[(1S)-1-methoxyethyl]pyridine (0.60 g, 2.8 mmol), Pd(dppf)Cl.sub.2 (0.39 g, 0.5 mmol), and K.sub.3PO.sub.4 (1.2 g, 6.0 mmol) in 1,4-dioxane (50 mL) and H.sub.2O (10 mL) under an atmosphere of N.sub.2 was heated to 70° C. and stirred for 2 h. The mixture was diluted with H.sub.2O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (3×50 mL), dried over anhydrous Na.sub.2SO.sub.4, and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (1.5 g, 74% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.40H.sub.49N.sub.5O.sub.6 695.4; found 696.5.

(610) Step 15.

(611) To a solution of tert-butyl ((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl) pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl) carbamate (20 g, 28.7 mmol) and Cs.sub.2CO.sub.3 (18.7 g, 57.5 mmol) in DMF (150 mL) at 0° C. was added a solution of ethyl iodide (13.45 g, 86.22 mmol) in DMF (50 mL). The resulting mixture was stirred overnight at 35° C. and was then diluted with H.sub.2O (500 mL). The mixture was extracted with EtOAc (2×300 mL) and the combined organic layers were washed with brine (3×100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography to give tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)- benzenacycloundecaphane-4-yl)carbamate (4.23 g, 18.8% yield) and the atropisomer (5.78 g, 25.7% yield) as solids. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.53N.sub.5O.sub.6 724.4; found 724.6.

(612) Step 16.

(613) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (1.3 g, 1.7 mmol) in TFA (10 mL) and DCM (20 mL) was stirred at 0° C. for 2 h. The mixture was concentrated under reduced pressure to afford (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (1.30 g, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.37H.sub.45N.sub.5O.sub.4 623.3; found 624.4.

(614) Step 17.

(615) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (258 mg, 0.41 mmol) and 2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-3-methylbutanoic acid (162 mg, 0.5 mmol) in DMF (4 mL) at 0° C. was added a mixture of HATU (188 mg, 0.5 mmol) and DIPEA (534 mg, 4.14 mmol) in DMF (2 mL). The mixture was stirred at 0° C. for 1 h, then diluted with H.sub.2O (30 mL) and extracted with EtOAc (30 mL×3). The combined organic layers were concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methylbutanamide (250 mg, 64%

yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.54H.sub.71N.sub.7O.sub.7 929.5; found 930.5; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.90-8.79 (m, 1H), 8.54-8.21 (m, 1H), 8.15-7.91 (m, 2H), 7.88-7.67 (m, 2H), 7.65-7.52 (m, 2H), 7.47-7.15 (m, 2H), 5.80-5.52 (m, 1H), 4.53-4.23 (m, 5H), 4.23-3.93 (m, 3H), 3.90-3.76 (m, 2H), 3.75-3.58 (m, 3H), 3.57-3.44 (m, 1H), 3.38 (s, 1H), 3.29-3.26 (m, 2H), 3.21-2.85 (m, 8H), 2.82-2.65 (m, 3H), 2.51-2.30 (m, 1H), 2.24-2.03 (m, 1H), 1.99-1.87 (m, 1H), 1.86-1.69 (m, 6H), 1.67-1.57 (m, 2H), 1.57-1.39 (m, 4H), 1.45-1.05 (m, 2H), 1.04-0.96 (m, 3H), 0.96-0.88 (m, 3H), 0.88-0.79 (m, 3H), 0.79-0.63 (m, 3H), 0.56 (s, 1H).
(616) Step 18. 2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methylbutanamide (180 mg, 0.194 mmol) was purified by prep-HPLC to afford (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methylbutanamide (41.8 mg, 23.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.54H.sub.71N.sub.7O.sub.7 930.5; found 930.5; .sup.1H NMR (400 MHz, MeOD) δ 8.74 (d, J=4.0 Hz, 1H), 8.53-8.30 (m, 1H), 8.10-7.95 (m, 1H), 7.94-7.80 (m, 2H), 7.68 (t, J=8.0 Hz, 1H), 7.65-7.58 (m, 1H), 7.58-7.46 (m, 2H), 7.38-7.17 (m, 2H), 5.73-5.60 (m, 1H), 4.52-4.40 (m, 1H), 4.35-4.15 (m, 4H), 4.14-3.95 (m, 2H), 3.90-3.72 (m, 3H), 3.71-3.45 (m, 4H), 3.30-3.20 (m, 3H), 3.06-2.72 (m, 5H), 2.49-2.28 (m, 4H), 2.28-2.20 (m, 3H), 2.18-2.06 (m, 1H), 2.00-1.90 (m, 1H), 1.90-1.52 (m, 4H), 1.52-1.40 (m, 5H), 1.40-1.22 (m, 4H), 1.09-0.92 (m, 8H), 0.90-0.75 (m, 3H), 0.71-0.52 (m, 3H) and (2S)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methylbutanamide (51.2 mg, 28.4% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.54H.sub.71N.sub.7O.sub.7 930.5; found 930.3; .sup.1H NMR (400 MHz, MeOD) δ 8.74 (d, J=4.0 Hz, 1H), 8.28-8.20 (m, 0.6H), 8.11-7.98 (m, 1H), 7.97-7.80 (m, 2H), 7.73-7.48 (m, 4H), 7.46-7.36 (m, 0.4H), 7.33-7.26 (m, 1H), 7.25-7.13 (m, 1H), 5.79-5.66 (m, 1H), 4.54-4.43 (m, 1H), 4.42-4.01 (m, 7H), 3.90-3.75 (m, 2H), 3.73-3.48 (m, 4H), 3.27-3.12 (m, 3H), 3.08-2.99 (m, 1H), 2.96-2.85 (m, 2H), 2.84-2.69 (m, 2H), 2.69-2.49 (m, 6H), 2.41-2.29 (m, 1H), 2.15-2.05 (m, 1H), 1.95-1.85 (m, 1H), 1.84-1.71 (m, 1H), 1.71-1.38 (m, 11H), 1.14-1.00 (m, 3H), 1.00-0.71 (m, 9H), 0.70-0.56 (m, 3H).
Example A427. Synthesis of 3-((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)-N-((6.SUP.3.S,4S,Z)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)propanamide
(617) ##STR01262##
Step 1.
(618) To a mixture of tert-butyl prop-2-ynoate (5 g, 40 mmol) and [3-(3-hydroxyazetidin-1-yl)phenyl]methyl formate (4.1 g, 20 mmol) in DCM (150 mL) was added DMAP (9.8 g, 80 mmol). The mixture was stirred for 2 h, then diluted with H.sub.2O and washed with H.sub.2O (60 mL×3). The organic layer was dried over Na.sub.2SO.sub.4, filtered, the filtrate was concentrated under reduced pressure and the residue purified by silica gel column chromatography to give benzyl (E)-3-((3-(tert-butoxy)-3-oxoprop-1-en-1-yl)oxy)azetidine-1-carboxylate (6.6 g, 90% yield) as an oil. LCMS (ESI): m/z [M+Na] calc'd for C.sub.18H.sub.23NO.sub.5Na 356.2; found 356.2.
(619) Step 2.
(620) A mixture of benzyl (E)-3-((3-(tert-butoxy)-3-oxoprop-1-en-1-yl)oxy)azetidine-1-carboxylate (1.4 g, 4 mmol) and Pd/C (200 mg) in THF (10 mL) was stirred under an atmosphere of H.sub.2 (1 atmosphere) for 16 h. The mixture was filtered and the filtrate and was concentrated under reduced

pressure to give tert-butyl 3-(azetidin-3-yloxy)propanoate, which was used directly in the next step. LCMS (ESI): m/z [M+H] calc'd for C.sub.10H.sub.19NO.sub.3 201.1; found 202.2.
(621) Step 3.
(622) To a mixture of tert-butyl 3-(azetidin-3-yloxy)propanoate (300 mg, 1.5 mmol) and 4-(dimethylamino)-4-methylpent-2-ynoic acid (2.3 g, 15 mmol) in DMF (15 mL) at 5° C. was added DIPEA (1.9 g, 15 mmol) and T3P (4.77 g, 7.5 mmol) dropwise. The mixture was stirred at 5° C. for 2 h, then H.sub.2O and EtOAc (80 mL) were added. The organic and aqueous layers were separated and the organic layer was washed with H.sub.2O (20 mL×3), brine (30 mL), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to afford tert-butyl 3-((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)propanoate (60 mg, 12% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.18H.sub.30N.sub.2O.sub.4 338.2; found 339.2.
(623) Step 4.
(624) A mixture tert-butyl 3-((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)propanoate (70 mg, 0.21 mmol) in TFA/DCM (1:3, 2 mL) was stirred at 0-5° C. for 1 h, then concentrated under reduced pressure to give 3-({1-[4-(dimethylamino)-4-methylpent-2-ynoyl]azetidin-3-yl}oxy)propanoic acid (56 mg, 95% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.14H.sub.22N.sub.2O.sub.4 282.2; found 283.3.
(625) Step 5.
(626) To a mixture of 3-((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)propanoic acid (56 mg, 0.19 mmol), (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane- 5,7-dione (90 mg, 0.14 mmol) and DIPEA (200 mg, 1.9 mmol) in DMF (1 mL) at 0° C. was added HATU (110 mg, 0.38 mmol) portion-wise. The mixture was stirred at 0° C. for 1 h, then H.sub.2O added and the mixture extracted with EtOAx (150 mL×2). The combined organic layers were washed with H.sub.2O (150 mL) and brine (150 mL), then dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give 3-((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)propanamide (12.6 mg, 7.5% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.48H.sub.62N.sub.8O.sub.7S 894.5; found 895.3; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.73 (dd, J=4.8, 1.6 Hz, 1H), 8.57 (s, 1H), 8.28 (s, 0.3H), 7.84 (m, 1H), 7.71 (m, 1H), 7.52 (m, 3H), 5.76 (dd, J=30.2, 7.8 Hz, 1H), 4.40 (m, 4H), 4.32-4.12 (m, 4H), 4.06 (dd, J=12.4, 6.0 Hz, 1H), 3.97-3.86 (m, 1H), 3.79-3.66 (m, 4H), 3.46 (dd, J=14.8, 4.8 Hz, 1H), 3.41-3.33 (m, 3H), 3.29-3.19 (m, 1H), 3.17-3.05 (m, 1H), 2.79 (m, 1H), 2.73-2.50 (m, 3H), 2.49-2.43 (m, 3H), 2.38 (s, 3H), 2.21 (dd, J=12.6, 9.6 Hz, 1H), 1.95 (d, J=12.8 Hz, 1H), 1.86-1.73 (m, 1H), 1.61 (dd, J=12.6, 3.6 Hz, 1H), 1.51 (s, 2H), 1.46-1.43 (m, 4H), 1.38-1.27 (m, 3H), 1.01-0.86 (m, 6H), 0.44 (d, J=11.6 Hz, 3H).
Example A716. Synthesis of (3S)-1-acryloyl-N-((2S)-1-(((6.SUP.3.S,4S)-1-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.SUP.1.,2.SUP.2.,2.SUP.3.,2.SUP.6.,6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-decahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,1)-pyridinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide
(627) ##STR01263##
Step 1.
(628) To a solution of methyl (tert-butoxycarbonyl)-L-serinate (10 g, 45 mmol) in anhydrous MeCN (150 mL), was added DIPEA (17 g, 137 mmol). The reaction mixture was stirred at 45° C. for 2 h to give methyl 2-((tert-butoxycarbonyl)amino)acrylate in solution. LCMS (ESI): m/z [M+Na] calc'd

for $C_9H_{15}NO_4$ 201.1; found 224.1.
(629) Step 2.
(630) To a solution of methyl 2-((tert-butoxycarbonyl)amino)acrylate (12 g, 60 mmol) in anhydrous MeCN (150 mL) at 0° C., was added 4-DMAP (13 g, 90 mmol) and $(BoC)_2O$ (26 g, 120 mmol). The reaction was stirred for 6 h, then quenched with H_2O (100 mL) and extracted with DCM (200 mL×3). The combined organic layers were washed with brine (150 mL), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to give methyl 2-(bis(tert-butoxycarbonyl)amino)acrylate (12.5 g, 65% yield) as solid. LCMS (ESI): m/z [M+Na] calc'd for $C_{14}H_{23}NO_6$ 301.2; found 324.1.
(631) Step 3.
(632) To a mixture of 5-bromo-1,2,3,6-tetrahydropyridine (8.0 g, 49 mmol) in MeOH (120 mL) under an atmosphere of Ar was added methyl 2-{bis[(tert-butoxy)carbonyl]amino}prop-2-enoate (22 g, 74 mmol). The mixture was stirred for 16 h, then concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl 2-(bis(tert-butoxycarbonyl)amino)-3-(5-bromo-3,6-dihydropyridin-1(2H)-yl)propanoate (12 g, 47% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{19}H_{31}BrN_2O_6$ 462.1; found 463.1.
(633) Step 4.
(634) To a mixture of methyl 2-(bis(tert-butoxycarbonyl)amino)-3-(5-bromo-3,6-dihydropyridin-1(2H)-yl)propanoate (14 g, 30 mmol) in 1,4-dioxane (30 mL) and H_2O (12 mL) was added LiOH (3.6 g, 151 mmol). The mixture was heated to 35° C. and stirred for 12 h, then 1M HCl was added and the pH adjusted to ˜3-4. The mixture was extracted with DCM (300 mL×2) and the combined organic layers were dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure to give 3-(5-bromo-3,6-dihydropyridin-1(2H)-yl)-2-((tert-butoxycarbonyl)amino)propanoic acid (10 g, 85% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{13}H_{21}BrN_2O_4$ 348.1; found 349.0.
(635) Step 5.
(636) To a mixture of 3-(5-bromo-3,6-dihydropyridin-1(2H)-yl)-2-((tert-butoxycarbonyl)amino)propanoic acid (10 g, 30 mmol), DIPEA (12 g, 93 mmol) and methyl (3S)-1,2-diazinane-3-carboxylate (5.4 g, 37 mmol) in DMF (100 mL) at 0° C. under an atmosphere of Ar was added HATU (13 g, 34 mmol). The mixture was stirred at 0° C. for 2 h, then H_2O added and the mixture extracted with EtOAc (300 mL×2). The combined organic layers were dried over anhydrous Na_2SO_4, filtered, the filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give methyl (3S)-1-(3-(5-bromo-3,6-dihydropyridin-1(2H)-yl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (9.0 g, 55% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{19}H_{31}BrN_4O_5$ 474.1; found 475.1.
(637) Step 6.
(638) A mixture of methyl (3S)-1-(3-(5-bromo-3,6-dihydropyridin-1(2H)-yl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (9.0 g, 18 mmol), K_2CO_3 (4.5 g, 32 mmol), $Pd(dppf)Cl_2$.DCM (1.4 g, 2 mmol), 3-(1-ethyl-2-{2-[(1S)-1-methoxyethyl]pyridin-3-yl}-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl)-2,2-dimethylpropan-1-ol (9.8 g, 20 mmol) in 1,4-dioxane (90 mL) and H_2O (10 mL) under an atmosphere of Ar was heated to 75° C. and stirred for 2 h. H_2O was added and the mixture was extracted with EtOAc (200 mL×3). The combined organic layers were dried over Na_2SO_4, filtered, the filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (3S)-1-(2-((tert-butoxycarbonyl)amino)-3-(5-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)-3,6-dihydropyridin-1(2H)-yl)propanoyl)hexahydropyridazine-3-carboxylate (4.0 g, 25% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{42}H_{60}N_6O_7$ 760.5; found 761.4.

(639) Step 7.

(640) To a mixture of methyl (3S)-1-(2-((tert-butoxycarbonyl)amino)-3-(5-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)-3,6-dihydropyridin-1(2H)-yl)propanoyl)hexahydropyridazine-3-carboxylate (4.1 g, 5.0 mmol) in THF (35 mL) at 0° C. was added LiOH (0.60 g, 27 mmol). The mixture was stirred at 0° C. for 1.5 h, then 1M HCl added to adjust pH to ˜6-7 and the mixture extracted with EtOAc (200 mL×3). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and the filtrate was concentrated under reduced pressure to give (3S)-1-(2-((tert-butoxycarbonyl)amino)-3-(5-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)-3,6-dihydropyridin-1 (2H)-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (3.6 g, 80% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.41H.sub.58N.sub.6O.sub.7 746.4; found 747.4.

(641) Step 8.

(642) To a mixture of (3S)-1-(2-((tert-butoxycarbonyl)amino)-3-(5-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)-3,6-dihydropyridin-1(2H)-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (3.6 g, 5.0 mmol) and DIPEA (24 g, 190 mmol) in DCM (700 mL) under an atmosphere of Ar was added EDCl.Math.HCl (28 g, 140 mmol) and HOBT (6.5 g, 50 mmol). The mixture was heated to 30° C. and stirred for 16 h at 30° C., then concentrated under reduced pressure. The residue was diluted with EtOAc (200 mL) and washed with H.sub.2O (200 mL×2), brine (200 mL), dried over Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.1 2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-decahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,1)-pyridinacycloundecaphane-4-yl)carbamate (1.45 g, 40% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.41H.sub.56N.sub.6O.sub.6 728.4; found 729.4.

(643) Step 9.

(644) To a mixture of tert-butyl ((6.sup.3S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.1 2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-decahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,1)-pyridinacycloundecaphane-4-yl)carbamate (130 mg, 0.20 mmol) in DCM (1.0 mL) at 0° C. was added TFA (0.3 mL). The mixture was warmed to room temperature and stirred for 2 h, then concentrated under reduced pressure to give (6.sup.3S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-2.sup.1 2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5, 6.sup.6-decahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(5,1)-pyridinacycloundecaphane-5,7-dione, which was used directly in the next step directly without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.36H.sub.48N.sub.6O.sub.4 628.4; found 629.4.

(645) Step 10.

(646) To a mixture of ((6.sup.3S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-2.sup.1 2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-decahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,1)- pyridinacycloundecaphane-5,7-dione (130 mg, 0.2 mmol), DIPEA (270 mg, 2.0 mmol) and (2S)-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanoic acid (118 mg, 0.40 mmol) in DMF (3.0 mL) at 0° C. under an atmosphere of Ar was added HATU (87 mg, 0.30 mmol) in portions. The mixture was stirred at 0° C. for 1 h, then diluted with H.sub.2O extracted with EtOAc (30 mL×2). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered, the filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give (3S)-1-acryloyl-N-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.1 2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-

decahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,1)-pyridinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide (17.2 mg, 10% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.50H.sub.68N.sub.6O.sub.4 892.5; found 893.5; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.74 (d, J=4.4 Hz, 1H), 7.93-7.90 (m, 1H), 7.56-7.51 (m, 3H), 7.43 (d, J=4.4 Hz, 1H), 6.63-6.53 (m, 2H), 6.33-6.23 (m, 2H), 5.83-5.70 (m, 1H), 4.73-4.70 (d, J=11.0 Hz, 1H), 4.48-4.45 (d, J=13.0 Hz, 1H), 4.12-4.10 (m, 3H), 3.86-3.81 (m, 4H), 3.79-3.75 (m, 1H), 3.72-3.69 (m, 3H), 3.57-3.47 (m, 2H), 3.21-3.09 (m, 1H), 3.07-3.04 (q, 4H), 3.02-2.95 (m, 3H), 2.86-2.82 (m, 3H), 2.66-2.48 (m, 2H), 2.29-2.17 (m, 4H), 2.11-1.98 (m, 2H), 1.95-1.91 (m, 1H), 1.45 (d, J=6.2 Hz, 3H), 1.23-1.16 (m, 2H), 1.09-1.04 (m, 1H), 0.97-0.93 (m, 3H), 0.92-0.81 (m, 5H), 0.67-0.63 (m, 3H).

Example A663. The synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)405yridine405-3-yl)oxy)methyl)-N-((6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)405yridine-3-yl)-10,10-dimethyl-5,7-dioxo-2.SUP.1.,2.SUP.2.,2.SUP.3.,2.SUP.6.,6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-decahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)- pyridazina-2(5,1)-pyridinacycloundecaphane-4-yl)-3-methylbutanamide

(647) ##STR01264##

(648) To a mixture of (6.sup.3S,4S)-4-amino-1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)405yridine-3-yl)-10,10-dimethyl-2.sup.1 2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.2 6.sup.3 6.sup.4 6.sup.5 6.sup.6-decahydro-1.sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(5,1)- pyridinacycloundecaphane-5,7-dione (100 mg, 0.16 mmol),®-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)405yridine405-3-yl)oxy)methyl)-3-methylbutanoic acid (80 mg, 0.24 mmol) and DIPEA (825 mg, 6.4 mmol) in DMF (2 mL) at 0° C., was added HATU (95 mg, 0.24 mmol). The reaction mixture was stirred at 0° C. for 1 h, then poured into H.sub.2, (60 mL), extracted with EtOAc (80 mL×2). The combined organic layers were washed with H.sub.2S (80 mL) and brine (80 mL, dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel chromatography to afford (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)405yridine405-3-yl)oxy)methyl)-N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)405yridine-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.1 2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.26.sup.3 6.sup.4,6.sup.5,6.sup.6-decahydro-1.sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(5,1)-pyridinacycloundecaphane-4-yl)-3-methylbutanamide (55 mg, 36% yield) as solid. .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.76-8.70 (m, 1H), 8.49 (dd, J=4.3, 1.4 Hz, 0.1H), 7.93-7.87 (m, 1H), 7.58-7.50 (m, 3H), 7.41 (dd, J=8.8, 3.2 Hz, 1H), 6.26 (d, J=16.8 Hz, 1H), 5.96 (t, J=9.6 Hz, 1H), 4.47 (d, J=12.8 Hz, 1H), 4.39-4.28 (m, 2H), 4.21-3.97 (m, 5H), 3.96-3.70 (m, 5H), 3.68-3.54 (m, 3H), 3.51-3.35 (m, 1H), 3.11 (d, J=22.7 Hz, 3H), 3.00-2.67 (m, 5H), 2.46-2.30 (m, 7H), 2.24 (s, 3H), 2.11 (d, J=12.4 Hz, 1H), 1.92 (d, J=13.2 Hz, 1H), 1.85-1.60 (m, 3H), 1.45 (d, J=7.8 Hz, 6H), 1.32 (d, J=16.0 Hz, 3H), 1.12 (dt, J=24.5, 6.8 Hz, 3H), 0.95 (m, 6H), 0.76 (in, 6H). LCMS (ESI): m/z [M+H] calc'd for C.sub.53H.sub.74N.sub.8O.sub.7 934.6; found 935.5.

Example A646. The synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-4-yl)-3-methylbutanamide

(649) ##STR01265##

Step 1.

(650) A mixture of tert-butyl ((6.sup.3S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-2.sup.1,2.sup.2,2.sup.3,2.sup.6,6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-decahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,1)-pyridinacycloundecaphane-4-yl)carbamate (0.2 g, 0.28 mmol) and Pd/C (0.2 g, 2 mmol) in MeOH (10 mL) was stirred at 25° C. for 16 h under an H.sub.2

atmosphere. The reaction mixture was filtered through Celite, concentrated under reduced pressure to afford tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-4-yl)carbamate as solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.41H.sub.58N.sub.6O.sub.6 730.4; found 731.4.

(651) Step 2.

(652) To a solution of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-4-yl)carbamate (150 mg, 0.2 mmol) in DCM (1.5 mL) at 0° C. was added TFA (0.5 mL). The reaction mixture was stirred at 20° C. for 1 h, then concentrated under reduced pressure to afford (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-5,7-dione as solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.36H.sub.50N.sub.6O.sub.4 630.4; found 631.4.

(653) Step 3.

(654) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-5,7-dione (240 mg, 0.4 mmol), DIPEA (982 mg, 2 mmol) and (R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-3-methylbutanoic acid (148 mg, 0.45 mmol) in DMF (4 mL) at 0° C. under argon atmosphere, was added HATU (173 mg, 0.46 mmol) in portions. The reaction mixture was stirred at 0° C. under an argon atmosphere for 1 h, then quenched with H.sub.2O at 0° C. The resulting mixture was extracted with EtOAc (30 mL×2). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by reverse phase chromatography to afford (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-4-yl)-3-methylbutanamide (150 mg, 38% yield) as solid. .sup.1H NMR (400 MHz, CD3OD) δ 8.72 (d, J=4.8 Hz, 1H), 7.88 (d, J=8.0 Hz, 1H), 7.53-7.49 (m, 2H), 7.42 (d, J=8.4 Hz, 1H), 7.18 (d, J=8.4 Hz, 1H), 5.95-5.91 (m, 1H), 4.52-4.49 (m, 1H), 4.37-4.25 (m, 3H), 4.18-4.15 (m, 2H), 3.99-3.98 (m, 2H), 3.90-3.86 (m, 1H), 3.76-3.68 (m, 2H), 3.55-3.50 (m, 2H), 3.39-3.36 (m, 2H), 3.20 (s, 3H), 3.02 (s, 3H), 2.89-2.79 (m, 3H), 2.62-2.50 (m, 2H), 2.36 (s, 3H), 2.35-2.30 (m, 1H), 2.26 (s, 3H), 2.20-1.15 (m, 1H), 1.97-1.93 (m, 3H), 1.81-1.76 (m, 4H), 1.64-1.61 (m, 2H), 1.46-1.43 (m, 6H), 1.36 (d, J=14.8 Hz, 3H), 1.02 (s, 3H), 0.94 (m, 6H), 0.81 (s, 3H), 0.65 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.53H.sub.76N.sub.8O.sub.7 936.6; found 937.5.

Example A740. Synthesis of (3S)-1-acryloyl-N-((2S)-1-(((2.SUP.3.S,6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide

(655) ##STR01266##

(656) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-5,7-dione (140 mg, 0.20 mmol), DIPEA (570 mg, 4.4 mmol) and (2S)-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanoic acid (124 mg, 0.40 mmol) in DMF (3.0 mL) at 0° C. under an atmosphere of

Ar was added HATU (100 mg, 0.30 mmol) in portions. The mixture was stirred at 0° C. for 1 h, then H.sub.2O was added and the mixture extracted with EtOAc (2×30 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give (3S)-1-acryloyl-N-((2S)-1-(((2.sup.3S,6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(3,1)-piperidinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide (41 mg, 20% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.50H.sub.70N.sub.8O.sub.7 894.5; found 895.5; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.72 (d, J=4.8 Hz, 1H), 7.87 (d, J=7.6 Hz, 1H), 7.51-7.49 (m, 2H), 7.42-7.37 (m, 1H), 7.18-7.14 (m, 1H), 6.64-6.54 (m, 1H), 6.30-6.23 (m, 1H), 5.77-5.70 (m, 2H), 4.65-4.60 (m, 1H), 4.50-4.40 (m, 1H), 4.27-4.16 (m, 2H), 4.00-3.95 (m, 2H), 3.83-3.78 (m, 2H), 3.73-3.60 (m, 4H), 3.51-3.36 (m, 3H), 3.22-3.19 (m, 4H), 3.07 (d, J=6.8 Hz, 2H), 2.99 (d, J=12.0 Hz, 3H), 2.90-2.78 (m, 2H), 2.75-2.64 (m, 3H), 2.20-2.10 (m, 4H), 2.02-1.93 (m, 3H), 1.87-1.64 (m, 4H), 1.45 (d, J=4.8 Hz, 3H), 1.06-1.00 (m, 4H), 0.97-0.89 (m, 3H), 0.83-0.79 (m, 3H), 0.66 (s, 3H).

Example A534. (2S)-2-((S)-7-(4-(dimethylamino)-4-methylpent-2-ynoyl)-1-oxo-2,7-diazaspiro[4.4]nonan-2-yl)-N-((6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methylbutanamide

(657) ##STR01267##

Step 1.

(658) To a mixture of 1-tert-butyl 3-methyl pyrrolidine-1,3-dicarboxylate (20.0 g, 87.2 mmol) in THF (150 mL) at −78° C. under an atmosphere of nitrogen was added 1M LiHMDS in THF (113.4 mL, 113.4 mmol). After stirring at −78° C. for 40 min, allyl bromide (13.72 g, 113.4 mmol) was added and the mixture was allowed to warm to room temperature and stirred for 4 h. The mixture was cooled to 0° C., saturated NaCl (30 mL) was added and the mixture extracted with EtOAc. The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 1-(tert-butyl) 3-methyl 3-allylpyrrolidine-1,3-dicarboxylate (17 g, 72% yield) as an oil. .sup.1H NMR (300 MHz, CDCl.sub.3) δ 5.80-5.60 (m, 1H), 5.16-5.02 (m, 2H), 3.71 (s, 4H), 3.42 (d, J=9.3 Hz, 2H), 3.27 (t, J=11.2 Hz, 1H), 2.42 (d, J=7.6 Hz, 2H), 2.38-2.24 (m, 1H), 2.05 (s, 1H), 1.85 (dt, J=14.3, 7.5 Hz, 1H), 1.46 (s, 10H), 1.27 (t, J=7.1 Hz, 1H).

(659) Step 2.

(660) To a mixture of 1-(tert-butyl) 3-methyl 3-allylpyrrolidine-1,3-dicarboxylate (4.0 g, 14.9 mmol) and 2,6-dimethylpyridine (3.18 g, 29.7 mmol) in 1,4-dioxane (200 mL) and H.sub.2O (100 mL) at 0° C. was added K.sub.2OsO.sub.4 2H.sub.2O (0.11 g, 0.3 mmol) in portions. The mixture was stirred for 15 min at 0° C., then NaIO.sub.4 (6.35 g, 29.7 mmol) was added in portions. The mixture was stirred at room temperature for 3 h at room temperature, then cooled to 0° C. and saturated aqueous Na.sub.2S.sub.2O.sub.3 (50 mL) added. The mixture was extracted with EtOAc (3×100 mL) and the combined organic layers were washed with 2 M HCl, then dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure to give 1-(tert-butyl) 3-methyl 3-(2-oxoethyl)pyrrolidine-1,3-dicarboxylate (4 g, 52% yield) as an oil. .sup.1H NMR (300 MHz, CDCl.sub.3) δ 5.80-5.60 (m, 1H), 5.16-5.04 (m, 2H), 3.72 (s, 3H), 3.41 (s, 3H), 3.28 (d, J=11.0 Hz, 1H), 2.44 (s, 2H), 2.31 (d, J=9.1 Hz, 1H), 1.85 (dt, J=12.7, 7.5 Hz, 1H), 1.69 (s, 1H), 1.47 (s, 10H).

(661) Step 3.

(662) To a mixture of 1-(tert-butyl) 3-methyl 3-(2-oxoethyl)pyrrolidine-1,3-dicarboxylate (6.30 g, 23.2 mmol), in MeOH (70 mL) at 0° C. was added benzyl (2S)-2-amino-3-methylbutanoate (7.22 g, 34.8 mmol) and ZnCl.sub.2 (4.75 g, 34.8 mmol). The mixture was warmed to room temperature and

stirred for 30 min, then cooled to 0° C. and NaCNBH.sub.3 (2.92 g, 46.4 mmol) was added in portions. The mixture was warmed to room temperature and stirred for 2 h, then cooled to 0° C. and saturated aqueous NH.sub.4Cl added. The mixture was extracted with EtOAc (3×200 mL) and the combined organic layers were washed with brine (150 mL), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 1-(tert-butyl) 3-methyl 3-(2-(((S)-1-(benzyloxy)-3-methyl-1-oxobutan-2-yl)amino)ethyl)pyrrolidine-1,3-dicarboxylate (6.4 g, 54% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.25H.sub.38N.sub.2O.sub.6 462.3; found 463.4.
(663) Step 4.
(664) To a mixture of 1-(tert-butyl) 3-methyl 3-(2-(((S)-1-(benzyloxy)-3-methyl-1-oxobutan-2-yl)amino)ethyl)pyrrolidine-1,3-dicarboxylate (4.50 g, 9.7 mmol) in toluene (50 mL) was added DIPEA (12.57 g, 97.3 mmol) and DMAP (1.19 g, 9.7 mmol). The resulting mixture was heated to 80° C. and stirred for 24 h, then concentrated under reduced pressure and the residue was purified by preparative-HPLC, then by chiral-HPLC to give tert-butyl (R)-7-((S)-1-(benzyloxy)-3-methyl-1-oxobutan-2-yl)-6-oxo-2,7-diazaspiro[4.4]nonane-2-carboxylate (1.0 g, 32% yield) and tert-butyl (S)-7-((S)-1-(benzyloxy)-3-methyl-1-oxobutan-2-yl)-6-oxo-2,7-diazaspiro[4.4]nonane-2-carboxylate (1.0 g, 32% yield) and as a an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.24H.sub.34N.sub.2O.sub.5 430.5; found 431.2 and LCMS (ESI): m/z [M+H] calc'd for C.sub.24H.sub.34N.sub.2O.sub.5 430.3; found 431.2.
(665) Step 5.
(666) A mixture of tert-butyl (R)-7-((S)-1-(benzyloxy)-3-methyl-1-oxobutan-2-yl)-6-oxo-2,7-diazaspiro[4.4]nonane-2-carboxylate (4.0 g) and 10% Pd/C (1 g) in MeOH (40 mL) was stirred at room temperature under an atmosphere of H.sub.2. The mixture was filtered through a pad of Celite pad and the filtrate was concentrated under reduced pressure to give (S)-2-((R)-7-(tert-butoxycarbonyl)-1-oxo-2,7-diazaspiro[4.4]nonan-2-yl)-3-methylbutanoic acid (4.9 g) as a solid. LCMS (ESI): m/z [M−H] calc'd for C.sub.17H.sub.28N.sub.2O.sub.5 340.2; found 339.3.
(667) Step 6.
(668) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (500 mg, 0.8 mmol) in DCM at 0° C. were added DIPEA (829 mg, 6.4 mmol), ((S)-2-((R)-7-(tert-butoxycarbonyl)-1-oxo-2,7-diazaspiro[4.4]nonan-2-yl)-3-methylbutanoic acid (273 mg, 0.8 mmol) and HATU (396 mg, 1.0 mmol) in portions over 1 min. The mixture was allowed to warm to room temperature and stirred 2 h, then concentrated under reduced pressure and the residue was purified by preparative-TLC to give tert-butyl (5R)-7-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-6-oxo-2,7-diazaspiro[4.4]nonane-2-carboxylate (500 mg, 64% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.54H.sub.71N.sub.7O.sub.8 945.5; found 946.5.
(669) Step 7.
(670) To a mixture of tert-butyl (5R)-7-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-6-oxo-2,7-diazaspiro[4.4]nonane-2-carboxylate (1.0 g, 1.06 mmol) in DCM (10 mL) at 0° C. was added TFA (3 mL) dropwise. The mixture was warmed to room temperature and stirred for 1 h, then concentrated under reduced pressure to give (2S)—N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methyl-2-((S)-1-oxo-2,7-diazaspiro[4.4]nonan-

2-yl)butanamide (1.3 g). LCMS (ESI): m/z [M−H] calc'd for C.sub.49H.sub.63N.sub.7O.sub.6 846.1; found 845.5.
(671) Step 8.
(672) To a mixture of (2S)—N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methyl-2-((S)-1-oxo-2,7-diazaspiro[4.4]nonan-2-yl)butanamide (500 mg, 0.59 mmol) and DIPEA (764 mg, 5.9 mmol) in DMF (5 mL) at 0° C. were added 4-(dimethylamino)-4-methylpent-2-ynoic acid (110 mg, 0.71 mmol) and HATU (292 mg, 0.77 mmol) in portions. The mixture was warmed to room temperature and stirred for 1 h, then H.sub.2O (10 mL) was added and the mixture extracted with EtOAc (10 mL×3). The combined organic layers were washed with brine (10 mL), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give (2S)-2-((S)-7-(4-(dimethylamino)-4-methylpent-2-ynoyl)-1-oxo-2,7-diazaspiro[4.4]nonan-2-yl)-N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1 .sup.1H-8-oxa-1 (5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methylbutanamide (177 mg, 28.94% yield) as a white solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.57H.sub.74N.sub.8O.sub.7 982.6; found 983.8; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.76 (dd, J=4.7, 1.7 Hz, 1H), 8.52 (d, J=7.9 Hz, 1H), 7.99 (d, J=1.7 Hz, 1H), 7.83 (d, J=10.2 Hz, 2H), 7.74-7.58 (m, 3H), 7.53 (dd, J=7.7, 4.8 Hz, 1H), 7.24 (t, J=7.6 Hz, 1H), 7.12 (d, J=7.6 Hz, 1H), 5.32 (d, J=9.7 Hz, 2H), 4.33-4.20 (m, 4H), 4.03 (dd, J=15.0, 8.6 Hz, 2H), 3.88-3.82 (m, 1H), 3.63 (dq, J=20.5, 10.3 Hz, 4H), 3.42-3.34 (m, 2H), 3.21 (s, 1H), 3.13 (d, J=2.8 Hz, 3H), 2.87 (s, 2H), 2.83-2.72 (m, 2H), 2.69-2.62 (m, 1H), 2.21 (d, J=22.6 Hz, 6H), 2.12-1.76 (m, 7H), 1.75-1.47 (m, 2H), 1.46-1.28 (m, 9H), 0.99-0.89 (m, 6H), 0.79-0.71 (m, 6H), 0.52 (s, 3H).
Example A341. Synthesis of (3S)-1-acryloyl-N-((2S)-1-(((6.SUP.3.S,4S)-2.SUP.5.-(difluoromethyl)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide
(673) ##STR01268## ##STR01269## ##STR01270##
Step 1.
(674) To a mixture of 3-bromo-5-iodobenzaldehyde (4.34 g, 14.0 mmol) in DCM at 0° C. under an atmosphere of N.sub.2 was added BAST (6.8 g, 30.7 mmol) and EtOH (129 mg, 2.8 mmol) dropwise. The mixture was heated with microwave heating at 27° C. for 14 h. H.sub.2O (500 mL) was added and the mixture was extracted with DCM (200 mL×3), the combined organic layers were concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give 1-bromo-3-(difluoromethyl)-5-iodobenzene (3.2 g, 65% yield) as a solid. .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 8.16 (p, J=1.2 Hz, 1H), 7.94 (p, J=1.3 Hz, 1H), 7.81 (p, J=1.3 Hz, 1H), 7.00 (t, J=55.3 Hz, 1H).
(675) Step 2.
(676) A mixture of Zn (2.28 g, 34.8 mmol) and 12 (442 mg, 1.74 mmol) in DMF (20 mL) under an atmosphere of Ar was stirred at 50° C. for 0.5 h. To this mixture was added a solution of methyl (methyl (R)-2-((tert-butoxycarbonyl)amino)-3-iodopropanoate (2.39 g, 7.25 mmol) in DMF (20 mL) and the mixture was stirred at 50° C. for 2 h. After cooling, the mixture was added to 1-bromo-3-(difluoromethyl)-5-iodobenzene (2.90 g, 8.7 mmol), Pd.sub.2(dba).sub.3 (239 mg, 0.26 mmol) and tri-2-furylphosphine (162 mg, 0.7 mmol) in DMF (20 mL). The mixture was heated to 70° C. and stirred for 2 h, then H.sub.2O (200 mL) was added and the mixture extracted with EtOAc (200 mL×3). The combined organic layers were concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (S)-3-(3-bromo-5-

(difluoromethyl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoate (560 mg, 19% yield) as a solid. .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 7.65 (d, J=10.0 Hz, 2H), 7.47 (s, 1H), 7.36 (d, J=8.4 Hz, 1H), 7.00 (t, J=55.6 Hz, 1H), 4.25 (td, J=9.6, 4.7 Hz, 1H), 3.64 (s, 3H), 3.11 (dd, J=13.6, 4.9 Hz, 1H), 3.00-2.80 (m, 1H), 1.32 (s, 9H).
(677) Step 3.
(678) To a mixture of methyl (S)-3-(3-bromo-5-(difluoromethyl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoate (650 mg, 1.6 mmol) in THF (1.5 mL) at 0° C. under an atmosphere of N.sub.2 was added LiOH (114 mg, 4.8 mmol) in H.sub.2O (1.50 mL). The mixture was stirred at 0° C. for 1 h, then acidified to pH 5 with 1M HCl. The mixture was extracted with DCM/MeOH (10/1) (100 mL×3) and the combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure to give (S)-3-(3-bromo-5-(difluoromethyl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoic acid (500 mg), which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.15H.sub.18BrF.sub.2NO.sub.4 393.0; found 392.1.
(679) Step 4.
(680) To a mixture of methyl (3S)-1,2-diazinane-3-carboxylate (475 mg, 3.3 mmol) in DCM (10 mL) at 0° C. under an atmosphere of N.sub.2 were added N-methylmorpholine (3.34 g, 33.0 mmol) and (S)-3-(3-bromo-5-(difluoromethyl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoic acid (650 mg, 1.7 mmol) and HOBt (45 mg, 0.33 mmol) and EDCI (632 mg, 3.3 mmol). The mixture was warmed to room temperature and stirred for 16 h, then diluted with DCM (100 mL) and H.sub.2O. The organic and aqueous layer was separated and the aqueous layer was extracted with DCM (100 mL×3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (S)-1-((S)-3-(3-bromo-5-(difluoromethyl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (510 mg, 56% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.21H.sub.28BrF.sub.2N.sub.3O.sub.5 519.1; found 520.3.
(681) Step 5.
(682) To a mixture of 4,4,5,5-tetramethyl-2-(tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (488 mg, 1.92 mmol) and methyl (S)-1-((S)-3-(3-bromo-5-(difluoromethyl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (500 mg, 0.96 mmol) in 1,4-dioxane (5 mL) was added Pd(dppf)Cl.sub.2 (70 mg, 0.07 mmol) and KOAc (236 mg, 2.4 mmol) in portions. The mixture was heated to 90° C. and stirred for 4 h then diluted with H.sub.2O (100 mL). The mixture was extracted with DCM (100 mL×3) and the combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(difluoromethyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (423 mg, 73% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.27H.sub.40BF.sub.2N.sub.3O.sub.7 567.3; found 568.2.
(683) Step 6.
(684) To a mixture of methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(difluoromethyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (260 mg, 0.47 mmol), (S)-3-(5-bromo-1-ethyl-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-indol-3-yl)-2,2-dimethylpropan-1-ol and Pd(dppf)Cl.sub.2 (34 mg, 0.05 mmol) in 1,4-dioxane (3 mL) and H.sub.2O (0.6 mL) was added K.sub.2CO.sub.3 (163 mg, 1.12 mmol). The mixture was heated to 60° C. and stirred for 16 h, then diluted with H.sub.2O (100 mL) and extracted with DCM (100 mL×3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(difluoromethyl)-5-

(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (350 mg, 78% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.44H.sub.57F.sub.2N.sub.5O.sub.7 805.4; found 806.6.
(685) Step 7.
(686) To a mixture of methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(difluoromethyl)-5-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (350 mg, 0.43 mmol) in THF (2.8 mL) at 0° C. was added LiOH H.sub.2O (54 mg, 1.3 mmol) in H.sub.2O (0.7 mL). The mixture was warmed to room temperature and stirred for 2 h, then acidified to pH 5 with 1M HCl and extracted with EtOAc (50 mL×3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered and the filtrate was concentrated under reduced pressure to give (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(difluoromethyl)-5-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylic acid (356 mg) was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.43H.sub.55F.sub.2N.sub.5O.sub.7 791.4; found 792.6.
(687) Step 8.
(688) To a mixture of S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(difluoromethyl)-5-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylic acid (356 mg, 0.45 mmol) and DIPEA (1.74 g, 13.5 mmol) in DCM were added EDCI (2.41 g, 12.6 mmol) and HOBt (304 mg, 2.3 mmol). The mixture was stirred for 16 h then H.sub.2O was added and the mixture extracted with EtOAc (200 mL×3). The combined organic layers were washed with brine (50 mL×4), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl ((6.sup.3S,4S)-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (202 mg, 51% yield). LCMS (ESI): m/z [M+H] calc'd for C.sub.43H.sub.53F.sub.2N.sub.5O.sub.6 773.4; found 774.6.
(689) Step 9.
(690) To a mixture of tert-butyl ((6.sup.3S,4S)-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (202 mg, 0.26 mmol) in DCM (2 mL) at 0° C. was added TFA (1.0 mL) dropwise. The mixture was stirred at 0° C. for 1.5 h, then concentrated under reduced pressure and dried azeotropically with toluene (3 mL×3) to give (6.sup.3S,4S)-4-amino-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione, which was used directly in the next without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.38H.sub.45F.sub.2N.sub.5O.sub.4 673.3; found 674.5.
(691) Step 10.
(692) To a mixture of (6.sup.3S,4S)-4-amino-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (202 mg, 0.3 mmol) and (2S)-2-[(tert-butoxycarbonyl)(methyl)amino]-3-methylbutanoic acid (139 mg, 0.6 mmol) in THF under an atmosphere of Ar were added DIPEA (581 mg, 4.5 mmol), EDCI (86 mg, 0.45 mmol) and HOBt (61 mg, 0.45 mmol). The mixture was stirred for 16 h, then H.sub.2O (100 mL) added and the mixture extracted with EtOAc (200 mL×3). The combined organic layers were washed with brine (50 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica

gel column chromatography to give tert-butyl ((2S)-1-(((6.sup.3S,4S)-2.sup.5-(difluoromethyl)-11-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (135 mg, 46% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.49H.sub.64F.sub.2N.sub.6O.sub.7 886.5; found 887.6.

(693) Step 11.

(694) To a mixture of tert-butyl ((2S)-1-(((6.sup.3S,4S)-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (130 mg, 0.15 mmol) in DCM at 0° C. under an atmosphere of N.sub.2 was added TFA (1.0 mL) dropwise. The mixture was stirred at 0° C. for 1.5 h, then concentrated under reduced pressure and dried azeotropically with toluene (3 mL×3) to give (2S)—N-((6.sup.3S,4S)-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methyl-2-(methylamino)butanamide (130 mg), which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.44H.sub.56F.sub.2N.sub.6O.sub.5 786.4; found 787.6.

(695) Step 12.

(696) To a mixture of (2S)—N-((6.sup.3S,4S)-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methyl-2-(methylamino)butanamide (130 mg, 0.17 mmol) and (3S)-1-(prop-2-enoyl)pyrrolidine-3-carboxylic acid (56 mg, 0.33 mmol) in MeCN (1.5 mL) at 0° C. under an atmosphere of N.sub.2 were added DIPEA (427 mg, 3.3 mmol) and CIP (69 mg, 0.25 mmol). The mixture was stirred at 0° C. for 1 h, then H.sub.2O (100 mL) was added and the mixture extracted with EtOAc (200 mL×3). The combined organic layers were washed with brine (50 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give (3S)-1-acryloyl-N-((2S)-1-(((6.sup.3S,4S)-2.sup.5-(difluoromethyl)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide (58 mg, 36% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.52H.sub.65F.sub.2N.sub.7O.sub.7 937.4; found 938.1; .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.78 (dd, J=4.8, 1.7 Hz, 1H), 8.43-8.21 (m, 1H), 8.02 (s, 2H), 7.93-7.81 (m, 2H), 7.76 (dd, J=9.3, 3.9 Hz, 1H), 7.66 (d, J=8.7 Hz, 1H), 7.56 (dd, J=7.7, 4.8 Hz, 1H), 7.34 (d, J=5.4 Hz, 1H), 7.20-6.86 (m, 1H), 6.80-6.40 (m, 1H), 6.15 (ddt, J=16.8, 4.9, 2.4 Hz, 1H), 5.90-5.60 (m, 1H), 5.59-5.19 (m, 2H), 4.71 (dd, J=10.7, 3.1 Hz, 1H), 4.40-4.17 (m, 3H), 4.12-3.90 (m, 3H), 3.85-3.71 (m, 1H), 3.61 (tdd, J=23.4, 9.9, 4.3 Hz, 6H), 3.40-3.30 (m, 2H), 3.11 (d, J=6.8 Hz, 3H), 3.08-2.90 (m, 2H), 2.87 (s, 2H), 2.84 (s, 3H), 2.69-2.30 (d, J=16.5 Hz, 1H), 2.30-1.79 (m, 5H), 1.75-1.45 (m, 2H), 1.40 (d, J=6.1 Hz, 3H), 1.05-0.85 (m, 6H), 0.85-0.66 (m, 6H), 0.57 (d, J=11.8 Hz, 3H).

Example A741. Synthesis of (3S)-1-acryloyl-N-((2S)-1-(((2.SUP.3.S,6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-piperidinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide

(697) ##STR01271##

Step 1.

(698) A solution of tert-butyl (3R)-3-(hydroxymethyl) piperidine-1-carboxylate (10 g, 46.45 mmol) in DCM (200 mL) at 0° C., was added PPh_3 (15.8 g, 60.4 mmol), Imidazole (4.7 g, 69.7 mmol) and I_2 (14.1 g, 55.74 mmol). The reaction suspension was stirred at 20° C. for 17 h, then concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford tert-butyl (3R)-3-(iodomethyl) piperidine-1-carboxylate (10 g, 66% yield) as oil. LCMS (ESI): m/z [M+H] calc'd for $C_{11}H_{20}INO_2$ 325.1; found no mass.

(699) Step 2.

(700) To a mixture of 3-isopropyl-2,5-dimethoxy-3,6-dihydropyrazine (10.8 g, 58.9 mmol) in THF (150 mL) at −60° C. under an atmosphere of N_2 was added n-BuLi (47 mL, 2.5 M in hexane, 117.7 mmol) dropwise. The mixture was warmed to 0° C. and was stirred for 2 h, then re-cooled to −60° C., and a solution of tert-butyl (3R)-3-(iodomethyl)piperidin-1-yl formate (9.60 g, 29.4 mmol) in THF (50 mL) was slowly added dropwise. The mixture was stirred at −60° C. for 2 h then warmed to room temperature and stirred for 2 h. Saturated NH_4Cl (150 mL) was slowly added and the mixture extracted with EtOAc (150 mL×2). The combined organic layers were washed with brine (200 mL), dried over anhydrous Na_2SO_4 and filtered. The filtrate was reduced under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl (3S)-3-{[(2S)-5-isopropyl-3,6-dimethoxy-2,5-dihydropyrazin-2-yl]methyl}piperidin-1-yl formate (5.3 g, 46% yield) as a gum. LCMS (ESI): m/z [M+H] calc'd for $C_{20}H_{35}N_3O_4$ 381.5; found 382.3.

(701) Step 3.

(702) A mixture of tert-butyl (3S)-3-{[(2S)-5-isopropyl-3,6-dimethoxy-2,5-dihydropyrazin-2-yl]methyl}piperidin-1-yl formate (5.30 g, 13.9 mol) in MeCN (4 mL) was added 1M HCl (27.7 mL, 27.7 mmol) dropwise. The mixture was stirred for 2 h, then saturated $NaHCO_3$ until ˜pH 7-8, then extracted with DCM (30 mL×2). The combined organic layers was dried over anhydrous Na_2SO_4, filtered and concentrated under reduced pressure to give methyl (S)-tert-butyl 3-((S)-2-amino-3-methoxy-3-oxopropyl)piperidine-1-carboxylate (4.3 g, 95% yield) as an oil, which was used in next step without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{14}H_{26}N_2O_4$ 286.2; found 287.3.

(703) Step 4.

(704) To a mixture of methyl (S)-tert-butyl 3-((S)-2-amino-3-methoxy-3-oxopropyl)piperidine-1-carboxylate (4.30 g, 15.0 mmol) in EtOAc (30 mL) and H_2O (20 mL) at −10° C. was added $NaHCO_3$ (3.77 g, 44.88 mmol). The mixture was stirred at −10° C. for 10 min, then a solution of benzyl chloroformate (3.83 g, 22.44 mmol) was added dropwise. The mixture was warmed to 0° C. and stirred for 1 h, then H_2O (50 mL) was added and the mixture extracted with EtOAc (50 mL×2). The combined organic layers were dried over anhydrous Na_2SO_4, filtered, the filtrate was concentrated under reduced pressure to give tert-butyl (3S)-3-[(2S)-2-{[(benzyloxy)carbonyl] amino}-3-methoxy-3-oxopropyl]piperidine-1-carboxylate (4.0 g, 60% yield) as a gum. LCMS (ESI): m/z [M−Boc+H] calc'd for $C_{17}H_{24}N_2O_4$ 320.2; found 321.3.

(705) Step 5.

(706) To a mixture of tert-butyl (3S)-3-[(2S)-2-{[(benzyloxy)carbonyl] amino}-3-methoxy-3-oxopropyl]piperidine-1-carboxylate (1.0 g, 2.38 mmol) in EtOAc (8 mL) was added 2M HCl in EtOAc (11.9 mL, 23.8 mmol). The mixture was stirred for 2 h, then saturated $NaHCO_3$ added until ˜pH 8-9, and the mixture extracted with DCM (30 mL×3). The combined organic layer was dried over anhydrous Na_2SO_4, filtered and concentrated under reduced pressure to give (2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-piperidin-3-yl]propanoate (740 mg, 91% yield) as a gum. LCMS (ESI): m/z [M+H] calc'd for $C_{17}H_{24}N_2O_4$ 320.2; found 321.2.

(707) Step 6.

(708) To a mixture of (3-{3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1

S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)boranediol (5.47 g, 8.43 mmol) and methyl (2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-piperidin-3-yl]propanoate (2.70 g, 8.43 mmol) in DCM (70 mL) was added Cu(OAc).sub.2 (6.06 g, 16.86 mmol) and pyridine (2.0 g, 25.3 mmol). The mixture was stirred under an atmosphere of 02 for 48 h, then diluted with DCM (200 mL) and washed with H.sub.2O (150 mL×2). The organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered, concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl 2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-(3-{3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)piperidin-3-yl]propanoate (3.6 g, 42% yield) as a solid. LCMS (ESI): m/z [M/2+H] calc'd for C.sub.56H.sub.70N.sub.4O.sub.6Si 462.3; found 462.3.
(709) Step 7.
(710) To a mixture of methyl 2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-(3-{3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)piperidin-3-yl]propanoate (3.60 g, 3.57 mmol) in THF (60 mL) and H.sub.2O (30 mL) was added LiOH (342 mg, 14.28 mmol). The mixture was stirred for 2 h, then diluted with H.sub.2O (150 mL), then 1M HCl was added slowly until ˜pH 3-4 and the mixture extracted with EtOAc (200 mL×3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered and the filtrate concentrated under reduced pressure to give 2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-(3-{3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)piperidin-3-yl]propanoic acid (3.3 g, 85% yield) as a solid, which was used directly in the next step without further purification. LCMS (ESI): m/z [M/2+H] calc'd for C.sub.55H.sub.68N.sub.4O.sub.6Si 455.3; found 455.3.
(711) Step 8.
(712) To a mixture of methyl 2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-(3-{3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)piperidin-3-yl]propanoate (3.30 g, 2.91 mmol) in DMF (40 mL) was added methyl (3S)-1,2-diazinane-3-carboxylate (0.42 g, 2.91 mmol), HATU (2.21 g, 5.82 mmol) and DIPEA (2.26 g, 17.46 mmol). The mixture was stirred for 3 h, then poured into ice-H.sub.2O and extracted with EtOAc (120 mL×2). The combined organic layers were washed with saturated NaHCO.sub.3 (150 mL), brine (150 mL), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (3S)-1-[(2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-(3-{3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)piperidin-3-yl]propanoyl]-1,2-diazinane-3-carboxylate (2.9 g, 95% yield) as a gum. LCMS (ESI): m/z [M/2+H] calc'd for C.sub.61H.sub.78N.sub.6O.sub.7Si 518.3; found 518.3.
(713) Step 9.
(714) To a mixture of methyl (3S)-1-[(2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-(3-{3-[(tert-butyldiphenylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)piperidin-3-yl]propanoyl]-1,2-diazinane-3-carboxylate (1.70 g, 1.64 mmol) was added a mixture of 1M TBAF in THF (19.68 mL, 19.68 mmol) and AcOH (1.18 g, 19.68 mmol). The reaction was heated to 60° C. and stirred for 22 h, then diluted with EtOAc (80 mL) and washed with saturated NaHCO.sub.3 (80 mL), H.sub.2O (60 mL×2) and brine (60 mL). The organic layer was dried over anhydrous Na.sub.2SO.sub.4, filtered, the filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give methyl (3S)-1-[(2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl]piperidin-3-yl]propanoyl]-1,2-diazinane-3-carboxylate (1.0 g, 73% yield) as a solid. LCMS (ESI): m/z [M/2+H] calc'd for C.sub.45H.sub.60N.sub.6O.sub.7 399.2; found 399.4.
(715) Step 10.
(716) To a mixture m methyl (3S)-1-[(2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-[1-ethyl-3-(3-

hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl]piperidin-3-yl]propanoyl]-1,2-diazinane-3-carboxylate (1.0 g, 1.1 mmol) in 1,2-dichloroethane (10 mL) was added Me.sub.3SnOH (1.42 g 7.84 mmol). The mixture was heated to 65° C. and stirred for 10 h, then filtered and the filtrate was concentrated under reduced pressure to give (3S)-1-[(2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl]piperidin-3-yl]propanoyl]-1,2-diazinane-3-carboxylic acid (1.0 g, 99% yield) as a gum. The product was used in the next step without further purification. LCMS (ESI): m/z [M/2+H] calc'd for C.sub.44H.sub.58N.sub.6O.sub.7 392.2; found 392.3.
(717) Step 11.
(718) To a mixture of (3S)-1-[(2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(3S)-1-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl]piperidin-3-yl]propanoyl]-1,2-diazinane-3-carboxylic acid (1.0 g, 1.1 mmol) in DCM (30 mL) at 0° C. was added HOBT (1.51 g, 11.2 mmol), N-(3-dimethylaminopropyl)-N′-ethylcarbodiimide HCl (6.44 g, 33.6 mmol) and DIPEA (5.79 g, 44.8 mmol). The mixture was warmed to room temperature and stirred for 6 h, then diluted with H.sub.2O and extracted with EtOAc (100 mL×3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered, the filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give benzyl N-[(6S,8S,14S)-22-ethyl-21-{2-[(1S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-2,10,22,28-tetraazapentacyclo [18.5.2.1{circumflex over ( )}{2,6}.1{circumflex over ( )}{10, 14}.0{circumflex over ( )}{23,27}]nonacosa-1(26),20,23(27),24-tetraen-8-yl]carbamate (340 mg, 36% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.44H.sub.56N.sub.6O.sub.6 383.2; found 383.3.
(719) Step 12.
(720) A mixture of benzyl N-[(6S,8S,14S)-22-ethyl-21-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-2,10,22,28-tetraazapentacyclo [18.5.2.1{circumflex over ( )}{2,6}.1{circumflex over ( )}{10, 14}.0{circumflex over ( )}{23,27}]nonacosa-1(26),20,23(27),24-tetraen-8-yl]carbamate (250 mg, 0.33 mmol), Pd/C (100 mg) and NH.sub.4Cl (353 mg, 6.6 mmol) in MeOH (5 mL) was stirred under an atmosphere of H.sub.2 for 4 h. The mixture was filtered through Celite and the filtrate was concentrated under reduced pressure. The residue was dissolved in DCM (30 mL) and washed with saturated NaHCO.sub.3 (20 mL), H.sub.2O (20 mL) and brine (20 mL). The organic layer was dried over Na.sub.2SO.sub.4, filtered, and the filtrate concentrated under reduced pressure to give (6S,8S,14S)-8-amino-22-ethyl-21-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-16-oxa-2,10,22,28-tetraazapentacyclo[18.5.2.1{2,6}.1{10,14}.0{23,27}]nonacosa-1(26),20,23(27),24-tetraene-9,15-dione, which was used in next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.36H.sub.50N.sub.6O.sub.4 631.4; found 631.4.
(721) Step 13.
(722) To a mixture of (6S,8S,14S)-8-amino-22-ethyl-21-{2-[(1S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-16-oxa-2,10,22,28-tetraazapentacyclo[18.5.2.1{circumflex over ( )}{2,6}.1{circumflex over ( )}{10,14}.0{circumflex over ( )}{23,27}]nonacosa-1(26),20,23(27),24-tetraene-9,15-dione (300 mg, 0.48 mmol), (2S)-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanoic acid (136 mg, 0.48 mmol) and DIPEA (620 mg, 4.8 mmol) in DMF (5 mL) at 0° C. was added HATU (183 mg, 0.48 mmol). The mixture was stirred at 0-5° C. for 1 h, then diluted with EtOAc (50 mL), washed with H.sub.2O (50 mL×2), brine (50 mL), dried over Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (2S)—N-[(6S,8S,14S)-22-ethyl-21-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}-18,18-dimethyl-9,15-dioxo-16-oxa-2,10,22,28-tetraazapentacyclo[18.5.2.1{circumflex over ( )}{2,6}.1{circumflex over ( )}{10,14}.0{circumflex over ( )}{23,27}]nonacosa-1(26),20,23(27),24-tetraen-8-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide (90 mg, 20% yield) as a

solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.50H.sub.70N.sub.8O.sub.7 895.5; found 895.4; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.71 (dd, J=4.8, 1.5 Hz, 1H), 7.86 (dd, J=7.7, 1.5 Hz, 1H), 7.51 (dd, J=7.7, 4.8 Hz, 1H), 7.36 (dd, J=8.9, 1.9 Hz, 1H), 7.22 (d, J=12.1 Hz, 1H), 7.09 (dd, J=8.9, 1.9 Hz, 1H), 6.59 (dt, J=16.9, 9.9 Hz, 1H), 6.26 (ddd, J=16.8, 5.0, 1.9 Hz, 1H), 5.80-5.67 (m, 1H), 5.59-5.46 (m, 1H), 4.93 (d, J=12.4 Hz, 1H), 4.66 (dd, J=11.1, 6.4 Hz, 1H), 4.45 (d, J=12.6 Hz, 1H), 4.28-4.19 (m, 1H), 4.13 (dd, J=14.5, 7.2 Hz, 1H), 4.02-3.87 (m, 1H), 3.87-3.36 (m, 11H), 3.16 (s, 2H), 3.10 (d, J=3.4 Hz, 2H), 2.76 (dd, J=26.9, 13.5 Hz, 3H), 2.61 (s, 1H), 2.35-1.97 (m, 5H), 1.78 (dd, J=25.4, 22.1 Hz, 10H), 1.45 (d, J=6.2 Hz, 3H), 1.04 (d, J=6.2 Hz, 3H), 0.95 (dd, J=6.5, 1.8 Hz, 3H), 0.83 (d, J=6.6 Hz, 3H), 0.72 (d, J=31.8 Hz, 6H).

Example A715. Synthesis of benzyl ((2.SUP.3.S,6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-piperidinacycloundecaphane-4-yl)carbamate

(723) ##STR01272##

(724) To a solution of ((2.sup.3S,6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-piperidinacycloundecaphane-5,7-dione (50 mg, 0.08 mmol), (R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-3-methylbutanoic acid (26 mg, 0.08 mmol) and DIPEA (31 mg, 0.24 mmol) in DMF (1 mL) at 0° C., was added HATU (30 mg, 0.08 mmol). The reaction mixture was stirred at 0-5° C. for 1 h, then diluted with EtOAc (20 mL), washed with H.sub.2O (20 mL×2) and brine (20 mL). The organic phase was separated and dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to give (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.sup.3S,6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-piperidinacycloundecaphane-4-yl)-3-methylbutanamide as solid. .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.71 (d, J=4.7 Hz, 1H), 8.24 (s, 1H), 8.10 (dd, J=26.7, 7.8 Hz, 1H), 7.86 (d, J=7.7 Hz, 1H), 7.51 (dd, J=7.7, 4.8 Hz, 1H), 7.39 (dd, J=8.9, 3.1 Hz, 1H), 7.26 (d, J=16.6 Hz, 1H), 7.11 (d, J=8.8 Hz, 1H), 5.61 (s, 1H), 4.50-4.28 (m, 3H), 4.27-4.07 (m, 3H), 3.98 (ddd, J=25.6, 13.4, 5.1 Hz, 2H), 3.84-3.72 (m, 2H), 3.62 (dd, J=10.7, 4.8 Hz, 2H), 3.55 (d, J=7.1 Hz, 2H), 3.47 (d, J=6.5 Hz, 2H), 3.16 (s, 3H), 3.03-2.91 (m, 1H), 2.76 (dd, J=28.7, 15.2 Hz, 3H), 2.62 (s, 1H), 2.40 (t, J=7.0 Hz, 3H), 2.33 (dd, J=14.3, 5.0 Hz, 4H), 2.05 (d, J=11.6 Hz, 1H), 1.99-1.64 (m, 10H), 1.64-1.55 (m, 1H), 1.51-1.42 (m, 6H), 1.37 (d, J=12.3 Hz, 3H), 1.05 (s, 3H), 0.94 (ddd, J=9.3, 6.7, 2.0 Hz, 6H), 0.76 (d, J=3.8 Hz, 3H), 0.69 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.53H.sub.76N.sub.8O.sub.7 936.6; found 937.4.

Example A347. Synthesis of (2S)—N-[(7S,13S)-21-ethyl-20-{2-[(1S)-1-methoxyethyl]pyridin-3-yl}-17,17-dimethyl-8,14-dioxo-15-oxa-4-thia-9,21,25,27,28-pentaazapentacyclo[17.5.2.1{circumflex over ( )}{2,5}.1{circumflex over ( )}{9,13}.0{circumflex over ( )}{22,26}]octacosa-1(25),2,5(28),19,22(26),23-hexaen-7-yl]-3-methyl-2-{N-methyl-1-[(3S)-1-(prop-2-enoyl)pyrrolidin-3-yl]formamido}butanamide

(725) ##STR01273## ##STR01274## ##STR01275##

Step 1.

(726) To a mixture of (5-chloro-1H-pyrrolo[3,2-b]pyridin-3-yl)methanol (3.5 g, 19 mmol), and ((1-methoxy-2-methylprop-1-en-1-yl)oxy)trimethylsilane (6.7 g, 38 mmol) in THF (50 ml) at 0° C. was added TMSOTf (3.8 g, 17 mmol) dropwise. The mixture was stirred at 0-5° C. for 2 h, then diluted with EtOAc (100 mL) and washed with saturated NaHCO.sub.3 (50 mL) and brine (50 mL×2). The organic layer was dried over Na.sub.2SO.sub.4, filtered and the filtrate was concentrated under reduced pressure. The residue was purified by silica gel column chromatography to give methyl 3-(5-chloro-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (3.0 g, 59% yield) as a solid.

LCMS (ESI): m/z [M+H] calc'd for $C_{13}H_{15}ClN_2O_2$ 266.1; found 267.1.
(727) Step 2.
(728) To a mixture of methyl 3-(5-chloro-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (3.0 g, 11 mmol) in anhydrous THF (50 mL) at 0° C. was added AgOTf (4.3 g, 17 mmol) and 12 (2.9 g, 11 mmol). The mixture was stirred at 0° C. for 2 h, then saturated Na_2SO_3 (20 mL) and EtOAc (50 mL) added. The mixture was filtered and the filtrate was washed with brine (50 mL), dried over Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl 3-(5-chloro-2-iodo-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (2.3 g, 52% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{13}H_{18}ClIN_2O_2$ 392.0; found 393.0.
(729) Step 3.
(730) To a mixture of methyl 3-(5-chloro-2-iodo-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (2.3 g, 5.9 mmol) and 2-(2-(2-methoxyethyl)phenyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (1.6 g, 7.1 mmol) and K_2CO_3 (2.4 g, 18 mol) in 1,4-dioxane (25 mL) and H_2O (5 mL) under an atmosphere of N_2 was added Pd(dppf)Cl_2.DCM (480 mg, 0.59 mmol). The mixture was heated to 70° C. and for 4 h, then diluted with EtOAc (200 mL) and washed with brine (25 mL), dried over Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (S)-3-(5-chloro-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (2.0 g, 84% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{21}H_{24}ClN_3O_3$ 401.2; found 402.2.
(731) Step 4.
(732) A mixture of ethyl 3-(5-chloro-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2-methylpropanoate (2.0 g, 5.0 mmol), Cs_2CO_3 (3.3 g, 10 mmol) and Etl (1.6 g, 10 mmol) in DMF (30 mL) was stirred for 10 h. The mixture was diluted with EtOAc (100 mL) and washed with brine (20 mL×4), dried over Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give two diastereomers of methyl (S)-3-(5-chloro-1-ethyl-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (0.7 g, 32% yield; 0.6 g, 28% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{23}H_{28}ClN_3O_3$ 429.2; found 430.2.
(733) Step 5.
(734) To a mixture of methyl (S)-3-(5-chloro-1-ethyl-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropanoate (1.9 g, 4.4 mmol) in anhydrous THF (20 mL) at 0° C. was added $LiBH_4$ (200 mg, 8.8 mmol). The mixture was heated to 60° C. and stirred for 4 h, then saturated NH_4Cl (20 mL) and EtOAc (50 mL) added. The aqueous and organic layers were separated and the organic layer was washed with brine (30 mL), dried over Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (S)-3-(5-chloro-1-ethyl-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropan-1-ol (1.5 g, 85% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{22}H_{28}ClN_3O_2$ 401.2; found 402.2.
(735) Step 6.
(736) To a mixture of (S)-3-(5-chloro-1-ethyl-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-3-yl)-2,2-dimethylpropan-1-ol (550 mg, 1.37 mmol), (S)-(2-(2-((tert-butoxycarbonyl)amino)-3-methoxy-3-oxopropyl)thiazol-4-yl)boronic acid (907.4 mg, 2.74 mmol, 2 eq) and K_2CO_3 (568 mg, 4.11 mmol) in 1,4-dioxane (25 mL) and H_2O (5 mL) under an atmosphere of N_2 was added Pd(dppf)Cl_2.DCM (89 mg, 0.14 mmol). The mixture was heated to 70° C. and stirred for 4 h, then H_2O (50 mL) added and the mixture extracted with EtOAc (100 mL×3). The combined organic layers were washed with brine (50 mL), dried over Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue

was purified by silica gel column chromatography to give methyl (S)-2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-5-yl)thiazol-2-yl)propanoate (440 mg, 22% yield) as a solid, which was used directly in the next step. LCMS (ESI): m/z [M+H] calc'd for $C_{34}H_{45}N_5O_6S$ 651.3; found 652.3.

(737) Step 7.

(738) To a mixture of (2S)-methyl 2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-5-yl)thiazol-2-yl)propanoate (280 mg, 0.43 mmol) in MeOH (4 mL) was added a solution of LiOH (51 mg, 2.2 mmol) in H_2O (2 mL). The mixture was stirred for 5 h, then pH adjusted to ˜3-4 by addition of 1M HCl. The mixture was diluted with H_2O (30 mL) and extracted with EtOAc (15 mL×3). The combined organic layers were washed with brine (10 mL), dried over anhydrous Na_2SO_4, filtered and the filtrate was concentrated under reduced pressure to give (2S)-2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-5-yl)thiazol-2-yl)propanoic acid (280 mg) as solid, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{33}H_{43}N_5O_6S$ 637.3; found 638.3.

(739) Step 8.

(740) To a mixture of (2S)-2-((tert-butoxycarbonyl)amino)-3-(4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-pyrrolo[3,2-b]pyridin-5-yl)thiazol-2-yl)propanoic acid (274 mg, 0.43 mmol) and methyl (3S)-1,2-diazinane-3-carboxylate (280 mg, 0.64 mmol) in DMF (3 mL) at 0-5° C. was added a solution of HATU (245 mg, 0.64 mmol) and DIPEA (555 mg, 4.3 mmol) in DMF (2 mL). The mixture was stirred for 1 h, then diluted with EtOAc (20 mL) and H_2O (20 mL). The aqueous and organic layers were partitioned and the organic layer was washed with H_2O (20 mL×3), brine (20 mL), dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (3S)-1-[(2S)-2-{[(tert-butoxy)carbonyl]amino}-3-{4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}pyrrolo[3,2-b]pyridin-5-yl]-1,3-thiazol-2-yl}propanoyl]-1,2-diazinane-3-carboxylate (230 mg, 70% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{39}H_{53}N_7O_7S$ 763.4; found 764.3.

(741) Step 9.

(742) To a mixture of methyl (3S)-1-[(2S)-2-{[(tert-butoxy)carbonyl]amino}-3-{4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}pyrrolo[3,2-b]pyridin-5-yl]-1,3-thiazol-2-yl}propanoyl]-1,2-diazinane-3-carboxylate (230 mg, 0.3 mmol) in DCE (3 mL) under an atmosphere of N_2 was added Me_3SnOH (300 mg). The mixture was heated to 65° C. and stirred for 16 h, then concentrated under reduced pressure. The residue was diluted with EtOAc (20 mL), washed with H_2O (20 mL) and brine (10 mL), dried over Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure to give (3S)-1-[(2S)-2-{[(tert-butoxy)carbonyl]amino}-3-{4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}pyrrolo[3,2-b]pyridin-5-yl]-1,3-thiazol-2-yl}propanoyl]-1,2-diazinane-3-carboxylic acid (200 mg) as a foam, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{38}H_{51}N_7O_7S$ 749.4; found 750.3.

(743) Step 10.

(744) To a mixture of (3S)-1-[(2S)-2-{[(tert-butoxy)carbonyl]amino}-3-{4-[1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}pyrrolo[3,2-b]pyridin-5-yl]-1,3-thiazol-2-yl}propanoyl]-1,2-diazinane-3-carboxylic acid (245 mg, 0.32 mmol) in DCM (50 mL) at 0-5° C. were added HOBT (432 mg, 3.2 mmol), EDCI HCl (1.8 g, 9.6 mmol) and DIPEA (1.65 g, 12.8 mmol). The mixture was warmed to room temperature and stirred for 16 h, then concentrated under reduced pressure. The residue was diluted with EtOAc (20 mL) and H_2O (20 mL) and the

aqueous and organic layers were partitioned. The organic layer was washed with H_2O (30 mL×3), brine (30 mL), dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-TLC to give tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-pyrrolo[3,2-b]pyridina-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (100 mg, 43% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{38}H_{49}N_7O_6S$ 731.4; found 732.3.

(745) Step 11.

(746) A mixture of tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-pyrrolo[3,2-b]pyridina-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (80 mg, 0.11 mmol) in DCM (0.6 mL) and TFA (0.2 mL) was stirred for 1 h. The mixture was concentrated under reduced pressure to give (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-pyrrolo[3,2-b]pyridina-6(1,3)-pyridazinacycloundecaphane-5,7-dione (72 mg, 95% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{33}H_{41}N_7O_4S$ 631.3; found 632.3.

(747) Step 12.

(748) To a mixture of (6.sup.3S,4S,Z)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-pyrrolo[3,2-b]pyridina-6(1,3)-pyridazinacycloundecaphane-5,7-dione (120 mg, 0.39 mmol) and DIPEA (335 mg, 2.6 mmol) in DMF (1 mL) at 0° C. was added HATU (60 mg, 0.16 mmol). The mixture was stirred at 0° C. for 1 h, then diluted with H_2O (110 mL) and extracted with EtOAc (80 mL×2). The combined organic layers were washed with H_2O (100 mL), brine (100 mL), dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-TLC to give (3S)-1-acryloyl-N-((2S)-1-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-pyrrolo[3,2-b]pyridina-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide (1.8 mg, 2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{47}H_{61}N_9O_7S$ 895.4; found 896.3; ^1H NMR (400 MHz, CD_3OD) δ 8.72 (d, J=4.5 Hz, 1H), 7.98-7.77 (m, 3H), 7.72 (dd, J=12.0, 8.6 Hz, 1H), 7.54 (dd, J=7.6, 4.8 Hz, 1H), 6.67-6.54 (m, 1H), 6.26 (m, 1H), 5.79-5.58 (m, 2H), 4.83-4.75 (m, 1H), 4.39-4.16 (m, 4H), 4.02 (dd, J=28.0, 10.6 Hz, 2H), 3.89-3.65 (m, 6H), 3.50 (m, 4H), 3.34 (d, J=6.2 Hz, 3H), 3.12 (d, J=4.0 Hz, 2H), 3.00 (s, 1H), 2.73 (m, 1H), 2.48-2.37 (m, 1H), 2.31-2.07 (m, 4H), 1.88 (d, J=11.2 Hz, 1H), 1.71 (d, J=12.8 Hz, 1H), 1.44 (m, 7H), 0.97 (dd, J=6.2, 4.4 Hz, 3H), 0.92-0.84 (m, 8H), 0.41 (d, J=6.2 Hz, 3H).

Example 647. Synthesis of 1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-N-((2S)-1-(((2.SUP.2.S,6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-4-fluoro-N-methylpiperidine-4-carboxamide

(749) ##STR01276## ##STR01277## ##STR01278## ##STR01279##

Step 1.

(750) A mixture of 1-[(tert-butoxy)carbonyl]-4-fluoropiperidine-4-carboxylic acid (2.0 g, 8.1 mmol) in DCM (20 mL) was added oxalic dichloride (1.34 g, 10.5 mmol) and DMF (30 mg, 0.4 mmol). The resulting solution was stirred at room temperature for 1 h. Et_3N (3.2 g, 3.2 mmol) and (2S)-3-methyl-2-(methylamino)butanoic acid (1.25 g, 9.5 mmol) were added and the mixture was

stirred at room temperature for 1 h. H_2O (100 mL) was added and the mixture was extracted with EtOAc (50 mL×3). The combined organic layers were concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl (S)-4-((1-(tert-butoxy)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-4-fluoropiperidine-1-carboxylate (1.34 g, 45% yield) as a solid. LCMS (ESI): m/z [M+Na] calc'd for $C_{21}H_{37}FN_2O_5Na$ 439.3; found 439.3.

(751) Step 2.

(752) A mixture of tert-butyl (S)-4-((1-(tert-butoxy)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-4-fluoropiperidine-1-carboxylate (290 mg, 0.70 mmol) in DCM (4 mL) and TFA (2 mL) was stirred at room temperature for 2 h, then concentrated under reduced pressure to give N-(4-fluoropiperidine-4-carbonyl)-N-methyl-L-valine, which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{12}H_{21}FN_2O_3$ 260.2; found 261.2.

(753) Step 3.

(754) To a solution of the tert-butyl N-(4-fluoropiperidine-4-carbonyl)-N-methyl-L-valinate (1.7 g, 5.3 mmol), sodium 4-(dimethylamino)-4-methylpent-2-ynoate (1.67 g, 9.4 mmol) and Et_3N (2.73 g, 36.9 mmol) in DMF (20 mL) stirred at 5° C. was added T3P (4.11 g, 10.7 mmol, 50 wt % in EtOAc). The reaction mixture was stirred at 5° C. for 1 h. The resulting mixture was quenched with H_2O (100 mL) and extracted with EtOAc (50 mL×3). The combined organic layers were concentrated and purified by silica gel column chromatography to give tert-butyl N-(1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-4-fluoropiperidine-4-carbonyl)-N-methyl-L-valinate (1.6 g, 74.0% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{24}H_{40}FN_3O_4$ 453.3; found 454.2.

(755) Step 4.

(756) To a solution of tert-butyl N-(1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-4-fluoropiperidine-4-carbonyl)-N-methyl-L-valinate (50 mg, 0.11 mmol) in DCM (2 mL) was added TFA (1 mL). The reaction mixture was stirred at 20° C. for 2 h, then concentrated under reduced pressure to afford crude N-(1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-4-fluoropiperidine-4-carbonyl)-N-methyl-L-valine. It was used for the next step directly without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{20}H_{32}FN_3O_4$ 397.2; found 398.3.

(757) Step 5.

(758) To a solution of tert-butyl (2R)-2-(hydroxymethyl)morpholin-4-yl formate (50 g, 230 mmol) in EtOAc (1 L) was added TEMPO (715 mg, 4.6 mmol) and $NaHCO_3$ (58 g, 690 mmol) at 20° C. The mixture was cooled to −50° C., then TCCA (56 g, 241 mmol) in EtOAc (100 mL) was added dropwise over 30 min. The reaction mixture was warmed to 5° C. for 2 h, then quenched with 10% $Na_2S_2O_3$ (200 mL) and stirred for 20 min. The resulting mixture was filtered and the organic phase was separated from filtrate. The aqueous phase was extracted with EtOAc (100 mL×2). The combined organic layers were washed with H_2O (100 mL) and brine (100 mL), and dried over anhydrous Na_2SO_4. The organic layer was concentrated under reduced pressure to afford tert-butyl (2R)-2-formylmorpholin-4-yl formate (50 g, crude) as an oil.

(759) Step 6.

(760) To a solution of tert-butyl (2R)-2-formylmorpholin-4-yl formate (49 g, 153 mmol) and methyl 2-{[(benzyloxy)carbonyl]amino}-2-(dimethoxyphosphoryl)acetate (60 g, 183 mmol) in CAN (300 mL) was added tetramethylguanidine (35 g, 306 mmol) at 0-10° C. The reaction mixture was stirred at 10° C. for 30 min then warmed to 20° C. for 2 h. The reaction mixture was diluted with DCM (200 mL) and washed with Citric acid (10%, 200 mL) and 10% $NaHCO_3$ aqueous solution (200 mL). The organic phase was concentrated under reduced pressure, and purified by silica gel column chromatography to afford tert-butyl (S,Z)-2-(2-(((benzyloxy)carbonyl)amino)-3-methoxy-3-oxoprop-1-en-1-yl)morpholine-4-carboxylate (36 g, 90% yield) as solid. LCMS (ESI): m/z [M+Na] calc'd for $C_{21}H_{28}N_2O_4$ 420.2; found: 443.1.

(761) Step 7.

(762) To a solution of tert-butyl (S,Z)-2-(2-(((benzyloxy)carbonyl)amino)-3-methoxy-3-oxoprop-1-en-1-yl)morpholine-4-carboxylate (49 g, 0.12 mol) in MeOH (500 mL) was added (S,S)-Et-DUPHOS-Rh (500 mg, 0.7 mmol). The mixture was stirred at 25° C. under an H_2 (60 psi) atmosphere for 48 h. The reaction was concentrated and purified by chromatography to give tert-butyl (S)-2-((S)-2-(((benzyloxy)carbonyl)amino)-3-methoxy-3-oxopropyl)morpholine-4-carboxylate (44 g, 89.8% yield) as solid. LCMS (ESI): m/z [M+Na] calc'd for $C_{21}H_{30}N_2O_7$ 422.2; found: 445.2.

(763) Step 8.

(764) To a stirred solution of tert-butyl (S)-2-((S)-2-(((benzyloxy)carbonyl)amino)-3-methoxy-3-oxopropyl)morpholine-4-carboxylate (2.2 g, 5.2 mmol) in EtOAc (2 mL) was added HCl/EtOAc (25 mL) at 15° C. The reaction was stirred at 15° C. for 2 h, then concentrated under reduced pressure to afford methyl (S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-morpholin-2-yl)propanoate (1.51 g, 90.4% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{16}H_{22}N_2O_5$ 322.1; found 323.2.

(765) Step 9.

(766) To a solution of 3-(5-bromo-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-3-yl)-2,2-dimethylpropan-1-ol (100 g, 0.22 mol) and 1H-imidazole (30.6 g, 0.45 mol) in DCM (800 mL) was added TBSCI (50.7 g, 0.34 mol) in DCM (200 mL) at 0° C. The reaction was stirred at 25° C. for 2 h. The resulting solution was washed with H_2O (300 mL×3) and brine (200 mL×2), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified with silica gel column chromatography to give (S)-5-bromo-3-(3-((tert-butyldimethylsilyl)oxy)-2,2-dimethylpropyl)-1-ethyl-2-(2-(1-methoxyethyl)pyridin-3-yl)-1H-indole (138 g, 90% yield) as an solid. LCMS (ESI): m/z [M+H] calc'd for $C_{29}H_{43}BrN_2O_2Si$ 558.2; found 559.2.

(767) Step 10.

(768) To a stirred solution of Intermediate 1 (50 g, 89.3 mmol) in dioxane (500 mL) was added methyl (2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(2S)-morpholin-2-yl]propanoate from step 1 (31.7 g, 98.2 mmol), RuPhos (16.7 g, 35.7 mmol), Di-mu-chlorobis(2-amino-1,1-biphenyl-2-yl-C,N)dipalladium(II) (2.8 g, 4.4 mmol) and cesium carbonate (96 g, 295 mmol) followed by RuPhos-Pd-G2 (3.5 g, 4.4 mmol) at 105° C. under an N_2 atmosphere. The reaction mixture was stirred for 6 h at 105° C. under an N_2 atmosphere. The resulting mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified by prep-TLC chromatography to afford methyl (2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(2S)-4-(3-{3-[(tert-butyldimethylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)morpholin-2-yl]propanoate (55 g, 73% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{45}H_{64}N_4O_7Si$ 800.5; found 801.5.

(769) Step 11.

(770) To a solution of methyl (2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(2S)-4-(3-{3-[(tert-butyldimethylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)morpholin-2-yl]propanoate (10 g, 12 mmol) in THF (270 mL) was added LiOH (1.3 g, 31 mmol) in H_2O (45 mL) at 20° C. The reaction was stirred at 20° C. for 2 h, then treated with 1N HCl to adjust pH to 4-5 at 0˜5° C. The resulting mixture was extracted with EtOAc (50 mL×2). The combined organic layers were washed with brine and dried over anhydrous Na_2SO_4. The organic phase was then concentrated under reduced pressure to afford (2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(2S)-4-(3-{3-[(tert-butyldimethylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)morpholin-2-yl]propanoic acid (9.5 g, 97% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{44}H_{62}N_4O_7Si$ 786.4; found 787.4.

(771) Step 12.

(772) To a stirred solution of (2S)-2-{[(benzyloxy)carbonyl]amino}-3-[(2S)-4-(3-{3-[(tert-butyldimethylsilyl)oxy]-2,2-dimethylpropyl}-1-ethyl-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}indol-5-yl)morpholin-2-yl]propanoic acid (10 g, 12.7 mmol) in DMF (150 mL), was added methyl (S)-hexahydropyridazine-3-carboxylate (2 g, 14 mmol), then cooled to 0° C., DIPEA (32.8 g, 254 mmol) was added followed by HATU (9.7 g, 25.4 mmol) at 0˜5° C. The reaction mixture was stirred at 0˜5° C. for 1 h. The resulting mixture was diluted with EtOAc (500 mL) and H_2O (200 mL). The organic layer was separated and washed with H_2O (100 mL×2) and brine (100 mL), dried over anhydrous sodium sulfate. The solution was filtered and concentrated under reduced pressure, and the residue was purified by silica gel column chromatography to afford methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldimethylsilyl)oxy)-2,2-dimethylpropyl)-1-ethyl-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (8 g, 70% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{50}H_{72}N_6O_8Si$ 912.5; found 913.4.

(773) Step 13.

(774) A solution of methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldimethylsilyl)oxy)-2,2-dimethylpropyl)-1-ethyl-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (8.5 g, 9 mmol) in THF (8 mL) was added a mixture of tetrabutylammonium fluoride (1M in THF, 180 mL, 180 mmol) and AcOH (11 g, 200 mmol) at 20° C. The reaction mixture was stirred at 75° C. for 3 h. The resulting mixture was diluted with EtOAc (150 mL) and washed with H_2O (20 mL×6). The organic phase was concentrated under reduced pressure to give methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (7.4 g, 100% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{44}H_{58}N_6O_8$ 799.4; found 798.4.

(775) Step 14.

(776) To a solution of methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (8 g, 10 mmol) in THF (200 mL) was added lithium hydroxide (600 mg, 25 mmol) in H_2O (30 mL). The reaction mixture was stirred at 20° C. for 1 h, then treated with 1N HCl to adjust pH to 4-5 at 0˜5° C., and extracted with EtOAc (500 mL×2). The organic phase was washed with brine, and concentrated under reduced pressure to afford (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (8 g, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{43}H_{56}N_6O_8$ 784.4; found 785.4.

(777) Step 15.

(778) To a stirred solution of (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (8 g, 10.2 mmol) and DIPEA (59 g, 459 mmol) in DCM (800 mL) was added EDCI (88 g, 458 mmol) and HOBT (27.6 g, 204 mmol) at 25° C. under an argon atmosphere. The reaction mixture was stirred at 25° C. for 16 h. The resulting mixture was concentrated under reduced pressure, and the residue was purified by silica gel column chromatography to afford benzyl ((2^2S,6^3S,4S)-1^1-ethyl-1^2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6^1,6^2,6^3,6^4,6^5,6^6-hexahydro-1^1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (5 g, 66% yield) as a solid; LCMS (ESI): m/z [M+H] calc'd for $C_{43}H_{54}N_6O_7$ 766.4; found 767.4.

(779) Step 16.

(780) To a solution of benzyl ((2^2S,6^3S,4S)-1^1-ethyl-1^2-(2-((S)-1-

methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)carbamate (400 mg, 0.5 mmol) in MeOH (20 mL) was added Pd/C (200 mg) and ammonium acetate (834 mg, 16 mmol) at 20° C. under an H.sub.2 atmosphere and the mixture was stirred for 2 h. Then resulting mixture was filtered and concentrated under reduced pressure. The residue was redissolved in DCM (20 mL) and washed with H.sub.2O (5 mL×2), then concentrated under reduced pressure to afford (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (320 mg, 97% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.35H.sub.48N.sub.6O.sub.5 632.4; found 633.3.
(781) Step 17.
(782) To a solution of the (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (50 mg, 0.079 mmol), N-(1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-4-fluoropiperidine-4-carbonyl)-N-methyl-L-valine (47 mg, 0.12 mmol) in DMF (2 mL) stirred at 0° C. was added HATU (36 mg, 0.09 mmol) and DIPEA (153 mg, 1.2 mmol) dropwise. The reaction was stirred at 0° C. for 1 h. The resulting mixture was purified by reverse phase to afford 1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-N-((2S)-1-(((2.sup.2S,6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-4-fluoro-N-methylpiperidine-4-carboxamide (11.9 mg, 13.9% yield) as a solid. .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.71 (dd, J=4.8, 1.7 Hz, 1H), 7.86 (d, J=7.8 Hz, 1H), 7.51 (dd, J=7.8, 4.8 Hz, 1H), 7.39 (d, J=8.9 Hz, 1H), 7.15-7.04 (m, 2H), 5.67 (d, J=8.8 Hz, 1H), 4.62 (d, J=11.2 Hz, 1H), 4.46 (d, J=12.4 Hz, 1H), 4.39-4.27 (m, 2H), 4.23 (d, J=6.1 Hz, 1H), 4.17-4.08 (m, 1H), 3.93 (s, 2H), 3.86 (s, 1H), 3.84-3.76 (m, 2H), 3.74-3.65 (m, 2H), 3.63-3.51 (m, 2H), 3.27-3.23 (m, 1H), 3.22-3.11 (m, 6H), 3.0-2.89 (m, 2H), 2.85-2.75 (m, 2H), 2.74-2.55 (m, 2H), 2.36 (d, J=8.2 Hz, 6H), 2.32-2.21 (m, 2H), 2.20-2.02 (m, 5H), 1.92 (d, J=12.5 Hz, 2H), 1.69 (dd, J=43.8, 12.6 Hz, 2H), 1.46 (dt, J=8.0, 4.9 Hz, 9H), 1.03 (d, J=3.5 Hz, 3H), 0.90 (dd, J=48.3, 6.5 Hz, 6H), 0.77 (d, J=3.0 Hz, 3H), 0.69 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.55H.sub.78FN.sub.9O.sub.8 1011.6; found 1012.5.
Example A375. Synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.SUP.2.S,6.SUP.3.S,4S)-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.SUP.1.-(2,2,2-trifluoroethyl)-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)-3-methylbutanamide
(783) ##STR01280## ##STR01281## ##STR01282##
Step 1.
(784) To a mixture of 5-5bromo-3-{3-[(tert-butyldiphenylsiyl)oxy]-2,2-dimethylpropyl}-2-{2-[(1 S)-1-methoxyethyl]pyridin-3-yl}-1H-indole (10.0 g, 15.2 mmol) in anhydrous DMF (120 mL) at 0° C. under an atmosphere of N.sub.2 was added NaH, 60% dispersion in oil (1.2 g, 30.4 mmol) and 2,2,2-trifluoroethyl trifluoromethanesulfonate (35.4 g, 152 mmol). The mixture was stirred at 0° C. for 1 h, then saturated NH.sub.4Cl (30 mL) added and the mixture extracted with EtOAc (100 mL×3). The combined organic layers were washed with brine, dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (S)-5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (8 g) as an oil and the other atropisomer (6 g, 48% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.44BrF.sub.3N.sub.2O.sub.2Si 736.2; found 737.1.

(785) Step 2.

(786) To a mixture of (S)-5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (7.2 g, 9.7 mmol) in toulene (80 mL) under an atmosphere of N.sub.2 was added 4,4,4′,4′,5,5,5′,5′-octamethyl-2,2′-bi(1,3,2-dioxaborolane) (2.7 g, 10.6 mmol), KOAc (1.9 g, 19.4 mmol) and Pd(dppf)Cl.sub.2 DCM (0.8 g, 0.1 mmol). The mixture was heated to 90° C. and stirred for 8 h, then saturated NH.sub.4Cl (30 mL) added and the mixture extracted with EtOAc (40 mL×3). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (S)-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (6.1 g, 64% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.45H.sub.56BF.sub.3N.sub.2O.sub.4Si 784.4; found 785.3.

(787) Step 3.

(788) To a mixture of (S)-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (33 g, 42 mmol) in THF (120 mL) and MeOH (330 mL) at 0° C. under an atmosphere of N.sub.2 was added MeB(OH).sub.2 (50.4 g, 841 mmol), then a mixture of NaOH (33.6 g, 841 mmol) in H.sub.2O (120 mL). The mixture was warmed to room temperature and stirred for 16 h, then concentrated under reduced pressure. H.sub.2O (500 mL) was added to the residue and the mixture extracted with EtOAc (300 mL×3). The combined organic layers were washed with brine (300 mL), H.sub.2O (300 mL), then concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give (S)-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)boronic acid (20 g, 68% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.46BF.sub.3N.sub.2O.sub.4Si 702.3; found 703.3.

(789) Step 4.

(790) Note: Three reactions were run in parallel—the yield reflects the sum of the products.

(791) A mixture of (S)-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)boronic acid (1.85 g, 5.6 mmol) and methyl (S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-morpholin-2-yl)propanoate in DCM (150 mL) under air was added pyridine (1.35 g, 16.9 mmol) and Cu(OAc).sub.2 (2.0 g, 11.3 mmol). The mixture was stirred for 48 h, then concentrated under reduced pressure. H.sub.2O (300 mL) was added to the residue and the mixture was extracted with EtOAc (300 mL×2). The combined organic layers were washed with brine (100 mL), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was silica gel column chromatography to give methyl (S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoate (9.2 g, 55% yield) as a solid. LCMS (ESI): m/z [M/2+H] calc'd for C.sub.55H.sub.65F.sub.3N.sub.4O.sub.7Si 490.2; found 490.3.

(792) Step 5.

(793) To a mixture of methyl (S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoate (10.8 g, 11.0 mmol) in THF (50 mL) was added LiOH (528 mg, 22 mmol) in H.sub.2O (10 mL). The mixture was stirred for 1 h, then cooled to 0-5° C. and acidified to pH˜7 using 2N HCl (10 mL). The mixture was extracted with DCM (100 mL×2) and the combined organic layers were dried over Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure to give (S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoic acid (10.6 g, 100% yield) as a solid. LCMS (ESI): m/z [M/2+H] calc'd for C.sub.54H.sub.63F.sub.3N.sub.4O.sub.7Si 483.2; found

483.3.

(794) Step 6.

(795) To a mixture of (S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoic acid (10.6 g, 11.0 mmol) and methyl (3S)-1,2-diazinane-3-carboxylate (15.8 g, 22.0 mmol) in DMF (150 mL) at 0° C. was added DIPEA (28.4 g, 220 mmol) and HATU (8.4 g, 22.0 mmol). The mixture was stirred at 0˜5° C. for 1 h, then EtOAc (500 mL) was added and the mixture was washed with H_2O (200 mL×2), brine (100 mL), dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (11 g, 90% yield) as a solid. LCMS (ESI): m/z [M/2+H] calc'd for $C_{60}H_{73}F_3N_6O_8Si$ 546.3; found 546.3.

(796) Step 7.

(797) To a mixture of methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (11.0 g, 10.1 mmol) in THF (10 mL) was added a mixture of AcOH (21.2 g, 353 mmol) and 1M TBAF in THF (300 mL, 300 mmol). The mixture was heated to 80° C. and stirred for 16 h, then concentrated under reduced pressure. EtOAc (800 mL) was added to the residue and the mixture was washed with H_2O (80 mL×6), concentrated under reduced pressure and the residue was purified by preparative-HPLC to give methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (7.9 g, 91% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{44}H_{55}F_3N_6O_8$ 852.4; found 853.3.

(798) Step 8.

(799) To a mixture of methyl (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (7.9 g, 9.3 mmol) in THF (50 mL) was added LiOH (443 mg, 18.5 mmol) in H_2O (10 mL). The mixture was stirred for 1 h, then cooled to 0-5° C. and acidified to ˜pH 7 with 2N HCl (9 mL). The mixture was extracted with DCM (100 mL×2) and the combined organic layers were dried over Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure to give (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (7.6 g, 98% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{43}H_{53}F_3N_6O_8$ 838.4; found 839.3.

(800) Step 9.

(801) To a mixture of (S)-1-((S)-2-(((benzyloxy)carbonyl)amino)-3-((S)-4-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)morpholin-2-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (7.6 g, 9.0 mmol) and DIPEA (52.3 g, 405 mmol) in DCM (800 mL) under an atmosphere of Ar was added EDCI (77.6 g, 405 mmol) and HOBT (12 g, 90 mmol). The mixture was stirred for 16 h, then concentrated under reduced pressure. The residue was diluted with EtOAc (500 mL), washed with H_2O (100 mL×2) and filtered. The organic layer was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give benzyl (($2^2S,6^3S$,4S)-1^2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1^1-(2,2,2-

trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (6.1 g, 74% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.43H.sub.51F.sub.3N.sub.6O.sub.7 820.4; found 821.3.

(802) Step 10.

(803) To a mixture of benzyl ((2.sup.2S,6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (700 mg, 0.85 mmol) in MeOH (30 mL) was added 10% Pd on C (317 mg) and NH.sub.4Cl (909 mg). The mixture was stirred under an atmosphere of H.sub.2 (1 atm) for 16 h, then filtered through Celite and the filter cake was washed with MeOH (150 mL). The filtrate was concentrated under reduced pressure, DCM (20 mL) was added to the residue and the mixture was washed with saturated NaHCO.sub.3 (20 mL×3). The organic layer was concentrated under reduced pressure to give (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-5,7-dione (660 mg, 95% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.35H.sub.45F.sub.3N.sub.6O.sub.5 686.3; found 687.3; .sup.1H NMR (400 MHz, CDCl.sub.3) δ 8.80 (dd, J=4.7, 1.7 Hz, 1H), 7.66 (d, J=7.4 Hz, 1H), 7.43-7.30 (m, 2H), 7.12-7.01 (m, 2H), 4.90-4.83 (m, 1H), 4.68 (d, J=12.5 Hz, 1H), 4.57 (dd, J=16.2, 8.1 Hz, 1H), 4.24 (q, J=6.1 Hz, 1H), 4.08 (d, J=10.6 Hz, 1H), 3.97-3.82 (m, 4H), 3.80-3.68 (m, 2H), 3.55 (d, J=11.6 Hz, 1H), 3.21 (d, J=9.4 Hz, 1H), 2.93 (dd, J=19.9, 9.3 Hz, 3H), 2.66 (t, J=11.6 Hz, 1H), 2.47 (d, J=14.5 Hz, 1H), 2.19-2.04 (m, 4H), 1.96 (d, J=13.6 Hz, 2H), 1.80-1.71 (m, 2H), 1.66-1.59 (m, 1H), 1.47 (d, J=6.1 Hz, 3H), 0.88 (s, 3H), 0.42 (s, 3H).

(804) Step 11.

(805) To a mixture of (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (300 mg, 0.4 mmol), (2R)-2-[({1-[4-(dimethylamino)-4-methylpent-2-ynoyl]azetidin-3-yl}oxy)methyl]-3-methylbutanoic acid (157 mg, 0.48 mmol) and DIPEA (569.0 mg, 0.4 mmol) in DMF (5 mL) at 0° C. was added HATU (217 mg, 0.57 mmol). The mixture was stirred at 0° C. for 0.5 h, then diluted with H.sub.2O and extracted with EtOAc (20 mL×3). The combined organic layers were washed with brine (20 mL×3), dried over Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-TLC to give (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.sup.2S,6.sup.3S,4S)-12-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1′-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (200 mg, 46% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.52H.sub.71F.sub.3N.sub.8O.sub.8 992.5; found 993.4; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.74 (m, 1H), 8.00 (m, 1H), 7.85 (d, J=7.7 Hz, 1H), 7.60-7.43 (m, 2H), 7.14 (t, J=9.5 Hz, 2H), 5.63 (s, 1H), 5.06 (m, 1H), 4.64 (s, 1H), 4.52-4.31 (m, 3H), 4.27-4.05 (m, 3H), 3.97 (m, 1H), 3.92-3.66 (m, 6H), 3.59 (m, 2H), 3.46 (m, 2H), 3.25 (d, J=5.3 Hz, 3H), 3.07-2.89 (m, 2H), 2.86-2.59 (m, 3H), 2.38-2.32 (m, 3H), 2.28 (s, 3H), 2.13 (m, 2H), 2.03-1.51 (m, 6H), 1.50-1.41 (m, 6H), 1.38 (d, J=7.5 Hz, 3H), 0.98 (t, J=8.7 Hz, 6H), 0.89 (t, J=6.4 Hz, 3H), 0.54 (d, J=8.4 Hz, 3H).

Example A722. Synthesis of 1-acryloyl-N-((2S)-1-(((6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-4-fluoro-N-methylpiperidine-4-carboxamide

(806) ##STR01283##
Step 1.
(807) To a mixture of N-(4-fluoropiperidine-4-carbonyl)-N-methyl-L-valine (190 mg, 0.73 mmol) and NaHCO.sub.3 (306 mg, 3.6 mmol) in DCM (2 mL) and H.sub.2O (1 mL) at −10° C. was added prop-2-enoyl chloride (132 mg, 1.45 mmol). The mixture was stirred at 0-5° C. for 1 h, then diluted with DCM (20 mL) and washed with H.sub.2O (20 mL×2), brine (20 mL) and the organic layer dried over Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give N-(1-acryloyl-4-fluoropiperidine-4-carbonyl)-N-methyl-L-valine (120 mg, 52% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.15H.sub.23FN.sub.2O.sub.4 314.2; found 315.2.
(808) Step 2.
(809) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (153 mg, 0.24 mmol), N-(1-acryloyl-4-fluoropiperidine-4-carbonyl)-N-methyl-L-valine (106 mg, 0.34 mmol) in DMF (2 mL) at 5° C. was added HATU (110 mg, 0.29 mmol) and DIPEA (468 mg, 3.6 mmol) dropwise. The mixture was stirred at 5° C. for 1 h, then purified by preparative-HPLC to give 1-acryloyl-N-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-4-fluoro-N-methylpiperidine-4-carboxamide (69.5 mg, 29% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.50H.sub.69FN.sub.8O.sub.8 928.5; found 929.4; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.71 (d, J=3.2 Hz, 1H), 7.86 (d, J=7.7 Hz, 1H), 7.51 (dd, J=7.7, 4.8 Hz, 1H), 7.39 (d, J=8.9 Hz, 1H), 7.18-7.02 (m, 2H), 6.80 (dd, J=16.8, 10.7 Hz, 1H), 6.23 (d, J=16.8 Hz, 1H), 5.77 (d, J=10.6 Hz, 1H), 5.67 (d, J=6.5 Hz, 1H), 4.61 (d, J=11.1 Hz, 1H), 4.44 (t, J=15.1 Hz, 2H), 4.23 (q, J=6.1 Hz, 1H), 4.18-4.10 (m, 1H), 4.09-4.01 (m, 1H), 3.99-3.83 (m, 3H), 3.83-3.65 (m, 4H), 3.58-3.46 (m, 2H), 3.27 (s, 1H), 3.21-3.11 (m, 6H), 3.00-2.91 (m, 2H), 2.85-2.75 (m, 2H), 2.73-2.64 (m, 1H), 2.62-2.54 (m, 1H), 2.36-2.21 (m, 2H), 2.19-2.01 (m, 5H), 1.92 (d, J=12.7 Hz, 2H), 1.79-1.57 (m, 2H), 1.44 (d, J=6.2 Hz, 3H), 1.04 (t, J=6.3 Hz, 3H), 0.98-0.81 (m, 6H), 0.76 (s, 3H), 0.68 (s, 3H).
Example A377. Synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.SUP.2.S,6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide
(810) ##STR01284##
Step 1.
(811) (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.sup.2S,6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl) pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide was synthesized in a manner similar to (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.sup.2S,6.sup.3S,4S)-12-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide except (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione was substituted with (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-

dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione and 3-({1-[4-(dimethylamino)-4-methylpent-2-ynoyl]azetidin-3-yl}oxy)propanoic acid was substituted with (2R)-2-[({1-[4-(dimethylamino)-4-methylpent-2-ynoyl]azetidin-3-yl}oxy)methyl]-3-methylbutanoic acid to give the desired product (25.6 mg, 26% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.52H.sub.74N.sub.8O.sub.8 938.6; found 939.5; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.72 (dd, J=4.8, 1.5 Hz, 1H), 7.97 (dd, J=20.0, 6.8 Hz, 1H), 7.87 (dd, J=5.8, 2.6 Hz, 1H), 7.54-7.51 (m, 1H), 7.41 (d, J=8.9 Hz, 1H), 7.14 (dd, J=36.0, 10.4 Hz, 2H), 5.65 (s, 1H), 4.49-4.33 (m, 3H), 4.27-4.08 (m, 4H), 3.96 (d, J=8.6 Hz, 2H), 3.87 (dd, J=10.8, 3.6 Hz, 2H), 3.79 (dd, J=10.8, 7.7 Hz, 3H), 3.69-3.58 (m, 3H), 3.43 (dd, J=23.9, 11.7 Hz, 2H), 3.17 (d, J=21.9 Hz, 3H), 3.00-2.95 (m, 1H), 2.70 (t, J=14.0 Hz, 8H), 2.34-2.24 (m, 1H), 2.05 (d, J=34.4 Hz, 3H), 1.92-1.82 (m, 2H), 1.69-1.62 (m, 5H), 1.57 (d, J=11.5 Hz, 3H), 1.45 (d, J=6.2 Hz, 3H), 1.33 (d, J=12.6 Hz, 1H), 1.05-0.93 (m, 10H), 0.80 (d, J=9.8 Hz, 3H), 0.64 (d, J=12.2 Hz, 2H).

Example A643. Synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)piperidin-4-yl)oxy)methyl)-N-((2.SUP.2.S,6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide

(812) ##STR01285## ##STR01286##

Step 1.

(813) A mixture of 1-(1-methylphenyl)piperidin-4-yl methanesulfonate (2 g, 7.4 mmol) and tert-butyl (2R)-2-(hydroxymethyl)-3-methylbutanoate (1.39 g, 7.4 mmol) was stirred at 120° C. for 1 h, then purified by silica gel column chromatography to give tert-butyl (R)-2-(((1-benzylpiperidin-4-yl)oxy)methyl)-3-methylbutanoate (800 mg, 28% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.22H.sub.35NO.sub.3 361.3; found 362.3.

(814) Step 2.

(815) A mixture of tert-butyl (R)-2-(((1-benzylpiperidin-4-yl)oxy)methyl)-3-methylbutanoate (700 mg, 1.9 mmol), 10% wet Pd/C (411 mg, 3.9 mmol) and 20% wet Pd(OH).sub.2/C (542 mg, 3.9 mmol) in THF (30 mL) was stirred under an atmosphere of H.sub.2 (15 psi) for 16 h. The mixture was filtered and the filtrate was concentrated under reduced pressure to give tert-butyl (R)-3-methyl-2-((piperidin-4-yloxy)methyl)butanoate (440 mg, 80% yield) as a an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.15H.sub.29NO.sub.3 271.2; found 272.2.

(816) Step 3.

(817) To a mixture of tert-butyl (R)-3-methyl-2-((piperidin-4-yloxy)methyl)butanoate (440 mg, 1.6 mmol) 4-(dimethylamino)-4-methylpent-2-ynoic acid (3.77 g, 24.3 mmol) and DIPEA (2.09 g, 16.2 mmol) in DMF (50 mL) at 0° C. was added T3P (2.57 g, 8.1 mmol). The mixture was stirred at 0° C. for 1 h, then poured into H.sub.2O (50 mL) and extracted with EtOAc (50 mL×3). The combined organic layers were washed with brine, concentrated under reduced pressure and the residue purified by silica gel column chromatography to give tert-butyl (R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)piperidin-4-yl)oxy)methyl)-3-methylbutanoate (190 mg, 27% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.40N.sub.2O.sub.4 408.3; found 409.4.

(818) Step 4.

(819) To a mixture of tert-butyl (R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)piperidin-4-yl)oxy)methyl)-3-methylbutanoate (180 mg, 0.47 mmol) in DCM (2 mL) was added TFA (1 mL). The mixture was stirred at room temperature for 1 h, then concentrated under reduced pressure to give (R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)piperidin-4-yl)oxy)methyl)-3-methylbutanoic acid (170 mg, 98% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.19H.sub.32N.sub.2O.sub.4 352.2; found 353.2.

(820) Step 5.

(821) (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)piperidin-4-yl)oxy)methyl)-N-

((2.sup.2S,6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide was synthesized in a manner similar to (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.sup.2S,6.sup.3S,4S)-12-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide except (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione was substituted with (2.sup.2S,6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-morpholina-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione and 3-({1-[4-(dimethylamino)-4-methylpent-2-ynoyl]azetidin-3-yl}oxy)propanoic acid was substituted with (2R)-2-[({1-[4-(dimethylamino)-4-methylpent-2-ynoyl]piperidin-4-yl}oxy)methyl]-3-methylbutanoic acid. (101 mg, 42% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.54H.sub.78N.sub.8O.sub.8 966.6; found 969.5; .sup.1H NMR (400 MHz, CD.sub.3OD) δ 8.70 (d, J=4.8 Hz, 1H), 7.81-7.76 (m, 1H), 7.56-7.46 (m, 1H), 7.43-7.34 (m, 1H), 7.24-7.01 (m, 2H), 5.66-5.54 (m, 1H), 4.50-4.40 (m, 1H), 4.31-4.22 (m, 1H), 4.19-4.08 (m, 1H), 4.02-3.82 (m, 4H), 3.80-3.53 (m, 10H), 3.47-3.34 (m, 2H), 3.26-3.15 (m, 3H), 2.98-2.57 (m, 5H), 2.37-2.30 (m, 3H), 2.27-2.18 (m, 4H), 2.15-2.02 (m, 2H), 2.00-1.80 (m, 4H), 1.78-1.71 (m, 2H), 1.68-1.55 (m, 3H), 1.49-1.37 (m, 6H), 1.35-1.28 (m, 3H), 1.05-0.92 (m, 9H), 0.85-0.72 (m, 3H), 0.68-0.51 (m, 3H).
Example A328. Synthesis of two atropisomers of (3S)-1-acryloyl-N-((2S)-1-(((6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide
(822) ##STR01287## ##STR01288## ##STR01289## ##STR01290##
Step 1.
(823) To a mixture of 3-bromo-5-iodo-2-[(1S)-1-methoxyethyl]pyridine (2.20 g, 6.4 mmol) and tert-butyl piperazine-1-carboxylate (1.20 g, 6.4 mmol) in toluene (50 mL) under an atmosphere of Ar were added tBuONa (0.74 g, 7.7 mmol) and portion-wise addition of Pd.sub.2(dba).sub.3 (0.59 g, 0.64 mmol), followed by portion-wise addition of Xantphos (0.74 g, 1.3 mmol). The mixture was heated to 100° C. and stirred for 16 h then H.sub.2O added and the mixture extracted with EtOAc (400 mL×3). The combined organic layers were washed with brine (150 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give tert-butyl 4-[5-bromo-6-[(1S)-1-methoxyethyl]pyridin-3-yl]piperazine-1-carboxylate (1.7 g, 61% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.17H.sub.26BrN.sub.3O.sub.3 399.1; found 400.1.
(824) Step 2.
(825) A mixture of tert-butyl 4-[5-bromo-6-[(1 S)-1-methoxyethyl]pyridin-3-yl]piperazine-1-carboxylate (1.76 g, 4.4 mmol) and 4,4,4′,4′,5,5,5′,5′-octamethyl-2,2′-bi(1,3,2-dioxaborolane) (1.67 g, 6.6 mmol) in toluene (18 mL) under an atmosphere of Ar were added KOAc (0.95 g, 9.7 mmol) and Pd(PPh.sub.3).sub.2Cl.sub.2 (0.31 g, 0.44 mmol) in portions. The mixture was heated to 80° C. and stirred for 16 h, then diluted with H.sub.2O and the mixture extracted with EtOAc (500 mL×3). The combined organic layers were washed with brine (100 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl 4-[6-[(1 S)-1-methoxyethyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-3-yl]piperazine-1-carboxylate (1.4 g, 68%

yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.38BN.sub.3O.sub.5 447.4; found 448.2.
(826) Step 3.
(827) To a mixture of tert-butyl ((6.sup.3S,4S)-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (1.0 g, 1.5 mmol) in DCM (10 mL) at 0 ′C under an atmosphere of N.sub.2 was added TFA (5.0 mL, 67.3 mmol) in portions. The mixture was stirred at 0° C. for 1 h then concentrated under reduced pressure and dried azeotropically with toluene (3 mL×3) to give (6.sup.3S,4S)-4-amino-1.sup.2-iodo-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (1.0 g), which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.27H.sub.31IN.sub.4O.sub.3 586.1; found 587.3.
(828) Step 4.
(829) To a mixture of (6.sup.3S,4S)-4-amino-1.sup.2-iodo-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione (1.0 g, 1.7 mmol) in DMF (15 mL) at 0° C. under an atmosphere of N.sub.2 were added DIPEA (2.20 g, 17.0 mmol) and (2S)-2-[[(benzyloxy)carbonyl](methyl)amino]-3-methylbutanoic acid (0.90 g, 3.4 mmol) in portions, followed by COMU (1.10 g, 2.6 mmol) in portions over 10 min. The mixture was stirred at 0° C. for 1.5 h, then diluted with H.sub.2O and the mixture was extracted with EtOAc (300 mL×3). The combined organic layers were washed with brine (100 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give benzyl ((2S)-1-(((6.sup.3S,4S)-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)- pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (790 mg, 53% yield) as a solid.
(830) Step 5.
(831) To a mixture of benzyl ((2S)-1-(((6.sup.3S,4S)-1.sup.2-iodo-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (480 mg, 0.58 mmol) and tert-butyl 4-[6-[(1S)-1-methoxyethyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-3-yl]piperazine-1-carboxylate (309 mg, 0.69 mmol) in 1,4-dioxane (8.0 mL) and H.sub.2O (1.6 mL) under an atmosphere of Ar were added K.sub.2CO.sub.3 (199 mg, 1.4 mmol) and Pd(dppf)Cl.sub.2 (42 mg, 0.06 mmol) in portions. The mixture was heated to 70° C. and stirred for 16 h, then diluted with H.sub.2O and extracted with EtOAc (200 mL×3). The combined organic layers were washed with brine (150 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by silica gel column chromatography to give tert-butyl 4-(5-((6.sup.3S,4S)-4-((S)-2-(((benzyloxy)carbonyl)(methyl)amino)-3-methylbutanamido)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)- benzenacycloundecaphane-1.sup.2-yl)-6-((S)-1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (335 mg, 51% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.58H.sub.74N.sub.8O.sub.9 1026.6; found 1027.4.
(832) Step 6.
(833) To a mixture of tert-butyl 4-(5-((6.sup.3S,4S)-4-((S)-2-(((benzyloxy)carbonyl)(methyl)amino)-3-methylbutanamido)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-1.sup.2-yl)-6-((S)-1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (335 mg, 0.33 mmol) in DMF (5 mL) at 0° C. under an atmosphere of

N.sub.2 were added Cs.sub.2CO.sub.3 (234 mg, 0.72 mmol) and iodoethane (102 mg, 0.65 mmol) in portions. The mixture was warmed to room temperature and stirred for 16 h, then diluted with H.sub.2O and the mixture extracted with EtOAc (100 mL×3). The combined organic layers were washed with brine (50 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-TLC to give tert-butyl 4-(5-((6.sup.3S,4S)-4-((S)-2-(((benzyloxy)carbonyl)(methyl)amino)-3-methylbutanamido)-1.sup.1-ethyl-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-1.sup.2-yl)-6-((S)-1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (320 mg, 84% yield) as a light yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.60H.sub.78N.sub.8O.sub.9 1054.6; found 1055.8.
(834) Step 7.
(835) A mixture of tert-butyl 4-(5-((6.sup.3S,4S)-4-((S)-2-(((benzyloxy)carbonyl)(methyl)amino)-3-methylbutanamido)-1.sup.1-ethyl-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-1.sup.2-yl)-6-((S)-1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (320 mg) in xx M HCl in 1,4-dioxane (3.0 mL) at 0° C. under an atmosphere of N.sub.2 was stirred at room temperature for 2 h, then concentrated under reduced pressure to give benzyl ((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)-5-(piperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate, which was used directly in the next step further purification. LCMS (ESI): m/z [M+H] calc'd for C.sub.55H.sub.70N.sub.8O.sub.7 954.5; found 955.3.
(836) Step 8.
(837) To a mixture of benzyl ((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)-5-(piperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (320 mg, 0.34 mmol) and HCHO (60 mg, 2.0 mmol) in MeOH (3.0 mL) at 0° C. under an atmosphere of N.sub.2 were added NaCNBH.sub.3 (42 mg, 0.67 mmol) and AcOH (60 mg, 1.0 mmol) in portions. The mixture was warmed to room temperature and stirred for 2 h, then diluted with H.sub.2O and the mixture extracted with DCM/MeOH (5:1) (200 mL×3). The combined organic layers were washed with brine (100 mL×3), dried over anhydrous Na.sub.2SO.sub.4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give benzyl ((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (160 mg, 59% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.56H.sub.72N.sub.8O.sub.7 968.6; found 969.6.
(838) Step 9.
(839) To a mixture of benzyl ((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (160 mg, 0.17 mmol) in toluene (10 mL) and MeOH (1.0 mL) was added Pd/C (130 mg, 1.2 mmol) in portions. The mixture was evacuated and re-filled with H.sub.2 (×3), then stirred under an atmosphere of H.sub.2 for 16 h. The mixture was filtered and the filtrate was concentrated under reduced pressure to give (2S)—N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-

methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)- benzenacycloundecaphane-4-yl)-3-methyl-2-(methylamino)butanamide (140 mg), which was used directly in the next step without further purification. LCMS (ESI): m/z [M+H] calc'd for $C_{48}H_{66}N_8O_5$ 834.5; found 835.5.

(840) Step 10.

(841) To a mixture of (2S)—N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)-3-methyl-2-(methylamino)butanamide (140 mg, 0.17 mmol) in ACN (2.0 mL) at 0° C. under an atmosphere of N_2 were added DIPEA (433 mg, 3.35 mmol), (3S)-1-(prop-2-enoyl)pyrrolidine-3-carboxylic acid (57 mg, 0.34 mmol) in portions and CIP (70 mg, 0.25 mmol) in portions over 10 min. The mixture was stirred at 0° C. for 1.5 h, then H_2O added and the mixture extracted with EtOAc (150 mL×3). The combined organic layers were washed with brine (100 mL×3), dried over anhydrous Na_2SO_4 and filtered. The filtrate was concentrated under reduced pressure and the residue was purified by preparative-HPLC to give two atropisomers of (3S)-1-acryloyl-N-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide (40 mg, 24% yield) as a solid and (20 mg, 12% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{56}H_{75}N_9O_7$ 985.6; found 986.7; ^1H NMR (400 MHz, DMSO-d_6) 8.47 (t, J=2.1 Hz, 1H), 8.00 (d, J=4.7 Hz, 1H), 7.78-7.59 (m, 3H), 7.58-7.48 (m, 1H), 7.42-7.30 (m, 1H), 7.23 (dq, J=8.0, 4.0, 3.5 Hz, 1H), 7.15-7.03 (m, 1H), 6.75-6.50 (m, 1H), 6.18 (dt, J=16.8, 2.7 Hz, 1H), 5.70 (tt, J=9.3, 2.7 Hz, 1H), 5.48-5.23 (m, 1H), 5.06 (dd, J=31.1, 12.3 Hz, 1H), 4.74 (dd, J=11.0, 4.3 Hz, 1H), 4.33-4.15 (m, 2H), 4.01 (ddd, J=36.1, 12.6, 7.6 Hz, 2H), 3.91-3.56 (m, 6H), 3.52-3.39 (m, 2H), 3.31-3.28 (m, 2H), 3.24 (d, J=5.7 Hz, 4H), 3.06 (s, 4H), 2.93 (d, J=9.8 Hz, 2H), 2.81 (d, J=5.4 Hz, 3H), 2.47-2.43 (m, 4H), 2.22 (s, 4H), 2.09 (tq, J=12.0, 7.4, 6.6 Hz, 3H), 1.81 (s, 1H), 1.74 (d, J=11.7 Hz, 1H), 1.56 (d, J=11.7 Hz, 1H), 1.20 (dd, J=6.3, 1.5 Hz, 3H), 1.10 (td, J=7.2, 2.4 Hz, 3H), 1.00-0.86 (m, 6H), 0.86-0.72 (m, 3H), 0.54 (d, J=3.5 Hz, 3H) and LCMS (ESI): m/z [M−H] calc'd for $C_{56}H_{75}N_9O_7$ 985.6; found 984.4; ^1H NMR (400 MHz, DMSO-d_6) δ 8.46 (d, J=3.0 Hz, 2H), 7.98 (s, 1H), 7.89-7.83 (m, 1H), 7.76-7.57 (m, 3H), 7.24 (s, 2H), 7.07 (s, 1H), 6.70-6.58 (m, 1H), 6.17 (d, J=16.5 Hz, 1H), 5.73-5.67 (m, 1H), 5.36-5.30 (m, 1H), 4.31-3.97 (m, 6H), 3.83-3.77 (m, 2H), 3.74-3.49 (m, 6H), 3.48-3.41 (m, 1H), 3.40-3.37 (m, 2H) 3.28-3.24 (m, 4H), 3.07 (s, 3H), 2.88-2.82 (m, 1H), 2.80-2.64 (m, 7H), 2.49-2.44 (m, 4H), 2.22 (s, 3H), 2.04 (d, J=26.1 Hz, 3H), 1.85-1.79 (m, 1H), 1.67-1.55 (m, 2H), 1.35 (d, J=6.1 Hz, 3H), 1.27-1.22 (m, 1H), 1.05-0.93 (m, 4H), 0.89 (d, J=6.9 Hz, 2H), 0.79 (d, J=12.4 Hz, 5H), 0.73 (d, J=6.5 Hz, 1H), 0.56 (s, 3H).

Example A542. The synthesis of 1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-4-fluoro-N-((2S)-1-(((6.SUP.3.S,4S)-1.SUP.2.-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.SUP.1.-(2,2,2-trifluoroethyl)-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpiperidine-4-carboxamide

(842) ##STR01291## ##STR01292## ##STR01293## ##STR01294##

Step 1.

(843) To a solution of (S)-3-bromo-5-iodo-2-(1-methoxyethyl)pyridine (15 g, 43.86 mmol), and benzyl piperazine-1-carboxylate (8.7 g, 39.48 mmol) in toluene (150 mL) at 0° C., were added cesium carbonate (71.46 g, 219.32 mmol), BINAP (0.55 g, 0.88 mmol) and palladium acetate (0.49 g, 2.19 mmol) in portions. The reaction mixture was stirred at 90° C. for 12 h under an argon

atmosphere. The resulting mixture was cooled down to room temperature, filtered and the filter cake was washed with EtOAc (150 mL×3). The filtrate was concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford benzyl (S)-4-(5-bromo-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (16 g, 84% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{44}H_{58}N_6O_7$ 433.1; found 434.0.

(844) Step 2.

(845) To a stirred solution of benzyl (S)-4-(5-bromo-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (22.7 g, 52.26 mmol), bis(pinacolato)diboron (19.91 g, 78.4 mmol) in toluene (230 mL) at 0° C., were added potassium acetate (12.82 g, 130.66 mmol) and $Pd(dppf)Cl_2$.Math.DCM (4.26 g, 5.23 mmol) in portions. The reaction mixture was stirred at 90° C. for 6 h under an argon atmosphere. The resulting mixture was filtered and the filter cake was washed with EtOAc (200 mL×3). The filtrate was concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford benzyl (S)-4-(6-(1-methoxyethyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-3-yl)piperazine-1-carboxylate (14.7 g, 58% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{26}H_{36}BN_3O_5$ 481.3; found 482.3.

(846) Step 3.

(847) To a stirred solution of 5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-iodo-1H-indole (17.46 g, 27 mmol) in 1,4-dioxane (150 mL) and H_2O (30 mL) at 0° C., were added potassium carbonate (9.33 g, 67.51 mmol) and $Pd(dppf)Cl_2$ DCM (2.2 g, 2.7 mmol) in portions, followed by benzyl (S)-4-(6-(1-methoxyethyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl) pyridin-3-yl) piperazine-1-carboxylate (13 g, 27 mmol). The reaction mixture was stirred at 70° C. for 12 h under an argon atmosphere. The resulting mixture was cooled to room temperature and quenched with H_2O, then extracted with EtOAc (200 mL×3). The organic phase was dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford benzyl (S)-4-(5-(5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-1H-indol-2-yl)-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (20 g, 84.7% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{49}H_{57}BN_4O_4Si$ 873.2; found 873.3.

(848) Step 4.

(849) To a mixture of benzyl (S)-4-(5-(5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-1H-indol-2-yl)-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (19 g, 21.74 mmol) and Cs_2CO_3 (49.58 g, 152.17 mmol) in DMF (190 mL) at 0° C. under argon atmosphere, was dropwise added 2,2,2-trifluoroethyl trifluoromethanesulfonate (50.46 g, 217.39 mmol). The reaction mixture was stirred at room temperature for 12 h under an argon atmosphere, then quenched with H_2O, extracted with EtOAc (200 mL×3). The combined organic layers were washed with brine (200 mL×3), dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford benzyl (S)-4-(5-(5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-2-yl)-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (17.6 g, 84.7% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{51}H_{58}BF_3N_4O_4Si$ 954.2; found 955.3.

(850) Step 5.

(851) To a solution of benzyl (S)-4-(5-(5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-2-yl)-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (18 g, 18.83 mmol), was added TBAF in THF (180.0 mL) at 0° C. The reaction mixture was stirred at 40° C. for 12 h under an argon atmosphere, then quenched with cold H_2O. The resulting mixture was extracted with EtOAc (200 mL×3). The combined organic layers were washed with brine (200 mL×3), dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford benzyl (S)-4-(5-(5-bromo-3-(3-hydroxy-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-2-

yl)-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (7.8 g, 57.7% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{35}H_{40}BrF_3N_4O_4$ 716.2; found 717.1.

(852) Step 6.

(853) A solution of benzyl (S)-4-(5-(5-bromo-3-(3-hydroxy-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-2-yl)-6-(1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (1 g, 1.39 mmol) in 1,4-dioxane (10 mL) and H_2O (2 mL), was added methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)propanoyl)hexahydropyridazine-3-carboxylate (1.08 g, 2.09 mmol), potassium carbonate (481.47 mg, 3.48 mmol) and $Pd(dtbpf)Cl_2$ (181.64 mg, 0.28 mmol) in portions at 0° C. The reaction mixture was stirred at 70° C. for 3 h under an argon atmosphere. The resulting mixture was cooled to room temperature, then quenched with H_2O and extracted with EtOAc (50 mL×3). The combined organic layers were washed with brine (20 mL×3), dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by silica gel chromatography to afford methyl (S)-1-((S)-3-(3-(2-(5-(4-((benzyloxy)carbonyl)piperazin-1-yl)-2-((S)-1-methoxyethyl)pyridin-3-yl)-3-(3-hydroxy-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (1.1 g, 77% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{55}H_{68}F_3N_7O_9$ 1027.5; found 1028.3.

(854) Step 7.

(855) To a solution of methyl (S)-1-((S)-3-(3-(2-(5-(4-((benzyloxy)carbonyl)piperazin-1-yl)-2-((S)-1-methoxyethyl)pyridin-3-yl)-3-(3-hydroxy-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (1.1 g, 1.07 mmol) in THF (8 mL) and H_2O (2 mL) at 0° C., was dropwise added LiOH (2.2 mL, 1M aqueous) under an argon atmosphere. The reaction mixture was stirred for 2 h then concentrated under reduced pressure. The residue was acidified to pH 5 with citric acid (1M) and extracted with EtOAc (20 mL×3). The combined organic layers were concentrated under reduced pressure. The residue was purified by reverse phase chromatography to afford (S)-1-((S)-3-(3-(2-(5-(4-((benzyloxy)carbonyl)piperazin-1-yl)-2-((S)-1-methoxyethyl)pyridin-3-yl)-3-(3-hydroxy-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylic acid (750 mg, 69% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{54}H_{66}F_3N_7O_9$ 1013.5; found 1014.3.

(856) Step 8.

(857) To a solution of (S)-1-((S)-3-(3-(2-(5-(4-((benzyloxy)carbonyl)piperazin-1-yl)-2-((S)-1-methoxyethyl)pyridin-3-yl)-3-(3-hydroxy-2,2-dimethylpropyl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)phenyl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylic acid (0.75 g, 0.74 mmol) in DCM (75 mL) at 0° C., were added in portions HOBT (0.5 g, 3.7 mmol), DIPEA (3.82 g, 29.58 mmol), and EDCI (4.25 g, 22.19 mmol) at 0° C. The reaction mixture was stirred at room temperature for 12 h under an argon atmosphere. The resulting mixture was concentrated under reduced pressure and extracted with EtOAc (100 mL×3). The combined organic layers were washed with brine (50 mL×3), dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by silica gel chromatography to afford benzyl 4-(5-((6^3S,4S)-4-((tert-butoxycarbonyl)amino)-10,10-dimethyl-5,7-dioxo-1^1-(2,2,2-trifluoroethyl)-6^1,6^2,6^3,6^4,6^5,6^6-hexahydro-1^1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)- benzenacycloundecaphane-1^2-yl)-6-((S)-1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (0.5 g, 67.9% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{54}H_{64}F_3N_7O_8$ 995.5; found 996.3.

(858) Step 9.

(859) To a mixture of benzyl 4-(5-((6^3S,4S)-4-((tert-butoxycarbonyl)amino)-10,10-dimethyl-5,7-dioxo-1^1-(2,2,2-trifluoroethyl)-6^1,6^2,6^3,6^4,6^5,6^6-hexahydro-

1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)- benzenacycloundecaphane-1.sup.2-yl)-6-((S)-1-methoxyethyl)pyridin-3-yl)piperazine-1-carboxylate (500 mg, 0.5 mmol) in MeOH (15 mL) at 0° C., was added paraformaldehyde (135.64 mg, 1.5 mmol), and Pd/C (750 mg) in portions. The reaction mixture was stirred at room temperature for 12 h under a hydrogen atmosphere. The resulting mixture was filtered and the filter cake was washed with EtOAc (50 mL×5). The filtrate was concentrated under reduced pressure. The residue was purified by silica gel chromatography to afford tert-butyl ((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (350 mg, 79.6% yield) as an solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.47H.sub.60F.sub.3N.sub.7O.sub.8 875.5; found 876.5.

(860) Step 10.

(861) To a solution of tert-butyl ((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)carbamate (300 mg, 0.34 mmol) in DCM (2 mL) at 0° C., was dropwise added HCl in 1,4-dioxane (1 mL, 4M, 4 mmol). The reaction mixture was stirred at room temperature for 2 h, then concentrated under reduced pressure to give (6.sup.3S,4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione hydrochloride (350 mg, crude) as solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.52F.sub.3N.sub.7O.sub.4 775.4; found 766.4.

(862) Step 11.

(863) To a solution of (6.sup.3S,4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-5,7-dione hydrochloride (150 mg, 0.19 mmol) and N-(1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-4-fluoropiperidine-4-carbonyl)-N-methyl-L-valine (154 mg, 0.39 mmol) in DMF (2 mL) at 0° C., was dropwise added a mixture of DIPEA (1 g, 7.72 mmol) and HATU (110 mg, 0.29 mmol) in DMF (0.2 mL). The reaction mixture was stirred at 0° C. for 2 h under an argon atmosphere, then quenched with H.sub.2O. The resulting mixture was extracted with EtOAc (20 mL×3). The combined organic phase was washed with brine (10 mL×3), dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by reverse phase chromatography to afford 1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-4-fluoro-N-((2S)-1-(((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)-5-(4-methylpiperazin-1-yl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-benzenacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpiperidine-4-carboxamide (39.5 mg, 17% yield) as solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.48 (d, J=2.9 Hz, 1H), 8.32 (t, J=7.3 Hz, 1H), 7.97 (s, 1H), 7.82-7.69 (m, 3H), 7.66 (t, J=7.8 Hz, 1H), 7.33-7.07 (m, 3H), 5.50 (dd, J=16.7, 8.6 Hz, 1H), 5.33 (t, J=9.2 Hz, 1H), 5.16 (d, J=12.2 Hz, 1H), 4.94-4.80 (m, 1H), 4.64 (d, J=10.8 Hz, 1H), 4.33-4.16 (m, 3H), 4.12-4.02 (m, 2H), 3.71-3.50 (m, 3H), 3.25 (s, 3H), 3.21-3.16 (m, 3H), 3.14-3.05 (m, 1H), 2.96 (t, J=4.7 Hz, 4H), 2.84 (s, 1H), 2.82-2.72 (m, 2H), 2.59-2.53 (m, 1H), 2.47-2.40 (m, 4H), 2.22 (d, J=2.9 Hz, 9H), 2.18-2.12 (m, 2H), 2.11-1.99 (m, 3H), 1.87-1.78 (m, 1H), 1.74-1.62 (m, 1H), 1.59-1.48 (m, 1H), 1.40-1.32 (m, 9H), 1.01 (t, J=7.7 Hz, 1H), 0.89 (s, 5H), 0.83 (d, J=6.3 Hz, 1H), 0.77 (d, J=6.6 Hz, 2H), 0.38 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.44H.sub.58N.sub.6O.sub.7 1154.6; found 1155.7.

Example A735. Synthesis of 2-acryloyl-N-((2S)-1-(((6.SUP.3.S,4S,Z)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methyl-5-oxa-2,9-diazaspiro[3.5]nonane-9-carboxamide

(864) ##STR01295##

Step 1.

(865) To a stirred mixture of BTC (425.2 mg, 1.448 mmol) in DCM (10 mL) was added dropwise pyridine (1.04 g, 13.16 mmol) and tert-butyl 5-oxa-2,9-diazaspiro[3.5]nonane-2-carboxylate (1 g, 4.39 mmol), the reaction mixture was stirred at room temperature for 2 h. The resulting mixture was concentrated under reduced pressure to give crude tert-butyl 9-(chlorocarbonyl)-5-oxa-2,9-diazaspiro[3.5]nonane-2-carboxylate.

(866) Step 2.

(867) To a stirred solution of tert-butyl 9-(chlorocarbonyl)-5-oxa-2,9-diazaspiro[3.5]nonane-2-carboxylate (2.5 g, crude) in MeCN (20 mL) were added dropwise pyridine (1.04 g, 13.16 mmol) and benzyl (2S)-3-methyl-2-(methylamino)butanoate (970.66 mg, 4.38 mmol) at room temperature. The reaction mixture was stirred at 80° C. for 12 h and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford tert-butyl (S)-9-((1-(benzyloxy)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-5-oxa-2,9-diazaspiro[3.5]nonane-2-carboxylate (783 mg, 37.6% yield, two steps) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{25}H_{37}N_3O_6$ 475.3; found 476.3.

(868) Step 3.

(869) A solution of tert-butyl (S)-9-((1-(benzyloxy)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-5-oxa-2,9-diazaspiro[3.5]nonane-2-carboxylate (783 mg, 1.65 mmol) and 10 wt % palladium on carbon (226.29 mg) in THF (10 mL) was stirred for 2 h at 50° C. under a hydrogen atmosphere. The resulting mixture was cooled to room temperature, filtered and the filter cake was washed with MeCN (10 mL×3). The filtrate was concentrated under reduced pressure to give N-(2-(tert-butoxycarbonyl)-5-oxa-2,9-diazaspiro[3.5]nonane-9-carbonyl)-N-methyl-L-valine (591 mg, 98.8% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{18}H_{31}N_3O_6$ 385.2; found 386.3.

(870) Step 4.

(871) To a stirred solution of intermediate 2 (731 mg, 1.16 mmol) and DIPEA (2.25 g, 17.38 mmol) in MeCN (50 mL) was added CIP (644.31 mg, 2.32 mmol) and N-(2-(tert-butoxycarbonyl)-5-oxa-2,9-diazaspiro[3.5]nonane-9-carbonyl)-N-methyl-L-valine (446.68 mg, 1.16 mmol) at room temperature. The reaction mixture was stirred for 2 h then concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford tert-butyl 9-(((2S)-1-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-5-oxa-2,9-diazaspiro[3.5]nonane-2-carboxylate (752 mg, 65% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{52}H_{71}N_9O_9S$ 997.5; found 996.6.

(872) Step 5.

(873) To a stirred solution of tert-butyl 9-(((2S)-1-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-5-oxa-2,9-diazaspiro[3.5]nonane-2-carboxylate (752 mg, 0.75 mmol) in DCM (40 mL) was added TFA (10 mL) in portions at room temperature. The reaction mixture was stirred for 2 h, then concentrated under reduced pressure. To the residue was added saturated aqueous sodium bicarbonate (100 mL) and DCM (100 mL). The aqueous layer was separated and extracted

with DCM (100 mL×2). The combined organic phase was dried over anhydrous sodium sulfate, filtrated and concentrated under reduced pressure to afford N-((2S)-1-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methyl-5-oxa-2,9-diazaspiro[3.5]nonane-9-carboxamide (587 mg, 87% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.47H.sub.63N.sub.9O.sub.7S 897.5; found 898.4.

(874) Step 6.

(875) A stirred solution of N-((2S)-1-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methyl-5-oxa-2,9-diazaspiro[3.5]nonane-9-carboxamide (586 mg, 0.65 mmol) in MeCN (10 mL) was added acrylic acid (47 mg, 0.65 mmol), DIPEA (421 mg, 3.26 mmol), CIP (362 mg, 1.3 mmol). The reaction mixture was stirred for 12 h and concentrated under reduced pressure. The residue was purified by reverse phase chromatography to afford 2-acryloyl-N-((2S)-1-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(4,2)-thiazola-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methyl-5-oxa-2,9-diazaspiro[3.5]nonane-9-carboxamide (194 mg, 27% yield) as a solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.78 (dd, J=4.8, 1.7 Hz, 1H), 8.48 (d, J=1.6 Hz, 2H), 7.85-7.65 (m, 3H), 7.65-7.42 (m, 2H), 6.30 (mm, 1H), 6.10 (m, J=17.0, 1H), 5.78-5.50 (m, J=10.3, 2H), 5.10 (dd, 1H), 4.40-3.80 (m, 14H), 3.60-3.10 (m, 10H), 2.94 (d, J=14.5 Hz, 1H), 2.85 (s, 4H), 2.42 (dd, 1H), 2.07 (dd, 2H), 1.80 (s, 2H), 1.55 (s, 3H), 1.32 (d, 3H), 0.95-0.75 (m, 12H), 0.33 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.50H.sub.65N.sub.9O.sub.8S 951.5; found 952.6.

Example A720. Synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.SUP.3.S,4S,Z)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.SUP.1.,10,10-trimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H,2.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)-3-methylbutanamide

(876) ##STR01296## ##STR01297## ##STR01298## ##STR01299##

Step 1.

(877) To a stirred solution of methyl 1-methyl-1,2,4-triazole-3-carboxylate (7.0 g, 49.60 mmol) in CCl.sub.4 (70. mL) was added NBS (13.24 g, 74.40 mmol) and AIBN (11.40 g, 69.44 mmol) in portions at 25° C. under an argon atmosphere. The resulting mixture was stirred for 24 h at 80° C. The resulting mixture was filtered, the filtrate was cooled to 20° C. and kept at 20° C. for 30 min. The resulting mixture was filtered. The filter cake was washed with H.sub.2O (3×50 mL) and pet. ether (3×100 mL). The filter cake was dried under reduced pressure. This resulted in methyl 5-bromo-1-methyl-1,2,4-triazole-3-carboxylate (10 g, crude) as a light yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.5H.sub.6BrN.sub.3O.sub.2 219.0; found 219.9.

(878) Step 2.

(879) To a stirred solution of methyl 5-bromo-1-methyl-1,2,4-triazole-3-carboxylate (10.0 g, 45.50 mmol) in MeOH (150.0 mL) and H.sub.2O (30.0 mL) was added NaBH.sub.4 (6.88 g, 181.80 mmol) in portions at −5° C. under a nitrogen atmosphere. The resulting mixture was stirred for 2 h at 0˜10° C. Desired product could be detected by LCMS. The reaction was quenched with brine (100 mL) at 0° C. The resulting mixture was extracted with pet. ether (100 mL). The aqueous layer was separated and filtered. The filter cake was washed with MeOH (2×50 mL). The filtrate was concentrated under reduced pressure to afford (5-bromo-1-methyl-1,2,4-triazol-3-yl)methanol (6 g, crude) as a light yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.4H.sub.6BrN.sub.3O 191.98; found 192.0.

(880) Step 3.
(881) A solution of (5-bromo-1-methyl-1,2,4-triazol-3-yl)methanol (6.0 g) and HBr in AcOH (144.0 mL) was stirred for overnight at 80° C. The mixture was neutralized to pH 9 with saturated NaHCO.sub.3 (aq.). The resulting mixture was extracted with EtOAc (3×60 mL). The combined organic layers were washed with brine (3×100 mL), dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure. This resulted in 5-bromo-3-(bromomethyl)-1-methyl-1,2,4-triazole (6 g, crude) as a white solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.4H.sub.5Br.sub.2N.sub.3 253.89; found 253.8.
(882) Step 4.
(883) To a stirred mixture of 5-bromo-3-(bromomethyl)-1-methyl-1,2,4-triazole (6.0 g, 23.54 mmol) and tert-butyl 2-[(diphenylmethylidene)amino]acetate (6.95 g, 23.54 mmol) in toluene (42 mL) and DCM (18.0 mL) was added (2R,4R,5S)-1-(anthracen-9-ylmethyl)-5-ethenyl-2-[(S)-(prop-2-en-1-yloxy)(quinolin-4-yl)methyl]-1-azabicyclo[2.2.2]octan-1-ium bromide (1.43 g, 2.35 mmol) in portions at 0° C. under argon atmosphere. The resulting mixture was stirred and KOH (60 mL) in H.sub.2O was added. The resulting mixture was stirred for 24 h at −10° C. under an argon atmosphere. Desired product could be detected by LCMS. The reaction was quenched with sat. NH.sub.4Cl (aq.) at 0° C. The resulting mixture was extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (1×200 mL), dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by Prep-TLC to afford tert-butyl (2S)-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)-2-[(diphenylmethylidene)amino]propanoate (5 g, 38.6% yield) as a yellow oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.21H.sub.21BrN.sub.4O.sub.2 469.12; found 469.1.
(884) Step 5.
(885) To a stirred solution of tert-butyl (2S)-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)-2-[(diphenylmethylidene)amino]propanoate (5.0 g, 10.65 mmol) in DCM (50.0 mL) was added TFA (25.0 mL) dropwise at 0° C. under argon atmosphere. The resulting mixture was stirred for 16 h at room temperature under an argon atmosphere. The resulting mixture was concentrated under reduced pressure to afford (2S)-2-amino-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)propanoic acid (6 g, crude) as a brown oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.6H.sub.9BrN.sub.4O.sub.2 249.00; found 249.0.
(886) Step 6.
(887) To a stirred solution of (2S)-2-amino-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)propanoic acid (6.0 g, 24.09 mmol) in THF (36.0 mL) was added NaHCO.sub.3 (10.14 g, 120.69 mmol), Boc.sub.2O (7.89 g, 36.14 mmol) in portions at 0° C. under an argon atmosphere. The resulting mixture was stirred for 16 h at room temperature. Desired product could be detected by LCMS. The resulting mixture was concentrated under reduced pressure. The mixture was purified by reverse phase chromatography to afford (2S)-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)-2-[(tert-butoxycarbonyl)amino]propanoic acid (3 g, 33.9% yield) as a white solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.1H.sub.17BrN.sub.4O.sub.4 349.05; found 349.0.
(888) Step 7.
(889) To a stirred solution of 2-[[(2M)-1-ethyl-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]methyl]-2-methylpropyl (3S)-1,2-diazinane-3-carboxylate (1.0 g, 1.65 mmol) in DMF (10.0 mL) was added DIPEA (4.28 g, 33.08 mmol), (2S)-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)-2-[(tert-butoxycarbonyl)amino]propanoic acid (0.69 g, 1.98 mmol) and HATU (0.75 g, 1.99 mmol) in portions at 0° C. The resulting mixture was stirred for 2 h at 20° C. under an argon atmosphere. Desired product could be detected by LCMS. The resulting mixture was quenched with H.sub.2O (100 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (3×100 mL), dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by reverse phase chromatography to afford 2-[[(2M)-1-ethyl-2-[2-[(1 S)-1-

methoxyethyl]pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]methyl]-2-methylpropyl (3S)-1-[(2S)-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)-2-[(tert-butoxycarbonyl)amino]propanoyl]-1,2-diazinane-3-carboxylate (800 mg, 46.5% yield) as a light yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.45H.sub.64BBrN.sub.8O.sub.8 935.42; found 935.2.
(890) Step 8.
(891) To a stirred solution of 2-[[(2M)-1-ethyl-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]methyl]-2-methylpropyl (3S)-1-[(2S)-3-(5-bromo-1-methyl-1,2,4-triazol-3-yl)-2-[(tert-butoxycarbonyl)amino]propanoyl]-1,2-diazinane-3-carboxylate (800.0 mg, 0.86 mmol) in dioxane (10.0 mL) were added K.sub.3PO.sub.4 (0.45 g, 2.12 mmol), XPhos (122.26 mg, 0.27 mmol), XPhos Pd G3 (0.22 g, 0.27 mmol) and H.sub.2O (2.0 mL) at room temperature. The resulting mixture was stirred for 3 h at 75° C. under an argon atmosphere. Desired product could be detected by LCMS. The resulting mixture was extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (3×60 mL), dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by reverse phase chromatography to afford tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)carbamate (400 mg, 56.8% yield) as a light yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.52N.sub.8O.sub.6 729.41; found 729.3.
(892) Step 9.
(893) To a solution of tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)carbamate (400.0 mg, 0.56 mmol) in DCM (1 mL) was added TFA (0.5 mL). The reaction was stirred for 1 h at room temperature under an argon atmosphere. After concentration, the mixture was neutralized to pH 8 with saturated NaHCO.sub.3 (aq., 20 mL). The mixture was extracted with DCM (3×20 mL). The organic layers were dried over Na.sub.2SO.sub.4 and concentrated to afford (6.sup.3S,4S,2)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-5,7-dione (500 mg, crude) as a light yellow solid. ESI-MS m/z=629.3 [M+H].sup.+; Calculated MW:628.3. LCMS (ESI): m/z [M+H] calc'd for C.sub.34H.sub.44N.sub.8O.sub.4 629.36; found 629.3.
(894) Step 10.
(895) To a stirred solution of (6.sup.3S,4S,2)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-5,7-dione (170.0 mg, 0.27 mmol) and (R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-3-methylbutanoic acid (114.68 mg, 0.32 mmol) in DMF (5 mL) were added DIPEA (698.86 mg, 5.41 mmol) and HATU (123.36 mg, 0.32 mmol) dropwise at 0° C. under an air atmosphere. The resulting mixture was stirred for 2 h at 0° C. The resulting mixture was diluted with 25 mL H.sub.2O. The resulting mixture was extracted with EtOAc (3×25 mL). The combined organic layers were washed with brine (3×25 mL), dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure to afford (2R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)-3-methylbutanamide (180 mg, crude) as an

off-white oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.56H.sub.69N.sub.9O.sub.6 964.54; found 964.4.
(896) Step 11.
(897) To a stirred solution of (2R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)-3-methylbutanamide (180.0 mg, 0.19 mmol) and Pd/C (90.0 mg, 0.85 mmol) in MeOH (10 mL) was added Boc.sub.2O (81.48 mg, 0.37 mmol) at room temperature under a hydrogen atmosphere. The resulting mixture was stirred overnight at room temperature. The resulting mixture was filtered, the filter cake was washed with MeOH (3×10 mL). The filtrate was concentrated under reduced pressure to afford tert-butyl 3-((2R)-2-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)carbamoyl)-3-methylbutoxy)azetidine-1-carboxylate (80 mg, 47.7% yield) as an off-white solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.48H.sub.67N.sub.9O.sub.8 898.52; found 898.4.
(898) Step 12.
(899) To a stirred solution of tert-butyl 3-((2R)-2-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)carbamoyl)-3-methylbutoxy)azetidine-1-carboxylate in DCM (2 mL) was added TFA (1.0 mL) dropwise at 0° C. under an air atmosphere. The resulting mixture was stirred for 1 h at 0° C. The resulting mixture was concentrated under reduced pressure to afford (2R)-2-((azetidin-3-yloxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)-3-methylbutanamide (85 mg, crude) as a yellow green oil.
(900) Step 13.
(901) To a stirred solution of (2R)-2-((azetidin-3-yloxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)-3-methylbutanamide (80.0 mg, 0.10 mmol) and 4-(dimethylamino)-4-methylpent-2-ynoic acid (38.90 mg, 0.25 mmol) in DMF (2 mL) were added DIPEA (518.27 mg, 4.01 mmol) and COMU (51.52 mg, 0.12 mmol) in portions at 0° C. The reaction mixture was stirred under an air atmosphere for 2 h. The crude product (150 mg) was purified by reverse phase chromatography to afford (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-2.sup.1,10,10-trimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H,2.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(5,3)-triazolacycloundecaphane-4-yl)-3-methylbutanamide (15.3 mg, 16.3% yield) as an off-white solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.77 (dd, J=4.8, 1.7 Hz, 1H), 8.15 (d, J=1.7 Hz, 1H), 8.07 (d, J=8.1 Hz, 1H), 7.85-7.78 (m, 1H), 7.70 (d, J=8.6 Hz, 1H), 7.58-7.48 (m, 2H), 5.82 (s, 1H), 4.95 (d, J=11.7 Hz, 1H), 4.41-4.30 (m, 5H), 4.30 (d, J=8.2 Hz, 2H), 4.25 (d, J=5.6 Hz, 4H), 4.10 (td, J=17.1, 16.1, 9.1 Hz, 2H), 3.99-3.82 (m, 3H), 3.71-3.60 (m, 1H), 3.54-3.43 (m, 3H), 3.39 (s, 2H), 3.22 (d, J=1.6 Hz, 1H), 2.92 (d, J=13.6 Hz, 1H), 2.86-2.77 (m, 2H), 2.45 (s, 6H), 2.37 (q, J=7.7 Hz, 1H), 2.17 (d, J=6.6 Hz, 2H), 2.03 (d, J=10.2 Hz, 2H), 1.78-1.66 (m, 3H), 1.47 (t, J=10.9 Hz, 6H), 1.35-1.28 (m, 12H), 0.32 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.70N.sub.10O.sub.7 935.55; found 935.3.
Example A692. Synthesis of (3S)-1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-N-((2S)-1-

(((6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(5,2)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide

(902) ##STR01300## ##STR01301## ##STR01302##

Step 1.

(903) To a solution of 1,3-oxazol-2-ylmethanol (5.0 g, 50.46 mmol) in THF (75 mL), were added imidazole (8.59 mg, 0.13 mmol), and TBSCI (11.41 mg, 0.08 mmol) at 0° C. The resulting solution was stirred for 5 h then concentrated under reduced pressure. The crude material was purified by silica gel column chromatography to afford 2-[[(tert-butyldimethylsilyl)oxy]methyl]-1,3-oxazole (10 g, 92.8% yield) as colorless oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.10H.sub.19NO.sub.2Si 214.13; found 214.3.

(904) Step 2.

(905) To a solution of 2-[[(tert-butyldimethylsilyl)oxy]methyl]-1,3-oxazole (10.0 g, 46.87 mmol) in THF (150.0 mL, 1851.45 mmol) at −78° C. was added n-BuLi (22.4 mL, 56.25 mmol) over 10 min and stirred for 30 min at −78° C. under an argon atmosphere. Then the solution of Br.sub.2 (3.6 mL, 70.31 mmol) in THF (10 mL) was added over 10 min to the solution at −78° C. The resulting solution was slowly warmed to room temperature and stirred for 2 h. The resulting mixture was diluted with NH.sub.4Cl/H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure and purified by silica gel column chromatography to afford 5-bromo-2-[[(tert-butyldimethylsilyl)oxy]methyl]-1,3-oxazole (5.3 g, 38.6% yield) as yellow oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.10H.sub.18BrNO.sub.2Si 292.04; found 292.0.

(906) Step 3.

(907) To a solution of 5-bromo-2-[[(tert-butyldimethylsilyl)oxy]methyl]-1,3-oxazole (4.0 g, 13.69 mmol) in DCM (60.0 mL) was added PBr.sub.3 (7.41 g, 27.37 mmol) at 0° C. under an argon atmosphere. The resulting solution was stirred for 4 h then diluted with NaHCO.sub.3/H.sub.2O (30 mL). The mixture was extracted with EtOAc (3×40 mL). The organic layers were concentrated under reduced pressure and purified by silica gel column chromatography to afford 5-bromo-2-(bromomethyl)-1,3-oxazole (2.5 g, 75.7% yield) as yellow oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.4H.sub.3Br.sub.2NO 239.87; found 241.9.

(908) Step 4.

(909) A mixture of 5-bromo-2-(bromomethyl)-1,3-oxazole (9.0 g, 37.36 mmol), Cat:200132-54-3 (2.26 g, 3.74 mmol), DCM (45.0 mL), toluene (90.0 mL), KOH (20.96 g, 373.63 mmol), H.sub.2O (42 mL),and tert-butyl 2-[(diphenylmethylidene)amino]acetate (13.24 g, 44.82 mmol) at 0° C. was stirred for 4 h then diluted with H.sub.2O (30 mL). The mixture was extracted with DCM (3×40 mL). The organic layers were concentrated under reduced pressure and purified by reverse phase column chromatography to afford tert-butyl (2S)-3-(5-bromo-1,3-oxazol-2-yl)-2-[(diphenylmethylidene)amino]propanoate (4.8 g, 28.2% yield) as a yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.23BrN.sub.2O.sub.3 455.10; found 457.1.

(910) Step 5.

(911) A mixture of tert-butyl (2S)-3-(5-bromo-1,3-oxazol-2-yl)-2-[(diphenylmethylidene)amino]propanoate (1.20 g, 2.64 mmol), DCM (10.0 mL, 157.30 mmol), and TFA (5.0 mL, 67.32 mmol) at 0° C. was stirred for 2 h then concentrated under reduced pressure to afford (S)-3-(5-bromooxazol-2-yl)-2-((2,2,2-trifluoroacetyl)-14-azaneyl)propanoic acid (0.5 g, 81.3% yield) as a yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.6H.sub.7BrN.sub.2O.sub.3 234.97; found 237.0.

(912) Step 6.

(913) A mixture of (S)-3-(5-bromooxazol-2-yl)-2-((2,2,2-trifluoroacetyl)-14-azaneyl)propanoic acid (500.0 mg, 2.13 mmol), Boc.sub.2O (928.56 mg, 4.26 mmol), dioxane (2.50 mL), H.sub.2O (2.50

mL), and NaHCO.sub.3 (714.84 mg, 8.51 mmol) at 0° C. was stirred for 3 h. The resulting solution was purified by reverse phase column chromatography to afford (2S)-3-(5-bromo-1,3-oxazol-2-yl)-2-[(tert-butoxycarbonyl)amino]propanoic acid (0.65 g, 91.1% yield) as a yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.11H.sub.15BrN.sub.2O.sub.5 335.02; found 334.8.

(914) Step 7.

(915) To a solution of (2S)-3-(5-bromo-1,3-oxazol-2-yl)-2-[(tert-butoxycarbonyl)amino]propanoic acid (500.0 mg, 1.49 mmol) and 3-[(2M)-1-ethyl-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]-2,2-dimethylpropan-1-ol (808.16 mg, 1.64 mmol) in DMF (5.0 mL) and H.sub.2O (1.0 mL) were added K.sub.3PO.sub.4 (791.67 mg, 3.73 mmol) and Pd(dppf)Cl.sub.2 (109.16 mg, 0.15 mmol). The resulting mixture was stirred for 2 h at 70° C. under an argon atmosphere. The mixture was purified by reverse phase column chromatography to afford (2S)-2-[(tert-butoxycarbonyl)amino]-3-[5-[(2M)-1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-1,3-oxazol-2-yl]propanoic acid (600 mg, 64.79% yield) as a light brown solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.34H.sub.44N.sub.4O.sub.7 621.33; found 621.3.

(916) Step 8.

(917) To a stirred mixture of methyl (3S)-1,2-diazinane-3-carboxylate (627.10 mg, 4.350 mmol) and (2S)-2-[(tert-butoxycarbonyl)amino]-3-[5-[(2M)-1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-1,3-oxazol-2-yl]propanoic acid (900.0 mg, 1.45 mmol) in DCM (10.0 mL) were added HATU (661.54 mg, 1.74 mmol) and DIPEA (3747.71 mg, 29.00 mmol) at 0° C. The resulting mixture was stirred for 2 h. Desired product could be detected by LCMS. The resulting mixture was concentrated under reduced pressure and the crude material was purified by silica gel column chromatography to afford methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[5-[(2M)-1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-1,3-oxazol-2-yl]propanoyl]-1,2-diazinane-3-carboxylate (900 mg, 83.11%) as a brown yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.40H.sub.54N.sub.6O.sub.8 747.41; found 747.2.

(918) Step 9.

(919) To a stirred mixture of methyl (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[5-[(2M)-1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-1,3-oxazol-2-yl]propanoyl]-1,2-diazinane-3-carboxylate (2000.0 mg, 2.68 mmol) in THF (18. mL) and H.sub.2O (6.0 mL) was added LiOH.Math.H.sub.2O (337.10 mg, 8.03 mmol) at 0° C. The resulting mixture was stirred for 2 h. Desired product could be detected by LCMS. The reaction was quenched with H.sub.2O at 0° C. and adjusted to pH 6 with 1N HCl solution. The resulting mixture was extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (1×10 mL), dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure to afford (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[5-[(2M)-1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-1,3-oxazol-2-yl]propanoyl]-1,2-diazinane-3-carboxylic acid (1300 mg, 66.2% yield) as a yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.52N.sub.6O.sub.8 733.39; found 733.3.

(920) Step 10.

(921) To a stirred mixture of (3S)-1-[(2S)-2-[(tert-butoxycarbonyl)amino]-3-[5-[(2M)-1-ethyl-3-(3-hydroxy-2,2-dimethylpropyl)-2-[2-[(1 S)-1-methoxyethyl]pyridin-3-yl]indol-5-yl]-1,3-oxazol-2-yl]propanoyl]-1,2-diazinane-3-carboxylic acid (1.2 g, 1.64 mmol) and DIPEA (8.5 g, 65.50 mmol) in DCM (120.0 mL) were added HOBT (1.8 g, 13.10 mmol) and EDCI (7.8 g, 40.93 mmol) at 0° C. The resulting mixture was stirred for 2 h. Desired product could be detected by LCMS. The mixture was concentrated under reduced pressure and purified by silica gel column chromatography to afford tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (660 mg, 56.4%

yield) as a brown yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.50N.sub.6O.sub.7 715.38; found 715.3.

(922) Step 11.

(923) A mixture of tert-butyl ((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (20.0 mg, 0.028 mmol) and TFA (3.0 mL) in DCM (6.0 mL) at 0° C. was stirred for 2 h. Desired product could be detected by LCMS. The resulting mixture was concentrated under reduced pressure to afford (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (160 mg, crude) as a yellow green solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.34H.sub.42N.sub.6O.sub.5 615.33; found 615.2.

(924) Step 12.

(925) To a stirred mixture of (6.sup.3S,4S)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (180.0 mg, 0.293 mmol) and (2S)-2-[(tert-butoxycarbonyl)(methyl)amino]-3-methylbutanoic acid (135.45 mg, 0.59 mmol) in DMF (2.0 mL) were added HATU (133.60 mg, 0.35 mmol) and DIPEA (756.86 mg, 5.86 mmol) at 0° C. The resulting mixture was stirred for 2 h. Desired product could be detected by LCMS. The mixture was purified by reverse phase column chromatography to afford tert-butyl ((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1 (5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (160 mg, 66% yield) as a brown yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.45H.sub.61N.sub.7O.sub.8 828.47; found 828.4.

(926) Step 13.

(927) To a stirred mixture of tert-butyl ((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1 (5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamate (160.0 mg, 0.19 mmol) in DCM (2.0 mL) was added TFA (1.0 mL) at 0° C. The resulting mixture was stirred for 2 h. Desired product could be detected by LCMS. The resulting mixture was concentrated under reduced pressure to afford (2S)—N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methyl-2-(methylamino)butanamide (130 mg, crude) as a yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.40H.sub.53N.sub.7O.sub.6 728.41; found 728.5.

(928) Step 14.

(929) To a stirred mixture of ((2S)—N-((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)-3-methyl-2-(methylamino)butanamide (130.0 mg, 0.18 mmol) and (3S)-1-[4-(dimethylamino)-4-methylpent-2-ynoyl]pyrrolidine-3-carboxylic acid (180.25 mg, 0.72 mmol) in DMF (2.0 mL) were added DIPEA (461.64 mg, 3.57 mmol) and HATU (135.81 mg, 0.36 mmol) at 0° C. The resulting mixture was stirred for 2 h. Desired product could be detected by LCMS. The mixture was purified by reverse phase column chromatography to afford (3S)-1-(4-(dimethylamino)-4-methylpent-2-ynoyl)-N-((2S)-1-(((6.sup.3S,4S)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,2)-oxazola-1(5,3)-

indola-6(1,3)-pyridazinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methylpyrrolidine-3-carboxamide (55.3 mg, 31.3% yield) as a solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.77 (dd, J=4.8, 1.7 Hz, 1H), 8.09-8.00 (m, 1H), 7.92 (s, 1H), 7.88-7.80 (m, 1H), 7.62 (d, J=8.7 Hz, 1H), 7.59-7.49 (m, 2H), 7.39-7.30 (m, 1H), 5.69 (p, J=8.8 Hz, 1H), 5.44 (d, J=12.1 Hz, 1H), 4.67 (d, J=10.7 Hz, 1H), 4.30-4.15 (m, 3H), 3.99 (dt, J=13.2, 6.4 Hz, 3H), 3.89-3.79 (m, 1H), 3.62 (ddd, J=30.5, 18.6, 11.4 Hz, 5H), 3.39 (dd, J=9.4, 3.8 Hz, 2H), 3.21-3.13 (m, 1H), 3.08 (d, J=15.0 Hz, 3H), 2.99-2.74 (m, 6H), 2.26-2.18 (m, 5H), 2.16 (s, 2H), 2.14-1.94 (m, 3H), 1.86-1.68 (m, 2H), 1.57 (q, J=9.2, 5.8 Hz, 1H), 1.44-1.27 (m, 9H), 0.94 (d, J=6.6 Hz, 4H), 0.89 (dd, J=6.5, 2.5 Hz, 2H), 0.80 (d, J=6.3 Hz, 2H), 0.77-0.69 (m, 4H), 0.58 (d, J=20.4 Hz, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.53H.sub.71N.sub.9O.sub.8 962.55; found 962.5.
Example A675. Synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.SUP.3.S,4S,Z)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide
(930) ##STR01303## ##STR01304## ##STR01305##
Step 1.
(931) A mixture of 2-bromo-4-(ethoxycarbonyl)-1,3-oxazol-5-ylium (6.83 g, 31.19 mmol), EtOH (100.0 mL) and NaBH.sub.4 (4.72 g, 124.76 mmol) at 0° C. was stirred for 6 h at 0° C. under an air atmosphere. The reaction was quenched with H.sub.2O at 0° C. The resulting mixture was extracted with EtOAc (3×100 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure to afford (2-bromo-1,3-oxazol-4-yl)methanol (4.122 g, 74.3% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.4H.sub.4BrNO.sub.2 177.95; found 178.0.
(932) Step 2.
(933) A mixture of (2-bromo-1,3-oxazol-4-yl)methanol (4.30 g, 24.16 mmol), DCM (50 mL) and phosphorus tribromide (9809.39 mg, 36.24 mmol) at 0° C. was stirred overnight at 0° C. under an air atmosphere. The reaction was quenched by the addition of NaHCO.sub.3 (aq.) at 0° C. The resulting mixture was extracted with EtOAc (3×100 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure to afford 2-bromo-4-(bromomethyl)-1,3-oxazole (3.28 g, 56.4% yield) as a liquid. LCMS (ESI): m/z [M+H] calc'd for C.sub.4H.sub.3Br.sub.2NO 239.87; found 239.9.
(934) Step 3.
(935) A mixture of 2-bromo-4-(bromomethyl)-1,3-oxazole (3280.0 mg, 13.62 mmol), KOH (9M, 10 mL), 30 mL mixture of toluene/DCM (7/3) and tert-butyl 2-[(diphenylmethylidene)amino]acetate (5228.74 mg, 17.70 mmol) at −16° C. was stirred overnight under an air atmosphere. The resulting mixture was extracted with EtOAc (3×50 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure to afford tert-butyl (2S)-3-(2-bromo-1,3-oxazol-4-yl)-2-[(diphenylmethylidene)amino]propanoate (8.33 g, 80.6% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.23H.sub.23BrN.sub.2O.sub.3 455.10; found 455.1.
(936) Step 4.
(937) A mixture of tert-butyl (2S)-3-(2-bromo-1,3-oxazol-4-yl)-2-[(diphenylmethylidene)amino]propanoate (4100.0 mg, 9.0 mmol) and citric acid (1 N) (40.0 mL, 0.21 mmol), in THF (40 mL) at room temperature was stirred overnight under an air atmosphere. The reaction was quenched by the addition of HCl (aq.) (100 mL) at 0° C. The aqueous layer was extracted with EtOAc (3×100 mL). K.sub.2CO.sub.3 (aq.) (200 mL) was added to the resulting mixture and extracted with EtOAc (3×100 mL). The organic layer was concentrated under reduced pressure to afford tert-butyl (2S)-2-amino-3-(2-bromo-1,3-oxazol-4-yl)propanoate (1730 mg, 66% yield) as a dark yellow solid. LCMS (ESI): m/z [M+H] calc'd for

$C_{10}H_{15}BrN_2O_3$ 291.03; found 291.0.
(938) Step 5.
(939) A mixture of tert-butyl (2S)-2-amino-3-(2-bromo-1,3-oxazol-4-yl)propanoate (1780.0 mg, 6.11 mmol), TFA (10.0 mL) and DCM (10.0 mL) at 0° C. was stirred for overnight under an air atmosphere. The resulting mixture was concentrated under reduced pressure to afford (2S)-2-amino-3-(2-bromo-1,3-oxazol-4-yl)propanoic acid (1250 mg, 87% yield) as a dark yellow solid. LCMS (ESI): m/z [M+H] calc'd for $C_6H_7BrN_2O_3$ 234.97; found 234.9.
(940) Step 6.
(941) A mixture of di-tert-butyl dicarbonate (4178.58 mg, 19.15 mmol), THF (10 mL), H_2O (10 mL),(2S)-2-amino-3-(2-bromo-1,3-oxazol-4-yl)propanoic acid (1500.0 mg, 6.38 mmol) and $NaHCO_3$ (3216.74 mg, 38.29 mmol) at room temperature was stirred overnight under an air atmosphere. The reaction was quenched with H_2O at room temperature. The resulting mixture was concentrated under reduced pressure. The resulting mixture was extracted with EtOAc (3×100 mL). The aqueous layer was acidified to pH 6 with 1 M HCl (aq.). The resulting mixture was extracted with EtOAc (3×100 mL). The combined organic layers were dried over anhydrous Na_2SO_4. After filtration, the filtrate was concentrated under reduced pressure to afford (2S)-3-(2-bromo-1,3-oxazol-4-yl)-2-[(tert-butoxycarbonyl)amino]propanoic acid (920 mg, 43.0% yield) as an oil. LCMS (ESI): m/z [M+H] calc'd for $C_{11}H_{15}BrN_2O_5$ 335.02; found 335.0.
(942) Step 7.
(943) A mixture of 3-(2-bromo-1,3-oxazol-4-yl)-2-[(tert-butoxycarbonyl)amino]propanoic acid (850.0 mg, 2.54 mmol), methyl 1,2-diazinane-3-carboxylate (1.88 g, 13.04 mmol), DIPEA (1966.68 mg, 15.22 mmol), DCM (30.0 mL) and HATU (1446.48 mg, 3.80 mmol) at 0° C. was stirred for 3 h under an air atmosphere. The resulting mixture was extracted with DCM (3×50 mL). The combined organic layers were dried over anhydrous Na_2SO_4. After filtration, the filtrate was concentrated under reduced pressure. The resulting mixture was purified by reverse flash chromatography to afford methyl 1-[3-(2-bromo-1,3-oxazol-4-yl)-2-[(tert-butoxycarbonyl)amino]propanoyl]-1,2-diazinane-3-carboxylate (610 mg, 52.1% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{17}H_{25}BrN_4O$ 461.10; found 461.0.
(944) Step 8.
(945) A mixture of methyl 1-[3-(2-bromo-1,3-oxazol-4-yl)-2-[(tert-butoxycarbonyl)amino]propanoyl]-1,2-diazinane-3-carboxylate (570.0 mg, 1.24 mmol), LiOH (2.0 mL, 1 M aq.) and THF (2.0 mL) at 0° C. was stirred for 3 h under an air atmosphere. The resulting mixture was concentrated under reduced pressure. The resulting mixture was extracted with EtOAc (3×50 mL). The combined aqueous layers were acidified to pH 5 with 1N HCl (aq.). The aqueous phase was extracted with EtOAc (3×50 mL). The combined organic layers were dried over anhydrous Na_2SO_4. After filtration, the filtrate was concentrated under reduced pressure to afford 1-[3-(2-bromo-1,3-oxazol-4-yl)-2-[(tert-butoxycarbonyl)amino]propanoyl]-1,2-diazinane-3-carboxylic acid (500 mg, 90.5% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{16}H_{23}BrN_4O$ 447.09; found 446.8.
(946) Step 9.
(947) A mixture of 1-[3-(2-bromo-1,3-oxazol-4-yl)-2-[(tert-butoxycarbonyl)amino]propanoyl]-1,2-diazinane-3-carboxylic acid (450.0 mg, 1.01 mmol), 3-[(2M)-1-ethyl-2-[2-[(1S)-1-methoxyethyl]pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)indol-3-yl]-2,2-dimethylpropan-1-ol (743.19 mg, 1.51 mmol), DMAP (24.58 mg, 0.20 mmol), DCM (15.0 mL) and DCC (311.37 mg, 1.51 mmol) at 0° C. was stirred for 3 h under an air atmosphere. The resulting mixture was extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous Na_2SO_4. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by Prep-TLC to afford 3-(1-ethyl-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-indol-3-yl)-2,2-dimethylpropyl(S)-1-((S)-3-

(2-bromooxazol-4-yl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (330 mg, 35.6% yield) as a white solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.45H.sub.62BBrN.sub.6O.sub.9 921.39; found 921.4.

(948) Step 10.

(949) A mixture of 3-(1-ethyl-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-indol-3-yl)-2,2-dimethylpropyl(S)-1-((S)-3-(2-bromooxazol-4-yl)-2-((tert-butoxycarbonyl)amino)propanoyl)hexahydropyridazine-3-carboxylate (290.0 mg, 0.33 mmol), K.sub.3PO.sub.4 (206.64 mg, 0.97 mmol), X-Phos (30.94 mg, 0.07 mmol), XPhos Pd G3 (54.93 mg, 0.07 mmol), dioxane (5. mL) and H.sub.2O (1.0 mL) at 70° C. was stirred for 4 h under an argon atmosphere. The resulting mixture was extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by Prep-TLC to afford tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (130 mg, 56.0% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.50N.sub.6O.sub.7 715.38; found 715.3.

(950) Step 11.

(951) A mixture of tert-butyl ((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (120.0 mg), DCM (2.0 mL) and TFA (0.2 mL) at room temperature was stirred for 6 h under an air atmosphere. The resulting mixture was concentrated under reduced pressure. to afford (6.sup.3S,4S,2)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (90 mg, 87.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.34H.sub.42N.sub.6O.sub.5 615.33; found 615.3.

(952) Step 12.

(953) A mixture of (6.sup.3S,4S,2)-4-amino-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (200.0 mg, 0.33 mmol), (2R)-2-([[1-(diphenylmethyl)azetidin-3-yl]oxy]methyl)-3-methylbutanoic acid (172.49 mg, 0.49 mmol), DIPEA (420.48 mg, 3.25 mmol), DMF (3.0 mL) and HATU (148.44 mg, 0.39 mmol) at 0° C. was stirred for 3 h under an air atmosphere. The resulting mixture was extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by Prep-TLC to afford (2R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (154 mg, 77.0% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.56H.sub.67N.sub.7O.sub.7 950.52; found 950.6.

(954) Step 13.

(955) A mixture of (2R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (240.0 mg, 0.25 mmol), (Boc).sub.2O (165.37 mg, 0.76 mmol), MeOH (5.0 mL) and Pd(OH).sub.2 (72.0 mg, 0.51 mmol) at room temperature was stirred overnight under an H.sub.2 atmosphere. The resulting mixture was filtered, the filter cake was washed with MeOH (3×5 mL). The filtrate was concentrated under reduced pressure. The residue was purified by Prep-TLC to afford tert-butyl 3-((2R)-2-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-

dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamoyl)-3-methylbutoxy)azetidine-1-carboxylate (150 mg, 67.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.48H.sub.65N.sub.7O.sub.9 884.49; found 884.2.

(956) Step 14.

(957) A mixture of tert-butyl 3-((2R)-2-(((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)- pyridazinacycloundecaphane-4-yl)carbamoyl)-3-methylbutoxy)azetidine-1-carboxylate (150.0 mg), DCM (2.0 mL) and TFA (0.40 mL) at 0° C. was stirred for 3 h under an air atmosphere. The resulting mixture was concentrated under reduced pressure to afford (2R)-2-((azetidin-3-yloxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (120 mg, 90.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.43H.sub.57N.sub.7O.sub.7 784.44; found 784.2.

(958) Step 15.

(959) A mixture of (2R)-2-((azetidin-3-yloxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (130.0 mg, 0.17 mmol), sodium 4-(dimethylamino)-4-methylpent-2-ynoate (44.07 mg, 0.25 mmol), DMF (3.0 mL), DIPEA (64.29 mg, 0.50 mmol) and COMU (106.46 mg, 0.25 mmol) at 0° C. was stirred for 3 h under an air atmosphere. The resulting mixture was extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4. After filtration, the filtrate was concentrated under reduced pressure. The crude product was purified by reverse phase chromatography to afford (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.1-ethyl-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(2,4)-oxazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (25 mg, 16.4% yield) as a solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.77 (dd, J=4.8, 1.8 Hz, 1H), 8.55 (s, 1H), 8.14 (dd, J=8.3, 2.9 Hz, 1H), 7.82 (d, J=7.6 Hz, 1H), 7.76-7.65 (m, 3H), 7.54 (dd, J=7.7, 4.7 Hz, 1H), 7.54-7.02 (m, 1H), 5.72 (td, J=7.4, 3.4 Hz, 1H), 4.97 (d, J=11.9 Hz, 1H), 4.46-4.24 (m, 6H), 4.18-4.04 (m, 2H), 3.94 (dd, J=32.9, 7.8 Hz, 1H), 3.77-3.63 (m, 2H), 3.57 (s, 1H), 3.49 (s, 2H), 3.21 (s, 3H), 2.90 (d, J=14.6 Hz, 1H), 2.87-2.79 (m, 1H), 2.72 (td, J=15.5, 14.6, 3.1 Hz, 2H), 2.46 (s, 1H), 2.43-2.26 (m, 6H), 2.11-1.99 (m, 1H), 1.82-1.66 (m, 2H), 1.56-1.37 (m, 11H), 0.89 (dt, J=12.3, 7.7 Hz, 12H), 0.35 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.68N.sub.8O.sub.8 921.52; found 921.5.

Example A607. The synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((2.SUP.3.S,6.SUP.3.S,4S)-1.SUP.1.-ethyl-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-6.SUP.1.,6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(1,3)-piperidinacycloundecaphane-4-yl)-3-methylbutanamide

(960) ##STR01306## ##STR01307## ##STR01308##

Step 1.

(961) A mixture of Zn (44.18 g, 675.41 mmol) and I.sub.2 (8.58 g, 33.77 mmol) in DMF (120 mL) was stirred for 30 min at 50° C. under an argon atmosphere, followed by the addition of methyl (2R)-2-[(tert-butoxycarbonyl) amino]-3-iodopropanoate (72.24 g, 219.51 mmol) in DMF (200 mL). The reaction mixture was stirred at 50° C. for 2 h under an argon atmosphere. Then a mixture of 2,6-dibromo-pyridine (40 g, 168.85 mmol) and Pd(PPh.sub.3).sub.4 (39.02 g, 33.77 mmol) in DMF (200

mL) was added. The resulting mixture was stirred at 75° C. for 2 h, then cooled down to room temperature and extracted with EtOAc (1 L×3). The combined organic layers were washed with H_2O (1 L×3), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford methyl (2S)-3-(6-bromopyridin-2-yl)-2-[(tert-butoxycarbonyl) amino] propanoate (41 g, 67% yield) as oil. LCMS (ESI): m/z [M+H] calc'd for $C_{14}H_{19}BrN_2O_4$ 358.1; found 359.1.

(962) Step 2.

(963) To a solution of 3-[(2M)-2-[2-[(1 S)-1-methoxyethyl] pyridin-3-yl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-(2,2,2-trifluoroethyl) indol-3-yl]-2,2-dimethylpropan-1-ol (45.0 g, 82.35 mmol) in dioxane (400 mL) and H_2O (80 mL), were added potassium carbonate (28.45 g, 205.88 mmol), methyl (2S)-3-(6-bromopyridin-2-yl)-2-[(tert-butoxycarbonyl) amino] propanoate (35.5 g, 98.8 mmol), Pd(dtbpf)Cl_2 (5.37 g, 8.24 mmol) at room temperature. The reaction mixture was stirred at 70° C. for 2 h under a nitrogen atmosphere. The resulting mixture was extracted with EtOAc (500 mL×3). The combined organic layers were washed with H_2O (300 mL×3), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel chromatography to afford methyl (S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoate (48 g, 83% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{37}H_{45}F_3N_4O_6$ 698.3; found 699.4.

(964) Step 3.

(965) A solution of methyl (S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoate (52 g, 74.42 mmol) in THF (520 mL), was added LiOH (74.41 mL, 223.23 mmol) at 0° C.

(966) The reaction mixture was stirred at room temperature for 3 h. The resulting mixture was acidified to pH 5 with HCl (aq.) and extracted with EtOAc (1 L×3). The combined organic layers were washed with H_2O (1 L×3), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure to afford (S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoic acid (50 g, 98% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{36}H_{43}F_3N_4O_6$ 684.3; found 685.1.

(967) Step 4.

(968) To a solution of (S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoic acid (55 g, 80.32 mmol) in DCM (600 mL), were added DIPEA (415.23 g, 3212.82 mmol), and HATU (45.81 g, 120.48 mmol) at 0° C. The reaction mixture was stirred at room temperature for 12 h and then quenched with H_2O. The resulting mixture was extracted with EtOAc (1 L×3). The combined organic layers were washed with H_2O (1 L), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford methyl 1-((S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (63 g, 96% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for $C_{42}H_{53}F_3N_6O_7$ 810.4; found 811.3.

(969) Step 5.

(970) A solution of methyl 1-((S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoyl)hexahydropyridazine-3-carboxylate (50 g, 61.66 mmol) in THF (500 mL) and 3M LiOH (61.66 mL, 184.980 mmol) at 0° C. was stirred at room temperature for 3 h, then acidified to pH 5 with HCl (aq.). The resulting mixture was extracted with EtOAc (800 mL×3). The

combined organic layers were washed with H.sub.2O (800 mL), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure to afford 1-((S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (48 g, 97% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.41H.sub.51F.sub.3N.sub.6O.sub.7 796.3; found 797.1.

(971) Step 6.

(972) To a solution of 1-((S)-2-((tert-butoxycarbonyl)amino)-3-(6-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)pyridin-2-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (50 g, 62.74 mmol) in DCM (10 L) at 0° C., were added DIPEA (243.28 g, 1882.32 mmol), EDCI (360.84 g, 1882.32 mmol) and HOBT (84.78 g, 627.44 mmol). The reaction mixture was stirred at room temperature for 3 h, quenched with H.sub.2O and concentrated under reduced pressure. The residue was extracted with EtOAc (2 L×3). The combined organic layers were washed with H.sub.2O (2 L×3), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford tert-butyl ((4S)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-4-yl)carbamate (43.6 g, 89% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.41H.sub.49F.sub.3N.sub.6O.sub.6 778.3; found 779.3.

(973) Step 7.

(974) To a solution of tert-butyl ((4S)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-4-yl)carbamate (300 mg) in DCM (10 mL), was added TFA (3 mL) at 0° C. The reaction mixture was stirred at room temperature for 1 h. The resulting mixture was diluted with toluene (10 mL) and concentrated under reduced pressure three times to afford (4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-5,7-dione (280 mg, crude) as oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.36H.sub.41F.sub.3N.sub.6O.sub.4 679.2; found 678.3.

(975) Step 8.

(976) To a solution of (4S)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1′-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane- 5,7-dione (140 mg, 0.21 mmol) in MeCN (2 mL), were added DIPEA (266.58 mg, 2.06 mmol), N-(4-(tert-butoxycarbonyl)-1-oxa-4,9-diazaspiro[5.5]undecane-9-carbonyl)-N-methyl-L-valine (127.94 mg, 0.31 mmol) and CIP (114.68 mg, 0.41 mmol) at 0° C. The reaction mixture was stirred at room temperature for 1 h. The resulting mixture was concentrated under reduced pressure. The residue was extracted with EtOAc (10 mL×3). The combined organic layers were washed with H.sub.2O (10 mL×3), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel chromatography to afford tert-butyl 9-(((2S)-1-(((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl) pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-4-yl) amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-1-oxa-4,9-diazaspiro [5.5] undecane-4-carboxylate (170 mg, 76% yield) as solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.56H.sub.74F.sub.3N.sub.9O.sub.9 1073.5; found 1074.6.

(977) Step 9.

(978) To a solution of tert-butyl 9-(((2S)-1-(((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl) pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-4-yl) amino)-3-methyl-1-oxobutan-2-yl)(methyl)carbamoyl)-1-oxa-4,9-diazaspiro [5.5] undecane-4-carboxylate (160 mg, 0.15 mmol) in DCM (5 mL) at 0° C., was dropwise added TFA (1.5 mL). The reaction mixture was stirred at 0° C. for 1 h. The resulting mixture was diluted with toluene (10 mL) and concentrated under reduced pressure three times to afford N-((2S)-1-(((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl) pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-4-yl) amino)-3-methyl-1-oxobutan-2-yl)-N-methyl-1-oxa-4,9-diazaspiro [5.5]undecane-9-carboxamide (150 mg, crude) as oil. LCMS (ESI): m/z [M+H] calc'd for C.sub.51H.sub.65F.sub.3N.sub.9O.sub.7 973.5; found 974.4.
(979) Step 10.
(980) To a solution of N-((2S)-1-(((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl) pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-4-yl) amino)-3-methyl-1-oxobutan-2-yl)-N-methyl-1-oxa-4,9-diazaspiro [5.5] undecane-9-carboxamide (150 mg, 0.15 mmol) in DMF (3 mL), were added DIPEA (199.01 mg, 1.54 mmol), acrylic acid (16.64 mg, 0.23 mmol) and COMU (98.39 mg, 0.23 mmol) at 0° C. The reaction mixture was stirred at room temperature for 1 h and concentrated under reduced pressure. The residue was extracted with EtOAc (10 mL×3). The combined organic layers were washed with H.sub.2O (10 mL×3), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by silica gel chromatography and reverse phase chromatography to afford 4-acryloyl-N-((2S)-1-(((6.sup.3S,4S)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-1(5,3)-indola-6(1,3)-pyridazina-2(2,6)-pyridinacycloundecaphane-4-yl)amino)-3-methyl-1-oxobutan-2-yl)-N-methyl-1-oxa-4,9-diazaspiro[5.5]undecane-9-carboxamide (53 mg, 33% yield) as solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.79 (dd, J=4.8, 1.9 Hz, 2H), 8.09 (d, J=31.4 Hz, 1H), 7.96 (d, J=8.9 Hz, 1H), 7.90-7.72 (m, 3H), 7.64 (t, J=7.8 Hz, 1H), 7.56 (dd, J=7.8, 4.7 Hz, 1H), 7.03 (s, 1H), 6.86 (dd, J=16.6, 10.4 Hz, 1H), 6.20 (d, J=14.0 Hz, 1H), 5.73 (d, J=12.2 Hz, 2H), 5.47 (s, 1H), 5.35 (d, J=11.8 Hz, 1H), 4.68 (s, 1H), 4.28 (d, J=12.5 Hz, 1H), 4.13 (d, J=6.4 Hz, 1H), 3.92 (s, 1H), 3.84-3.64 (m, 6H), 3.59 (d, J=14.0 Hz, 3H), 3.50 (s, 3H), 3.10 (s, 5H), 3.02 (d, J=13.3 Hz, 2H), 2.85 (d, J=12.1 Hz, 3H), 2.08-1.89 (m, 2H), 1.81 (s, 1H), 1.75-1.51 (m, 5H), 1.39 (d, J=6.1 Hz, 4H), 1.24 (s, OH), 0.91-0.66 (m, 10H), 0.53 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.54H.sub.68F.sub.3N.sub.9O.sub.8 1027.5; found 1028.1.
Example A590. The synthesis of (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.SUP.3.S,4S,Z)-1.SUP.2.-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1′-(2,2,2-trifluoroethyl)-6.SUP.1., 6.SUP.2.,6.SUP.3.,6.SUP.4.,6.SUP.5.,6.SUP.6.-hexahydro-1.SUP.1.H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide
(981) ##STR01309## ##STR01310## ##STR01311## ##STR01312## ##STR01313##
Step 1.
(982) To a mixture of ethyl 2-ethoxy-2-iminoacetate (25.0 g, 172.23 mmol) and EtOH (250.0 mL) at 0° C. was added ammonium chloride (9.21 g, 172.23 mmol) in portions then stirred for 4 h at room temperature under an argon atmosphere. The resulting mixture was concentrated under reduced pressure and washed with Et.sub.2O (3×200 mL). The organic layers were combined and concentrated under reduced pressure. This resulted in ethyl 2-amino-2-iminoacetate hydrochloride (20 g, crude) as a light yellow solid. LCMS (ESI): m/z [M+H] calc'd for

C.sub.4H.sub.8N.sub.2O.sub.2 117.07; found 116.9.
(983) Step 2.
(984) To a mixture of ethyl 2-amino-2-iminoacetate hydrochloride (13.30 g, 87.17 mmol), H.sub.2O (50.0 mL) and Et.sub.2O (100.0 mL) at 0° C. was added sodium hypochlorite pentahydrate (7.79 g, 104.60 mmol) dropwise. The resulting mixture was stirred for 3 h under an argon atmosphere. The mixture was extracted with Et.sub.2O (3×200 mL). The resulting solution was washed with brine (3×100 mL). The organic phase was dried over anhydrous Na.sub.2SO.sub.4 and concentrated under reduced pressure to afford ethyl (Z)-2-amino-2-(chloroimino) acetate (7 g, crude) as a light yellow solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.4H.sub.7ClN.sub.2O.sub.2 151.03; found 150.8.
(985) Step 3.
(986) To a solution of ethyl (Z)-2-amino-2-(chloroimino) acetate (8.40 g, 55.792 mmol) and MeOH (130.0 mL) at 0° C. was added potassium thiocyanate (5.42 g, 55.79 mmol) in portions. The resulting mixture was stirred for 4 h at room temperature under an argon atmosphere. The reaction was quenched with H.sub.2O/Ice. The mixture was extracted with EtOAc (5×100 mL). The resulting organic phase was dried over anhydrous Na.sub.2SO.sub.4 and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford ethyl 5-amino-1,2,4-thiadiazole-3-carboxylate (2.3 g, 23.8% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.5H.sub.7N.sub.3O.sub.2S 174.03; found 173.8.
(987) Step 4.
(988) To a solution of ethyl 5-amino-1,2,4-thiadiazole-3-carboxylate (5.80 g, 33.49 mmol), MeCN (90.0 mL) and CuBr.sub.2 (11.22 g, 50.23 mmol) at 0° C. was added 2-methyl-2-propylnitrit (6.91 g, 66.98 mmol) dropwise under an argon atmosphere. The mixture was stirred for 30 min. The mixture was then stirred for 4 h at 50° C. The mixture was cooled to 0° C. and quenched with H.sub.2O/Ice. The mixture was extracted with EtOAc (3×100 mL). The resulting organic phase was dried over anhydrous Na.sub.2SO.sub.4 and concentrated under reduced pressure. The residue was purified by silica gel chromatography to afford ethyl 5-bromo-1,2,4-thiadiazole-3-carboxylate (6.2 g, 78.1% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.5H.sub.5BrN.sub.2O.sub.2S 236.93; found 237.1.
(989) Step 5.
(990) To a solution of (S)-3-(5-bromo-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-3-yl)-2,2-dimethylpropan-1-ol (16.60 g, 33.24 mmol), DCM (170.0 mL) and imidazole (5.66 g, 83.10 mmol) at 0° C. was added tert-butyl-chlorodiphenylsilane (11.88 g, 43.21 mmol) dropwise. The resulting mixture was stirred for 3 h at room temperature under an argon atmosphere. The reaction was quenched with H.sub.2O/Ice. The mixture was extracted with EtOAc (3×200 mL). The organic layer was washed with brine (3×100 mL), dried over anhydrous Na.sub.2SO.sub.4 and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford (S)-5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (22 g, 89.7% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.39H.sub.44BrF.sub.3N.sub.2O.sub.2Si 737.24; found 737.0.
(991) Step 6.
(992) To a solution of (S)-5-bromo-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (28.0 g, 37.95 mmol), toluene (270.0 mL), KOAc (9.31 g, 94.88 mmol) and bis(pinacolato)diboron (19.27 g, 75.90 mmol) at 0° C. was added Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (6.18 g, 7.59 mmol) in portions. The resulting mixture was stirred for 3 h at 90° C. under an argon atmosphere. The mixture was cooled to room temperature and quenched with H.sub.2O/Ice. The mixture was extracted with EtOAc (3×200 mL). The resulting organic phase was dried over anhydrous Na.sub.2SO.sub.4 and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford (S)-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-5-(4,4,5,5-

tetramethyl-1,3,2-dioxaborolan-2-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (28.2 g, 94.7% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{45}H_{56}BF_3N_2O_4Si$ 785.41; found 785.4.

(993) Step 7.

(994) To a solution of (S)-3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-(2,2,2-trifluoroethyl)-1H-indole (19.60 g, 24.97 mmol), 1,4-dioxane (200 mL), H_2O (40 mL), ethyl 5-bromo-1,2,4-thiadiazole-3-carboxylate (5.92 g, 24.97 mmol) and K_3PO_4 (13.25 g, 62.43 mmol) at 0° C. was added Pd(dtbpf)Cl_2 (1.63 g, 2.50 mmol) in portions. The resulting mixture was stirred for 1.5 h at 75° C. under an argon atmosphere. The mixture was cooled to 0° C. and quenched with H_2O/Ice and extracted with EtOAc (3×200 mL). The resulting organic phase was dried over anhydrous Na_2SO_4 and concentrated under reduced pressure.

(995) The residue was purified by silica gel column chromatography to afford ethyl (S)-5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazole-3-carboxylate (14 g, 68.8% yield) as a yellow solid. LCMS (ESI): m/z [M+H] calc'd for $C_{44}H_{49}F_3N_4O_4SSi$ 815.33; found 815.2.

(996) Step 8.

(997) To a solution of ethyl (S)-5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazole-3-carboxylate (13.60 g, 16.69 mmol) and EtOH (140.0 mL) at 0° C. was added $NaBH_4$ (3.16 g, 83.43 mmol) in portions. The resulting mixture was stirred for 3 h then quenched with H_2O/Ice. The resulting mixture was washed with brine (3×100 mL) and extracted with EtOAc (3×200 mL). The combined organic layers were dried over anhydrous Na_2SO_4 and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford (S)-5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-ol (9.7 g, 75.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{42}H_{47}F_3N_4O_3SSi$ 773.32; found 773.3.

(998) Step 9.

(999) To a solution of (S)-5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-ol (9.70 g, 12.55 mmol), DCM (100.0 mL) and CBr_4 (8.32 g, 25.10 mmol) at 0° C. was added PPh_3 (6.58 g, 25.10 mmol) in DCM (20.0 mL) dropwise. The resulting mixture was stirred for 2 h under an argon atmosphere then quenched with H_2O/Ice. The mixture was extracted with DCM (3×200 mL). The resulting organic phase was dried over anhydrous Na_2SO_4 and concentrated under reduced pressure. The residue was purified by reverse flash chromatography to afford (S)-3-(bromomethyl)-5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazole (9.5 g, 90.6% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for $C_{42}H_{46}BrF_3N_4O_2SSi$ 835.23; found 834.9.

(1000) Step 10.

(1001) To a stirred solution of (S)-3-(bromomethyl)-5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-(1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazole (9.40 g, 11.25 mmol), toluene (84.0 mL), DCM (36.0 mL), tert-butyl 2-[(diphenylmethylidene)amino]acetate (3.32 g, 11.25 mmol) and O-Allyl-N-(9-anthracenylmethyl)cinchonidinium bromide (0.68 g, 1.13 mmol) at 0° C. was added 9M KOH aqueous (94.0 mL) dropwise. The resulting mixture was stirred overnight under an argon atmosphere. The mixture was extracted with EtOAc (3×200 mL) and the organic phase was concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford tert-butyl (S)-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-

methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)-2-((diphenylmethylene)amino)propanoate (9 g, 76.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.61H.sub.66F.sub.3N.sub.5O.sub.4SSi 1050.46; found 1050.8.

(1002) Step 11.

(1003) To a solution of tert-butyl (S)-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)-2-((diphenylmethylene)amino)propanoate (8.0 g, 7.62 mmol) and DCM (40.0 mL) at 0° C. solution was added TFA (40.0 mL) dropwise. The resulting mixture was stirred overnight at room temperature then concentrated under reduced pressure. The residue was basified to pH 8 with NaHCO.sub.3. The mixture was extracted with EtOAc (3×200 mL). The organic phase was concentrated under reduced pressure. The residue was purified by reverse flash chromatography to afford (S)-2-amino-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoic acid (5 g, 79.1% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.44H.sub.50F.sub.3N.sub.5O.sub.4SSi 830.34; found 830.2.

(1004) Step 12.

(1005) To a solution of (S)-2-amino-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoic acid (4.70 g, 5.66 mmol), DCM (50.0 mL) and Et.sub.3N (2.86 g, 28.31 mmol) at 0° C. was added (Boc).sub.2O (1.36 g, 6.23 mmol) dropwise. The resulting mixture was stirred for 3 h at room temperature under an argon atmosphere then concentrated under reduced pressure and purified by reverse flash chromatography to afford (S)-2-((tert-butoxycarbonyl)amino)-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoic acid (5 g, 94.9% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.49H.sub.58F.sub.3N.sub.5O.sub.6SSi 930.39; found 930.3.

(1006) Step 13.

(1007) To a mixture of (S)-2-((tert-butoxycarbonyl)amino)-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoic acid (5.30 g, 5.70 mmol), DMF (60.0 mL), methyl 1,2-diazinane-3-carboxylate (1.64 g, 11.40 mmol) and DIPEA (22.09 g, 170.94 mmol) at 0° C. was added HATU (2.82 g, 7.41 mmol) in DMF (5 mL) dropwise. The resulting mixture was stirred for 3 h at room temperature under an argon atmosphere. The reaction was then quenched with H.sub.2O/Ice. The mixture was extracted with EtOAc (3×100 mL) and the organic phase was washed with brine (3×100 mL). The resulting mixture was dried over anhydrous Na.sub.2SO.sub.4 and concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoyl)hexahydropyridazine-3-carboxylate (5.6 g) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.55H.sub.68F.sub.3N.sub.7O.sub.7SSi 1056.47; found 1056.2.

(1008) Step 14.

(1009) A mixture of methyl (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(5-(3-(3-((tert-butyldiphenylsilyl)oxy)-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoyl)hexahydropyridazine-3-carboxylate (5.60 g, 5.30 mmol) and TBAF in THF (56.0 mL) was stirred overnight at 40° C. under an argon atmosphere. The reaction was quenched with sat. NH.sub.4Cl (aq.). The mixture was extracted with EtOAc (3×100 mL) and the organic phase was concentrated under reduced pressure. The residue was purified by reverse flash chromatography to afford (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(5-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-

trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (4.1 g, 96.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.38H.sub.48F.sub.3N.sub.7O.sub.7S 804.34; found 804.3.

(1010) Step 15.

(1011) To a solution of (S)-1-((S)-2-((tert-butoxycarbonyl)amino)-3-(5-(3-(3-hydroxy-2,2-dimethylpropyl)-2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-1-(2,2,2-trifluoroethyl)-1H-indol-5-yl)-1,2,4-thiadiazol-3-yl)propanoyl)hexahydropyridazine-3-carboxylic acid (4.0 g, 5.0 mmol) and DCM (450.0 mL) at 0° C. were added DIPEA (51.45 g, 398.08 mmol), HOBt (6.72 g, 49.76 mmol) and EDCI (57.23 g, 298.55 mmol) in portions. The resulting mixture was stirred for 16 h at room temperature under an argon atmosphere. The reaction was quenched with H.sub.2O/Ice and extracted with EtOAc (3×30 mL). The organic phase was concentrated under reduced pressure. The residue was purified by silica gel column chromatography to afford tert-butyl ((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (1.7 g, 43.5% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.38H.sub.46F.sub.3N.sub.7O.sub.6S 786.33; found 786.3.

(1012) Step 16.

(1013) To a solution of ((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamate (300.0 mg, 0.38 mmol) and DCM (2.0 mL) at 0° C. was added TFA (1.0 mL) dropwise. The resulting mixture was stirred for 2 h at room temperature then concentrated under reduced pressure. The residue was basified to pH 8 with saturated NaHCO.sub.3 (aq.). The mixture was extracted with EtOAc (3×20 mL). The organic phase was concentrated under reduced pressure to afford (6.sup.3S,4S,Z)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (270 mg, crude) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.33H.sub.38F.sub.3N.sub.7O.sub.4S 686.27; found 686.1.

(1014) Step 17.

(1015) To a solution of (6.sup.3S,4S,Z)-4-amino-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-5,7-dione (160.0 mg, 0.23 mmol), (R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-3-methylbutanoic acid (123.70 mg, 0.35 mmol) and DMF (2.0 mL) at 0° C. were added DIPEA (603.09 mg, 4.660 mmol) and COMU (119.91 mg, 0.28 mmol) in DMF (0.5 mL). The resulting mixture was stirred for 2 h at room temperature under an argon atmosphere. The reaction was quenched with H.sub.2O/Ice and extracted with EtOAc (3×20 mL). The organic phase was washed with brine (3×10 mL) and concentrated under reduced pressure. The residue was purified by Prep-TLC to afford (2R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (160 mg, 67.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.55H.sub.63F.sub.3N.sub.8O.sub.6S 1021.46; found 1021.4.

(1016) Step 18.

(1017) To a solution of (2R)-2-(((1-benzhydrylazetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-

2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (160.0 mg, 0.16 mmol) and MeOH (5.0 mL) at 0° C. was added (Boc).sub.2O (85.49 mg, 0.39 mmol) dropwise followed by Pd/C (320.0 mg) in portions. The resulting mixture was stirred overnight at room temperature under a hydrogen atmosphere. The resulting mixture was filtered and the filter cake was washed with EtOAc (3×20 mL). The filtrate was concentrated under reduced pressure. The residue was purified by Prep-TLC to afford tert-butyl 3-((2R)-2-(((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamoyl)-3-methylbutoxy)azetidine-1-carboxylate (80 mg, 53.5% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.47H.sub.61F.sub.3N.sub.8O.sub.8S 955.44; found 955.2.

(1018) Step 19.

(1019) To a solution of tert-butyl 3-((2R)-2-(((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1′-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)carbamoyl)-3-methylbutoxy)azetidine-1-carboxylate (120.0 mg, 0.13 mmol) and DCM (0.80 mL) at 0° C. was added TFA (0.4 mL) dropwise and the resulting mixture was stirred for 2 h at room temperature. The mixture was basified to pH 8 with saturated NaHCO.sub.3 (aq.). The mixture was extracted with EtOAc (3×10 mL) and concentrated under reduced pressure. The residue was purified by reverse flash chromatography to afford (2R)-2-((azetidin-3-yloxy)methyl)-N-((6.sup.3S,4S,2)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (40 mg, 37.2% yield) as a solid. LCMS (ESI): m/z [M+H] calc'd for C.sub.42H.sub.53F.sub.3N.sub.8O.sub.6S 855.38; found 855.3.

(1020) Step 20.

(1021) To a solution of (2R)-2-((azetidin-3-yloxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1′-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (32.0 mg, 0.037 mmol),4-(dimethylamino)-4-methylpent-2-ynoic acid (11.62 mg, 0.074 mmol) and DMF (0.50 mL) at 0° C. were added DIPEA (193.49 mg, 1.48 mmol) and COMU (19.23 mg, 0.044 mmol) in DMF (0.1 mL) dropwise. The resulting mixture was stirred for 2 h at room temperature under an argon atmosphere. The reaction was quenched with H.sub.2O/Ice and extracted with EtOAc (3×20 mL). The organic phase was washed with brine (3×10 mL) and concentrated under reduced pressure. The crude product (60 mg) was purified by reverse phase chromatography to afford (2R)-2-(((1-(4-(dimethylamino)-4-methylpent-2-ynoyl)azetidin-3-yl)oxy)methyl)-N-((6.sup.3S,4S,Z)-1.sup.2-(2-((S)-1-methoxyethyl)pyridin-3-yl)-10,10-dimethyl-5,7-dioxo-1.sup.1-(2,2,2-trifluoroethyl)-6.sup.1,6.sup.2,6.sup.3,6.sup.4,6.sup.5,6.sup.6-hexahydro-1.sup.1H-8-oxa-2(5,3)-thiadiazola-1(5,3)-indola-6(1,3)-pyridazinacycloundecaphane-4-yl)-3-methylbutanamide (11.7 mg, 30.9% yield) as a solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.80 (dd, J=4.7, 1.8 Hz, 1H), 8.58 (s, 1H), 8.30 (d, J=8.9 Hz, 1H), 7.92 (d, J=8.6 Hz, 1H), 7.84-7.69 (m, 2H), 7.56 (dd, J=7.8, 4.8 Hz, 1H), 5.78 (t, J=8.6 Hz, 2H), 5.10 (d, J=12.1 Hz, 1H), 4.91 (dd, J=16.9, 8.8 Hz, 1H), 4.31 (d, J=6.6 Hz, 6H), 4.05 (dd, J=16.3, 6.6 Hz, 2H), 3.87 (d, J=6.3 Hz, 1H), 3.75 (d, J=11.3 Hz, 1H), 3.61 (d, J=11.0 Hz, 1H), 3.53 (d, J=9.8 Hz, 1H), 3.45 (s, 3H), 3.25 (s, 1H), 3.09 (d, J=10.4 Hz, 1H), 3.01 (d, J=14.5 Hz, 1H), 2.79 (s, 1H), 2.45-2.35 (m, 2H), 2.20 (d, J=5.4 Hz, 6H), 2.15 (d, J=12.0 Hz, 1H), 1.81 (d, 2H), 1.74-1.65 (m, 1H), 1.54 (s, 1H), 1.41-1.30 (m, 9H), 1.24 (s, 1H), 0.94 (s, 3H), 0.90-0.81 (m, 5H), 0.29 (s, 3H). LCMS (ESI): m/z [M+H] calc'd for C.sub.50H.sub.64F.sub.3N.sub.9O.sub.7S 992.47; found 992.5.

(1022) The following table of compounds (Table 3) were prepared using the aforementioned methods or variations thereof, as is known to those of skill in the art.

(1023) TABLE-US-00004 TABLE 3 Exemplary Compounds Prepared by Methods of the Present Invention LCMS (ESI): m/z Ex# [M + H] Found A1 944.5 A2 888.6 A3 903.2 A4 876.3 A5 903.4 A6 945.8 A7 1058.1 A8 960.8 A9 921.7 A10 928.8 A11 936.4 A12 942.5 A13 932.5 A14 917.5 A15 961.5 A16 890.6 A17 875.4 A18 918.5 A19 975.6 A20 934.5 A21 903.3 A22 903.6 A23 844.6 A24 887.6 A25 921.4 A26 930.4 A27 915.6 A28 890.6 A29 903.5 A30 902.7 A31 901.4 A32 1003.4 A33 917.4 A34 942.7 A35 956.1 A36 1005.5 A37 917.5 A38 959.5 A39 942.9 A40 915.7 A41 917.5 A42 888.6 A43 942.5 A44 904.35 A45 846.35 A46 982.5 A47 960.1 A48 905.3 A49 981.8 A50 916.9 A51 946.7 A52 916.5 A53 921.6 A54 947.7 A55 916.5 A56 904.5 A57 930.5 A58 895.5 A59 944.7 A60 931.5 A61 847.7 A62 871.5 A63 930.4 A64 893.7 A65 874.7 A66 876.5 A67 970.5 A68 915.4 A69 927.3 A70 876.4 A71 933.5 A72 1003.5 A73 903.6 A74 885.5 A75 904.7 A76 904.5 A77 860.6 A78 1018.5 A79 947.5 A80 933.3 A81 917.8 A82 908.7 A83 890.6 A84 890.6 A85 891.5 A86 885.5 A87 947.7 A88 945.5 A89 944.8 A90 898 A91 890.5 A92 903.4 A93 901.4 A94 881.5 A95 987.8 A96 945.7 A97 931.8 A98 943.8 A99 849.7 A100 942.8 A101 893.4 A102 904.4 A103 947.5 A104 931.7 A105 915.6 A106 902.5 A107 945.7 A108 902.3 A109 906.5 A110 793.4 A111 946.7 A112 905.14 A113 849.6 A114 888.4 A115 888.4 A116 986.5 A117 879.5 A118 881.6 A119 874.5 A120 929.7 A121 943.6 A122 888.8 A123 959.3 A124 924.6 A125 962.5 A126 986.5 A127 946.6 A128 903.4 A129 1001.2 A130 930.5 A131 959.5 A132 940.5 A133 969.7 A134 990.5 A135 933.5 A136 875.4 A137 889.5 A138 929.4 A139 930.4 A140 943.7 A141 943.8 A142 901.7 A143 852.3 A144 886.7 A145 916.7 A146 963.4 A147 864.4 A148 885.5 A149 947.5 A150 902.7 A151 904.5 A152 885.5 A153 949.6 A154 907.5 A155 944.5 A156 943.7 A157 943.5 A158 904.5 A159 904.5 A160 858.5 A161 947.4 A162 973.5 A163 989.4 A164 947.5 A165 960.3 A166 907.7 A167 918.5 A168 907.4 A169 859.4 A170 874.3 A171 867.5 A172 883.6 A173 900.5 A174 848.4 A175 959.5 A176 944.5 A177 935.4 A178 930.4 A179 849.5 A180 849.5 A181 918.8 A182 1002.2 A183 961.5 A184 1003.5 A185 928.5 A186 992.5 A187 945.5 A188 960.4 A189 1000.6 A190 934.3 A191 985.5 A192 971.5 A193 871.6 A194 918.5 A195 927.4 A196 972.3 A197 896.4 A198 916.6 A199 916.6 A200 987.4 A201 993.7 A202 896.4 A203 987.6 A204 987.6 A205 933.5 A206 930.5 A207 930.5 A208 958.8 A209 1015.7 A210 906.5 A211 905.5 A212 901.3 A213 969.5 A214 933.6 A215 890.3 A216 916.4 A217 930.5 A218 983.5 A219 917.4 A220 996.5 A221 915.4 A222 930.05 A223 920.01 A224 983.5 A225 982.7 A226 873.7 A227 887.7 A228 911 A229 896.7 A230 905.5 A231 912.4 A232 849.4 A233 905.4 A234 1053.3 A235 835.5 A236 882.3 A237 1015.4 A238 971.5 A239 869.4 A240 884.4 A241 987.4 A242 877.4 A243 863.5 A244 1029.6 A245 1029.5 A246 501.4 A247 985.5 A248 896.4 A249 984.5 A250 927.4 A251 985.4 A252 927.3 A253 990.4 A254 933.3 A255 930.7 A256 890.4 A257 947.4 A258 990.3 A259 933.3 A260 918.4 A261 975.5 A262 919.4 A263 976.5 A264 990.5 A265 933.5 A266 945.5 A267 987.5 A268 973.5 A269 487.3 A270 959.4 A271 861.3 A272 847.6 A273 861.6 A274 930.5 A275 916.3 A276 892.8 A277 895.7 A278 879.5 A279 895.4 A280 954.6 A281 916.6 A282 952.5 A283 931.4 A284 889.5 A285 839.4 A286 877.4 A287 894.7 A288 879.3 A289 908.7 A290 971.6 A291 978.5 A292 987.5 A293 921.4 A294 902.8 A295 916.4 A296 930.8 A297 938.4 A298 952.6 A299 966.7 A300 919.6 A301 907.7 A302 895.8 A303 895.4 A304 897.5 A305 968.6 A306 968.7 A307 952.6 A308 952.7 A309 933.6 A310 926.8 A311 944.7 A312 897.5 A313 976.5 A314 893.3 A316 930.5 A317 982.6 A318 883.5 A319 916.5 A320 886.6 A321 900.6 A322 1004.4 A323 976.5 A324 969.5 A325 1033.7 A326 997.5 A327 906.6 A328 986.7 A329 890.5 A330 895.6 A331 879.5 A332 942.5 A333 919.6 A334 930.7 A335 924.5 A336 927.4 A337 501.4 A338 919.4 A339 971.6 A340 895.7 A341 938.5 A342 906.8 A343 992.4 A344 901.40 A345 897.40 A346 942.50 A347 896.30 A348 1,002.30 A349 924.6 A350 968.2 A351 997.6 A352 952.7 A353 966.3 A354 966.2 A355 855.6 A356 910.4 A357 1091.0 A358 902.5 A359 904.7 A360 924.5 A361 967.5 A362 936.6 A363 961.4 A364 946.5 A365 995.6 A366 921.5 A367 950.6 A368 924.5 A369 924.5 A370 1087.3 A371 902.5 A372 992.1 A373 902.6 A374 1,034.40 A375 993.50 A376 984.40 A377 939.50 A378 897.50 A379 1005.6 A380 949.4 A381 921.5 A382 938.5 A383 938.5

A384 924.5 A385 938.3 A386 938.4 A387 938.5 A388 938.5 A389 936.5 A390 924.5 A391 1011.6 A392 958.0 A393 922.4 A394 950.3 A395 936.5 A396 924.4 A397 924.7 A398 914.6 A399 902.5 A400 956.6 A401 955.9 A402 940.8 A403 948.40 A404 980.40 A405 911.50 A406 938.7 A407 922.5 A408 922.5 A409 922.6 A410 1046.9 A411 990.7 A412 912.7 A413 914.7 A414 983.5 A415 950.5 A416 924.6 A417 936.7 A418 938.5 A419 922.3 A420 936.7 A421 924.5 A422 910.5 A423 1028.9 A424 934.6 A425 1012.5 A426 912.5 A427 895.30 A428 951.30 A429 911.50 A430 938.5 A431 952.3 A432 924.2 A433 924.2 A434 924.7 A435 910.5 A436 924.6 A437 912.5 A438 898.5 A439 1016.6 A440 990.5 A441 939.6 A442 889.2 A443 913.5 A444 913.5 A445 890.1 A446 957.6 A447 880.5 A448 888.4 A449 920.5 A450 897.2 A451 902.6 A452 888.4 A453 901.3 A454 1044.6 A455 949.2 A456 1059.5 A457 916.7 A458 916.2 A459 902.3 A460 874.5 A461 957.9 A462 854.3 A463 910.7 A464 964.5 A465 964.5 A466 1033.7 A467 874.5 A468 879.6 A469 902.6 A470 900.6 A471 900.6 A472 964.6 A473 854.3 A474 914.4 A475 914.40 A476 983.6 A477 907.5 A478 900.5 A479 993.6 A480 902.2 A481 930.6 A482 974.6 A483 847.5 A484 847.4 A485 938.4 A486 893.5 A487 900.6 A488 888.5 A489 962.6 A490 896.5 A491 910.2 A492 910.5 A493 910.5 A494 886.3 A495 907.5 A496 900.4 A497 902.7 A498 902.6 A499 998.7 A500 897.5 A501 906.3 A502 949.2 A503 920.6 A504 920.8 A505 920.0 A506 879.3 A507 897.4 A508 942.4 A509 942.4 A510 899.3 A511 919.5 A512 900.5 A513 983.7 A514 983.7 A515 985.7 A516 985.7 A517 892.6 A518 920.6 A519 920.4 A520 990.6 A521 990.6 A522 945.3 A523 945.4 A524 914.5 A525 914.4 A526 1005.4 A527 972.4 A528 985.4 A529 987.4 A530 898.5 A531 912.3 A532 1013.6 A533 999.3 A534 983.4 A535 919.5 A536 961.7 A537 961.6 A538 999.7 A539 960.4 A540 1050.2 A541 1078.7 A542 1155.7 A543 952.6 A544 952.6 A545 981.2 A546 922.3 A547 922.5 A548 950.5 A549 997.4 A550 1025.6 A551 997.5 A552 950.4 A553 974.6 A554 957.5 A555 1054.4 A556 972.5 A557 952.4 A558 959.6 A559 1011.7 A560 1011.6 A561 922.5 A562 954.3 A563 978.5 A564 982.5 A565 1039.3 A566 1039.4 A567 1039.4 A568 1039.3 A569 997.4 A570 974.7 A571 1038.5 A572 1056.5 A573 1063.5 A574 990.6 A575 1004.5 A576 991.5 A577 1062.3 A578 1048.3 A579 963.3 A580 966.6 A581 966.6 A582 953.5 A583 938.5 A584 944.7 A585 944.6 A586 950.6 A587 962.6 A588 938.5 A589 1016.6 A590 992.5 A591 1008.7 A592 997.5 A593 1020.5 A594 1020.6 A595 1000.6 A596 966.7 A597 946.6 A598 979.6 A599 958.7 A600 887.4 A601 985.5 A602 1036.5 A603 989.9 A604 1006.1 A605 1059.7 A606 1032.6 A607 1028.1 A608 1027.1 A609 966.6 A610 945.6 A611 1021.5 A612 920.6 A613 956.1 A614 992.5 A615 1010.5 A616 1010.5 A617 990.5 A618 950.7 A619 950.4 A620 938.1 A621 966.1 A622 982.6 A623 999.6 A624 1019.5 A625 1019.5 A626 913.5 A627 913.4 A628 1046.5 A629 952.2 A630 967.6 A631 942.5 A632 942.6 A633 938.6 A634 938.6 A635 978.6 A636 916.6 A637 909.5 A638 909.5 A639 928.3 A640 942.0 A641 942.6 A642 1047.5 A643 966.6 A644 944.4 A645 986.5 A646 937.5 A647 1012.5 A648 944.5 A649 921.6 A650 1007.6 A651 964.2 A652 1903.1 A653 938.6 A654 938.6 A655 934.4 A656 1005.5 A657 960.7 A658 1003.3 A659 974.6 A660 925.5 A661 936.6 A662 936.6 A663 935.5 A664 1047.5 A665 1035.5 A666 1035.2 A667 936.3 A668 965.6 A669 1049.7 A670 1047.7 A671 964.6 A672 997.9 A673 1015.3 A674 975.5 A675 921.5 A676 966.5 A677 966.4 A678 950.5 A679 990.2 A680 943.6 A681 980.6 A682 968.6 A683 1000.6 A684 915.0 A685 977.6 A686 922.5 A687 936.4 A688 936.5 A689 1036.9 A690 979.6 A691 923.5 A692 962.5 A693 906.5 A694 896.1 A695 980.7 A696 978.4 A697 978.3 A698 964.3 A699 1003.3 A700 927.2 A701 993.7 A702 950.6 A703 985.5 A704 938.6 A705 925.5 A706 952.5 A707 1016.5 A708 903.2 A709 981.5 A710 967.5 A711 964.5 A712 937.9 A713 939.9 A714 952.5 A715 937.4 A716 893.5 A717 924.6 A718 936.6 A719 950.5 A720 935.3 A721 937.3 A722 929.4 A723 910.4 A724 985.4 A725 919.3 A726 1010.4 A727 1025.5 A728 924.6 A729 993.5 A730 993.2 A731 922.7 A732 928.6 A733 997.6 A734 936.5 A735 952.5 A736 922.2 A737 922.1 A738 924.5 A739 924.5 A740 895.5 A741 895.4 Blank = not determined

Matched Pair Analysis

(1024) FIGS. **1**A-**1**B compare the potency in two different cell-based assays of compounds of Formula BB of the present invention (points on the right) and corresponding compounds of Formula AA (points on the left) wherein a H is replaced with (S)Me. The y axes represent pERK EC50 (FIG.

1A) or CTG IC50 (FIG. **1**B) as measured in an H358 cell line. Assay protocols are below. The linked points represent a matched pair that differs only between H and (S)Me substitution. Each compound of Formula BB demonstrated reduced potency in cell assays compared to the corresponding compound of Formula AA.

(1025) Biological Assays

(1026) Potency Assay: pERK

(1027) The purpose of this assay is to measure the ability of test compounds to inhibit K-Ras in cells. Activated K-Ras induces increased phosphorylation of ERK at Threonine 202 and Tyrosine 204 (pERK). This procedure measures a decrease in cellular pERK in response to test compounds. The procedure described below in NCI-H358 cells is applicable to K-Ras G12C.

(1028) Note: This protocol may be executed substituting other cell lines to characterize inhibitors of other RAS variants, including, for example, AsPC-1 (K-Ras G12D), Capan-1 (K-Ras G12V), or NCI-H1355 (K-Ras G13C).

(1029) NCI-H358 cells were grown and maintained using media and procedures recommended by the ATCC. On the day prior to compound addition, cells were plated in 384-well cell culture plates (40 μl/well) and grown overnight in a 37° C., 5% CO.sub.2 incubator. Test compounds were prepared in 10, 3-fold dilutions in DMSO, with a high concentration of 10 mM. On the day of assay, 40 nL of test compound was added to each well of cell culture plate using an Echo550 liquid handler (LabCyte®). Concentrations of test compound were tested in duplicate. After compound addition, cells were incubated 4 hours at 37° C., 5% CO.sub.2. Following incubation, culture medium was removed and cells were washed once with phosphate buffered saline.

(1030) In some experiments, cellular pERK level was determined using the AlphaLISA SureFire Ultra p-ERK1/2 Assay Kit (PerkinElmer). Cells were lysed in 25 μL lysis buffer, with shaking at 600 RPM at room temperature. Lysate (10 μL) was transferred to a 384-well Opti-plate (PerkinElmer) and 5 μL acceptor mix was added. After a 2-hour incubation in the dark, 5 μL donor mix was added, plate was sealed, and incubated 2 hours at room temperature. Signal was read on an Envision plate reader (PerkinElmer) using standard AlphaLISA settings. Analysis of raw data was carried out in Excel (Microsoft) and Prism (GraphPad). Signal was plotted vs. the decadal logarithm of compound concentration, and IC.sub.50 was determined by fitting a 4-parameter sigmoidal concentration response model.

(1031) In other experiments, cellular pERK was determined by In-Cell Western. Following compound treatment, cells were washed twice with 200 μL tris buffered saline (TBS) and fixed for 15 minutes with 150 μL 4% paraformaldehyde in TBS. Fixed cells were washed 4 times for 5 minutes with TBS containing 0.1% Triton X-100 (TBST) and then blocked with 100 μL Odyssey blocking buffer (LI-COR) for 60 minutes at room temperature. Primary antibody (pERK, CST-4370, Cell Signaling Technology) was diluted 1:200 in blocking buffer, and 50 μL was added to each well and incubated overnight at 4° C. Cells were washed 4 times for 5 minutes with TBST. Secondary antibody (IR-800CW rabbit, LI-COR, diluted 1:800) and DNA stain DRAQ5 (LI-COR, diluted 1:2000) were added and incubated 1-2 hours at room temperature. Cells were washed 4 times for 5 minutes with TBST. Plates were scanned on a LI-COR Odyssey CLx Imager. Analysis of raw data was carried out in Excel (Microsoft) and Prism (GraphPad). Signal was plotted vs. the decadal logarithm of compound concentration, and IC.sub.50 was determined by fitting a 4-parameter sigmoidal concentration response model.

(1032) The following compounds exhibited a pERK EC50 of under 5 uM (H358 KRAS G12C): A48, A15, A272, A174, A163, A453, A447, A279, A240, A214, A225, A136, A226, A219, A228, A21, A12, A78, A424, A219, A378, A224, A4, A53, A187, A218, A213, A314, A220, A208, A24, A9, A126, A345, A46, A203, A210, A184, A 469, A366, A113, A328, A693, A639, A364, A100, A249, A486, A307, A347, A33, A210, A192, A285, A468, A185, A 612, A109, A284, A200, A2, A6, A606, A325, A139, A496, A393, A561, A125, A494, A547, A215, A258, A195, A259, A212, A637, A53, A63, A68, A178, A189, A205, A78, A254, A690, A563, A14, A19, A92, A576, A278, A331, A42, A6 7, A209, A350, A562, A652, A703, A623, A191, A241, A199, A193, A478, A251, A177, A222, A23, A59, A26, A211, A106, A279, A120, A7, A134, A521, A116, A467, A694, A729, A151, A110, A277, A340, A221, A723, A13, A442, A6 11, A50, A190, A553, A696, A211, A303, A613, A37, A146, A666, A688, A216, A390, A548, A238, A160, A183, A16 4, A451, A481, A524, A1, A186, A37, A635, A71, A269, A289, A489, A400, A731, A497, A568, A274, A253, A471, A 720, A241, A179, A180, A426, A117, A363, A716, A423, A217, A708, A227, A3, A12, A8, A381, A84, A408, A85, A1 71, A263, A473, A258, A564, A118, A103, A565, A641, A655, A47, A11, A392, A169, A487, A640, A206, A449, A35 8, A192, A148, A4, A41, A5, A18, A301, A10, A65, A554, A159, A264, A99, A79, A142, A143, A25, A98, A80, A101, A 730, A212, A359, A61, A441, A283, A413, A717, A145, A182, A62, A181, A233, A232, A634, A495, A34, A251, A53 9, A632, A54, A327, A37, A196, A607, A645, A35, A214, A225, A638, A40, A52, A268, A448, A575, A176, A593, A1 5, A17, A94, A170, A713, A93, A402, A64, A261, A399, A422, A214, A225, A625, A31, A119, A135, A281, A676, A7 09, A81, A32, A633, A39, A646, A662, A124, A732, A320, A81, A187, A354, A45, A570, A165, A66, A20, A455, A43 1, A270, A250, A457, A153, A404, A710, A541, A127, A373, A369, A557, A349, A598, A618, A60, A636, A499, A87, A156, A680, A477, A406, A330, A202, A535, A617, A737, A201, A302, A722, A209, A374, A631, A29, A555, A420, A380, A111, A306, A173, A628, A672, A51, A167, A588, A512, A194, A282, A412, A701, A583, A396, A678, A649, A27, A204, A626, A257, A614, A409, A172, A372, A353, A58, A728, A74, A619, A144, A183, A538, A445, A531, A3 60, A361, A459, A536, A344, A267, A574, A677, A530, A415, A30, A73, A152, A490, A702, A714, A483, A567, A43, A310, A319, A86, A321, A656, A739, A115, A130, A155, A608, A648, A168, A485, A738, A129, A650, A715, A488, A147, A121, A470, A115, A133, A510, A421, A309, A335, A387, A386, A734, A95, A430, A604, A458, A592, A384, A664, A197, A725, A89, A83, A586, A622, A305, A498, A668, A427, A630, A158, A644, A735, A70, A683, A352, A3 41, A719, A674, A70, A44, A501, A438, A698, A377, A417, A154, A433, A104, A184, A603, A280, A712, A237, A10 5, A394, A605, A517, A704, A566, A77, A356, A454, A600, A643, A112, A569, A529, A247, A463, A437, A718, A47 2, A461, A558, A48, A671, A395, A670, A681, A687, A382, A82, A686, A342, A436, A296, A16, A545, A533, A416, A149, A207, A371, A596, A675, A132, A419, A56, A579, A733, A573, A707, A597, A697, A75, A653, A362, A615, A 332, A69, A162, A128, A432, A654, A22, A397, A526, A582, A418, A91, A260, A97, A191, A55, A581, A375, A522, A108, A367, A610, A552, A571, A57, A543, A661, A138, A196, A246, A337, A446, A265, A96, A509, A123, A627, A 651, A682, A157, A572, A624, A691, A532, A462, A580, A695, A186, A316, A540, A590, A665, A244, A166, A587, A629, A595, A518, A519, A131, A502, A726, A452, A141, A181, A262, A338, A155, A389, A124, A275, A414, A546, A679, A425, A669, A28, A520, A88, A131, A589, A621, A182, A297, A594, A283, A194, A250, A336, A706, A252, A440, A107, A724, A525, A388, A175, A300, A333, A659, A346, A150, A476, A368, A528, A503, A504, A505, A684, A76, A736, A551, A383, A491, A492, A493, A410, A316, A295, A559, A511, A38, A140, A663, A334, A700, A692, A348, A584, A513, A657, A328, A515, A317, A135, A660, A351, A544, A281, A685, A602, A556, A385, A326, A464, A465, A403, A133, A299, A667, A255, A334, A256, A585, A642, A133, A443, A435, A560, A444, A439, A324, A12 0, A407, A527, A245, A370, A537, A247, A474, A475, A705, A323, A112, A298, A609, A673, A292, A599, A132, A1 45, A266, A601, A466, A549, A379, A727, A167, A711, A75, A76, A121, A357, A620, A316, A479, A290, A339, A32 2, A376, A456, A391, A291, A550, A343, A721, A689, A411, A578, A616, A534, A365, A658, A699, A577, A647, A5 91, A542, A279, A294.

(1033) Determination of Cell Viability in RAS Mutant Cancer Cell Lines

(1034) Protocol: CellTiter-Glo® Cell Viability Assay

(1035) Note—The following protocol describes a procedure for monitoring cell viability of K-Ras mutant cancer cell lines in response to a compound of the invention. Other RAS isoforms may be employed, though the number of cells to be seeded will vary based on cell line used.

(1036) The purpose of this cellular assay was to determine the effects of test compounds on the proliferation of three human cancer cell lines (NCI-H358 (K-Ras G12C), AsPC-1 (K-Ras G12D),

and Capan-1 (K-Ras G12V)) over a 5-day treatment period by quantifying the amount of ATP present at endpoint using the CellTiter-Glo® 2.0 Reagent (Promega).
(1037) Cells were seeded at 250 cells/well in 40 μL of growth medium in 384-well assay plates and incubated overnight in a humidified atmosphere of 5% CO.sub.2 at 37° C. On the day of the assay, 10 mM stock solutions of test compounds were first diluted into 3 mM solutions with 100% DMSO. Well-mixed compound solutions (15 μL) were transferred to the next wells containing 30 μL of 100% DMSO, and repeated until a 9-concentration 3-fold serial dilution was made (starting assay concentration of 10 μM). Test compounds (132.5 nL) were directly dispensed into the assay plates containing cells. The plates were shaken for 15 seconds at 300 rpm, centrifuged, and incubated in a humidified atmosphere of 5% CO.sub.2 at 37° C. for 5 days. On day 5, assay plates and their contents were equilibrated to room temperature for approximately 30 minutes. CellTiter-Glo® 2.0 Reagent (25 μL) was added, and plate contents were mixed for 2 minutes on an orbital shaker before incubation at room temperature for 10 minutes. Luminescence was measured using the PerkinElmer Enspire. Data were normalized by the following: (Sample signal/Avg. DMSO)*100. The data were fit using a four-parameter logistic fit.
(1038) Disruption of B-Raf Ras-Binding Domain (BRAF.sup.RBD) Interaction with K-Ras by Compounds of the Invention (Also Called a FRET Assay or an MOA Assay)
(1039) Note—The following protocol describes a procedure for monitoring disruption of K-Ras G12C (GMP-PNP) binding to BRAF.sup.RBD by a compound of the invention. This protocol may also be executed substituting other Ras proteins or nucleotides.
(1040) The purpose of this biochemical assay was to measure the ability of test compounds to facilitate ternary complex formation between a nucleotide-loaded K-Ras isoform and Cyclophilin A; the resulting ternary complex disrupts binding to a BRAF.sup.RBD construct, inhibiting K-Ras signaling through a RAF effector. Data is reported as IC50 values.
(1041) In assay buffer containing 25 mM HEPES pH 7.3, 0.002% Tween20, 0.1% BSA, 100 mM NaCl and 5 mM MgCl.sub.2, tagless Cyclophilin A, His6-K-Ras-GMPPNP, and GST-BRAF.sup.RBD were combined in a 384-well assay plate at final concentrations of 25 μM, 12.5 nM and 50 nM, respectively. Compound was present in plate wells as a 10-point 3-fold dilution series starting at a final concentration of 30 μM. After incubation at 25° C. for 3 hours, a mixture of Anti-His Eu-W1024 and anti-GST allophycocyanin was then added to assay sample wells at final concentrations of 10 nM and 50 nM, respectively, and the reaction incubated for an additional 1.5 hours. TR-FRET signal was read on a microplate reader (Ex 320 nm, Em 665/615 nm). Compounds that facilitate disruption of a K-Ras:RAF complex were identified as those eliciting a decrease in the TR-FRET ratio relative to DMSO control wells.
(1042) TABLE-US-00005 TABLE 4 Biological Assay Data for Representative Compounds of the Present Invention H358 Cell H358 pERK Viability FRET FRET (K-Ras G12C) (K-Ras G12C) (K-Ras G12C) (K-Ras G13C) Ex# EC50, uM IC50, uM IC50, uM IC50, uM A208 0.013 0.004 0.023 7.48 A193 0.067 0.054 0.295 10.2 A186 0.013 0.005 0.038 0.94 A89a 0.003 0.0008 0.009 0.82 A89b 0.028 0.011 0.034 0.58
(1043) TABLE-US-00006 TABLE 5 Additional H358 Cell Viability assay data (K-Ras G12C, IC50, uM): IC50* Examples + A104, A107, A108, A112, A120, A121, A123, A124, A128, A131, A131, A132, A133, A133, A135, A138, A140, A141, A145, A150, A155, A157, A162, A166, A167, A175, A181, A182, A183, A186, A191, A194, A196, A237, A244, A245, A246, A247, A250, A255, A256, A260, A266, A275, A28, A281, A283, A290, A291, A292, A295, A296, A298, A299, A300, A309, A316, A316, A316, A317, A322, A323, A324, A326, A328, A332, A333, A334, A334, A336, A337, A339, A343, A344, A346, A348, A351, A352, A356, A357, A365, A368, A370, A371, A375, A376, A377, A379, A38, A380, A382, A383, A384, A385, A388, A389, A391, A394, A395, A396, A397, A403, A407, A410, A411, A414, A416, A419, A425, A427, A435, A439, A440, A443, A444, A445, A446, A452, A454, A456, A461, A462, A463, A464, A465, A466, A472, A474, A475, A476, A479, A483, A485, A488, A491, A492, A493, A502, A503, A504, A505, A509, A511, A513, A515, A517,

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Additional Ras-Raf Disruption/FRET/MOA Assay Data (IC50, uM):

(1044) TABLE-US-00007 TABLE 6 KRAS G12S FRET data IC50* Examples + none ++ A028, A075, A076, A076, A087, A112, A145, A155, A167, A181, A183, A184, A194, A233, A255, A256, A260, A262, A264, A265, A266, A268, A270, A275, A292, A294, A295, A298, A299, A300, A319, A333, A334, A334, A338, A367, A382, A383, A385, A388, A389, A418, A433, A464, A465, A479, A491, A492, A493, A527, A537, A542, A577, A596, A598, A602, A609, A621, A629, A664, A665, A679, A712, A734, A736 +++ A004, A005, A012, A015, A016, A022, A027, A029, A032, A034, A037, A037, A041, A043, A044, A047, A048, A051, A052, A054, A055, A056, A057, A058, A064, A070, A070, A072, A074, A077, A079, A082, A083, A086, A088, A091, A093, A095, A096, A097, A101, A103, A105, A107, A123, A127, A128, A129, A131, A132, A135, A141, A147, A148, A149, A150, A153, A154, A158, A164, A166, A178, A182, A189, A193, A194, A196, A206, A207, A237, A241, A244, A245, A246, A247, A250, A251, A252, A257, A261, A261, A263, A267, A269, A280, A281, A281, A296, 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(1045) TABLE-US-00008 TABLE 7 KRAS G12D FRET data IC50* Examples + none ++ A183, A264, A265, A268, A319, A388, A464, A465, A664 +++ A028, A054, A075, A076, A076, A087, A107, A112, A123, A131, A145, A153, A155, A167, A184, A189, A194, A233, A247, A255, A256, A257, A260, A261, A262, A263, A266, A269, A270, A275, A292, A294, A295, A298, A300, A333, A334, A334, A338, A349, A353, A354, A367, A368, A382, A383, A384, A385, A386, A387, A389, A394, A407, A409, A417, A418, A432, A433, A436, A479, A491, A492, A493, A527, A529, A537, A542, A555, A577, A578, A594, A596, A598, A602, A603, A605, A608, A609, A612, A621, A622, A629, A633, A650, A651, A653, A654, A662, A665, A667, A669, A671, A678, A679, A681, A682, A683, A685, A696, A697, A698, A700, A701, A704, A709, A710, A711, A712, A719, A734, A736, A737 ++++ A005, A007, A012, A015, A016, A022, A029, A034, A037, A037, A038, A038, A039, A043, A044, A046, A047, A048, A051, A052, A055, A056, A057, A058, A067, A070, A070, A071, A072, A074, A077, A079, 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(1046) TABLE-US-00009 TABLE 8 KRAS G13C FRET data IC50* Examples + A088, A096, A097, A107, A131, A132, A155, A175, A247, A250, A251, A252, A253, A255, A258, A262, A270, A292, A294, A297, A298, A299, A334, A336, A338, A474, A475, A529, A542, A683 ++ A031, A037, A038, A046, A067, A069, A071, A075, A076, A076, A090, A108, A112, A116, A121, A123, A124, A131, A132, A133, A133, A135, A140, A141, A145, A150, A155, A163, A167, A172, A178, A182, A183, A194, A196, A218, A224, A233, A244, A245, A246, A247, A249, A250, A251, A254, A256, A257, A258, A259, A260, A261, A263, A264, A265, A266, A267, A268, A275, A280, A281, A283, A295, A296, A300, A305, A306, A319, A333, A334, A337, A379, A383, A385, A388, A389, A394, A396, A397, A417, A418, A421, A432, A433, A438, A452, A464, A465, A479, A491, A492, A493, A517, A525, A527, A537, A538, A539, A540, A541, A545, A546, A552, A553, A554, A555, A564, A570, A571, A572, A573, A574, A575, A577, A578, A591, A594, A595, A597, A600, A602, A603, A604, A605, A606, 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(1047) TABLE-US-00010 TABLE 9 KRAS G12V FRET data IC50* Examples + None ++ A028, A075, A076, A076, A087, A112, A132, A145, A155, A167, A183, A184, A194, A233, A245, A262, A264, A265, A266, A268, A275, A292, A295, A298, A300, A305, A319, A333, A334, A338, A368, A383, A388, A389, A418, A464, A465, A479, A491, A492, A493, A527, A537, A542, A602, A609, A621, A629, A665, A679, A734 +++ A004, A005, A012, A015, A016, A022, A027, A029, A034, A037, A037, A041, A043, A044, A047, A048, A051, A052, A054, A055, A056, A057, A058, A069, A070, A072, A074, A077, A079, A082, A086, A088, A091, A093, A095, A096, A097, A101, A103, A105, A107, A123, A127, A128, A129, A131, A135, A141, A147, A148, A149, A150, A153, A154, A155, A162, A164, A166, A178, A180, A181, A182, A183, A189, A193, A194, A196, A206, A207, A209, A237, A238, A241, A244, A246, A247, A250, A251, A252, A253, A255, A256, A257, A258, A260, A261, A263, A267, A269, A270, A280, A281, A281, A294, A296, A297, A299, A306, A334, A336, A337, A349, 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(1049) TABLE-US-00012 TABLE 11 KRAS G12C FRET data IC50* Examples + A038, A075, A076, A088, A095, A096, A097, A107, A108, A112, A116, A120, A121, A129, A130, A131, A131, A132, A133, A133, A135, A140, A141, A145, A157, A162, A167, A175, A176, A182, A183, A184, A194, A196, A209, A244, A245, A246, A247, A247, A255, A256, A257, A266, A268, A270, A290, A291, A292, A294, A298, A299, A300, A316, A316, A316, A317, A322, A323, A324, A326, A333, A334, A337, A339, A343, A346, A348, A351, A365, A375, A376, A377, A379, A380, A381, A388, A391, A403, A407, A411, A414, A425, A427, A474, A475, A477, A479, A509, A527, A529, A534, A542, A543, A544, A549, A550, A551, A556, A577, A578, A579, A591, A599, A600, A601, A609, A620, A626, A627, A628, A638, A642, A647, A658, A660, A663, A667, A673, A684, A689, A691, A692, A699, A705, A707, A711, A721, A726, A727, A735 ++ A004, A005, A012, A015, A016, A020, A022, A027, A028, A029, A031, A032, A037, A037, A038, A043, A044, A047, A048, A051, A052, A054, A055, A056, 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(1050) TABLE-US-00013 TABLE 12 KRAS G13D FRET data IC50* Examples + None ++ A075, A076, A112, A155, A183, A260, A261, A262, A263, A264, A265, A267, A268, A270, A294, A296, A298, A300, A319, A333, A338, A388, A464, A465, A527, A537, A542, A664 +++ A028, A054, A096, A105, A107, A123, A128, A131, A132, A141, A145, A149, A150, A153, A158, A164, A167, A181, A182, A184, A189, A194, A196, A207, A233, A244, A245, A246, A247, A250, A251, A252, A253, A255, A256, A257, A258, A261, A266, A269, A275, A280, A281, A281, A292, A295, A297, A299, A305, A306, A334, A334, A336, A337, A349, A350, A353, A354, A357, A361, A367, A368, A379, A382, A383, A384, A385, A386, A387, A389, A394, A395, A397, A406, A407, A410, A415, A416, A417, A418, A421, A432, A433, A436, A438, A463, A479, A490, A491, A492, A493, A518, A519, A529, A536, A538, A543, A544, A545, A555, A558, A573, A577, A578, A580, A581, A582, A591, A594, A596, A598, A602, A603, A605, A606, A608, A609, A611, A612, A620, A621, A622, A624, A629, A631, A633, A650, A651, A653, A654, A655, A659, A661, A662, A665, A667, A669, A670, A671, A672, A678, A679, A681, A682, A683, A684, A685, A686, A687, A690, A695, A697, A698, A699, A700, A701, A704, A709, A710, A711, A712, A718, A719, A729, A730, A734, A736, A737 ++++ A016, A038, A069, A075, A088, A095, A103, A104, A108, A110, A111, A112, A116, A117, A120, A120, A121, A121, A125, A127, A129, A130, A131, A132, A133, A133, A133, A134, A135, A138, A140, A142, A144, A144, A145, A146, A147, A148, A152, A154, A155, A156, A157, A159, A161, A162, A163, A166, A175, A178, A179, A180, A181, A182, A183, A187, A193, A194, A197, A206, A209, A212, A213, A232, A237, A238, A241, A241, A247, A249, A251, A254, A258, A259, A277, A278, A282, A290, A302, A314, A321, A326, A328, A330, A331, A332, A335, A339, A340, A341, A342, A344, A351, A356, A358, A359, A360, A363, A364, A365, A366, A369, A370, A372, A373, A380, A381, A390, A391, A393, A396, A399, A401, A408, A409, A411, A412, A413, A419, A420, A422, A424, A427, A430, A431, A435, A437, A439, A440, A443, A444, A445, A446, A449, A451, A452, A454, A455, A456, A457, A458, A459, A466, A468, A469, A470, A472, A473, A474, A475, A476, A485, A487, A488, A489, A499, A501, A502, A503, A504, A505, A509, A510, A511, A513, A515, A517, A520, A521, A522, A525, A534, A539, A540, A541, A546, A547, A548, A549, A550, A551, A552, A553, A554, A557, A559, A560, A561, A562, A563, A564, A566, A567, A570, A571, A572, A574, A575, A583, A586, A588, A593, A595, A597, A600, A601, A604, A607, A610, A616, A618, A619, A623, A625, A626, A630, A632, A634, A635, A636, A637, A638, A640, A641, A642, A645, A647, A652, A656, A658, A660, A668, A673, A676, A677, A680, A688, A689, A691, A693, A694, A696, A702, A706, A707, A713, A714, A717, A720, A722, A725, A726, A727, A728, A731, A732, A738, A739 +++++ A024, A065, A094, A102, A106, A109, A113, A114, A115, A115, A118, A119, A122, A124, A124, A126, A135, A136, A137, A139, A143, A148, A151, A160, A165, A167, A168, A169, A170, A171, A172, A173, A174, A176, A177, A184, A185, A186, A186, A187, A188, A190, A191, A191, A192, A192, A195, A196, A198, A199, A200, A201, A202, A203, A204, A205, A208, A209, A210, A210, A210, A210, A211, A211, A211, A211, A212, A213, A214, A214, A214, A225, A225, A225, A226, A227, A228, A229, A230, A231, A231, A231, A234, A235, A236, A239, A239, A240, A242, A243, A248, A250, A271, A272, A273, A274, A276, A279, A279, A279, A283, A283, A283, A284, A285, A286, A287, A288, A289, A291, A293, A301, A303, A304, A304, A304, A304, A307, A308, A309, A310, A311, A312, A313, A316, A316, A316, A317, A318, A318, A320, A322, A323, A324, A325, A327, A328, A329, A343, A345, A346, A347, A348, A352, A355, A362, A371, A374, A375, A376, A377, A378, A392, A398, A400, A402, A403, A404, A405, A414, A423, A425, A426, A428, A429, A441, A442, A447, A448, A450, A453, A460, A461, A462, A467, A471, A477, A478, A480, A481, A482, A483, A484, A486, A494, A495, A496, A497, A498, A500, A506, A507, A508, A512, A514, A516, A523, A524, A526, A528, A530, A531, A532, A533, A535, A556, A565, A568, A569, A576, A579, A584, A585, A587, A589, A590, A592, A599, A613, A614, A615, A617, A627, A628, A639, A643, A644, A646, A648, A649, A657, A663, A666, A674, A675, A692, A703, A705, A708, A715, A716, A721, A723, A724, A733, A735

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(1052) TABLE-US-00015 TABLE 14 NRAS G12C FRET data IC50* Examples + A038, A075, A076, A088, A095, A096, A107, A108, A112, A116, A120, A121, A123, A129, A130, A131, A131, A132, A133, A133, A135, A140, A141, A145, A155, A157, A162, A167, A175, A176, A182, A183, A184, A194, A196, A209, A244, A245, A246, A247, A247, A250, A255, A257, A266, A268, A270, A290, A292, A294, A297, A298, A299, A300, A316, A316, A316, A323, A324, A326, A333, A334, A336, A337, A339, A343, A348, A351, A365, A375, A376, A377, A380, A388, A391, A407, A411, A414, A427, A435, A474, A475, A479, A509, A527, A529, A534, A537, A542, A543, A544, A550, A551, A556, A577, A578, A579, A591, A600, A601, A609, A612, A616, A620, A626, A627, A628, A638, A642, A647, A658, A660, A663, A667, A673, A684, A689, A691, A692, A699, A705, A707, A711, A721, A727, A735 ++ A016, A054, A069, A103, A104, A105, A115, A121, A124, A126, A127, A128, A132, A133, A134, A135, A138, A142, A144, A148, A149, A150, A152, A153, A154, A155, A158, A164, A165, A166, A172, A178, A179, A181, A182, A183, A186, A187, A189, A191, A196, A200, A204, A207, A233, A237, A241, A241, A250, A251, A251, A252, A256, A260, A261, A262, A263, A264, A265, A267, A269, A275, A281, A291, A295, A296, A305, A317, A319, A322, A328, A332, A334, A338, A344, A346, A349, A350, A352, A353, A354, A361, A362, A363, A367, A368, A370, A371, A379, A381, A382, A383, A384, A385, A386, A387, A389, A394, A396, A397, A400, A403, A410, A417, A418, A419, A421, A423, A425, A432, A433, A436, A437, A438, A439, A440, A443, A444, A445, A448, A452, A454, A456, A458, A459, A463, A464, A465, A466, A476, A477, A478, A485, A489, A490, A491, A492, A493, A499, A501, A503, A504, A505, A510, A513, A515, A517, A518, A519, A525, A526, A528, A532, A533, A536, A538, A545, A549, A552, A558, A559, A560, A566, A567, A569, A572, A573, A574, A575, A580, A581, A582, A584, A585, A587, A589, A590, A592, A594, A595, A596, A597, A598, A599, A602, A603, A610, A614, A615, A617, A621, A622, A624, A629, A631, A632, A633, A634, A636, A637, A643, A644, A645, A646, A649, A650, A651, A653, A654, A657, A659, A661, A662, A664, A665, A668, A669, A671, A672, A674, A675, A676, A678, A679, A681, A682, A683, A685, A686, A695, A697, A698, A700, A701, A703, A704, A706, A709, A710, A712, A715, A719, A720, A722, A724, A726, A730, A733, A734, A736, A737 +++ A065, A075, A094, A109, A111, A112, A113, A115, A117, A118, A119, A120, A137, A139, A143, A144, A145, A146, A147, A151, A156, A159, A161, A167, A169, A173, A177, A180, A181, A184, A185, A188, A190, A192, A193, A194, A195, A197, A199, A201, A206, A209, A210, A211, A212, A214, A225, A226, A227, A232, A238, A249, A253, A254, A258, A258, A259, A261, A277, A278, A280, A281, A282, A283, A301, A302, A306, A310, A314, A320, A321, A327, A330, A331, A335, A340, A341, A342, A347, A356, A357, A358, A359, A360, A364, A366, A369, A372, A373, A374, A378, A390, A392, A393, A395, A399, A402, A404, A406, A408, A409, A412, A413, A415, A416, A420, A422, A424, A430, A431, A442, A446, A447, A449, A451, A455, A457, A461, A462, A467, A468, A469, A470, A471, A472, A473, A481, A483, A486, A487, A488, A494, A495, A497, A498, A502, A506, A507, A508, A511, A512, A520, A522, A524, A530, A531, A539, A540, A541, A546, A547, A548, A553, A554, A555, A557, A561, A562, A563, A564, A565, A568, A570, A571, A576, A583, A586, A588, A593, A604, A605, A606, A607, A608, A611, A618, A619, A623, A625, A630, A635, A639, A640, A641, A648, A652, A655, A656, A670, A677, A680, A687, A688, A690, A693, A694, A696, A702, A708, A713, A714, A716, A717, A718, A723, A725, A728, A729, A731, A732, A738, A739 ++++ A024, A102, A106, A110, A124, A125, A136, A160, A163, A168, A170, A171, A174, A186, A187, A191, A192, A202, A203, A205, A208, A210, A211, A212, A213, A228, A229, A235, A240, A242, A248, A274, A276, A279, A283, A284, A289, A303, A307, A309, A313, A325, A328, A345, A398, A401, A426, A441, A453, A460, A480, A484, A496, A514, A516, A521, A523, A535, A613, A666 +++++ A114, A122, A148, A198, A210, A210, A211, A211, A213, A214, A214, A225, A225, A230, A231, A231, A231, A234, A236, A239, A239, A243, A271, A272, A273, A279, A279, A283, A285, A286, A287, A288, A293, A304, A304, A304, A304, A308, A311, A312, A318, A318, A329, A355, A405, A428, A429, A450, A482, A500

(1053) TABLE-US-00016 TABLE 15 NRAS Q61R FRET data IC50* Examples + A388 ++ A367, A368, A382, A383, A385, A386, A387, A389, A407, A417, A491, A492, A493, A542, A577, A594, A596, A598, A602, A603, A609, A611, A612, A621, A629, A633, A655, A664, A665, A667, A669, A678, A679, A697, A704, A712, A719, A734, A736, A737 +++ A012, A022, A075, A076, A082, A087, A105, A148, A167, A181, A207, A262, A275, A281, A297, A299, A300, A314, A333, A338, A349, A350, A353, A354, A356, A357, A360, A361, A369, A379, A384, A393, A394, A416, A463, A502, A537, A538, A539, A541, A543, A544, A545, A546, A548, A554, A555, A570, A571, A572, A573, A575, A578, A580, A581, A582, A583, A586, A591, A593, A595, A597, A604, A605, A606, A607, A608, A610, A618, A619, A620, A622, A624, A625, A630, A631, A632, A634, A635, A636, A650, A651, A653, A654, A656, A659, A661, A662, A668, A670, A671, A672, A676, A677, A680, A681, A682, A683, A684, A685, A686, A687, A688, A690, A695, A696, A698, A699, A700, A701, A702, A706, A709, A710, A711, A713, A714, A717, A718, A728, A732, A738 ++++ A016, A064, A074, A077, A083, A086, A089, A104, A115, A166, A168, A193, A252, A277, A290, A328, A330, A332, A335, A339, A340, A341, A342, A343, A344, A347, A351, A352, A358, A359, A363, A364, A365, A366, A370, A372, A373, A380, A381, A390, A391, A392, A442, A461, A472, A474, A475, A481, A494, A495, A496, A498, A503, A504, A505, A522, A531, A534, A540, A547, A549, A550, A551, A552, A553, A557, A558, A559, A560, A561, A564, A566, A567, A574, A588, A600, A601, A623, A626, A638, A639, A640, A641, A647, A652, A658, A660, A673, A689, A691, A693, A694, A720, A722, A725, A727, A729, A730, A731, A739 +++++ A023, A025, A026, A045, A050, A053, A060, A061, A062, A080, A084, A085, A102, A113, A136, A143, A151, A160, A170, A171, A172, A202, A205, A210, A211, A226, A227, A240, A242, A243, A248, A278, A285, A301, A302, A303, A304, A304, A313, A316, A329, A331, A345, A346, A348, A355, A362, A371, A374, A375, A376, A377, A378, A403, A427, A447, A468, A482, A486, A507, A556, A562, A563, A565, A568, A569, A576, A579, A584, A585, A587, A589, A590, A592, A599, A613, A614, A615, A616, A617, A627, A628, A637, A642, A643, A644, A645, A646, A648, A649, A657, A663, A666, A674, A675, A692, A703, A705, A707, A708, A715, A716, A721, A723, A724, A726, A733, A735

(1054) TABLE-US-00017 TABLE 16 NRAS Q61K FRET data IC50* Examples + A388 ++ A262, A297, A299, A338, A349, A350, A353, A354, A367, A368, A369, A382, A383, A384, A385, A386, A387, A389, A394, A407, A416, A417, A463, A491, A492, A493, A542, A543, A544, A545, A577, A580, A582, A594, A596, A598, A602, A603, A609, A611, A612, A620, A621, A622, A629, A631, A633, A650, A651, A653, A654, A655, A661, A662, A664, A665, A667, A668, A669, A671, A678, A679, A681, A682, A683, A686, A687, A696, A697, A698, A701, A702, A704, A709, A710, A711, A712, A718, A719, A734, A736, A737, A739 +++ A012, A022, A074, A075, A076, A077, A082, A086, A087, A105, A148, A167, A181, A207, A275, A277, A281, A300, A314, A330, A333, A340, A356, A360, A361, A379, A390, A393, A502, A537, A538, A539, A541, A546, A547, A548, A554, A555, A561, A570, A571, A572, A573, A574, A575, A578, A581, A583, A586, A588, A593, A595, A597, A604, A605, A606, A607, A608, A610, A618, A619, A623, A624, A625, A630, A632, A634, A635, A636, A656, A660, A670, A672, A676, A677, A680, A684, A685, A688, A690, A694, A695, A699, A700, A706, A713, A714, A717, A728, A729, A730, A731, A732, A738 ++++ A016, A025, A045, A050, A060, A061, A064, A080, A083, A089, A104, A115, A143, A151, A166, A168, A172, A193, A202, A210, A211, A227, A252, A278, A290, A328, A331, A332, A335, A339, A341, A342, A343, A344, A347, A351, A352, A357, A358, A359, A363, A364, A365, A366, A370, A372, A373, A380, A381, A391, A427, A442, A461, A468, A472, A474, A475, A481, A494, A495, A496, A498, A503, A504, A505, A522, A531, A534, A540, A549, A550, A551, A552, A553, A557, A558, A559, A560, A562, A563, A564, A566, A567, A576, A590, A591, A600, A601, A626, A637, A638, A639, A640, A641, A647, A652, A659, A673, A689, A691, A693, A705, A720, A722, A725, A727, A735 +++++ A023, A026, A053, A062, A084, A085, A102, A113, A136, A160, A170, A171, A205, A226, A240, A242, A243, A248, A285, A301, A302, A303, A304, A304, A313, A316, A329, A345, A346, A348, A355, A362, A371, A374, A375, A376, A377, A378, A392, A403, A447, A482, A486, A507, A556, A565, A568, A569, A579, A584, A585, A587, A589, A592, A599, A613, A614, A615, A616, A617, A627, A628, A642, A643, A644, A645, A646, A648, A649, A657, A658, A663, A666, A674, A675, A692, A703, A707, A708, A715, A716, A721, A723, A724, A726, A733

(1055) TABLE-US-00018 TABLE 17 NRAS WT FRET data IC50* Examples + A388 ++ A075, A076, A087, A167, A262, A275, A297, A299, A300, A333, A338, A349, A367, A368, A382, A383, A385, A386, A387, A389, A407, A417, A491, A492, A493, A537, A542, A577, A594, A596, A598, A602, A603, A605, A609, A612, A621, A629, A633, A664, A665, A667, A669, A679, A697, A704, A709, A711, A712, A719, A734, A736, A737 +++ A012, A016, A022, A074, A077, A082, A086, A105, A148, A166, A181, A207, A252, A277, A281, A340, A350, A351, A353, A354, A356, A357, A358, A360, A361, A369, A379, A384, A393, A394, A416, A463, A502, A538, A539, A541, A543, A544, A545, A546, A547, A548, A550, A552, A554, A555, A557, A558, A564, A570, A571, A572, A573, A575, A578, A580, A581, A582, A583, A586, A591, A593, A595, A600, A604, A606, A607, A608, A610, A611, A618, A620, A622, A624, A625, A630, A631, A632, A634, A635, A636, A647, A650, A651, A653, A654, A655, A656, A659, A660, A661, A662, A668, A670, A671, A672, A676, A677, A678, A680, A681, A682, A683, A684, A685, A686, A687, A688, A690, A695, A696, A698, A699, A700, A701, A702, A710, A713, A714, A717, A718, A722, A728, A729, A730, A731, A732, A738, A739 ++++ A025, A045, A060, A061, A062, A064, A083, A089, A104, A115, A151, A168, A172, A193, A202, A210, A211, A227, A278, A290, A314, A328, A330, A331, A332, A335, A339, A341, A342, A343, A344, A347, A348, A352, A359, A362, A363, A364, A365, A366, A370, A371, A372, A373, A375, A376, A378, A380, A381, A390, A391, A392, A427, A447, A461, A468, A472, A474, A475, A481, A494, A495, A496, A498, A503, A504, A505, A507, A522, A531, A534, A540, A549, A551, A553, A556, A559, A560, A561, A562, A563, A566, A567, A574, A576, A579, A584, A585, A588, A589, A590, A597, A601, A616, A619, A623, A626, A637, A638, A639, A640, A641, A642, A645, A648, A652, A658, A673, A689, A691, A693, A694, A703, A705, A706, A707, A708, A716, A720, A721, A723, A725, A726, A727, A735 +++++ A023, A026, A050, A053, A080, A084, A085, A102, A113, A136, A143, A160, A170, A171, A205, A226, A240, A242, A243, A248, A285, A302, A303, A304, A304, A313, A316, A329, A345, A346, A355, A374, A377, A403, A442, A482, A486, A565, A568, A569, A587, A592, A599, A613, A614, A615, A617, A627, A628, A643, A644, A646, A649, A657, A663, A666, A674, A675, A692, A715, A724, A733

In Vitro Cell Proliferation Panels

(1056) Potency for inhibition of cell growth was assessed at CrownBio using standard methods. Briefly, cell lines were cultured in appropriate medium, and then plated in 3D methylcellulose. Inhibition of cell growth was determined by CellTiter-Glo® after 5 days of culture with increasing concentrations of compounds. Compound potency was reported as the 50% inhibition concentration (absolute IC50). The assay took place over 7 days. On day 1, cells in 2D culture were harvested during logarithmic growth and suspended in culture medium at 1×105 cells/ml. Higher or lower cell densities were used for some cell lines based on prior optimization. 3.5 ml of cell suspension was mixed with 6.5% growth medium with 1% methylcellulose, resulting in a cell suspension in 0.65% methylcellulose. 90 μl of this suspension was distributed in the wells of 2 96-well plates. One plate was used for day 0 reading and 1 plate was used for the end-point experiment. Plates were incubated overnight at 37 C with 5% CO.sub.2. On day 2, one plate (for t0 reading) was removed and 10 μl growth medium plus 100 μl CellTiter-Glo® Reagent was added to each well. After mixing and a 10 minute incubation, luminescence was recorded on an EnVision Multi-Label Reader (Perkin Elmer). Compounds in DMSO were diluted in growth medium such that the final, maximum concentration of compound was 10 μM, and serial 4-fold dilutions were performed to generate a 9-point concentration series. 10 μl of compound solution at 10 times final concentration was added to wells of the second plate. Plate was then incubated for 120 hours at 37 C and 5% CO.sub.2. On day 7 the plates were removed, 100 μl CellTiter-Glo® Reagent was added to each well, and after mixing and a 10 minute incubation, luminescence was recorded on an EnVision Multi-Label Reader (Perkin Elmer). Data was exported to GeneData Screener and modeled with a sigmoidal concentration response model in order to determine the IC50 for compound response.

(1057) Not all cell lines with a given RAS mutation may be equally sensitive to a RAS inhibitor targeting that mutation, due to differential expression of efflux transporters, varying dependencies on RAS pathway activation for growth, or other reasons. This has been exemplified by the cell line KYSE-410 which, despite8 having a KRAS G12C mutation, is insensitive to the KRAS G12C (OFF) inhibitor MRTX-849 (Hallin et al., Cancer Discovery 10:54-71 (2020)), and the cell line SW1573, which is insensitive to the KRAS G12C (OFF) inhibitor AMG51 (Canon et al., Nature 575:217-223 (2019)).

(1058) TABLE-US-00019 TABLE 18 IC50 values for various cancer cell lines with Compound B Cell Line Histotype Mutant IC50* NCI-H358 Lung KRAS G12C very sensitive MIA PaCa-2 Pancreas KRAS G12C very sensitive SW837 Intestine/Large/ KRAS G12C very Colorectum sensitive KYSE-410 HN/Esophagus KRAS G12C moderately sensitive NCI-H727 Lung KRAS G12V moderately sensitive OVCAR-5 Ovary KRAS G12V moderately sensitive Capan-2 Pancreas KRAS G12V moderately sensitive NCI-H747 Intestine/Large/ other KRAS (G13D) moderately Colorectum sensitive NCI-H441 Lung KRAS G12V moderately sensitive HEC-1-A Uterus KRAS G12D moderately sensitive NOZ Liver/Bile duct KRAS G12V moderately sensitive HCT116 Intestine/Large/ other KRAS (G13D) moderately Colorectum sensitive Calu-6 Lung other KRAS (Q61K) moderately sensitive HuCCT1 Liver/Bile duct KRAS G12D moderately sensitive NCI-H2009 Lung other KRAS (G12A) low sensitivity NCI-H1975 Lung other MAPK (EGFR low T790M, L858R) sensitivity SW1573 Lung KRAS G12C not sensitive AGS Stomach KRAS G12D low sensitivity AsPC-1 Pancreas KRAS G12D low sensitivity SNU-668 Stomach other KRAS (Q61K) low sensitivity HPAC Pancreas KRAS G12D low sensitivity NCI-H1838 Lung other MAPK (NF1 low mut) sensitivity NCI-H3122 Lung other MAPK (EML4- low ALK(E13, A20)) sensitivity NCI-H460 Lung other KRAS (Q61H) low sensitivity SW403 Intestine/Large/ KRAS G12V low Colorectum sensitivity A549 Lung other KRAS (G12S) low sensitivity CAL-62 HN/Thyroid other KRAS (G12R) low sensitivity DV-90 Lung other KRAS (G13D) low sensitivity OZ Liver/Bile duct other KRAS (Q61L) low sensitivity A-375 Skin BRAF V600E low sensitivity BxPC-3 Pancreas other MAPK (BRAF low V487_P492delinsA) sensitivity SW48 Intestine/Large/ not MAPK (PIK3CA low Colorectum G914R, EGFR sensitivity G719S) HCC1588 Lung KRAS G12D low sensitivity TOV-21G Ovary other KRAS (G13C) low sensitivity SW948 Intestine/Large/ other KRAS (Q61L) low Colorectum sensitivity LS513 Intestine/Large/ KRAS G12D low Colorectum sensitivity MeWo Skin other MAPK (NF1 low mut) sensitivity *Key: low sensitivity: IC50 ≥ 1 uM moderately sensitive: 1 uM > IC50 ≥ 0.1 uM very sensitive: IC50 < 0.1 uM

(1059) TABLE-US-00020 TABLE 19 Summary of IC50 results for various cancer cell lines with several compounds of the present invention (Compounds B and E-M) Cell Line Histotype Mutant E F G H I J K L B M SW837 Intestine/Large/ KRAS G12C V V V V V V V V V V Colorectum MIA Pancreas KRAS G12C V V V V V V V V V V PaCa-2 KYSE-410 HN/Esophagus KRAS G12C L L L L L L L L L L H358 Lung KRAS G12C V V V V V V V V V V H2122 Lung KRAS G12C V V V V V V V V V V H1373 Lung KRAS G12C V V V V V V V V V V H23 Lung KRAS G12C V V V V

V V V V V V H1792 Lung KRAS G12C V V V V V V V V V V AsPC-1 Pancreas KRAS G12D L L M L L M L L M L A375 Skin BRAF V600E L L L L L L L L L L H1975 Lung other MAPK M L M L L M L M M L (EGFR T790M, L858R) HCC1588 Lung KRAS G12D L L L L L L L L L L H441 Lung KRAS G12V M M M L M V L M V L *Key: (L) low sensitivity: IC50 ≥ 1 uM (M) moderately sensitive: 1 uM > IC50 ≥ 0.1 uM (V) very sensitive: IC50 < 0.1 uM

In Vivo NSCLC K-Ras G12C Xenograft Models

Compound A:

Methods:

(1060) The effects of a compound of the present invention, Compound A (H358 pERK K-Ras G12C EC50: 0.001 uM), on tumor cell growth in vivo were evaluated in the human non-small cell lung cancer NCI-H358 KRASG12C xenograft model using female BALB/c nude mice (6-8 weeks old). Mice were implanted with NCI-H358 tumor cells in 50% Matrigel (5×106 cells/mouse) subcutaneously in the flank. At the indicated tumor volume (dotted line, FIG. **2**A), mice were randomized to treatment groups to start the administration of test articles or vehicle. Compound A was administered by oral gavage daily at the dose of 100 mg/kg. Body weight and tumor volume (using calipers) was measured twice weekly until study endpoints. Spaghetti plot (FIG. **2**B) shows the tumor volume change in individual tumors during the course of treatment.

(1061) Results:

(1062) FIG. **2**A shows Compound A dosed at 100 mg/kg by daily oral gavage led to tumor regression in NCI-H358 KRASG12C xenograft model, which is a sensitive model to KRASG12C inhibition alone. The spaghetti titer plot (FIG. **2**B) displaying individual tumor growth is shown next to the tumor volume plot (FIG. **2**A). Over the treatment course of 28 days, Compound A drove tumor regression in all 10 animals bearing NCI-H358 KRASG12C tumors.

(1063) Compound B:

(1064) Methods:

(1065) The combinatorial effect of a compound of the present invention, Compound B (H358 pERK K-Ras G12C EC50: 0.003 uM), with cobimetinib on tumor cell growth in vivo were evaluated in the human non-small cell lung cancer NCI-H358 KRASG12C xenograft model using female BALB/c nude mice (6-8 weeks old). Mice were implanted with NCI-H358 tumor cells in 50% Matrigel (5×106 cells/mouse) subcutaneously in the flank. At indicated tumor volume (dotted line, FIG. **3**A), mice were randomized to treatment groups to start the administration of test articles or vehicle. Compound B was administered by intermittent (twice weekly) intravenous injection at the dose of 50 mg/kg. Cobimetinib was administered by daily oral gavage at 2.5 mg/kg. The combination of Compound B and cobimetinib at their respective single-agent dose and regimen was also tested. Body weight and tumor volume (using calipers) was measured twice weekly until study endpoints. End of study responses in individual tumors were plotted as a waterfall plot (FIG. **3**B), and the numbers indicate number of tumor regression in each group. Tumor regression is defined as greater than 10% reduction of tumor volume at the end of study relative to initial volume.

(1066) Results:

(1067) FIG. **3**A shows the combination of intermittent intravenous administration of Compound B at 50 mg/kg plus daily oral administration of cobimetinib at 2.5 mg/kg drove tumor regression, whereas each single agent led to tumor growth inhibition. End of study responses were shown as waterfall plots (FIG. **3**B), which indicate 6 out 10 mice had tumor regression in the combination group, whereas no tumor regressions recorded in each single agent group.

(1068) Compound C:

(1069) Methods:

(1070) The combinatorial effect of a compound of the present invention, Compound C (H358 pERK K-Ras G12C EC50: 0.007 uM), with a SHP2 inhibitor, RMC-4550, on tumor cell growth in vivo were evaluated in the human non-small cell lung cancer NCI-H358 KRASG12C xenograft model using female BALB/c nude mice (6-8 weeks old). Mice were implanted with NCI-H358 tumor cells

in 50% Matrigel (5×106 cells/mouse) subcutaneously in the flank. At indicated tumor volume (dotted line, FIG. **4**A), mice were randomized to treatment groups to start the administration of test articles or vehicle. Compound C was administered by once weekly intravenous injection at the dose of 60 mg/kg. SHP2 inhibitor was administered by daily oral gavage at 30 mg/kg. The combination of Compound C and SHP2 inhibitor at their respective single-agent dose and regimen was also tested. Body weight and tumor volume (using calipers) was measured twice weekly until study endpoints. End of study responses in individual tumors were plotted as a waterfall plot (FIG. **4**B), and the numbers indicate number of tumor regression in each group. Tumor regression is defined as greater than 10% reduction of tumor volume at the end of study relative to initial volume.
(1071) Results:
(1072) In FIG. **4**A, the combinatorial activity of once weekly intravenous administration of Compound C at 60 mg/kg plus daily oral administration of SHP2 inhibitor at 30 mg/kg is shown. The combination treatment had similar anti-tumor activity as the single agent SHP2 inhibitor, but the combination treatment led to 8 out of 10 mice with tumor regression, whereas single agent SHP2 inhibitor led to 5 out of 10 mice with tumor regressions. Single agent Compound C administered once weekly via intravenous injection led to tumor growth inhibition with one tumor regression.
(1073) Cell Proliferation Assay
(1074) Methods:
(1075) NCI-H358 cells were plated in 12-well tissue culture plates at a density of 100,000 cells/well in RPMI 1640 (10% FBS, 1% PenStrep) and cultured overnight at 37° C., 5% CO.sub.2. The following day, cells were treated with either trametinib (10 nM) or a compound of the present invention, Compound D (H358 pERK K-Ras G12C EC50: 0.024 uM), (17 nM). These concentrations represent the EC50 values from a 72-hour proliferation assay using the CellTiter-Glo® reagent (Promega). Additionally, cells were treated with the combination of trametinib and Compound D at the above indicated concentrations. The plate was placed in the Incucyte S.sub.3 live cell analysis system (37° C., 5% CO.sub.2) and confluence was measured by recording images at 6-hour intervals for a maximum of 40 days, or until wells reached maximal confluence. Media and drug were replaced at 3-4 day intervals. Data are plotted as % confluence over the time course of the experiment for each single agent and respective combination (FIG. **5**).
(1076) Results:
(1077) As shown in FIG. **5**, treatment of NCI-H358 cells with submaximal (EC50) concentrations of Compound D or MEK inhibitor results in a short period of growth inhibition, followed by proliferation. Cells reach maximal confluence in ˜10 days after the addition of drug. The combination of the MEK inhibitor, trametinib, with Compound D resulted in complete and sustained inhibition of cell growth throughout the duration of the assay.
(1078) While the invention has been described in connection with specific embodiments thereof, it will be understood that it is capable of further modifications and this application is intended to cover any variations, uses, or adaptations of the invention following, in general, the principles of the invention and including such departures from the present disclosure come within known or customary practice within the art to which the invention pertains and may be applied to the essential features set forth herein.
(1079) All publications, patents and patent applications are herein incorporated by reference in their entirety to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated by reference in its entirety.

Claims

1. A method of treating lung cancer in a subject in need thereof, wherein the lung cancer comprises a K-Ras G12C mutation, the method comprising administering to the subject a therapeutically effective amount of a compound having the following structure: ##STR01314## or a

pharmaceutically acceptable salt thereof.
2. The method of claim 1, wherein the lung cancer is non-small cell lung cancer.
3. The method of claim 1, wherein the method comprises administering to the subject a therapeutically effective amount of the compound having the following structure: ##STR01315##
4. A method of treating non-small cell lung cancer in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of a compound having the following structure: ##STR01316## or a pharmaceutically acceptable salt thereof.
5. The method of claim 4, wherein the method comprises administering to the subject a therapeutically effective amount of the compound having the following structure: ##STR01317##
6. The method of claim 2, wherein the method comprises administering to the subject a therapeutically effective amount of the compound having the following structure: ##STR01318##
7. The method of claim 2, wherein the method comprises administering to the subject a therapeutically effective amount of the pharmaceutically acceptable salt of the compound having the following structure: ##STR01319##
8. The method of claim 4, wherein the method comprises administering to the subject a therapeutically effective amount of the pharmaceutically acceptable salt of the compound having the following structure: ##STR01320##